

**ALGEBRAIC DYNAMICS ON COMPACT NORMAL
VARIETIES**

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To Professor De-Qi Zhang on the occasion of his 60th birthday

DECLARATION

I hereby declare that the thesis is my original work and
it has been written by me in its entirety.

I have duly acknowledged all the sources of information
which have been used in the thesis.

This thesis has also not been submitted for any degree
in any university previously.

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October 28, 2021

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Summary

Throughout this thesis, we work over an algebraically closed field of characteristic zero. We refer to [Deb01], [Har77], [KMM87], [KM98] and [Uen75] for standard notations and terminologies in algebraic geometry and birational geometry. By a variety, we mean that it is either an algebraic or an analytic variety. Besides the introductory Chapter 1, my thesis contains three parts.

The first part consists of Chapters 2 and 3 where we worked over the field $\mathbb{C}$ of complex numbers. In this part, we generalized several previous results from the algebraic setting to the analytic situation, and gave the characterizations for a complex variety to be a quasi-étale quotient of a complex torus (*Q-complex torus* for short). In Chapter 2, we studied an int-amplified endomorphism f on a compact Kähler space (i.e., $f^*\xi - \xi$ is a Kähler class for some Kähler class ξ) and established the f -equivariant minimal model program for terminal threefolds which finally reaches either a Q -complex torus or a Fano variety of Picard number one. In Chapter 3, we studied the normal compact (possibly singular) Kähler threefold X admitting the action of a free abelian group G of maximal dynamical rank, all the non-trivial elements of which are of positive entropy; in particular, if such X is not rationally connected, we showed that it is G -equivariantly bimeromorphic to a Q -complex torus. This part is based on my works [Zho19] and [Zho21a].

The second part consists of Chapters 4 ~ 6. In this part, we characterized toric varieties from dynamical viewpoints. We asked whether a rationally connected smooth projective variety X admitting an int-amplified endomorphism f is toric. In Chapter 4, we answered this question affirmatively if such f is further assumed to have totally invariant ramifications. In Chapter 5, we gave a positive answer to this question if X is a smooth Fano threefold. More generally, we showed that a smooth Fano threefold is either toric or a product of a smooth rational curve and a del Pezzo surface if and only if it admits a non-isomorphic surjective endomorphism. In Chapter 6, we gave a positive answer to this question if X is a smooth Fano fourfold admitting a conic bundle structure. This part is based on the joint works [MZ20b], [MZZ21] and [JZ20].

The third part consists of Chapters 7 and 8. In this part, we considered a self-morphism f on an algebraic variety X and the number of f^{-1} -periodic and f -periodic subvarieties. In Chapter 7, we assumed that such X is projective with only klt singularities and f is int-amplified; then we bounded the number c of f^{-1} -periodic prime divisors in terms of the Picard number $\rho(X)$ and the dimension $\dim(X)$, and we showed that once c achieves the optimal upper bound, X is then toric. In Chapter 8, we considered the set Σ_{f^∞} of f -periodic subvarieties of small dynamical degree, the subset S_{f^∞} of maximal elements in Σ_{f^∞} , the subset S_f of f -invariant elements in S_{f^∞} , and the subset P_f of prime divisors in S_f . When X is projective, we gave an optimal upper bound of P_{f^n} , in terms of the first dynamical degree $d_1(f^n)$, as $n \rightarrow \infty$. When X is an algebraic group with f being a translation of an isogeny, or a toric variety with f fixing the big torus, we gave an optimal upper bound of S_{f^n} , in terms of $d_1(f^n)$ and $\dim(X)$, as $n \rightarrow \infty$. This part consists of my single-authored work [Zho21c] and the joint work [MMS⁺21].

Key words and Phrases. algebraic group, automorphism, conic bundle, equivariant minimal model program, Fano fourfold, Fano threefold, Fano type, group action, int-amplified endomorphism, Kähler space, nef or pseudo-effective cone, polarized endomorphism, positive entropy, $\mathbb{Q}$ -abelian, $\mathbb{Q}$ -complex torus, rationally connected variety, (semi-)abelian variety, (small) dynamical degree, toric variety, (totally) periodic subvariety.

List of Symbols

$\mathbb{P}^n$	projective space of dimension n 60
$\mathbb{F}_d$	Hirzebruch surface $\mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(-d))$ of degree d 113, 124
k^*	algebraic torus over a field k 71, 88, 124, 198
$\sim$	linear equivalence 70, 87, 123, 162, 196
$\sim_{\mathbb{Q}}$	$\mathbb{Q}$ -linear equivalence 87, 162, 196
$\sim_{\mathbb{R}}$	$\mathbb{R}$ -linear equivalence 196
$\equiv$	numerical equivalence 70, 87, 123, 162, 196
$\equiv_w$	weak numerical equivalence 32, 161, 162, 196
Δ_{τ}	discriminant locus of a fibration τ , over which τ is not smooth 88, 124
δ_X	Lefschetz defect of a projective variety X 125
$X_{\min}$	minimal model of a variety X 52
$\kappa(X)$	Kodaira dimension of a variety X 19, 51
$\kappa(X, D)$	Iitaka dimension of a $\mathbb{Q}$ -Cartier divisor D on a variety X 70
alb_X	the Albanese map of X 38, 54, 215
$\text{Alb}(X)$	the Albanese torus of X 38, 54, 208, 214
$\text{Aut}(X)$	automorphism group of a variety X 49, 53
$\text{Bim}(X)$	bimeromorphic transformation group of an analytic variety X 53
$q(X)$	irregularity of a variety X 54, 161

$q^\circ(X)$	augmented irregularity of a variety X 54
$\text{Pic}(X)$	group of Cartier divisors of a variety X modulo $\sim$ 70, 87, 161, 196
$\text{Pic}^\circ(X)$	neutral connected component of $\text{Pic}(X)$ 70, 87, 161, 196
$\text{NS}(X)$	$\text{Pic}(X)/\text{Pic}^\circ(X)$, Néron-Severi group of a variety X 70, 87, 161, 196
$N^1(X)$	$\text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, space of $\mathbb{R}$ -Cartier divisors on a projective variety X modulo $\equiv$ 50, 70, 87, 123, 161, 196
$H_{\text{BC}}^{1,1}(X)$	Bott-Chern cohomology of a compact complex space X 18, 50, 53, 82
$\rho(X)$	$\dim_{\mathbb{R}} N^1(X)$, Picard number of a variety X 70, 87, 123, 162, 197
$\text{Cl}(X)$	group of Weil divisors of a projective variety X modulo $\sim$ 197
$\text{Cl}^\circ(X)$	subgroup of $\text{Cl}(X)$, algebraically equivalent to 0 197
$\text{Cl}^{\text{ns}}(X)$	Néron-Severi Weil divisor class group of a projective variety X 197
$\rho^{\text{ns}}(X)$	$\dim_{\mathbb{R}} \text{Cl}^{\text{ns}}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, the rank of $\text{Cl}^{\text{ns}}(X)$ 197
$N_r(X)$	space of r -cycles modulo $\equiv_w$ on a projective variety X 70, 161
$N_1(X)$	space of 1-cycles (resp. real closed currents of bidimension $(1, 1)$) modulo $\equiv$ when X is projective (resp. compact Kähler) 24, 53, 87, 123, 162
$\overline{\text{NA}}(X)$	subspace of $N_1(X)$ generated by classes of positive closed currents 25
$\overline{\text{NE}}(X)$	Mori cone, subspace of $\overline{\text{NA}}(X)$ generated by the currents of integration T_C , where $C \subseteq X$ is an irreducible curve 25, 70, 87, 123
$\text{Amp}(X)$	cone of ample classes in $N^1(X)$ 123, 162
$\text{Nef}(X)$	cone of nef classes in $N^1(X)$ 70, 87, 162
$\text{PE}^1(X)$	cone of pseudo-effective classes in $N^1(X)$ 70, 196
$\text{PE}_r(X)$	cone of pseudo-effective r -cycles in $N_r(X)$ 162
$\mathcal{K}(X)$	cone of Kähler classes in $H_{\text{BC}}^{1,1}(X)$ 26
$\overline{\mathcal{K}}(X)$	cone of nef classes in $H_{\text{BC}}^{1,1}(X)$ 26, 57
$\mathcal{E}(X)$	cone of pseudo-effective classes in $H_{\text{BC}}^{1,1}(X)$ 26
$\mathcal{E}^\circ(X)$	cone of big classes in $H_{\text{BC}}^{1,1}(X)$ 26
$\text{Exc}(\bullet)$	exceptional locus of a morphism $\bullet$ 48, 132
$h(g)$	topological entropy of an automorphism g 49
$N(G)$	null entropy subset of a group G of automorphisms 66

$\rho(\varphi)$	spectral radius of a linear transformation φ 32, 49
$d_1(f)$	first dynamical degree of a dominant self-map f 50, 197
$d_1(f _\pi)$	relative first dynamical degree of a dominant self-map f with respect to a dominant rational or meromorphic map π 197
Σ_{f^∞}	set of f -periodic (irreducible closed) subvarieties of a self-morphism f with small dynamical degree 198
S_{f^∞}	subset of maximal f -periodic subvarieties in Σ_{f^∞} 199
S_f	subset of S_{f^∞} consisting of all f -invariant elements 199
P_f	subset of S_{f^∞} consisting of all f -invariant prime divisors 199
P_{f^∞}	subset of (maximal) f -periodic prime divisors in Σ_{f^∞} 199
$P_{f^\infty}^{-1}$	subset of (maximal) f^{-1} -periodic prime divisors in Σ_{f^∞} 199
$\text{Fix}(f)$	set of fixed points of a self-morphism f 34, 199
$\text{Per}(f)$	set of periodic points of a self-morphism f 34, 199
$\text{TI}_f(X)$	set of f^{-1} -invariant prime divisors of a self-morphism $f : X \rightarrow X$ 159
MMP	minimal model program 18, 51, 67, 96, 159
SDD	small dynamical degree 191, 198
Q -abelian	quasi-étale quotient of an abelian variety 163, 208
Q -complex torus	quasi-étale quotient of a (not necessarily projective) complex torus 19, 36, 51

Introduction

It has been a long history and involved many people working on the classification of normal projective varieties and compact Kähler spaces. The minimal model program (*MMP* for short) is one of the cornerstones in this classification problem. For the projective situation, it has been fully developed in dimension ≤ 3 and there is a tremendous progress in higher dimensions (cf. [\[BCHM10\]](#)). Based on this, the equivariant minimal model program (*EMMP* for short) for polarized endomorphisms and int-amplified endomorphisms are established by Meng and Zhang ([\[MZ18a\]](#), [\[Men20\]](#); cf. [\[NZ10\]](#), [\[Zha10\]](#) and [\[CMZ20\]](#)). More generally, given a normal projective variety admitting an endomorphism f , one may ask whether they can run the MMP f -equivariantly.

During the last decade, the existence of the MMP in the Kähler setting has also been nicely confirmed in dimension three (cf. [\[HP15\]](#), [\[HP16\]](#), [\[CHP16\]](#) and [\[DH20\]](#)). Following this line, one of my working topics is to generalize Meng-Zhang's work to the complex analytic setting, and to establish the EMMP for Kähler spaces (cf. Section [1.1](#) for the threefold case). As an application, we characterize the quasi-étale quotient of a (not necessarily projective) complex torus from the dynamical viewpoints; see Section [1.2](#) for another description from the aspect of automorphism groups of maximal dynamical rank in dimension three.

Another direction of my research is to characterize smooth projective varieties via non-isomorphic surjective endomorphisms. For the surface case, such (possibly singular)

varieties have been fully studied by Nakayama (cf. [Nak08], [Nak20a] and [Nak20b]). In higher dimensions, in view of the endomorphism-descending (or lifting) property of the three typical fibrations: Albanese map, Kodaira fibration, and maximal rationally connected fibration (cf. [NZ09] and [Nak10]), rationally connected varieties are the essential cases in studying the non-isomorphic surjective endomorphisms on them. Further, to exclude the non-essential case of the product of a non-isomorphic surjective endomorphism and an automorphism, we restrict ourselves to the int-amplified endomorphisms (i.e., $f^*H - H$ is ample for some ample Cartier divisor H). For the surface case, Nakayama showed that a smooth rational surface admitting a non-isomorphic surjective endomorphism is toric (cf. [Nak02]), confirming a conjecture of Sato. In higher dimensions, as an analogue of Sato's conjecture, we ask whether a rationally connected smooth projective variety X admitting an int-amplified endomorphism f is toric. We answer this question affirmatively if one of the following holds: (1) f has totally invariant ramifications (cf. Section 1.3); (2) X is a smooth Fano threefold (cf. Section 1.4); or (3) X is a smooth Fano fourfold admitting a conic bundle structure (cf. Section 1.5).

For a normal projective variety admitting an int-amplified endomorphism f , it was shown by Meng and Zhang that there are only finitely many f^{-1} -invariant subvarieties (cf. [MZ20a, Corollary 3.8]). With this in mind, one of my projects is to bound the number c of totally periodic prime divisors and describe the geometric property when c achieves the optimal upper bound (cf. Section 1.6). However, different from the f^{-1} -periodic subvarieties, there may exist infinitely many f -periodic subvarieties in general (cf. [Fak03, Theorem 5.1]). Therefore, as another one of my joint works, for an algebraic variety admitting a self-morphism, we consider the periodic subvarieties with small dynamical degree and try to give an optimal upper bound for the number of such subvarieties (cf. Section 1.7).

In the rest of this introductory chapter, I will describe my work in more details and then conclude by discussing some open problems that I would like to pursue in the future.

1.1 Int-amplified endomorphisms of compact Kähler spaces

Let X be a compact Kähler space with only rational singularities. A surjective endomorphism f is said to be *int-amplified* if $f^*\xi - \xi = \eta$ for some Kähler classes ξ and η on X . A normal compact Kähler space Y is said to be a Q -complex torus if there exists a finite surjective morphism $T \rightarrow Y$ from a complex torus T , which is étale in codimension one.

Remark 1.1.1. In the paper [Zho19], the notion Q -torus should be understood as Q -complex torus (cf. [Nak99, Definition 3.6]).

In my work [Zho19]=Chapter 2, we generalize [Men20] to the analytic situation and characterize the Q -complex torus from the viewpoints of int-amplified endomorphisms.

Theorem 1.1.2 ([Zho19, Theorem 1.1]=Theorem 2.1.1). *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal compact Kähler space X with at worst rational singularities. Let $\varphi := f^*|_{H_{\text{BC}}^{1,1}(X)}$, where $H_{\text{BC}}^{1,1}(X)$ denotes the Bott-Chern cohomology space. Then the following assertions are equivalent.*

- (1) *The endomorphism f is int-amplified.*
- (2) *All the eigenvalues of φ are of modulus greater than 1.*
- (3) *There exists some big $(1, 1)$ -class $[\theta]$ such that $f^*[\theta] - [\theta]$ is big, i.e., it can be represented by a Kähler current T .*
- (4) *If C is a non-empty and non-zero φ -invariant convex cone in $H_{\text{BC}}^{1,1}(X)$, then $\emptyset \neq (\varphi - \text{id})^{-1}(C) \subseteq C$.*

Furthermore, if X is projective, then the two definitions of int-amplified endomorphisms coincide (cf. Definition 2.2.11 in the Kähler setting and [Men20] in the projective setting).

Theorem 1.1.3 ([Zho19, Theorem 1.4]=Theorem 2.1.4). *Let X be one of the following cases: (i) a compact Kähler manifold of any dimension, (ii) a normal compact Kähler threefold with only canonical singularities, or (iii) a normal $\mathbb{Q}$ -factorial compact Kähler surface. Suppose that X admits an int-amplified endomorphism $f : X \rightarrow X$ and the canonical divisor K_X is pseudo-effective. Then X is a Q -complex torus.*

In particular, applying the theory of the minimal model program for Kähler threefolds (cf. [HP15], [HP16], [CHP16] and [DH20]), we establish the f -equivariant minimal model program for compact Kähler threefolds.

Theorem 1.1.4 ([Zho19, Theorem 1.5]=Theorem 2.1.5). *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler terminal threefold. Then, replacing f by a positive power, there exist a $\mathbb{Q}$ -complex torus Y , a morphism $X \rightarrow Y$ and an f -equivariant minimal model program of X over Y :*

$$X = X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_r = Y$$

(i.e., $f = f_1$ descends to holomorphic f_i on each X_i), with every $X_i \dashrightarrow X_{i+1}$ a divisorial contraction, a flip or a Mori fibre contraction, of a K_{X_i} -negative extremal ray, such that:

- (1) If K_X is pseudo-effective, then $X = Y$ and it is a $\mathbb{Q}$ -complex torus.
- (2) If K_X is not pseudo-effective, then for each i , f_i is int-amplified and $X_i \rightarrow Y$ is an equi-dimensional (holomorphic) morphism with every fibre irreducible and rationally connected; further, $X_{r-1} \rightarrow X_r = Y$ is a Mori fibre contraction.

1.2 Compact Kähler threefolds admitting abelian groups of automorphisms

Let X be an n -dimensional compact Kähler manifold. For each automorphism $g \in \text{Aut}(X)$, the pull-back operation g^* naturally acts on the cohomology group $\oplus_{i \geq 0} H^{i,i}(X, \mathbb{C})$. The *topological entropy* $h(g)$ of g turns out to coincide with the logarithm of the spectral radius of $g^*|_{\oplus_{i \geq 0} H^{i,i}(X, \mathbb{C})}$ (cf. [DS04], [Gro03] and [Yom87]). We say that g is of *positive entropy* (resp. *null entropy*) if $h(g) > 0$ (resp. $h(g) = 0$).

Remark 1.2.1. Let $\rho_{p,q}(g)$ denote the spectral radius of $g^* : H^{p,q}(X, \mathbb{C}) \rightarrow H^{p,q}(X, \mathbb{C})$. As shown in [Din05, Proposition 5.8] (cf. [Din12, Proposition 3.5]),

$$\rho_{p,q}(g) \leq \sqrt{\rho_{p,p}(g)\rho_{q,q}(g)}.$$

Indeed, this property holds for general meromorphic self-maps on compact Kähler manifolds (cf. [Din05]). Therefore, the topological entropy $h(g)$ coincides with the logarithm of the spectral radius of $g^*| \oplus_{i \geq 0} H^i(X, \mathbb{C})$.

Zhang proved in [Zha09] (cf. [CWZ14, Theorem 1.5]) that the automorphism group $\text{Aut}(X)$ satisfies the *Tits alternative*, i.e., for any subgroup $G \subseteq \text{Aut}(X)$, either G contains a subgroup which is isomorphic to the non-abelian free group $\mathbb{Z} * \mathbb{Z}$, or G is virtually solvable (i.e., some subgroup of finite index is solvable). In addition, if $G \subseteq \text{Aut}(X)$ is virtually solvable, then it contains a finite-index subgroup G_1 such that the null entropy subset $N(G_1)$ is a normal subgroup of G_1 and the quotient $G_1/N(G_1)$ is a free abelian group of rank $r \leq n - 1$, where $r = r(G)$ is said to be the *dynamical rank* of G , which is independent of the choice of G_1 . This extends the previous result of Dinh and Sibony (cf. [DS04] and [Din12]), where G is assumed to be abelian.

In the singular setting, we consider a normal projective variety (resp. a compact Kähler space with at worst rational singularities) X and define the *first dynamical degree* $d_1(g)$ of an automorphism $g \in \text{Aut}(X)$ as the spectral radius of its natural pull-back g^* on the Néron-Severi group $\text{NS}(X) \otimes \mathbb{R}$ (resp. the Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$). We say that g is of *positive entropy* if $d_1(g) > 1$, otherwise it is of *null entropy*. Note that our definition here for $d_1(f)$ in the singular setting coincides with the usual one when X is smooth (cf. Remark 3.1.2). We refer to [DS17, §4] and references therein for a comprehensive exposition on dynamical degrees, topological and algebraic entropies.

Moreover, to classify the normal projective varieties admitting a free abelian group G of automorphisms of maximal dynamical rank, Zhang established the G -equivariant minimal model program (G -MMP for short) for projective varieties in his series of papers (cf. [Zha09], [Zha13], [Zha16]). However, in the non-algebraic situation, there are many problems on the construction of the G -MMP. Thanks to the particularity of terminal threefolds, in my work [Zho21a]=Chapter 3, we extend [Zha16, Theorem 1.1] to the compact Kähler threefolds and characterize the $\mathbb{Q}$ -complex torus in dimension three.

Theorem 1.2.2 ([Zho21a, Theorem 1.3]=Theorem 3.1.4; cf. Remark 3.3.2). *Let X be a normal $\mathbb{Q}$ -factorial compact Kähler threefold with only terminal singularities. Suppose*

that there is a free abelian automorphism group $G \cong \mathbb{Z}^r$ of positive entropy with $r = 2$ (maximal). Then the following assertions hold:

- (1) If K_X is not pseudo-effective, then X is rationally connected. In particular, X is a projective threefold.
- (2) If K_X is pseudo-effective, then the Kodaira dimension $\kappa(X) = 0$ and X is bimeromorphic to $X_{\min} := T/H$ for a finite group H acting freely outside a finite set of a complex 3-torus T . Moreover, G descends to an automorphism subgroup $G|_{X_{\min}} \subseteq \text{Aut}(X_{\min})$ and further lifts to $G_T \subseteq \text{Aut}(T)$ such that $G|_{X_{\min}} \cong G_T/H$.

1.3 Dynamical rigidity of rationally connected projective varieties

Let X be a smooth projective variety which is *rationally connected*, i.e., any two general points of X can be connected by a chain of rational curves; see [Cam92] and [KMM92]. In the joint work [MZ20b]=Chapter 4 with Sheng Meng, generalizing [Fak03, Question 4.4], we ask the following question.

Question 1.3.1 ([MZ20b, Question 1.2]=Question 4.1.2). Let X be a rationally connected smooth projective variety admitting an int-amplified endomorphism. Is X a toric variety?

A normal projective variety X of dimension n is a *toric variety* if X contains a *big torus* $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on the whole variety X . In this case, $B := X \setminus T$ is a divisor; the pair (X, B) is said to be a *toric pair* with B the *toric boundary*.

In a recent paper [Yos20] by Yoshikawa, rationally connected smooth projective varieties admitting int-amplified endomorphisms are shown to be of Fano type. Toric varieties being of Fano type, [Yos20] provides some evidence for a positive answer to our Question 1.3.1. We also note that toric varieties usually have lots of dynamically interesting symmetries (cf. [Nak02]), and our Question 1.3.1 here is a sort of a converse direction to

it. In general, given a non-isomorphic endomorphism on a rationally connected smooth projective variety (or even a Fano manifold) X , it is very hard to find a big torus on X ; hence a positive answer to Question 1.3.1 reveals a very deep symmetric essence shared by toric varieties and int-amplified endomorphisms. Joint with Sheng Meng, we show two situations from the aspects of geometry and cohomology in which Question 1.3.1 holds.

Theorem 1.3.2 ([MZ20b, Theorem 1.5]=Theorem 4.1.4). *Let X be a rationally connected smooth projective variety with D a reduced divisor. Then (X, D) is a toric pair if and only if X admits an int-amplified endomorphism f such that $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is étale.*

Theorem 1.3.3 ([MZ20b, Theorem 1.6]=Theorem 4.1.5). *Let X be a rationally connected smooth projective variety of dimension n . Then*

$$X \cong (\mathbb{P}^1)^{\times n}$$

if and only if X admits a surjective endomorphism f such that the eigenvalues of $f^|_{N^1(X)}$ (without counting multiplicities) are n distinct real numbers greater than 1.*

1.4 Non-isomorphic endomorphisms of Fano threefolds

We consider a smooth *Fano* threefold X (i.e., a smooth projective threefold with the anti-canonical divisor $-K_X$ being ample) admitting a non-isomorphic surjective endomorphism. When the Picard number $\rho(X) = 1$, it was proved in [ARVdV99] or [HM03] that $X \cong \mathbb{P}^3$. Joint with Sheng Meng and De-Qi Zhang, we describe the smooth Fano threefolds (of higher Picard number) admitting non-isomorphic surjective endomorphisms without using the classification in [MZZ21]=Chapter 5.

Theorem 1.4.1 ([MZZ21, Main Theorem]=Theorem 5.1.2). *A smooth Fano threefold is either toric or a product of $\mathbb{P}^1$ and a del Pezzo surface if and only if it admits a non-isomorphic surjective endomorphism.*

As an important intermediate step to prove Theorem 1.4.1, we show the following theorem, which gives a positive answer to Question 1.3.1 for smooth Fano threefolds.

Theorem 1.4.2 ([MZZ21, Theorem 1.4]=Theorem 5.1.4). *Let X be a smooth Fano threefold. Then the following assertions are equivalent.*

- (1) X is a toric variety.
- (2) X admits a polarized endomorphism.
- (3) X admits an int-amplified endomorphism.

There are several applications arising from our Theorem 1.4.1. First, it implies that Fano threefolds have no “primitive” endomorphism unless they are toric ([MZZ21, Corollary 1.5]=Corollary 5.1.5). Here, by “primitive”, we mean that such endomorphism does not descend along some fibration. Second, since del Pezzo surfaces are all rational, our Theorem 1.4.1 recovers [Zha12, Theorem 1.2] which asserts that a smooth Fano threefold admitting a non-isomorphic surjective endomorphism is rational ([MZZ21, Corollary 1.6]=Corollary 5.1.6).

Starting from our Theorem 1.4.2, we try to give a more detailed description on the toric boundary. More precisely, we want to know whether all the totally periodic subvarieties are contained in a toric boundary.

Question 1.4.3 ([MZZ21, Question 1.8]=Question 5.1.8). *Let f be an int-amplified endomorphism of a rationally connected (or rational, or toric) smooth projective variety X . Let Σ be the union of f^{-1} -periodic (closed) subvarieties. Is X toric, and if so, is there a toric pair (X, Δ) such that $\Sigma \subseteq \Delta$?*

In the joint work [MZZ21], we study Question 1.4.3 for the case when Σ is a divisor and confirm the surface case and the case of splitting $\mathbb{P}^1$ -bundles over rational surfaces ([MZZ21, Theorems 3.2 and 3.3]=Theorems 5.3.2 and 5.3.3); see [JZ20, Theorem 3.1]=Theorem 6.3.1 which confirms the divisor case of smooth Fano threefolds.

1.5 Amplified endomorphisms of Fano fourfolds

We consider a smooth Fano fourfold admitting a *conic bundle structure*. Recall that a fibration $\tau: X \rightarrow Y$ of smooth projective varieties is a (*regular*) *conic bundle* if every

fibre of τ is a conic, i.e., it is isomorphic (as a scheme) to the zeros of a nontrivial section $\mathcal{O}_{\mathbb{P}^2}(2)$. Joint with Jia Jia, we extend Theorem 1.4.2 for the threefold case to the fourfold case admitting a conic bundle structure in [JZ20]=Chapter 6.

Theorem 1.5.1 ([JZ20, Theorem 1.4]=Theorem 6.1.3). *Let X be a smooth Fano fourfold admitting a conic bundle. Then the following assertions are equivalent.*

- (1) X is a toric variety.
- (2) X admits a polarized endomorphism.
- (3) X admits an (int-)amplified endomorphism.

In particular, X is rational if one of the above equivalent conditions holds.

During the proof of Theorem 1.5.1, we study Question 1.4.3 for the smooth Fano threefold case and show that the totally periodic divisors are all contained in the toric boundary; moreover, we take the first step towards the understanding of the totally periodic higher codimensional subvarieties ([JZ20, Section 3]=Section 6.3). We refer readers to Question 1.8.6 for our future work on the Fano fourfold.

1.6 Totally invariant divisors of int-amplified endomorphisms

We consider an int-amplified endomorphism $f : X \rightarrow X$ of a normal $\mathbb{Q}$ -factorial projective variety with only klt singularities. In my work [Zho21c]=Chapter 7, we bound the number c of f^{-1} -periodic prime divisors by running the f -equivariant minimal model program (cf. [Men20, Theorem 1.10]); see [Zha14, Theorem 1.3] for the polarized case. To simplify the notation, we denote by

$$\mathrm{TI}_f(X) = \{V \mid V \text{ is a prime divisor on } X \text{ such that } f^{-1}(V) = V\}$$

the set of f^{-1} -invariant (or f -totally invariant) prime divisors with $c = \#\mathrm{TI}_f(X)$.

Theorem 1.6.1 ([Zho21c, Theorem 1.1]=Theorem 7.1.1). *Let X be a projective variety of dimension n with only $\mathbb{Q}$ -factorial klt singularities, and $f : X \rightarrow X$ an int-amplified endomorphism. Let V_j ($1 \leq j \leq c$) be all the prime divisors in $\mathrm{TI}_f(X)$. Then we have (with $\rho := \rho(X)$):*

- (1) $c \leq n + \rho$. Furthermore, if $c \geq 1$, then the pair $(X, \sum V_j)$ is log canonical and X is uniruled.
- (2) Suppose that $c \geq n + \rho - 2$. Then either X is rationally connected and simply connected with respect to complex topology, or there is a fibration $X \rightarrow E$ onto an elliptic curve E such that every fibre is normal, irreducible, equi-dimensional and rationally connected. Furthermore, in the latter case, for some integer $t \geq 1$, f^t descends to an int-amplified endomorphism $g_E : E \rightarrow E$. In both cases, $(f^t)^*|_{\mathrm{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{Q}$.
- (3) Suppose that $c \geq n + \rho - 1$. Then X is rationally connected and simply connected with respect to complex topology. Furthermore, for some integer $t \geq 1$, $(f^t)^*|_{\mathrm{Pic}(X)}$ is diagonalizable over $\mathbb{Q}$.
- (4) Suppose that $c \geq n + \rho$. Then $c = n + \rho$, $K_X + \sum_{j=1}^{n+\rho} V_j \sim_{\mathbb{Q}} 0$ and f is étale outside $(\bigcup V_j) \cup f^{-1}(\mathrm{Sing}(X))$, where $\mathrm{Sing}(X)$ denotes the singular locus of X .

In addition, we show that if the cardinality c achieves the upper bound $\rho(X) + \dim(X)$, then $(X, \sum V_j)$ is a toric pair ([Zho21c, Corollary 1.2]=Corollary 7.1.2).

1.7 Invariant subvarieties with small dynamical degree

Let $f : X \rightarrow X$ be a dominant self-morphism of an algebraic variety X . An (irreducible) closed subvariety $V \subseteq X$ is f -periodic (resp. f -invariant) if $\overline{f^s(V)} = V$ for some $s > 0$ (resp. $\overline{f(V)} = V$). When X is projective or f is finite (and hence surjective), $\overline{f^s(V)} = V$ implies that V is f -periodic in the usual sense, i.e., $f^s(V) = V$. We say that V is f^{-1} -periodic if $f^{-s}(V) = V$ for some $s > 0$. Joint with Yohsuke Matsuzawa, Sheng Meng,

Takahiro Shibata and De-Qi Zhang, we consider the invariant subvarieties with small dynamical degree in [MMS⁺21]=Chapter 8.

An f -periodic closed subvariety V is said to be of *Small Dynamical Degree* (SDD for short) if the first dynamical degree $d_1(f^s|_V) < d_1(f^s)$ for some $s > 0$ such that $\overline{f^s(V)} = V$. Denote by Σ_{f^∞} the set of f -periodic (irreducible closed) subvarieties of small dynamical degree. For $V \in \Sigma_{f^\infty}$, we say that V is *maximal* if $V \subseteq V'$ and $V' \in \Sigma_{f^\infty}$ imply $V = V'$. Denote by S_{f^∞} the set of maximal elements in Σ_{f^∞} . Denote by S_f (resp. P_f) the subset of S_{f^∞} , consisting of all f -invariant elements (resp. prime divisors). Note that, in general, Σ_{f^∞} may have infinitely many non-maximal elements (cf. [Fak03, Theorem 5.1]).

Our first result in this section is to give an optimal bound for S_f when X is an algebraic group or a (not necessarily complete) toric variety.

Theorem 1.7.1 ([MMS⁺21, Theorem 1.5]=Theorem 8.1.5). *Let f be a dominant self-morphism of a variety X . Then*

$$\#S_{f^n} \leq (d_1(f))^{n \cdot \dim(X)}(1 + o(1))$$

holds as $n \rightarrow \infty$, if (X, f) fits one of the following cases.

- (1) *X is an algebraic group; and $f = R_a \circ g$ where g is an isogeny and R_a is the right multiplication map by some $a \in X$ (when X is semi-abelian, the self-morphism f is automatically of this form).*
- (2) *X is a toric variety with the big torus T being f -invariant.*

Our second result in this section is to give an optimal bound for P_f when X is a projective variety.

Theorem 1.7.2 ([MMS⁺21, Theorem 1.6]=Theorem 8.1.6). *Let f be a surjective endomorphism of a projective variety. Then*

$$\#P_{f^n} \leq (d_1(f))^n(1 + o(1))$$

holds as $n \rightarrow \infty$.

Finally, we confirm the finiteness of S_f for projective surfaces and Q -abelian varieties. A normal projective variety X is said to be *Q -abelian* if there is a finite surjective morphism $A \rightarrow X$ from an abelian variety A , which is étale in codimension one.

Corollary 1.7.3 ([MMS⁺21, Corollaries 1.8 and 5.2]=Corollaries 8.1.8 and 8.4.2). *S_f is finite for any surjective endomorphism f of a projective surface or a Q -abelian variety.*

1.8 Future work

In this section, I will discuss and state some unsolved problems which I prepare to study in the future. We refer readers to the remarks at the end of each chapter, which also contain some related questions.

Let us begin with the following series of problems on the equivariant minimal model program for endomorphisms or automorphism groups.

1.8.1 Equivariant minimal model program for Kähler varieties

First, due to the appearance of [DH20], we believe that, the equivariant minimal model program for an int-amplified endomorphism on a terminal Kähler threefold can be further extended to the following.

Question 1.8.1 (cf. Remark 2.1.6). Let X be a compact Kähler threefold with mild singularities. Suppose that X admits an int-amplified endomorphism f . Can we run the f -equivariant minimal model program for X ?

The following question, which is the main obstacle to Question 1.8.1 and plays a significant role in the equivariant minimal model program, is known in the algebraic situation ([Men20, Theorem 5.2]; cf. [NZ10, Theorem 3.3]). We refer readers to [Zho19, Theorem 1.4]=Theorem 2.1.4 for the case of canonical Kähler threefolds. Indeed, Theorem 2.1.4 can be further extended to the case of klt Kähler threefolds by taking the global index one cover (cf. [DH20, Theorem 2.14] and [GK20, Section 8.C]).

Question 1.8.2. Let X be a compact Kähler space with pseudo-effective canonical divisor K_X . Let $f : X \rightarrow X$ be an int-amplified endomorphism. Is X a $\mathbb{Q}$ -complex torus?

Except for the equivariant minimal model program for endomorphisms, we are also interested in the automorphism group of maximal dynamical rank. Note that the equivariant minimal model program for a single automorphism may not exist in general (cf. [Les18, Theorem 7.1]). For a normal projective variety X which is not rationally connected, if X admits an abelian (or more generally, solvable) automorphism group G of maximal dynamical rank, then Zhang established the G -equivariant minimal model program (G -MMP for short) and showed that such X is G -birational to a $\mathbb{Q}$ -abelian variety (cf. [Zha16]). Hence, we pose the following question, which together with [Zha16, Theorem 1.1] completes the construction of the G -MMP.

Question 1.8.3. Let X be a rationally connected normal projective variety. Suppose that there is a free abelian automorphism group $G \cong \mathbb{Z}^r$ of positive entropy with $r = \dim(X) - 1$ (maximal). Can we run the G -equivariant minimal model program?

At the end of this subsection, we ask the following question in the Kähler setting. We believe that, it also relates to the effectivity of Kodaira's lemma for a (not necessarily rational) nef and big class on a compact Kähler space.

Question 1.8.4 (cf. Remark 3.1.7). Let X be a dlt compact Kähler threefold. Suppose that there is a free abelian automorphism group $G \cong \mathbb{Z}^r$ of positive entropy with $r = 2$ (maximal). Can we run the minimal model program G -equivariantly?

1.8.2 Conic bundles

To classify and understand Fano manifolds X of large Picard number, it is crucial to study conic bundle structures τ . When $\dim(X) = 3$, such τ takes a significant role in the classification of Fano threefolds (cf. [MM83]). When $\dim(X) = 4$, such τ characterizes Fano fourfolds with certain Lefschetz defect, all of such fourfolds being rational (cf. [MR19, Corollary 1.3]). In higher dimension, such τ is closely related to a Fano contraction of

relative dimension one (cf. [And85, Theorem 1.3], Lemma 6.2.6), where we expect that the higher dimensional special fibres do not appear under the dynamical assumption:

Question 1.8.5. Let $\pi : X \rightarrow Y$ be a Fano contraction from a smooth projective variety. Suppose that X admits an int-amplified endomorphism and π is of relative dimension one (i.e., $\dim(X) - \dim(Y) = 1$). Is π equi-dimensional (and hence a conic bundle)? If we further assume that X is rationally connected, will we have π is a conic bundle?

To show that a Fano manifold admitting a conic bundle structure onto a projective toric manifold is toric under the extra dynamical assumption, our rough idea is to establish the equivariant minimal model program which is a composite of a series of toric blow-ups followed by a splitting $\mathbb{P}^1$ -bundle onto a toric variety (cf. [Rom19a, Proposition 3.5]). Therefore, we want to know when a Fano manifold admits a conic bundle structure.

A positive answer to the following question together with [JZ20] would answer [MZ20b, Question 1.2] affirmatively for the fourfold case. The first part generalizes [ARVdV99] and [HM03] (cf. [MZZ21, Conjecture 1.9]), while the second part extends [MM83, §9].

Question 1.8.6. Let X be a smooth Fano fourfold admitting a non-isomorphic surjective endomorphism. If the Picard number $\rho(X) = 1$, will we have $X \cong \mathbb{P}^4$? Does there exist a positive integer N such that if $\rho(X) \geq N$, then X admits a conic bundle structure?

The classification of smooth Fano manifolds of Picard number one seems to be a difficult question. Besides, we don't even have a full classification of smooth Fano fourfolds so far. We refer readers to [Bat99] and [Sat00] for the list of toric Fano fourfolds.

1.8.3 Strictly nef divisors on singular varieties

The following project is independent of algebraic dynamics and has its own interest. Since it is also my present work, we briefly summarize the questions for the completeness of this thesis.

Recall that a $\mathbb{Q}$ -Cartier divisor L on a projective variety X is said to be *strictly nef*, if $L \cdot C > 0$ for every curve C on X . In general, a strictly nef divisor may not be

ample (cf. e.g., [Har70]). Hence, it is natural that people want to know how far strictly nef divisors are from being ample (cf. [CP91], [Mae93], [Ser95], [Ueh00], [CCP08] and [LOY19]).

Extending [Ser95, Question 0.1] to the singular case, we ask the following question.

Question 1.8.7 ([Zho21b, Question 1.4]). Let (X, Δ) be a projective klt pair, and L a strictly nef $\mathbb{Q}$ -divisor on X . Is $K_X + \Delta + tL$ ample for sufficiently large $t \gg 1$?

Extending (the second assertion of) [LOY19, Theorem 1.3] to the singular case, we ask the following question. In view of [Zha06a], Question 1.8.8 below is a special case of Question 1.8.7.

Question 1.8.8 (cf. [Zho21b, Questions 1.2]). Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being strictly nef. Is X rationally connected?

In my work [Zho21b] which is not contained in this thesis, Question 1.8.8 has a positive answer when $\dim(X) = 3$ (cf. [Zho21b, Theorem 1.7]). Recently, joint with Juanyong Wang, we have also got some evidence of a positive answer in arbitrary dimensions.

Parallel to the anti-canonical divisor case, we end up this section with the following question, which relates to the hyperbolic geometry and the abundance conjecture.

Question 1.8.9. Let (X, Δ) be a projective klt pair with the canonical divisor $K_X + \Delta$ being strictly nef. Is X hyperbolic? Is $K_X + \Delta$ ample?

Int-amplified endomorphisms of compact Kähler spaces

2.1 Introduction

In this chapter, we work over the field $\mathbb{C}$ of complex numbers. By the fundamental work of Meng and Zhang (cf. [Men20] and [MZ18a]) during the past several years, we have known the building blocks and characteristic properties of polarized and int-amplified endomorphisms of normal projective varieties. Also, Meng and Zhang showed that we can run the minimal model program equivariantly for mildly singular normal projective varieties and finally reach either a Q -abelian variety or a Fano variety of Picard number one. As an application, one can use the results to study the totally invariant divisors of polarized or int-amplified endomorphisms (cf. [Zha14, Theorem 1.3] and [Zho21c, Theorem 1.1]=Theorem 7.1.1).

Now, we pose a natural question. Does there exist such nice building blocks of non-isomorphic surjective endomorphisms on compact Kähler spaces? This is open since the minimal model program for higher-dimensional compact Kähler spaces is unknown.

A surjective endomorphism f on a normal compact Kähler space X is said to be *int-amplified*, if $f^*\xi - \xi = \eta$ for some Kähler classes ξ and η (cf. Definition 2.2.11). We first prove that when X is projective, the generalized definition here coincides with the

previous one as in [Men20]. Then, we study the normal compact Kähler space admitting an int-amplified endomorphism in terms of the Kähler cone and canonical divisor. We show that most of these properties are preserved when the objects are extended to the analytic setting. Moreover, we study the periodic points and totally invariant analytic subvarieties on some specific complex spaces such as Q -complex tori, i.e., quasi-étale finite quotients of complex tori. Finally, as a consequence of the existence of the minimal model program (MMP for short) for compact Kähler terminal threefolds (cf. [HP15] and [HP16]), we prove that every int-amplified endomorphism of a compact Kähler terminal threefold has an equivariant descending, when running the MMP.

We divide this chapter into two parts. First, we recall notation and terminology for a singular analytic space such as differential forms, currents and invariant cones. We spend some time showing that all of these objects possess nice properties parallel to normal projective cases. Second, we apply these common properties in Section 2.2 so as to extend previous results in [Men20]. Here, we gather them together for readers' convenience.

Theorem 2.1.1 below generalizes [Men20, Theorem 1.1]. The proof of the equivalent conditions is the same as [Men20] while the coincidence of these two definitions of int-amplified endomorphisms when X is projective is new (cf. Proof of Theorem 2.1.1).

Theorem 2.1.1 ([Zho19, Theorem 1.1]). *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal compact Kähler space X with at worst rational singularities. Let $\varphi := f^*|_{H_{\text{BC}}^{1,1}(X)}$, where $H_{\text{BC}}^{1,1}(X)$ denotes the Bott-Chern cohomology space. Then the following assertions are equivalent.*

- (1) *The endomorphism f is int-amplified.*
- (2) *All the eigenvalues of φ are of modulus greater than 1.*
- (3) *There exists some big $(1, 1)$ -class $[\theta]$ such that $f^*[\theta] - [\theta]$ is big, i.e., it can be represented by a Kähler current T .*
- (4) *If C is a non-empty and non-zero φ -invariant convex cone in $H_{\text{BC}}^{1,1}(X)$, then $\emptyset \neq (\varphi - \text{id})^{-1}(C) \subseteq C$.*

Furthermore, if X is projective, the two definitions of int-amplified endomorphisms coincide (cf. Definition 2.2.11 in the Kähler setting and [Men20] in the projective setting).

As an application of Theorem 2.1.1, we are able to show the following, which asserts that the composition of a sufficient number of iterations of int-amplified endomorphisms is still int-amplified. Note that, in general, the composition of two int-amplified endomorphisms may not be int-amplified.

Theorem 2.1.2 ([Zho19, Theorem 1.2]). *Let f and g be two surjective endomorphisms of a normal compact Kähler space X with at worst rational singularities. Suppose that f is int-amplified. Then both $f^i \circ g$ and $g \circ f^i$ are int-amplified for sufficiently large $i \gg 1$.*

We refer readers to [Uen75, Sections 4 and 6] for the definitions of Cartier divisor, linear system and Kodaira dimension on a complex analytic variety. Also, the differential forms and currents will be discussed in the next section.

The following theorem is crucial for the equivariant minimal model program of compact Kähler threefolds (cf. [Men20, Theorem 2.5] for the normal projective case).

Theorem 2.1.3 ([Zho19, Theorem 1.3]). *Let X be a normal compact Kähler space with at worst rational singularities. Suppose that X admits an int-amplified endomorphism f and the canonical divisor K_X is $\mathbb{Q}$ -Cartier. Then, $-K_X$ is pseudo-effective. In particular, the Kodaira dimension $\kappa(X) \leq 0$.*

As an application of Theorem 2.1.3 and the main theorem in [GK20], we get Theorem 2.1.4 below so that when running the MMP for compact Kähler threefolds, we can reduce the case of K_X being pseudo-effective to the $\mathbb{Q}$ -complex torus case.

Theorem 2.1.4 ([Zho19, Theorem 1.4]). *Let X be one of the following cases: (i) a compact Kähler manifold of any dimension, (ii) a compact Kähler threefold with only canonical singularities, or (iii) a normal $\mathbb{Q}$ -factorial compact Kähler surface. Suppose X admits an int-amplified endomorphism $f : X \rightarrow X$ and the canonical divisor K_X is pseudo-effective. Then X is a $\mathbb{Q}$ -complex torus, i.e., a quasi-étale quotient of a complex torus.*

We generalize [Men20, Theorem 1.10] to the following equivariant MMP (cf. [MZ18a, Section 6] for the technical details). We will highlight and compare the differences between the cases for projective varieties and compact Kähler spaces in Section 2.8.

Theorem 2.1.5 ([Zho19, Theorem 1.5]). *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler threefold with only terminal singularities. Then, replacing f by a positive power, there exist a $\mathbb{Q}$ -complex torus Y , a morphism $X \rightarrow Y$ and an f -equivariant minimal model program of X over Y :*

$$X = X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_r = Y$$

(i.e., $f = f_1$ descends to holomorphic f_i on each X_i), with every $X_i \dashrightarrow X_{i+1}$ a divisorial contraction, a flip or a Mori fibre contraction, of a K_{X_i} -negative extremal ray, such that:

- (1) If K_X is pseudo-effective, then $X = Y$ and it is a $\mathbb{Q}$ -complex torus.
- (2) If K_X is not pseudo-effective, then for each i , f_i is int-amplified and $X_i \rightarrow Y$ is an equi-dimensional (holomorphic) morphism with every fibre irreducible and rationally connected; further, $X_{r-1} \rightarrow X_r = Y$ is a Mori fibre contraction.

Remark 2.1.6. We remark here that, after the appearance of [Zho19], the minimal model program for Kähler threefolds has been further developed. The joint work [DH20] of Hacon and Das has extended the previous joint work of Höring and Peternell (cf. [HP15] and [HP16]) by weakening the condition on singularities. Hence, we believe that with the same strategy, Theorem 2.1.5 can be further extended to the case of klt singularities (cf. Question 1.8.1).

Finally, we pose the question below. If X is assumed to be projective, the answer is affirmative by Fakhruddin's very motivating result (cf. [Fak03, Theorem 5.1]).

Question 2.1.7. Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler space. Does there exist a periodic point of f ? If so, is the set of f -periodic points $\text{Per}(f)$ Zariski dense in X ?

The proofs of Theorems 2.1.1 ~ 2.1.3 are in Section 2.3, the proof of Theorem 2.1.4 is in Section 2.6, and the proof of Theorem 2.1.5 is in Section 2.8.

This chapter is based on my work [Zho19].

2.2 Preliminaries

Let X be a normal compact complex space. We refer to [AG06] and [Uen75] for basic notation and terminology of complex spaces. We shall recall the differential forms (resp. currents) and their pull-backs (resp. push-forwards) in the singular setting.

Let $f : X \rightarrow Y$ be a (proper) surjective morphism between normal compact complex spaces. Here, a morphism is a holomorphic map in complex geometry. The proper morphism f is said to be *finite* (resp. *generically finite*) if f has discrete fibres (resp. is finite outside a nowhere dense analytic closed subspace of Y). An *endomorphism* of X is a surjective holomorphic self-map $f : X \rightarrow X$.

The following idea will be used more than once.

Main Reduction: Let $f : X \rightarrow Y$ be a surjective morphism between normal compact complex spaces. Since Y is compact, by Hironaka desingularization theorem, there exists a modification $h : \tilde{Y} \rightarrow Y$ obtained by a finite sequence of blowing-up along smooth subspaces, such that $\tilde{Y}$ is a complex manifold. By [AG06, Chapter I, Lemma 7.1], $h : \tilde{Y} \rightarrow Y$ lifts to $g' : X' \rightarrow X$ which is obtained by a finite sequence of blowing-up such that $f \circ g'$ factors through $\tilde{Y}$ with the induced morphism $f' : X' \rightarrow \tilde{Y}$. Taking a resolution $\sigma : \tilde{X} \rightarrow X'$, we get $g := g' \circ \sigma : \tilde{X} \rightarrow X$, $\tilde{f} := f' \circ \sigma : \tilde{X} \rightarrow \tilde{Y}$ and the commutative diagram $f \circ g = h \circ \tilde{f}$, such that all of the morphisms are surjective.

By the classical differential geometry, the morphism $\tilde{f}$ admits very nice properties. Hence, to detect good properties on f , we need to fix our attention to the resolution h especially to a single blow-up.

Remark 2.2.1. Note that there are different choices of $\tilde{X}$. For example, we can also choose $\tilde{X}$ to be the resolution of an irreducible component of the product $X \times_Y \tilde{Y}$ dominating X and also get a commutative diagram.

2.2.1 Differential forms and currents

In this subsection, we refer to [Dem85] for the definitions and properties below.

Let X be a normal compact complex space of dimension n . Then, X is locally a closed analytic subset of an open subset of $\mathbb{C}^N$, i.e., for any $x \in X$, there exists an open neighbourhood U of x and an open subset $\Omega \subseteq \mathbb{C}^N$ such that $j : U \rightarrow \Omega$ is a closed embedding. A differential form ω of type (p, q) on X is a differential form on the smooth locus X_{reg} such that for any $x \in X$, with the closed embedding $x \in U \subseteq \Omega \subseteq \mathbb{C}^N$, there exists a differential form α on Ω such that $\omega|_{X_{\text{reg}}} = j^*\alpha|_{X_{\text{reg}}}$ locally. We denote by $\mathcal{C}^{p,q}(X)$ the space of differential (p, q) -forms.

A current T of bidegree (p, q) is defined as an element in the dual space of differential (smooth) $(n-p, n-q)$ -forms. We denote by $\mathcal{D}^{p,q}(X)$ the space of (p, q) -currents with bidimension $(n-p, n-q)$ on X . It is well-defined, as well as the closedness and positivity.

Let $f : X \rightarrow Y$ be a (proper) morphism between two normal compact complex spaces of dimensions m and n , respectively. We refer the readers to [HP16] for the pull-back of differential forms by pulling back forms locally. Then, one can define the push-forward of currents f_* by setting $\langle f_*T, \alpha \rangle := \langle T, f^*\alpha \rangle$ for each $\alpha \in \mathcal{C}^{p,q}(X)$. For any (p, p) -current T and (q, q) -form α on X , we denote by $T \wedge \alpha$ the $(p+q, p+q)$ -current such that for each $(n-p-q, n-p-q)$ -form β on X , $\langle T \wedge \alpha, \beta \rangle = \langle T, \alpha \wedge \beta \rangle$. In addition, since each differential (p, q) -form α on X can also be naturally regarded as a (p, q) -current, we write $[\alpha]$ to represent a (p, q) -current, i.e., for each $(n-p, n-q)$ -form β , $\langle [\alpha], \beta \rangle := \int_{X_{\text{reg}}} \alpha \wedge \beta$.

We recall the following well-known projection formula which will be heavily used in this chapter.

Proposition 2.2.2 (Projection formula). *Let $f : X \rightarrow Y$ be a (proper) surjective morphism between normal compact complex spaces with $\dim(X) = m$ and $\dim(Y) = n$. Then for any (l, l) -current T on X and differential (k, k) -form β on Y , we have $f_*(T \wedge f^*\beta) = f_*T \wedge \beta$ in the sense of currents. In particular, if $m = n$ (and hence f is generically finite), then $f_*[f^*\beta] = (\deg(f)) \cdot [\beta]$ as currents.*

Proof. For any differential $(n - k - l, n - k - l)$ -form γ on Y , we have

$$\langle f_*(T \wedge f^*\beta), \gamma \rangle = \langle T \wedge f^*\beta, f^*\gamma \rangle = \langle T, f^*(\beta \wedge \gamma) \rangle = \langle f_*T, \beta \wedge \gamma \rangle = \langle f_*T \wedge \beta, \gamma \rangle,$$

which proves the first statement. Now we take two steps to prove the second claim.

Step 1. From **Main Reduction**, we get a commutative diagram $h \circ \tilde{f} = f \circ g$ such that $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ is a generically finite morphism of complex manifolds and $\deg(f) = \deg(\tilde{f})$. Then, it follows from the Gysin morphism that $\tilde{f}_*\tilde{f}^*\alpha = (\deg(f)) \cdot \alpha$ for each differential form α on $\tilde{Y}$ (cf. [Voi07, Remark 7.29]).

Step 2. For any $(n - k, n - k)$ -form γ on Y , we have

$$\langle f_*[f^*\beta], \gamma \rangle = \langle [f^*\beta], f^*\gamma \rangle = \int_{X_{\text{reg}}} f^*(\beta \wedge \gamma) = \int_{\tilde{X}} g^*f^*(\beta \wedge \gamma).$$

The third equality is because a proper closed subvariety (with strictly lower dimension) of a compact complex manifold has Lebesgue measure zero. Then,

$$\int_{\tilde{X}} g^*f^*(\beta \wedge \gamma) = \int_{\tilde{X}} \tilde{f}^*h^*(\beta \wedge \gamma) = (\deg(\tilde{f})) \cdot \int_{\tilde{Y}} h^*(\beta \wedge \gamma).$$

Note that the right hand side is equal to $(\deg(f)) \cdot \int_{Y_{\text{reg}}} (\beta \wedge \gamma)$ which is nothing but $(\deg(f)) \cdot \langle [\beta], \gamma \rangle$. Then, our second statement holds. $\square$

2.2.2 Classes and cones

In this subsection, we recall the notation and terminology which will be heavily used in this chapter. First we refer the readers to [BEG13, Definition 4.6.2] or [HP16, Definition 3.1] for the definition of Bott-Chern cohomology $H_{\text{BC}}^{1,1}(X)$ for singular spaces.

On the one hand, for any holomorphic map $f : X \rightarrow Y$ between normal compact complex spaces, the pull-back of differential forms f^* induces a well-defined morphism $f^* : H_{\text{BC}}^{1,1}(Y) \rightarrow H_{\text{BC}}^{1,1}(X)$: for any $c \in H_{\text{BC}}^{1,1}(Y)$, we can write $c = \omega + dd^c u$ where ω is a $(1, 1)$ -form with local potentials and u is a global smooth function (cf. [HP16, Remark 3.2]). Then, the pull-back f^*c is written as the form $f^*\omega + dd^c(u \circ f)$ and it is easy to see that $f^*\omega$ is a $(1, 1)$ -form with local potentials since the real parts of holomorphic functions of Y pull back to real parts of holomorphic functions of X . Hence, f^*c defines an element in $H_{\text{BC}}^{1,1}(X)$.

On the other hand, if X has at worst rational singularities in the Fujiki class $\mathcal{C}$, then it follows from the Leray sequence that there is an injection: $H_{\text{BC}}^{1,1}(X) \hookrightarrow H^2(X, \mathbb{R})$ (cf. [HP16, Equation (3) in Page 223, Remark 3.7]). Hence, one can define the intersection product on the Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$ via the cup-product for $H^2(X, \mathbb{R})$. In addition, this embedding implies that $H_{\text{BC}}^{1,1}(X)$ is a finite dimensional vector space.

Remark 2.2.3. Let X be a (smooth) compact Kähler manifold. By Hodge theory, there is a natural isomorphism from the Bott-Chern cohomology group to the Dolbeault group with real coefficients $H_{\text{BC}}^{1,1}(X) \cong H^{1,1}(X, \mathbb{R}) := H^{1,1}(X, \mathbb{C}) \cap H^2(X, \mathbb{R})$.

To make connections with the normal projective setting, let $\mathbf{N}_1(X)$ be the vector space of real closed currents of bi-degree $(n-1, n-1)$ modulo the following equivalence: $T_1 \equiv T_2$ if and only if: $T_1(\eta) = T_2(\eta)$ for all real closed $(1, 1)$ -forms η with local potentials (cf. [HP16, Definition 3.8]).

Remark 2.2.4. Let $f : X \rightarrow Y$ be a (proper) morphism between normal compact complex spaces and T_C (or denoted by $[C]$) a *current of integration* (cf. [HP16, Section 3.B]) of an irreducible curve C on X . We claim that: if $f(C)$ is a curve, then $f_*[C] = (\deg(f|_C)) \cdot [f(C)]$; if C is contracted by f , then $f_*[C] = 0$. This coincides with the push-forward of cycles in the projective setting.

Proof. We follow **Main Reduction**. First, we prove that for any resolution $h : \tilde{Y} \rightarrow Y$ with $E \subseteq \tilde{Y}$ a contracted subvariety, $h_*[E] = 0$. Indeed, for the local map $h' : \tilde{Y} \rightarrow Y \hookrightarrow \mathbb{C}^N$ and each differential (k, k) -form γ on Y with $\gamma = i^*\gamma_0$ and $k = \dim(E)$,

$$\langle h_*[E], \gamma \rangle = \langle [E], h^*\gamma \rangle = \langle [E], (h')^*\gamma_0 \rangle = 0.$$

The last equality is due to $\dim(h'(E)) < \dim(E)$ and h' is locally a holomorphic map between manifolds. Therefore, $h_*[E] = 0$. Second, suppose the curve C is contracted by f . Then, no matter $C \subseteq \text{Sing}(X)$ (the singular locus of X) or not, any analytic curve of $\tilde{X}$ dominating C must be contracted by either $\tilde{f}$ or h . Hence, $f_*[C] = 0$ when C is contracted by f .

From now on, suppose $f(C)$ is a curve. Then for any differential $(n-1, n-1)$ -form γ on Y , we apply **Main Reduction**. If $C \not\subseteq \text{Sing}(X)$, set $\tilde{C} \subseteq \tilde{X}$ to be the proper transform

of C ; if $C \subseteq \text{Sing}(X)$, let $\tilde{C} \subseteq \tilde{X}$ be any curve dominating C (which exists since the resolution g is projective). In both cases,

$$\langle f_*[C], \gamma \rangle = \int_{C_{\text{reg}}} f^* \gamma = \frac{1}{\deg(g|_{\tilde{C}})} \int_{\tilde{C}} \tilde{f}^* h^* \gamma = \deg(f|_C) \int_{f(C)} \gamma = \deg(f|_C) \cdot \langle [f(C)], \gamma \rangle.$$

Therefore, we complete the proof of our remark. $\square$

Let $\overline{\text{NA}}(X) \subseteq N_1(X)$ be the closed cone generated by the classes of positive closed $(n-1, n-1)$ -currents. We define the Mori cone $\overline{\text{NE}}(X) \subseteq \overline{\text{NA}}(X)$ as the closure of the cone generated by the currents of integration T_C , where $C \subseteq X$ is an irreducible curve.

From now on, we begin to discuss the positivity and cones in complex geometry. In the beginning, we refer the readers to [Dem12, Chapter I] and [Dem92, Introduction and main results] for the definition and basic properties of (strictly) plurisubharmonic (psh) functions.

Let X be a normal compact complex space of dimension m . Then, it follows from [Dem12, Chapter I, Theorem 5.8] that a smooth function φ is psh on X if and only if $dd^c \varphi \geq 0$ is a positive current.

A normal compact complex space X is said to be *Kähler* if there exists a Kähler form ω on X , i.e., a positive closed real $(1, 1)$ -form $\omega \in \mathcal{C}^{1,1}(X)$ such that the following holds. For every point $x \in \text{Sing}(X)$ (the singular locus of X), there exist an open neighbourhood $x \in U \subseteq X$, a closed embedding $i_U : U \subseteq \Omega$ into an open set $\Omega \subseteq \mathbb{C}^N$ and a strictly psh C^∞ -function $f : \Omega \rightarrow \mathbb{C}$ such that $\omega|_{U \cap X_{\text{reg}}} = (dd^c f)|_{U \cap X_{\text{reg}}}$.

Definition 2.2.5. Let (X, ω) be a normal compact Kähler space, where ω is a fixed Kähler form. A class $[\alpha] \in H_{\text{BC}}^{1,1}(X)$ (where α is a smooth form) is said to be

- (1) *Kähler*, if it contains a representative of a smooth Kähler form, i.e., there is a smooth function φ such that $\alpha + dd^c \varphi \geq \varepsilon \omega$ for some $\varepsilon > 0$;
- (2) *nef*, if for every $\varepsilon > 0$, there is a smooth function f_ε such that $\alpha + dd^c f_\varepsilon \geq -\varepsilon \omega$;
- (3) *pseudo-effective*, if it contains a positive current (that may not be smooth), i.e., there exists an almost psh function $\varphi = \psi + h$ where ψ is psh and h is smooth on X such that $\alpha + dd^c \varphi \geq 0$ in the sense of currents.

- (4) big, if it contains a Kähler current, i.e., there exists an almost psh function φ such that $\alpha + dd^c\varphi \geq \varepsilon\omega$ in the sense of currents for some $\varepsilon \geq 0$.

Let X be a normal compact Kähler space. We use the following to represent the cones in Definition 2.2.5 to make connections with the projective setting.

- $\mathcal{K}(X)$: the cone of Kähler classes in $H_{\text{BC}}^{1,1}(X)$;
- $\overline{\mathcal{K}}(X)$: the cone of nef classes in $H_{\text{BC}}^{1,1}(X)$;
- $\mathcal{E}(X)$: the cone of pseudo-effective classes in $H_{\text{BC}}^{1,1}(X)$;
- $\mathcal{E}^\circ(X)$: the cone of big classes in $H_{\text{BC}}^{1,1}(X)$.

Observe that $\mathcal{K}(X)$ is an open (convex) cone (cf. [GK20, Proposition 3.6]) and $\overline{\mathcal{K}}(X) \subseteq \mathcal{E}(X)$ are closed convex cones by the weak compactness of bounded sets in the space of currents (cf. [Dem92, Section 6]). Besides, $\overline{\mathcal{K}}(X)$ is the closure of $\mathcal{K}(X)$ (cf. [HP16, Remark 3.12]). Further, $\mathcal{E}^\circ(X)$ is the interior part of $\mathcal{E}(X)$.

We formulate the following important proposition which states that all of these cones mentioned above are preserved under suitable morphisms.

Proposition 2.2.6 (Invariant cones). *Let $f : X \rightarrow Y$ be a (proper) surjective morphism between normal compact Kähler spaces. Then, for any $[\alpha] \in H_{\text{BC}}^{1,1}(Y)$, the following assertions hold.*

- (1) *If $[\alpha]$ is pseudo-effective, then so is $f^*[\alpha]$;*
- (2) *If $[\alpha]$ is nef, then so is $f^*[\alpha]$ and the converse direction holds when f is further assumed to be a modification of normal compact Kähler threefolds;*
- (3) *If f is further assumed to be finite and $[\alpha]$ is Kähler, then so is $f^*[\alpha]$;*
- (4) *If f is a modification and $[\alpha]$ is big, then so is $f^*[\alpha]$;*
- (5) *If f is generically finite and $[\alpha]$ is big, then so is $f^*[\alpha]$.*

Proof. The pseudo-effective cone is invariant under any holomorphic map since the pull-back of a psh function is still psh, which gives (1). Let ω_X and ω_Y be two Kähler forms on X and Y respectively such that $f^*\omega_Y \leq a\omega_X$ for some constant $a > 0$. For any $\varepsilon > 0$, since $[\alpha]$ is nef, there exists $g_\varepsilon \in \mathcal{C}^0(Y)$ such that $\alpha + dd^c g_\varepsilon \geq -\frac{\varepsilon}{a} \cdot \omega_Y$. Then

$$f^*\alpha + dd^c(g_\varepsilon \circ f) \geq -\frac{\varepsilon}{a} f^*\omega_Y \geq -\varepsilon\omega_X,$$

which proves one direction of (2). For the converse, see [HP16, Lemma 3.13].

For (3), see [GK20, Proposition 3.5]. Now, we begin to prove (4). For each big class $[\alpha]$ on Y , there exists a Kähler form ω_Y on Y and a representative T' of $[\alpha]$ such that $T' \geq \omega_Y$. Therefore, it is enough to prove that the pull-back of each Kähler class on Y is big on X under any modification. Let $f : X \rightarrow Y$ be a modification and ω_Y a Kähler form on Y . By Hironaka's Chow lemma (cf. [AG06, Theorem 7.8]), there exists a normal compact complex space Z with $p : Z \rightarrow Y$ and $q : Z \rightarrow X$ such that $p = f \circ q$, where p and q are compositions of finite sequences of blowing-up along smooth centers.

We claim that $p^*[\omega_Y]$ is a big class. Indeed, we only need to consider the case when p is a single blow-up. Note that a blow-up of Y is locally the restriction of a blow-up of $\mathbb{C}^N$. Denote by E the exceptional divisor of p . Then E is locally the restriction of the exceptional divisor E_i of the blow-up $p_i : \tilde{U}_i \rightarrow U_i$, where U_i is an open subset of $\mathbb{C}^N$. For the smooth manifold U_i , it follows from the calculation in [Voi07, Proposition 3.24 and Lemma 3.26] that for sufficiently large $C_i \gg 1$, $(C_i p_i^* \omega_{U_i} - E_i)|_Z$ is a positive $(1, 1)$ -form (also cf. [AG06, pp.116]). Since Y is compact, there exists a sufficiently large $C \gg 1$ such that $Cp^*\omega_Y - E$ is a positive $(1, 1)$ -form. Hence, $p^*[\omega_Y]$ is a big class.

Moreover, $f^*\omega_Y = q_*q^*f^*\omega_Y = q_*p^*\omega_Y$ as currents by projection formula. Since $p^*[\omega_Y]$ is big, there exists a current $T \in p^*[\omega_Y] = [p^*\omega_Y]$ such that $T \geq \omega_Z$ as currents for some Kähler form ω_Z on Z . So we have $q_*T \in [f^*\omega_Y] = f^*[\omega_Y]$ such that $q_*T \geq q_*\omega_Z$ as currents. In addition, for any Kähler form ω_X on X , there exists $a > 0$ such that $q^*\omega_X \leq a\omega_Z$. Hence, $q_*T \geq q_*\omega_Z \geq \frac{1}{a}\omega_X$ as currents and then $f^*[\omega_Y]$ is big on Y .

Finally, for (5), if f is a generically finite surjective morphism, then taking the Stein factorization of f (cf. [Uen75, Theorem 1.9]), we get a modification $f' : X \rightarrow Y'$ followed by a finite morphism $f'' : Y' \rightarrow Y$. For any big class $[\alpha]$ on Y , there exists a Kähler

current $T \in [\alpha]$ such that $T \geq \omega_Y$ as currents for some Kähler form ω_Y on Y . Then, the pull-back $(f'')^*T \geq (f'')^*\omega_Y$ and the class of the right hand side is Kähler by (3). Hence, $(f'')^*[\alpha]$ is big. Further, it follows from the assertion (4) that $f^*[\alpha] = (f')^*(f'')^*[\alpha]$ is also big, and (5) has thus been checked. $\square$

Recall that if $f : X \rightarrow Y$ is a surjective morphism between normal projective varieties, then it follows from the projection formula and Lefschetz hyperplane theorem that the pull-back operation gives an injection $f^* : \text{NS}_{\mathbb{R}}(Y) \rightarrow \text{NS}_{\mathbb{R}}(X)$ on the Néron-Severi groups of X and Y . However, in the analytic case, there is no ample divisor and Lefschetz hyperplane theorem may not be true. Nevertheless, we have the lemma below by requiring mild singularities.

Lemma 2.2.7. *Let $f : X \rightarrow Y$ be a surjective holomorphic map between normal compact Kähler spaces with only rational singularities. Then $f^* : H_{\text{BC}}^{1,1}(Y) \hookrightarrow H_{\text{BC}}^{1,1}(X)$ is an injection.*

Proof. We shall apply **Main Reduction** to prove our lemma. Take a resolution $h : \tilde{Y} \rightarrow Y$ and let $\tilde{X}$ be the resolution of an irreducible component of the product $X \times_Y \tilde{Y}$ dominating X with $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ and $g : \tilde{X} \rightarrow X$. Since $\tilde{f}$ is a surjective morphism between compact Kähler manifolds, the induced linear operation $\tilde{f}^* : H_{\text{BC}}^{1,1}(\tilde{Y}) \rightarrow H_{\text{BC}}^{1,1}(\tilde{X})$ is an injection (cf. [Voi07, Lemma 7.28] and Remark 2.2.3).

Since both g and h are resolutions, $H_{\text{BC}}^{1,1}(Y)$ (resp. $H_{\text{BC}}^{1,1}(X)$) embeds into $H_{\text{BC}}^{1,1}(\tilde{Y})$ (resp. $H_{\text{BC}}^{1,1}(\tilde{X})$); see [HP16, Lemma 3.3]. By the commutative diagram, the induced linear operation $f^*|_{H_{\text{BC}}^{1,1}(Y)}$ is an injection. $\square$

In particular, we get the following proposition by setting $X = Y$ in Lemma 2.2.7.

Proposition 2.2.8. *Let X be a normal compact Kähler space with at worst rational singularities and $f : X \rightarrow X$ a surjective endomorphism. Then, f induces an isomorphism of the Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$.*

At the end of this subsection, we state the fact on spanning cones below. Indeed, it follows immediately from the openness of the Kähler cone.

Lemma 2.2.9. *Let X be a normal compact Kähler space. Then, the pseudo-effective cone $\mathcal{E}(X)$ defined in Definition 2.2.5 spans the Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$.*

2.2.3 Polarized and int-amplified endomorphisms

In this short subsection, we come to define the key notion in this chapter. First, we prove the next lemma which asserts that for each holomorphic self-map of a compact Kähler space with mild singularities, the surjectivity will imply the finiteness.

Lemma 2.2.10. *Let $f : X \rightarrow X$ be a surjective holomorphic map of a normal compact Kähler space X . Suppose that X has at worst rational singularities. Then f is finite.*

Proof. Since X is compact, any holomorphic self-map of X is proper. Besides, f is a surjective self-map and thus generically finite. By taking the Stein factorization $X \rightarrow Y \rightarrow X$ and applying Hironaka's Chow lemma (cf. [AG06, Theorem 7.8]) for the modification $X \rightarrow Y$, we see that f is dominated by a composition of finite sequences of blow-ups followed by a finite morphism $Y \rightarrow X$, which is projective. Therefore, every positive dimensional fibre of f will be covered by curves.

Let $C \subseteq X$ be a curve lying in the fibre of f . We fix a Kähler class $[\omega]$ on X . On the one hand, since f is proper and surjective, it follows from Proposition 2.2.6 (2) that $f^*\overline{\mathcal{K}}(X) \subseteq \overline{\mathcal{K}}(X)$. On the other hand, since X has at worst rational singularities, by Proposition 2.2.8, $f^*|_{H_{\text{BC}}^{1,1}(X)}$ is invertible; hence it maps interior part of $\overline{\mathcal{K}}(X)$ to the interior part of $\overline{\mathcal{K}}(X)$. In particular, $f^*[\omega]$ is still a Kähler class.

Let $j : \tilde{C} \rightarrow C \hookrightarrow X$ be the composite morphism, where $\tilde{C}$ is the normalization of the curve C . Then $j^*f^*[\omega]$ is also a Kähler class by Proposition 2.2.6 (3). Therefore, $0 < \int_{\tilde{C}} j^*f^*\omega = \int_C f^*\omega = \langle [C], f^*\omega \rangle$ by the numerical characterization of Kähler classes (cf. [DP04]). However, the right hand side coincides with $\langle f_*[C], \omega \rangle$ by the projection formula and hence vanishes (cf. Proposition 2.2.2 and Remark 2.2.4), a contradiction. $\square$

Definition 2.2.11. Let $f : X \rightarrow X$ be a surjective endomorphism of a normal compact Kähler space X . We say that f is

- (1) polarized, if there exists a Kähler class $[\alpha] \in H_{\text{BC}}^{1,1}(X)$ such that $f^*[\alpha] = q[\alpha]$ for some integer $q > 1$;
- (2) amplified, if there exist a class $[\xi] \in H_{\text{BC}}^{1,1}(X)$ and a Kähler class $[\eta] \in H_{\text{BC}}^{1,1}(X)$ such that $f^*[\xi] - [\xi] = [\eta]$;
- (3) int-amplified, if there exist Kähler classes $[\xi]$ and $[\eta]$ such that $f^*[\xi] - [\xi] = [\eta]$.

Remark 2.2.12. In the following, we will focus on int-amplified endomorphisms. We leave the similar extension on polarized endomorphisms to readers.

2.3 Proofs of Theorems 2.1.1 ~ 2.1.3

In this section, we study the properties of int-amplified endomorphisms. Most of them are extensions of [Men20, Section 3] for the case when X is projective and the proofs are similar except we apply new results developed in Section 2.2.

To begin with this section, we prove Theorem 2.1.1. Here, we require X to have only rational singularities to make sure the induced linear operation $f^*|_{H_{\text{BC}}^{1,1}(X)}$ is invertible (cf. Proposition 2.2.8).

Proof of Theorem 2.1.1. It is clear that (4) $\Rightarrow$ (1) $\Rightarrow$ (3) by letting $C = \mathcal{K}(X)$. Suppose condition (3). Then, applying [Men20, Lemma 3.1] with $V = H_{\text{BC}}^{1,1}(X)$ and $C = \mathcal{E}(X)$, we get (2). Finally, (2) implies (4) is exactly [Men20, Proposition 3.2].

Next, we show the equivalence of two kinds of definitions when X is projective. Indeed, it is obvious that if f is int-amplified in the sense of ample divisors, then f is int-amplified in the sense of Kähler classes. For the converse direction, suppose X is projective and f is int-amplified in the sense of Kähler classes. Then by (2), each eigenvalue of the invertible linear operation $f^*|_{H_{\text{BC}}^{1,1}(X)}$ has modulus greater than 1. Since the Néron-Severi space $\text{NS}_{\mathbb{R}}(X)$ is a subspace of $H_{\text{BC}}^{1,1}(X)$, each eigenvalue of $f^*|_{\text{NS}_{\mathbb{R}}(X)}$ also has modulus greater than 1. So the converse direction holds by [Men20, Theorem 1.1]. $\square$

By Theorem 2.1.1 and Proposition 2.2.6, we have the following lemma, which shows that an endomorphism being int-amplified is equivalent to its power being int-amplified.

Lemma 2.3.1. *Let $f : X \rightarrow X$ be a surjective endomorphism of a compact Kähler space X with only rational singularities. Then the following assertions are equivalent.*

- (1) f is int-amplified.
- (2) f^s is int-amplified for any positive integer s .
- (3) f^s is int-amplified for some positive integer s .

In the following, we shall apply Theorem 2.1.1 to prove several important lemmas.

Lemma 2.3.2. *Let $\pi : X \rightarrow Y$ be a surjective morphism of normal compact Kähler spaces with at worst rational singularities. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be two surjective endomorphisms such that $g \circ \pi = \pi \circ f$. If f is int-amplified, then so is g .*

Proof. By Lemma 2.2.7, $\pi^* : H_{\text{BC}}^{1,1}(Y) \rightarrow H_{\text{BC}}^{1,1}(X)$ is injective. Then, each eigenvalue of $g^*|_{H_{\text{BC}}^{1,1}(Y)}$ is an eigenvalue of $f^*|_{H_{\text{BC}}^{1,1}(X)}$ and thus has modulus greater than one, since f is int-amplified. Therefore, our lemma follows from Theorem 2.1.1. $\square$

Lemma 2.3.3. *Let $\pi : X \rightarrow Y$ be a generically finite surjective morphism between normal compact Kähler spaces with at worst rational singularities. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be surjective endomorphisms such that $g \circ \pi = \pi \circ f$. If g is int-amplified, then so is f .*

Proof. Since g is int-amplified, there exist Kähler classes $[\xi]$ and $[\eta]$ on Y such that $g^*[\xi] = [\eta] + [\xi]$. Note that the Kähler cone $\mathcal{K}(Y)$ is contained in the big cone $\mathcal{E}^\circ(Y)$ and big classes pull back to big classes under generically finite morphisms (cf. Proposition 2.2.6). Then, $\pi^*[\xi]$ and $\pi^*[\eta]$ are big classes on X and the commutative diagram implies $\pi^*[\eta] = f^*\pi^*[\xi] - \pi^*[\xi]$. Hence, our lemma follows from Theorem 2.1.1. $\square$

Next, we follow the ideas of [Men20] to prove Theorems 2.1.2 and 2.1.3 below in our present setting. Both of them are consequences of Theorem 2.1.1.

Proof of Theorem 2.1.2. We fix a norm on $V := H_{\text{BC}}^{1,1}(X)$. Let $\phi_f := f^*|_V$ and $\phi_g := g^*|_V$. Since f is int-amplified, by Theorem 2.1.1, all the eigenvalues of ϕ_f are of modulus

greater than 1. Recall that the norm of a linear operator is given by $\|\phi_f\| := \max_{v \in V} \frac{\|\phi_f(v)\|}{\|v\|}$, and the right hand side is no less than the spectral radius $\rho(\varphi)$ of $\varphi := \phi_f$. Also, by Gelfand's formula, $\lim_{i \rightarrow \infty} \|\phi_f^{-i}\|^{\frac{1}{i}} = \rho(\phi_f^{-1}) < 1$, where $\rho(\phi_f^{-1})$ is the spectral radius of ϕ_f^{-1} . Then, there exists i_0 such that for all $i \geq i_0$, $\|\phi_f^{-i} \phi_g^{-1}\| \leq \|\phi_f^{-i}\| \cdot \|\phi_g^{-1}\| < 1$. So, for each $v \in V$, $\|\phi_g \phi_f^i(v)\| > \|v\|$. Let $h = f^i \circ g$ with $i \geq i_0$ and $\phi_h = h^*|_V$.

We claim that all the eigenvalues of ϕ_h are of modulus greater than 1. Indeed, let r be the smallest absolute value of eigenvalues of ϕ_h . Since X is Kähler, V has nonvoid interior. So it follows from the Perron-Frobenius theorem that V contains an eigenvector v whose eigenvalue is the spectral radius of ϕ_h^{-1} , i.e., $\|\phi_h^{-1}(v)\| = \frac{1}{r}\|v\|$. With v replaced by $\phi_h(v)$, we have $r > 1$ by the inequality: $r\|v\| = \|\phi_h(v)\| = \|\phi_g(\phi_f^i(v))\| > \|v\|$. Hence h is int-amplified. Similar argument works for $g \circ f^i$. So Theorem 2.1.2 holds. $\square$

Proof of Theorem 2.1.3. By the ramification divisor formula, $f^*(-K_X) - (-K_X) = R_\phi$, where the ramification divisor R_ϕ is effective and thus pseudo-effective in the sense of currents. By Theorem 2.1.1 (4), $-K_X$ is also pseudo-effective.

Suppose on the contrary that $\kappa(X) > 0$. Then, there exists a non-zero effective divisor D on X , such that $sK_X \sim D$ for some $s \in \mathbb{N}^+$. Note that D is not contained in the singular locus due to the normality of X . Thus, for any Kähler form ω on X ,

$$sK_X \wedge \omega^{n-1} = [D] \wedge \omega^{n-1} = \int_{D_{\text{reg}}} (\omega|_D)^{n-1} > 0,$$

a contradiction to $-K_X$ being pseudo-effective. Then, Theorem 2.1.3 is proved. $\square$

Let $f : X \rightarrow X$ be a surjective endomorphism of a compact Kähler space X of dimension n with only rational singularities (cf. [HP16, Remark 3.7]). Denote by

$$L^k(X) := \left\{ \sum [\alpha_1] \cup \cdots \cup [\alpha_k] \mid [\alpha_i] \in H_{\text{BC}}^{1,1}(X) \right\},$$

the subspace of $H^{2k}(X, \mathbb{R})$. In particular, $L^1(X) = H_{\text{BC}}^{1,1}(X)$. Let $B^k(X) := L^k(X) / \equiv_w$, where $[\alpha] \in L^k(X)$ is weakly numerically equivalent (denoted by $\equiv_w$) to zero if and only if for any $[\beta_{k+1}], \dots, [\beta_n] \in H_{\text{BC}}^{1,1}(X)$, $[\alpha] \cup [\beta_{k+1}] \cup \cdots \cup [\beta_n] = 0$. Moreover, for any $[\alpha], [\beta] \in L^k(X)$, $[\alpha] \equiv_w [\beta]$ if and only if $[\alpha] - [\beta] \equiv_w 0$.

Let $L_{\mathbb{C}}^k(X) = L^k(X) \otimes_{\mathbb{R}} \mathbb{C}$ and $B_{\mathbb{C}}^k(X) = B^k(X) \otimes_{\mathbb{R}} \mathbb{C}$. Note that the pull-back operation f^* on Bott-Chern cohomology space induces a linear operation on the subspace $L_{\mathbb{C}}^k(X) \subseteq H^{2k}(X, \mathbb{C})$ for each k . With the notation settled, it follows from Proposition 2.2.8 that f^* also gives a well-defined linear operation on the quotient space $B_{\mathbb{C}}^k(X)$.

We generalize [Men20, Lemma 3.6] to the following. We skip its proof and readers may refer to [Men20, Lemma 3.6] by applying Theorem 2.1.1 and Proposition 2.2.8. For its application, see Theorem 2.3.5 and Proposition 2.6.1.

Lemma 2.3.4. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler space of dimension n . Suppose that X has at worst rational singularities. Then, for each $0 < k < n$, all the eigenvalues of $f^*|_{B_{\mathbb{C}}^k(X)}$ are of modulus less than $\deg(f)$. In particular, $\lim_{i \rightarrow +\infty} \frac{(f^i)^*[x]}{(\deg(f))^i} \equiv_w 0$ for any $[x] \in L_{\mathbb{C}}^k(X)$.*

Theorem 2.3.5. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a positive dimensional compact Kähler space with only rational singularities. Then $\deg(f) > 1$.*

Proof. If $\dim(X) = 1$, then X is normal since X has only rational singularities. Therefore, X is smooth and projective. By Theorem 2.1.1, there exist two ample divisors D and H such that $f^*D - D = H$. Taking the degree of both sides, we see that $\deg(f) > 1$.

If $\dim(X) \geq 2$, then by Theorem 2.1.1, all the eigenvalues of $f^*|_{H_{\text{BC}}^{1,1}(X)}$ have modulus greater than 1. Since the eigenvectors of the quotient space $B_{\mathbb{C}}^1(X)$ come from the elements in $H_{\text{BC}}^{1,1}(X)$, by Lemma 2.3.4 for $k = 1$, our $\deg(f) > 1$. $\square$

2.4 Some dynamics of int-amplified endomorphisms

In this section, we discuss the int-amplified endomorphisms in terms of fixed points and totally invariant subvarieties. First, Lemma 2.4.1 follows immediately from [Var89, Proposition 1.3.1].

Lemma 2.4.1. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler space X . If $Z \subseteq X$ is an f -invariant analytic subvariety (i.e., $f(Z) = Z$), then the restriction $f|_Z$ is still int-amplified.*

Lemma 2.4.2. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a compact Kähler space X . Then, the set of periodic points $\text{Per}(f)$ is countable.*

Proof. Suppose $\text{Per}(f)$ is uncountable. Then, there exists some $s > 0$, such that the set P_s of all f^s -fixed points is infinite. Let Z be an irreducible component of the closure of P_s in X with $\dim(Z) > 0$. Then $f^s|_Z = \text{id}_Z$, which is absurd by Lemmas 2.3.1 and 2.4.1. $\square$

Unlike the projective case, for a compact Kähler space admitting an int-amplified endomorphism, it is still open about the density of periodic points. Nevertheless, the proposition below partially answers Question 2.1.7 for complex tori. We will show in Section 2.5 that it can be further extended to the Q -complex torus case.

Proposition 2.4.3. *Let $f : T \rightarrow T$ be an int-amplified endomorphism of a complex torus T . Then, there exists a fixed point of f .*

Proof. By Künneth formula, if $\lambda_1, \dots, \lambda_k$ (may not be distinct) are all the eigenvalues of $f^*|_{H^1(T, \mathbb{C})}$, then each eigenvalue of $f^*|_{H^i(T, \mathbb{C})}$ is of the form $\prod_j \lambda_j^{l_j}$, where $\sum l_j = i$.

Since f is int-amplified, each eigenvalue of $f^*|_{H^{1,1}(T, \mathbb{C})}$ has modulus greater than 1. By Hodge theory, all of the eigenvalues of $f^*|_{H^1(T, \mathbb{C})}$ (and hence $f^*|_{H^i(T, \mathbb{C})}$) are of modulus greater than 1. Then, by the Lefschetz fixed point theorem, the cardinality

$$\#\text{Fix}(f) = \sum_i (-1)^i \text{Trace}(f^*|_{H^i(T, \mathbb{C})}) = \sum_{k_1, \dots, k_i} (-\lambda_{k_1}) \cdots (-\lambda_{k_i}) = \prod_i (1 - \lambda_i),$$

with $|\lambda_i| > 1$ for each i . Then $\#\text{Fix}(f) \neq 0$ implies that f has a fixed point. $\square$

The following lemma holds due to the openness of the cone $\mathcal{K}_k(X)$, the set of classes of strictly positive closed (k, k) -forms (cf. [Dem12, Chapter III]).

Lemma 2.4.4. *Let (X, ω) be a compact Kähler manifold of dimension n . Then, for any closed (k, k) -current T , there exists $A > 0$ such that $A\omega^k - T$ is a positive (k, k) -current.*

When X is projective, one may use Bertini's theorem to prove that for any closed subvariety Z , there exists an effective cycle C such that the sum $Z + C$ is the intersection of ample divisors. However, in the analytic setting, we may not have ample divisors.

Nevertheless, Lemma 2.4.4 says that for any closed current, there exists a positive “complement” such that the sum is (numerically) a power of some Kähler form.

As an application of Lemma 2.4.4, we show the lemma below. Note that when X is smooth, $H_{\text{BC}}^{1,1}(X) \cong H^{1,1}(X, \mathbb{R})$ and hence we may regard each $(n-k, n-k)$ -current as a function on $L^k(X)$ (cf. Section 2.3) using the Poincaré duality and Stokes’ formula.

Lemma 2.4.5. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler space with at worst rational singularities of dimension n . Let ξ be a Kähler form on X . Then for any k -dimensional analytic proper subvariety $Z \subsetneq X$, which is not contained in the singular locus, $\lim_{i \rightarrow +\infty} \int_Z \frac{(f^i)^*(\xi^k)}{(\deg(f))^i} = 0$.*

Proof. Let $x_i := \frac{(f^i)^*(\xi^k)}{(\deg(f))^i} \in L^k(X)$. We take a resolution $\pi : \tilde{X} \rightarrow X$ and consider the pull-back $y_i := \pi^* x_i$, the class of which is a product of nef classes in $\tilde{X}$. Assume that outside a proper closed subset $E \subseteq \tilde{X}$, we have $\tilde{X} \setminus E \cong X \setminus \text{Sing}(X)$. Since $Z \not\subseteq \text{Sing}(X)$, we denote by $\tilde{Z}$ the proper transform of Z in $\tilde{X}$.

For any fixed Kähler form ω on X , there exists $c \gg 1$ such that $\eta := c \cdot \pi^* \omega - E'$ is a Kähler form on $\tilde{X}$ (cf. [Voi07, pp.81]), where $E' \geq 0$ is a positive linear combination of the components in E . We may identify E' with E in the following. Since $[\tilde{Z}]$ is a positive $(n-k, n-k)$ -current and $[y_i]$ is a product of nef classes, by Lemma 2.4.4, there exists $a > 0$ such that $\langle a \cdot \eta^{n-k} - [\tilde{Z}], [y_i] \rangle \geq 0$ for any i ; hence

$$\int_Z x_i = \int_{\tilde{Z}} y_i = \langle [\tilde{Z}], [y_i] \rangle \leq [y_i] \cup a \cdot \eta^{n-k} \leq [y_i] \cup (ac) \pi^* \omega \cup \eta^{n-k-1}.$$

Note that the last inequality is due to the nefness of $[y_i]$ and $[\eta]$ (and hence $[y_i] \cup E \cup \eta^{n-k-1} \geq 0$). Repeat the above process and apply the nefness of $[\pi^* \omega]$, we have

$$\int_Z x_i \leq a \cdot c^{n-k} [y_i] \cup (\pi^* \omega)^{n-k} = a \cdot c^{n-k} [x_i] \cup \omega^{n-k}. \quad (2.1)$$

With $\tilde{Z}$ replaced by $-\tilde{Z}$, the same arguments above together with Lemma 2.4.4 show that there exists $b > 0$ such that $\langle b \cdot \eta^{n-k} + [\tilde{Z}], [y_i] \rangle \geq 0$ for any i ; hence

$$-\int_Z x_i = -\int_{\tilde{Z}} y_i \leq [y_i] \cup b \cdot \eta^{n-k} \leq b \cdot c^{n-k} [x_i] \cup \omega^{n-k}. \quad (2.2)$$

By Lemma 2.3.4, $\lim_{i \rightarrow \infty} [x_i] \equiv_w 0$ in $L_{\mathbb{C}}^k(X)$. Taking the upper limits in both (2.1) and (2.2), we have $0 \leq \varliminf_{i \rightarrow \infty} \langle [Z], x_i \rangle \leq \overline{\lim}_{i \rightarrow \infty} \langle [Z], x_i \rangle \leq 0$ and our result follows. $\square$

Proposition 2.4.6. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler space X with at worst rational singularities of dimension n . Let Y be an analytic closed subvariety of X not contained in the singular locus such that $f^{-1}(Y) = Y$ and $\deg(f|_Y) = \deg(f)$. Then $X = Y$.*

Proof. Assume that $\dim(Y) = m < n$. We fix a Kähler form ξ on X . By Lemma 2.4.5, $\lim_{i \rightarrow \infty} \int_Y \frac{(f^i)^* \xi^m}{(\deg(f))^i} = 0$. However, the integration $\langle [Y], (f^i)^* \xi^m \rangle = (\deg(f|_Y))^i \langle [Y], \xi^m \rangle = (\deg(f))^i \langle [Y], \xi^m \rangle$ gives a contradiction (cf. [DP04]):

$$0 < \int_{Y_{\text{reg}}} \xi^m = \int_Y \xi^m = \frac{1}{(\deg(f))^i} \int_Y (f^i)^* \xi^m = \lim_{i \rightarrow \infty} \int_Y \frac{(f^i)^* \xi^m}{(\deg(f))^i} = 0.$$

Then, our proposition follows from this contradiction. $\square$

As an application, we end up this section with the following lemma.

Lemma 2.4.7. *Let $f : T \rightarrow T$ be an int-amplified endomorphism of a complex torus T . Let Z be a non-empty f^{-1} -periodic closed analytic subvariety of T . Then $Z = T$.*

Proof. Since f^s is int-amplified for any positive integer s (cf. Lemma 2.3.1), with f replaced by its power, we may assume Z is irreducible and $f^{-1}(Z) = Z$. It follows from the ramification divisor formula and the purity of branch loci that f is étale. Therefore we get $\deg(f|_Z) = \deg(f)$; hence $Z = T$ by Proposition 2.4.6. $\square$

2.5 The quotients of complex tori

In this section, we consider Q -complex tori admitting int-amplified endomorphisms. Recall that a finite surjective morphism $f : X \rightarrow Y$ between compact Kähler spaces is *quasi-étale* if f is étale in codimension 1. A compact Kähler space X is said to be a **Q -complex torus** if there is a quasi-étale morphism $T \rightarrow X$ from a complex torus T (cf. e.g., [HP11, Notation 2.1]). By the ramification divisor formula, any surjective endomorphism of a Q -complex torus is quasi-étale.

First, let us extend the results in [NZ10, Lemma 2.12] and [NZ09, Proposition 4.3] to the following analytic case. Readers may refer to [Bea83, Proposition 3] for its proof.

Proposition 2.5.1. *Let X be a $\mathbb{Q}$ -complex torus. Then there exists a quasi-étale cover $\tau : T \rightarrow X$ satisfying the following conditions:*

- (1) T is a complex torus;
- (2) τ is Galois;
- (3) If $\tau' : T' \rightarrow X$ is another quasi-étale cover from a torus T' , then there exists an étale morphism $\sigma : T' \rightarrow T$ such that $\tau' = \tau \circ \sigma$.

In particular, we call the quasi-étale cover τ in Proposition 2.5.1 the Albanese closure of X in codimension one.

As a consequence of Proposition 2.5.1, we have the lifting lemma below. Then, Corollary 2.5.3 follows immediately from Proposition 2.4.3 and Lemma 2.5.2.

Lemma 2.5.2. *Let $f : X \rightarrow X$ be a surjective endomorphism of a $\mathbb{Q}$ -complex torus X . Then there exist a complex torus T , a quasi-étale morphism $\tau : T \rightarrow X$ and an étale endomorphism $\sigma_T : T \rightarrow T$ such that $\tau \circ \sigma_T = f \circ \tau$.*

Proof. Note that f is quasi-étale by the ramification divisor formula. Let $\tau : T \rightarrow X$ be the Albanese closure as in Proposition 2.5.1. Since the composition $f \circ \tau$ is still quasi-étale, by Proposition 2.5.1 (3), such σ_T exists. $\square$

Corollary 2.5.3. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a $\mathbb{Q}$ -complex torus. Then, there exists a fixed point of f .*

Comparing with Lemma 2.4.7, we have the same argument for $\mathbb{Q}$ -complex tori. For the application of Lemma 2.5.4, we refer to the proof of Theorem 2.1.5 in Section 2.8.

Lemma 2.5.4. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a $\mathbb{Q}$ -complex torus X . Then, there is no f^{-1} -periodic proper subvariety of X .*

Proof. Let Z be a non-empty f^{-1} -periodic closed subset of X , i.e., $f^{-s}(Z) = Z$ for some $s > 0$. One needs to show that $Z = X$. Since X is a finite quotient of some complex torus T , it has only quotient and hence rational singularities (cf. [KM98, Proposition 5.15]). Also, it follows from the ramification divisor formula that f is quasi-étale. By Lemma 2.5.2, there exists a quasi-étale cover $\tau : T \rightarrow X$ from a complex torus T . Since f is quasi-étale, so is the composition $f \circ \tau$. Then, one gets a surjective endomorphism $\sigma_T : T \rightarrow T$ such that $f \circ \tau = \tau \circ \sigma_T$. By Lemma 2.3.3, σ_T is int-amplified. Further, the commutative diagram gives that $\sigma_T^{-s}(\tau^{-1}(Z)) = \tau^{-1}(Z)$. Since σ_T^s is also int-amplified, $\tau^{-1}(Z) = T$ (cf. Lemma 2.4.7) and hence $Z = X$. $\square$

In the following, we aim to show that each dominant meromorphic map from a normal compact Kähler space with mild singularities to a Q -complex torus is holomorphic. We first give a generalized version of [Uen75, Lemma 9.11] by weakening the condition. Readers are referred to [Kaw85, Lemma 8.1] for the projective case but the proofs are different. To prove Lemma 2.5.5, we need to recall the *Albanese torus* for a singular complex space (cf. [Gra18, Section 3]).

Lemma 2.5.5. *Let $f : X \dashrightarrow T$ be a dominant meromorphic map from a normal compact Kähler space X with only rational singularities to a torus T . Then, f is a morphism.*

Let X be a compact complex space. A holomorphic map $\alpha : X \rightarrow A$ to a complex torus A is said to be the *Albanese map* of X , and A is called the *Albanese torus* of X , if any map from X to a complex torus T factors uniquely through α . Note that the Albanese torus is unique up to isomorphism if it exists. Let $A = \text{Alb}(X)$ and $\alpha = \text{alb}_X$. If X is further assumed to have at worst rational singularities, then X admits an Albanese torus and $\text{Alb}(X) = \text{Alb}(\tilde{X})$, where $\pi : \tilde{X} \rightarrow X$ is a projective desingularization such that $\text{alb}_X \circ \pi = \text{alb}_{\tilde{X}}$ (cf. [Gra18, Theorem 3.4]).

Proof of Lemma 2.5.5. Let $\pi : \tilde{X} \rightarrow X$ be a projective resolution, $\tilde{f} : \tilde{X} \dashrightarrow T$ the induced dominant meromorphic map, $\text{alb}_X : X \rightarrow \text{Alb}(X)$ and $\beta : \text{Alb}(X) \dashrightarrow T$ the induced map (cf. [Gra18, Section 3]). Since $\tilde{f}$ is a dominant meromorphic map from a complex manifold to a torus, by [Uen75, Lemma 9.11], $\tilde{f}$ is a morphism. Moreover,

$\text{Alb}(X) = \text{Alb}(\tilde{X})$ and thus β is a morphism by [Uen75, Definition 9.6 and Theorem 9.7]. Therefore, f is a morphism. $\square$

We end up this section with the following proposition. The proof is the same as in the projective setting (cf. [MZ18a, Lemma 5.3] or [Men20, Lemma 4.4]) except that we apply Lemma 2.5.5 instead of [MZ18a, Lemma 5.1].

Proposition 2.5.6. *Let $\pi : X \dashrightarrow Y$ be a (proper) dominant meromorphic map from a normal compact Kähler space X to a $\mathbb{Q}$ -complex torus Y . Suppose X has only klt singularities and the normalization of the graph $\Gamma_{X/Y}$ is equi-dimensional over Y (this holds when the general fibre of π is connected, X admits an int-amplified endomorphism f , and f descends to some int-amplified endomorphism g on Y). Then, π is a morphism.*

In Proposition 2.5.6, the endomorphisms f and g are to make sure the graph $\Gamma_{X/Y}$ is equi-dimensional over the $\mathbb{Q}$ -complex torus Y , noting that the non equi-dimensional locus is g^{-1} -invariant in Y and hence empty (cf. [Men20, Lemma 4.3] and Lemma 2.5.4).

2.6 K_X pseudo-effective: Proof of Theorem 2.1.4

Proceeding from this section, we will gradually turn to threefolds. In this section, we consider the case when the canonical divisor is pseudo-effective. Now, we are ready to prove Theorem 2.1.4. We first divide the theorem into three parts (cf. Propositions 2.6.1, 2.6.2 and 2.6.5). Then, we put them together to show Theorem 2.1.4.

Proposition 2.6.1. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a compact Kähler manifold of dimension n with pseudo-effective K_X . Then X is a $\mathbb{Q}$ -complex torus.*

Proof. Since $-K_X$ is pseudo-effective by Theorem 2.1.3 and K_X is pseudo-effective by assumption, $K_X \equiv 0$ in the sense of currents. Then, f is étale by the ramification divisor formula and Purity of branch loci of Grauert-Remmert. Fix a Kähler class $[\omega]$ on X .

We claim that $c_2(X) \cdot [\omega]^{n-2} = 0$. Indeed, according to [GK20, Proposition 5.6], $(\deg(f))c_2(X) \cdot [\omega]^{n-2} = c_2(X) \cdot (f^*[\omega])^{n-2} = c_2(X) \cdot f^*([\omega]^{n-2})$. Then,

$$c_2(X) \cdot [\omega]^{n-2} = c_2(X) \cdot \frac{f^*([\omega]^{n-2})}{\deg(f)} = c_2(X) \cdot \frac{(f^i)^*([\omega]^{n-2})}{(\deg(f))^i} =: c_2(X) \cdot [x_i].$$

Here, $\lim_{i \rightarrow \infty} [x_i] \equiv_w 0$ by Lemma 2.3.4. Moreover, applying Lemma 2.4.4 for both $c_2(X)$ and $-c_2(X)$, there exist two positive closed $(2, 2)$ -currents T_1 and T_2 and a positive number A such that $c_2(X) + T_1 = A[\omega]^2$ and $-c_2(X) + T_2 = A[\omega]^2$ in the sense of currents. Then, we get the inequalities: $\overline{\lim}_{i \rightarrow \infty} c_2(X) \cdot [x_i] \leq 0$ and $\overline{\lim}_{i \rightarrow \infty} -c_2(X) \cdot [x_i] \leq 0$, which in turn force $c_2(X) \cdot [x_i]$ tends to 0 when $i \rightarrow \infty$. So we complete the proof of our claim.

Note that the vanishing of the first Chern class of X implies that we can find a Ricci flat metric (still denoted ω) in the Kähler class $[\omega]$ (cf. [Yau78]). The vanishing $\int_X c_2(X) \wedge \omega^{n-2} = 0$ implies that the full curvature tensor of ω is identically zero. By the uniformisation theorem, the universal cover of X is an affine space and X is a quotient of a complex torus T by a finite group G acting freely on T . $\square$

Proposition 2.6.2. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler threefold with at worst canonical singularities. Suppose that the canonical divisor K_X is pseudo-effective. Then X is a Q -complex torus.*

We refer the readers to [NZ10, Theorem 3.3] and [Men20, Theorem 5.2] for the projective version while our present proof is quite different.

Proof. To prove Proposition 2.6.2, we resort to [GK20, Theorem 1.1] by showing that $c_1(X) = 0 \in H^2(X, \mathbb{R})$ and $\tilde{c}_2(X) \cdot [\omega] = 0$ for some Kähler class $[\omega] \in H^2(X, \mathbb{R})$. Here, $c_1(X)$ is the first Chern class and $\tilde{c}_2(X) \in H^2(X, \mathbb{R})^\vee$ is the orbifold second Chern class of X (cf. [GK20, Section 5]). Since $-K_X$ is pseudo-effective by Theorem 2.1.3 and K_X is pseudo-effective by assumption, $K_X \equiv 0$ in the sense of currents, i.e., $c_1(X)_{\mathbb{R}} = 0$. Therefore, the ramification divisor $R_f = 0$ and f is thus quasi-étale. Let $d := \deg(f) > 1$ and we fix a Kähler class $[\omega]$ on X .

Now we prove that $\tilde{c}_2(X) \cdot [\omega] = 0$. Since f is quasi-étale, by [GK20, Proposition 5.6], we get $\tilde{c}_2(X) \cdot f^*[\omega] = d(\tilde{c}_2(X) \cdot [\omega])$. Then with the equality divided by d , we have

$$\tilde{c}_2(X) \cdot [\omega] = \tilde{c}_2(X) \cdot \frac{f^*[\omega]}{d} = \tilde{c}_2(X) \cdot \frac{(f^i)^*[\omega]}{d^i} =: \tilde{c}_2(X) \cdot [x_i],$$

where $\lim_{i \rightarrow \infty} [x_i] \equiv_w 0$ by Lemma 2.3.4. Note that $\tilde{c}_2(X) \in H^2(X, \mathbb{R})^\vee$ and we can regard it as an element in the dual of the Bott-Chern cohomology space and hence $\tilde{c}_2(X) \in N_1(X)$

(cf. [HP16, Proposition 3.9]). Besides, for any normal compact Kähler threefold X , the nef cone $\overline{\mathcal{K}}(X)$ and the cone $\overline{\text{NA}}(X)$ generated by positive closed currents of bidimension $(1, 1)$ are dual (cf. [HP16, Proposition 3.15]). Therefore, $[\omega]^2$ is an interior point of $\overline{\text{NA}}(X)$ because as a linear function on $H_{\text{BC}}^{1,1}(X)$, $[\omega]^2$ is strictly positive on $\overline{\mathcal{K}}(X) \setminus \{0\}$ by Hodge-Riemann theorem (also cf. [DS04, Théorème 3.1]). Therefore, there exists $A > 0$ such that $A[\omega]^2 \geq \tilde{c}_2(X)$ and $A[\omega]^2 \geq -\tilde{c}_2(X)$ in the sense of currents. Hence, with the same argument as in Proof of Proposition 2.6.1, $\tilde{c}_2(X) \cdot [\omega] = \lim_{i \rightarrow \infty} \tilde{c}_2(X) \cdot [x_i] = 0$.

Then, by the criterion in [GK20, Theorem 1.1], our X is a $\mathbb{Q}$ -complex torus. $\square$

Lemma 2.6.3. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler surface with at worst klt singularities. Suppose that the canonical divisor K_X is pseudo-effective. Then X is a $\mathbb{Q}$ -complex torus.*

Proof. With the same arguments as in the previous proof (cf. Proposition 2.6.2), $K_X \equiv 0$ in the sense of currents, i.e., $c_1(X)_{\mathbb{R}} = 0$. Therefore, it follows from the ramification divisor formula that f is quasi-étale. Also, $\tilde{c}_2(X) = 0$ according to [GK20, Proposition 5.6]. Hence, X is a $\mathbb{Q}$ -complex torus (cf. [GK20, Proposition 7.2]). $\square$

Lemma 2.6.4. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler non-uniruled surface with K_X $\mathbb{Q}$ -Cartier. Then X is a $\mathbb{Q}$ -complex torus.*

Proof. By Theorem 2.1.3, $-K_X$ is pseudo-effective. Let $\pi : \tilde{X} \rightarrow X$ be a projective resolution. Since X and hence $\tilde{X}$ are non-uniruled Kähler surfaces, $K_{\tilde{X}}$ is pseudo-effective by the classification of surfaces. Further, the equality $\pi_* K_{\tilde{X}} = K_X$ gives that K_X is also pseudo-effective. Therefore, $K_X \equiv 0$.

Since $\dim(X) = 2$, with the same proof as in [HMZ14, Lemma 2.4], X has only canonical singularities. Then Lemma 2.6.3 gives us that X is a $\mathbb{Q}$ -complex torus. $\square$

Proposition 2.6.5. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler surface X such that the canonical divisor K_X is pseudo-effective. Then X is a $\mathbb{Q}$ -complex torus.*

Proof. Since K_X is pseudo-effective, with the same reason as in previous proofs, $K_X \equiv 0$. Then, it follows from [Fuj21, Theorem 1.2] that K_X is semi-ample and $K_X \sim_{\mathbb{Q}} 0$. If

X is non-uniruled, this is the case in Lemma 2.6.4. Therefore, we consider the case of X being uniruled. Suppose X is a uniruled surface. Then, X is bimeromorphic to a ruled surface which is algebraic. Thus, X is Moishezon by [AG06, Theorem 7.14]. Taking the global index-one log-canonical cover $\tilde{X} \rightarrow X$ associated with $K_X \sim_{\mathbb{Q}} 0$, we have the following results (cf. [Nak08, Lemma 3.2.7] or [Nak20b, Lemma 4.21] and Lemma 2.3.3): (1) $\tilde{X}$ is a normal Moishezon and Kähler surface; (2) $\tilde{X} \rightarrow X$ is quasi-étale; (3) $\tilde{X}$ has only rational double points; (4) $K_{\tilde{X}} \sim 0$; and (5) f lifts to an int-amplified endomorphism $\tilde{f}$ of $\tilde{X}$. Then it follows from [Nak08, Proposition 7.3.1] or [Nak20a, Proposition 5.2] that $\tilde{X}$ and hence X are $\mathbb{Q}$ -complex tori (cf. [GK20, Lemma 7.4]). $\square$

Proof of Theorem 2.1.4. It is a direct consequence of Propositions 2.6.1, 2.6.2 and 2.6.5, which together prove the cases: manifolds, threefolds and surfaces respectively. $\square$

It is natural to consider whether Theorem 2.1.4 can be generalized to arbitrary dimensional spaces. For the normal projective setting, the answer is affirmative (cf. [Men20, Theorem 5.2]). In our present context, it is still an open question (cf. Question 1.8.2).

We end up this section by assuming the following hypothesis when discussing the minimal model program for the case when K_X is pseudo-effective (cf. [HP16]).

Hyp A: X is a normal $\mathbb{Q}$ -factorial compact Kähler threefold with only terminal singularities. Also, the canonical divisor K_X is pseudo-effective.

2.7 K_X not pseudo-effective

In this short section, let X be a normal compact Kähler threefold with only canonical singularities. By a theorem of Brunella [Bru06] and [HP18, Theorem 4.2], the canonical divisor K_X is not pseudo-effective if and only if X is covered by rational curves. We shall deal with the uniruled case in the following. Let $\pi : \tilde{X} \rightarrow X$ be a projective resolution and $\tilde{X} \dashrightarrow Y$ the maximal rationally connected (MRC) fibration.

If $\dim(Y) = 0$, then $\tilde{X}$ is algebraic (cf. [Cam81]). We suppose that $\dim(Y) = 1$. Then, a general fibre F of the MRC fibration is smooth and rationally connected of dimension two in $\tilde{X}$. If $H^0(\tilde{X}, \Omega_{\tilde{X}}^2) \neq 0$, then there exists a global 2-form η on $\tilde{X}$ and hence

we get a non-zero holomorphic form on some rationally connected fibre F , a contradiction (cf. [Deb01, Corollary 4.18]). So, applying Hodge theory, we have $H^2(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$ and also $H^2(\tilde{X}, \mathbb{C}) = H^{1,1}(\tilde{X}, \mathbb{C})$. By [Gra18, Proposition 4.10], $\tilde{X}$ is projective and thus X is Moishezon (cf. [AG06, Theorem 7.14]).

In both cases, X is Moishezon and also projective by [Nam02, Theorem 1.6]. In conclusion, together with [Men20, Theorem 1.10], we get the following lemma.

Lemma 2.7.1. *Let X be a compact Kähler threefold with at worst canonical singularities. Suppose X is uniruled and the dimension of the base Y of the MRC fibration $\pi : X \dashrightarrow Y$ is at most 1. Then X is projective. In particular, if X further admits an int-amplified endomorphism, then there exists an f -equivariant MMP for X after iteration.*

From now on, when discussing the MMP for normal compact Kähler uniruled threefolds, we always assume the following hypothesis (cf. [HP15]).

Hyp B: X is a normal $\mathbb{Q}$ -factorial compact Kähler threefold with only terminal singularities. The canonical divisor K_X is not pseudo-effective and the base Y of the MRC fibration $\pi : X \dashrightarrow Y$ has dimension two.

2.8 Proof of Theorem 2.1.5

In this section, we always assume **Hyp A** in Section 2.6 or **Hyp B** in Section 2.7 when discussing the MMP for Kähler threefolds. Before proving Theorem 2.1.5, we refer to [HP16, Definitions 3.19, 4.3 and 7.1] for the basic notation and related properties of contraction morphisms, divisorial rays and small rays. Indeed, all of them are similar to those in projective settings. By [HP16, Remark 7.2], if the extremal ray R is small, then the corresponding Mori contraction contracts only finitely many curves. Moreover, [HP16] shows that if we assume **Hyp A** in Section 2.6 or **Hyp B** in Section 2.7, then after a contraction of either divisorial or small rays, the end product is still a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at worst terminal singularities. In addition, under the same assumption, [HP15] shows that if $\pi : X \rightarrow Y$ is a Mori fibre space, then Y is a normal $\mathbb{Q}$ -factorial compact Kähler surface with at worst klt singularities.

In what follows, we generalize [Zha10, Lemma 2.11] to the following lemma. We shall skip the proofs since they are the same as the proofs in [Zha10, Lemma 2.11] by applying Proposition 2.2.8 and [HP16, Proposition 3.9] to the present case.

Lemma 2.8.1. *Let X be a normal compact Kähler space with at worst rational singularities and $f : X \rightarrow X$ a surjective endomorphism. Let $R_\Gamma := \mathbb{R}_{\geq 0}[\Gamma] \subseteq \overline{\text{NA}}(X)$ (which is the closed cone in $N_1(X)$ generated by the classes of positive closed currents of bidimension $(1, 1)$) be an extremal ray (not necessarily K_X -negative) with Γ a positive closed current of bidimension $(1, 1)$. Then, we have:*

- (1) $R_{f_*\Gamma}$ is an extremal ray;
- (2) If C_1 is a curve such that $[f(C_1)] \in R_\Gamma$, then $R_{[C_1]}$ is an extremal ray;
- (3) Denote by Σ_Γ the set of curves whose classes are in R_Γ . Then $f(\Sigma_\Gamma) = \Sigma_{f_*\Gamma}$.
- (4) $\Sigma_\Gamma = f^{-1}(\Sigma_{f_*\Gamma}) := \{C \text{ a curve} \mid f(C) \in \Sigma_{f_*\Gamma}\}$.

The result below is known in the projective case (cf. [Men20, Lemma 8.1]) and we show that it also holds in the analytic setting. We rewrite the proof in the analytic version by highlighting the differences and skipping the same parts. The main difference is that, here, we may not have integral ample divisors in the analytic setting. Nevertheless, since every analytic subvariety outside the singular locus determines an integral homology class, the idea of the early proof is valid in the present case.

Lemma 2.8.2. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal compact Kähler space, which is of dimension $n \geq 1$ and has at worst rational singularities. Suppose $A \subseteq X$ is a closed subvariety with $f^{-i}f^i(A) = A$ for all $i \geq 0$. Then $M(A) := \{f^i(A) \mid i \in \mathbb{Z}\}$ is a finite set.*

Proof. Let $M_{\geq 0}(A) := \{f^i(A) \mid i \geq 0\}$. With the same reason as in the first step of the proof in [Men20, Lemma 8.1], we only need to prove $M_{\geq 0}(A)$ is finite.

We show that $M_{\geq 0}(A)$ is a finite set by induction on the codimension of A in X . If $A = X$, there is nothing to prove. Suppose that $k := \dim(A) < \dim(X)$ and $d :=$

$\deg(f) > 1$. Denote by Σ the union of $\text{Sing}(X)$, $f^{-1}(\text{Sing}(X))$ and the irreducible components in the ramification divisor R_f of f . Let $A_i := f^i(A)$ ($i \geq 0$). Note that $\dim(A_i) = \dim(A) = k$.

We claim that A_i is contained in Σ for infinitely many i . If not, with A replaced by some A_{i_0} , we may assume that A_i is not contained in Σ for all $i \geq 0$. Since $f^{-i-1}(A_{i+1}) = A$ and f^i is surjective, we have $f^{-1}(A_{i+1}) = A_i$. Besides, f is unramified on each A_i , and hence $\deg(f|_{A_i}) = \deg(f)$. Fix a Kähler form ξ on X . Then, by projection formula, $(\deg(f))(\xi^k \cdot [A_{i+1}]) = (\deg(f|_{A_i}))(\xi^k \cdot [A_{i+1}]) = (f^*\xi)^k \cdot [A_i]$. Substituting this expression repeatedly, we have

$$\xi^k \cdot [A_{i+1}] = \frac{((f^i)^*\xi)^k \cdot [A_1]}{d^i}. \quad (2.3)$$

On the one hand, all of these A_i are integral subvarieties of dimension k and not contained in the singular locus by assumption. Also, the homology classes of these $A_i \subseteq X$ form a discrete set (denoted by $\text{NE}_k(X)$), which is contained in the integral lattice $H_{2k}(X, \mathbb{Z})/\text{tors}$. In addition, ξ^k defines a norm on the closure of $\text{NE}_k(X)$ and ξ^k is strictly positive on $\overline{\text{NE}}_k(X) \setminus \{0\}$. **Thus, there exists a real constant $\mu > 0$ such that $\xi^k \cdot [A_i] \geq \mu$ for every analytic subvariety A_i of dimension k , which is not contained in the singular locus.** As a result, the LHS of Equation (2.3) is always no less than μ .

On the other hand, by Lemma 2.4.5, $\lim_{i \rightarrow \infty} \xi^k \cdot [A_{i+1}] = 0$. Taking the upper limit of Equation (2.3), we get a contradiction and prove the claim. The remaining proof is the same as [Men20, Lemma 8.1] after replacing [Men20, Lemma 2.3] by Lemma 2.4.1. $\square$

Let $f : X \rightarrow X$ be a surjective endomorphism of a compact Kähler manifold X . Suppose $\pi : X \dashrightarrow Y$ is a meromorphic map to a complex variety. In [HP11, Propositions 2.9 and 2.12], Höring and Peternell gave us nice criteria for f to descend as a (meromorphic) endomorphism of Y in terms of Albanese map, Iitaka fibration, rationally connected quotient or algebraic reduction.

Now, it is natural to ask whether there exists an equivariant descending for the contraction morphism of a K_X -negative extremal ray. Actually, it has an affirmative answer under the assumption of **Hyp A** in Section 2.6 or **Hyp B** in Section 2.7 (cf. Lemma 2.8.3).

The proof is the same as in [MZ18a, Lemma 6.2] for the projective setting after replacing [MZ18a, Lemma 2.13] by Lemma 2.8.1 and replacing [KM98, Corollary 3.17] by [HP16, Proposition 8.1].

Lemma 2.8.3. *Suppose either **Hyp A** in Section 2.6 or **Hyp B** in Section 2.7. Let f be a surjective endomorphism of X and $\pi : X \rightarrow Y$ a contraction of a K_X -negative extremal ray $R_\Gamma := \mathbb{R}_{\geq 0}[\Gamma]$ generated by a positive closed $(n-1, n-1)$ -current Γ . Suppose further that $E \subseteq X$ is an analytic subvariety such that $\dim(\pi(E)) < \dim(E)$ and $f^{-1}(E) = E$. Then, up to replacing f by its power, $f(R_\Gamma) = R_\Gamma$ (hence, for any curve C such that $[C] \in R_\Gamma$, its image $[f(C)]$ still lies in R_Γ). Thus, the contraction π is f -equivariant, i.e., f descends to a surjective endomorphism g on Y .*

Lemma 2.8.3 is also valid for any Kähler surface with only klt singularities. Besides, Lemma 2.8.3 can be applied when π is a Mori fibre space (with $E = X$): in that case, π is f^s -equivariant for some $s > 0$.

Proposition 2.8.4. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler threefold X with at worst terminal singularities. If $\pi : X \rightarrow Y$ is a divisorial contraction of a K_X -negative extremal ray $R_\Gamma := \mathbb{R}_{\geq 0}[\Gamma]$ generated by a positive closed $(n-1, n-1)$ -current Γ . Then f^s induces an int-amplified endomorphism of Y for some $s > 0$.*

Proof. Let $S := \bigcup_{C \subseteq X, [C] \in R_\Gamma} C$. By [HP16, Lemma 7.5], S is an irreducible Moishezon surface and $S \cdot C < 0$ for all curves C with $[C] \in R_\Gamma$. By Lemma 2.8.1, $f^{-i}f^i(S) = S$ for all $i \geq 0$, since f^i is surjective. Then, it follows from Lemma 2.8.2 that $M_{\geq 0}(S)$ is a finite set. So, we may assume $f^{-1}(S) = S$ with f replaced by its power. Also, π is f^s -equivariant for some $s > 0$ (cf. Lemma 2.8.3). By Lemma 2.3.2, the induced endomorphism on Y is int-amplified. $\square$

Indeed, without Lemma 2.8.2, the finiteness of the number of such S in Proposition 2.8.4 also comes from the fact that these Moishezon surfaces consist of the integral part of the Zariski decomposition for K_X (cf. [HP16, Proof of Lemma 7.5]).

The following result is the analytic version of [Zha10, Lemma 3.6] (cf. [Mor88, (0.4.1)] for the existence of flips of threefolds), the proof of which is the same since each contraction of compact Kähler threefolds is projective ($-K_X$ is π -ample).

Lemma 2.8.5. *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler threefold X with only terminal singularities and $X \dashrightarrow X'$ a flip with $X \rightarrow Y$ the corresponding flipping contraction of a K_X -negative extremal ray $R_\Gamma := \mathbb{R}_{\geq 0}[\Gamma]$ generated by some positive closed $(n-1, n-1)$ -current Γ . Suppose $R_{f_*\Gamma} = R_\Gamma$. Then the dominant meromorphic map $f^+ : X^+ \dashrightarrow X^+$ induced from f , is holomorphic. Both f and f^+ descend to the same endomorphism of Y .*

Proposition 2.8.6. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler threefold X with at worst terminal singularities and $\sigma : X \dashrightarrow X'$ a flip with $\pi : X \rightarrow Y$ the corresponding flipping contraction of a K_X -negative extremal ray $R_\Gamma := \mathbb{R}_{\geq 0}[\Gamma]$ generated by some positive closed current Γ of bidimension $(1, 1)$. Then there exists a K_X -flip of π , $\pi^+ : X^+ \rightarrow Y$ such that $\pi = \pi^+ \circ \sigma$ and for some $s > 0$, the commutativity is f^s -equivariant.*

Proof. The proof is the same as [MZ18a, Lemma 6.5] after replacing [MZ18a, Lemmas 2.13, 6.1, 6.2 and 6.6] by Lemmas 2.8.1, 2.8.2, 2.8.3 and 2.8.5. Also, Lemmas 2.3.2 and 2.3.3 show that the induced f^+ in Lemma 2.8.5 is int-amplified. $\square$

The next Theorem follows from Lemmas 2.8.3, 2.8.5 and Propositions 2.8.4, 2.8.6.

Theorem 2.8.7. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a $\mathbb{Q}$ -factorial compact Kähler terminal threefold X . Let $\pi : X \dashrightarrow Y$ be a dominant rational map which is either a divisorial contraction or a Fano contraction or a flip induced by a K_X -negative extremal ray R_Γ . Then there exists an int-amplified $g : Y \rightarrow Y$ such that $g \circ \pi = \pi \circ f$ with f replaced by its power.*

Proof of Theorem 2.1.5. First, if K_X is pseudo-effective, then $X = Y$ is a Q -complex torus by Proposition 2.6.2. Next, we consider the case when K_X is not pseudo-effective.

If the base of the MRC fibration $X \dashrightarrow Z$ has dimension ≤ 1 , then Theorem 2.1.5 follows from Lemma 2.7.1 and [Men20, Theorem 1.10], since X is projective.

Now, we assume **Hyp B** in Section 2.7. By [HP15, Theorem 1.1], we may run the MMP for finitely many steps: $X = X_1 \dashrightarrow \cdots \dashrightarrow X_j$ consisting of contractions of extremal rays and flips, such that X_j admits a Mori fibre space $X_j \rightarrow S = X_{j+1}$ onto a normal $\mathbb{Q}$ -factorial compact Kähler surface. Note that S is klt (cf. [HP15, Theorem 1.1]). Note also that S is bimeromorphic to the base of the MRC fibration of X ; hence S is non-uniruled and K_S is pseudo-effective by considering a resolution of S . By Proposition 2.6.5, S is a Q -complex torus, which is our end product. Replacing f by its power, the above sequence is f -equivariant by Theorem 2.8.7.

By Proposition 2.5.6, the composition $u_i : X_i \dashrightarrow Y$ is a morphism for each i . Also, for each small contraction $\pi_i : X_i \rightarrow Z_i$, we claim that **the meromorphic map $Z_i \dashrightarrow Y$ is also a morphism**. In the following, let $\text{Exc}(\bullet)$ denote the exceptional locus of $\bullet$. If the claim does not hold, then by the rigidity lemma, there exists a curve C on X_i with $[C] \in R$ such that $u_i(C)$ is a curve on Y . Here, R is a K_{X_i} -negative small ray and π_i is the small contraction of R . Recall that, under **Hyp B**, Y is a Q -complex torus. By Lemmas 2.8.1 and 2.8.2, after iteration, $f^{-1}(\text{Exc}(\pi_i)) = \text{Exc}(\pi_i)$. Since $\dim(\text{Exc}(\pi_i)) = 1$, with f further replaced by its power, $u_i(C)$ is totally invariant under the induced int-amplified endomorphism $Y \rightarrow Y$ (cf. [CMZ20, Lemma 7.5]), a contradiction to Lemma 2.5.4. Thus, $Z_i \rightarrow Y$ is also a morphism and the MMP is over Y .

By [MZ18a, Lemmas 2.6 and 5.2], $X_i \rightarrow Y$ is equi-dimensional with every fibre being irreducible and rationally connected. Also, K_{X_i} is not pseudo-effective for any $i < r$, otherwise X_i is a Q -complex torus for some $i < r$ and the MMP ends. Therefore, we complete the proof of Theorem 2.1.5. $\square$

We end up this chapter with the following remark as an application of Theorem 2.1.5.

Remark 2.8.8. If we start from a simply connected smooth compact Kähler threefold X and assume that such X admits an int-amplified endomorphism, then applying our Theorem 2.1.5, we can show that our X is projective and also rationally connected (cf. [Men20, Theorem 1.10], [CMZ20, Lemmas 7.4, 7.5], Lemma 2.5.4, and [MY19, Theorem 4.4]).

Compact Kähler threefolds admitting groups of automorphisms

3.1 Introduction

In this chapter, we work over the field $\mathbb{C}$ of complex numbers. Let X be a compact Kähler manifold of dimension n . For an automorphism $g \in \text{Aut}(X)$, its *topological entropy* $h(g)$, defined in the theory of dynamical systems, turns out to coincide with the logarithm of the spectral radius $\rho(\varphi)$ of the pull-back operator $\varphi := g^*$ acting on $\oplus_{0 \leq p \leq n} H^{p,p}(X, \mathbb{R})$ (cf. [Gro03], [Yom87] and Remark 1.2.1). We say that $g \in \text{Aut}(X)$ is of *positive entropy*, if the topological entropy is strictly positive, or equivalently, the spectral radius of g^* acting on $H^{p,p}(X, \mathbb{R})$ is strictly larger than 1 for some (and hence all) p with $1 \leq p \leq n - 1$. We say that $g \in \text{Aut}(X)$ is of *null entropy* if g is not of positive entropy. See the survey [Din12] and the references therein.

T.-C. Dinh and N. Sibony proved in [DS04] that every commutative subgroup $G \subseteq \text{Aut}(X)$ has the dynamical rank $\leq n - 1$, if all the non-trivial elements of G are of positive entropy. Subsequently, D.-Q. Zhang proved a theorem of Tits type for compact Kähler manifolds (cf. [Zha09, Theorem 1.1]), extending the classical Tits alternative, and also extending [DS04] to the solvable case.

Problem 3.1.1 below was first stated in [DS04] (cf. [Din12, Problem 1.5]). The interest

of studying these X when G has maximal rank is also the initial point of this chapter.

Problem 3.1.1. Classify compact Kähler manifolds X of dimension $n \geq 3$ admitting a free abelian group G of automorphisms of rank $n - 1$ which is of positive entropy.

In the singular setting, we consider a normal projective variety (resp. a compact Kähler space with at worst rational singularities) X and define the *first dynamical degree* $d_1(f)$ of an automorphism $f \in \text{Aut}(X)$ as the spectral radius $\rho(f^*)$ of its natural pull-back f^* on the Néron-Severi group $N^1(X)$ (resp. the Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$). We say that f is of *positive entropy* if $d_1(f) > 1$, otherwise it is of *null entropy*. Note that our definition here for $d_1(f)$ in the singular setting coincides with the usual one when X is smooth:

Remark 3.1.2. (1). For an n -dimensional normal projective variety X with an automorphism $f \in \text{Aut}(X)$, we take the f -equivariant resolution $\pi : \tilde{X} \rightarrow X$ (over a field of characteristic zero; cf. e.g. [Wlo09, Theorem 2.0.1]) and denote by $\tilde{f}$ the lifting of f to $\tilde{X}$. Taking an ample divisor H on X , we have

$$\begin{aligned} d_1(f) &= \rho(f^*|_{N^1(X)}) = \lim_{m \rightarrow \infty} ((f^m)^* H \cdot H^{n-1})^{1/m} \\ &= \lim_{m \rightarrow \infty} (\pi^*(f^m)^* H \cdot (\pi^* H)^{n-1})^{1/m} \\ &= \lim_{m \rightarrow \infty} ((\tilde{f}^m)^* \pi^* H \cdot (\pi^* H)^{n-1})^{1/m} = \rho(\tilde{f}^*|_{N^1(\tilde{X})}) = d_1(\tilde{f}). \end{aligned}$$

Here, the second and the fifth equalities are due to [NZ09, Proposition A.2], noting that $\pi^* H$ is a nef and big divisor on $\tilde{X}$ and lies in the interior of the pseudo-effective cone which spans $N^1(\tilde{X})$.

(2). For a compact Kähler space X with at worst rational singularities, if $f \in \text{Aut}(X)$ is an automorphism, then similarly, we take the f -equivariant resolution $\tilde{X} \rightarrow X$ and lift f to $\tilde{f}$. Replacing $N^1(X)$ in (1) by $H_{\text{BC}}^{1,1}(X)$ and replacing the ample divisor H in (1) by a Kähler class $[\omega]$ on X , we see that $d_1(f)$ coincides with $d_1(\tilde{f})$ by applying Remark 2.2.3, Propositions 2.2.6 and 2.2.8.

In the past decade, D.-Q. Zhang established the G -equivariant minimal model program for projective varieties, and Problem 3.1.1 has thus been intensively studied in his series of

papers (cf. [Zha09], [Zha13] and [Zha16]). Quite recently, F. Hu and S. Li characterized the sub-maximal dynamical rank case for projective varieties in [HL20]. Inspired by these previous works, we are interested in Problem 3.1.1 itself (without assuming the projectivity) and ask the following question.

Question 3.1.3. Let X be a normal compact Kähler space with mild singularities. Suppose that $\text{Aut}(X) \supseteq G := \mathbb{Z}^{\oplus r}$ with $r = \dim(X) - 1$ (maximal), and every non-trivial element of G is of positive entropy. Is X either rationally connected, or G -equivariantly bimeromorphic to a $\mathbb{Q}$ -complex torus?

Recall that a normal compact Kähler space X is said to be a **$\mathbb{Q}$ -complex torus**, if there exists a complex torus T and a finite surjective morphism $\pi : T \rightarrow X$ such that π is étale in codimension 1 (cf. e.g., [NZ10, Definition 2.13]).

Question 3.1.3 is related to the building blocks of the commutative subgroup G on Kähler spaces. For example, we don't know whether the automorphisms of Calabi-Yau manifolds would appear as a fundamental building block of G in the general non-algebraic situation. Question 3.1.3 also seems to have its own interest from the viewpoints of differential geometry, since the minimal model program (**MMP** for short) for Kähler spaces is less known in higher dimensions (cf. [HP15], [HP16] and [DH20] for Kähler 3-folds).

In this chapter, we study Problem 3.1.1 and answer Question 3.1.3 affirmatively for Kähler 3-folds, with the main result Theorem 3.1.4. Let us consider the hypothesis:

Hypothesis A: X is a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at worst terminal singularities; $\text{Aut}(X) \supseteq G := \mathbb{Z}^r$ with $r = \dim(X) - 1 = 2$ (maximal) and every non-trivial element of G is of positive entropy.

Theorem 3.1.4 ([Zho21a, Theorem 1.3]; cf. Remark 3.3.2). *Suppose that (X, G) satisfies Hypothesis A. Then the following assertions hold:*

- (1) *If K_X is not pseudo-effective, then X is rationally connected. In particular, X is a projective threefold.*
- (2) *If K_X is pseudo-effective, then the Kodaira dimension $\kappa(X) = 0$ and X is bimeromorphic to $X_{\min} := T/H$ for a finite group H acting freely outside a finite set of*

a complex torus T . Moreover, G descends to an automorphism subgroup $G|_{X_{\min}} \subseteq \text{Aut}(X_{\min})$ and further lifts to $G_T \subseteq \text{Aut}(T)$ such that $G|_{X_{\min}} \cong G_T/H$.

To prove Theorem 3.1.4, the main obstacle is to consider the key case $\kappa(X) = 0$ (cf. Proposition 3.2.11). Now, we briefly describe our strategy.

3.1.5 (Strategy for the proof of Theorem 3.1.4 when $\kappa(X) = 0$). First, we reduce X to its minimal model $X_{\min}$ equipped with the induced action $G|_{X_{\min}}$ (of the same rank) by bimeromorphic transformations. Note that elements in $G|_{X_{\min}}$ may not be biholomorphic.

Second, we study the Albanese closure $X'_{\min}$ of $X_{\min}$ (cf. Definition 3.2.5) and classify $X'_{\min}$ (cf. Proposition 3.2.4). Meanwhile, we lift $G|_{X_{\min}}$ to its Albanese closure $X'_{\min}$ and get a subgroup $G|_{X'_{\min}} \subseteq \text{Bim}(X'_{\min})$ (of the same rank) on $X'_{\min}$.

Finally, we lift $G|_{X'_{\min}}$ to its splitting cover $E \times F$, where E is a torus and F is a weak Calabi-Yau space (cf. Definition 3.2.2). We deduce that the group $G_{E \times F}$ to which G lifts actually consists of biholomorphic transformations (in $\text{Aut}(E \times F)$). Then, we show that there exists a finite index subgroup $G_1 \subseteq G_{E \times F}$ such that $(E \times F, G_1)$ satisfies Hypothesis A. So we can conclude our theorem by applying [Zha09, Lemma 2.10] to $E \times F \rightarrow E$.

The following special case is the key to proving Theorem 3.1.4 when X is minimal.

Proposition 3.1.6. *Suppose that (X, G) satisfies Hypothesis A. Suppose further that X is minimal, i.e., the canonical divisor K_X is nef. Then, X is a quasi-étale quotient of a complex torus T . Furthermore, the group G lifts to $G_T \subseteq \text{Aut}(T)$ on T .*

Remark 3.1.7. Initially, we tried to run the G -equivariant minimal model program (G -MMP for short) for X as in [Zha16], but we met lots of problems about the finiteness of $(K_X + \xi)$ -negative extremal rays. Here, $\xi = \sum \xi_i$ is the sum of common nef eigenvectors of G (cf. [DS04, Theorems 4.3 and 4.7]). Besides, for a (not necessarily rational) nef and big class η on a compact Kähler manifold, little was known about Kodaira's Lemma (cf. e.g., [KM98, Lemma 2.60]) for η and the classical base-point-free theorem (cf. [DH20, Conjecture 1.6 and Theorem 1.7]). Due to these difficulties mentioned, we will not (and cannot) run the G -MMP for X .

We refer to [DS04, Example 4.5] for concrete examples of complex tori with respect to Theorem 3.1.4. There is also a 3-dimensional $\mathbb{Q}$ -complex torus X being rational, which admits $\mathbb{Z}^2 \cong G \subseteq \text{Aut}(X)$ of positive entropy, though in this case, such X has worse singularities (cf. e.g., [Zha16, Example 1.6]).

In comparison, we pose the following question at the end of this section.

Question 3.1.8. Does there exist a pair (X, G) satisfying **Hypothesis A** such that X is not a $\mathbb{Q}$ -complex torus even after bimeromorphic change of models?

This chapter is based on my work [Zho21a].

3.2 Preliminaries, Proof of Proposition 3.1.6

Let X be a normal compact Kähler space. We refer to [KM98, Chapter 2] for different kinds of singularities. Let $H_{\text{BC}}^{1,1}(X)$ be the *Bott-Chern cohomology* (cf. [BEG13, Definition 4.6.2]) and $\mathbf{N}_1(X)$ the space of real closed currents of bidimension $(1, 1)$ modulo the equivalence relation: $T_1 \equiv T_2$ if and only if $T_1(\eta) = T_2(\eta)$ for all real closed $(1, 1)$ -forms η with local potentials (cf. [HP16, Section 2]).

Let $f : X \rightarrow Y$ be a (proper) surjective morphism (i.e., a holomorphic map) between normal compact complex spaces. Then f is said to be *finite* (resp. *generically finite*) if f has discrete fibres (resp. is finite outside a nowhere dense analytic closed subspace of Y). We say that f is *quasi-étale* if f is finite and étale in codimension one. Denote by $\mathbf{Bim}(X)$ (resp. $\mathbf{Aut}(X)$) the group of bimeromorphic (resp. biholomorphic) self-maps of X . Also, $\mathbf{Bim}(X)$ (resp. $\mathbf{Aut}(X)$) is called the *bimeromorphic transformation* (resp. *analytic automorphism*) group of X .

We shall apply the following theorem in Section 3.3 and recall it here for the convenience of readers. Note that the original proof (in the projective setting) still works in our present case for compact Kähler (terminal) threefolds.

Theorem 3.2.1 ([Kaw08, Theorem 1]). *Let $\tau : X \dashrightarrow X'$ be a bimeromorphic map between compact Kähler threefolds with at worst terminal singularities. Suppose X and X'*

are both minimal, i.e., the canonical divisors K_X and $K_{X'}$ are nef. Then τ is isomorphic in codimension one.

Let X be a normal compact Kähler space. We define the *augmented irregularity* $q^\circ(X)$ of X to be the supremum of the irregularities $q(X')$ when X' runs over the finite quasi-étale covers $X' \rightarrow X$ (cf. [NZ09, Section 4] and [Gra18, Definition 2.4]). Here, the irregularity $q(X)$ is defined to be $q(\tilde{X}') = h^1(\tilde{X}', \mathcal{O}_{\tilde{X}'})$, where $\tilde{X}' \rightarrow X'$ is a resolution. In general, the augmented irregularity can be infinite. For example, one can consider any finite étale cover of genus ≥ 2 curves and apply the Riemann-Hurwitz formula.

Definition 3.2.2 (cf. [NZ10, Definition 2.9]). A compact Kähler space X is said to be a *weak Calabi-Yau space* if X has only canonical singularities, $K_X \sim_{\mathbb{Q}} 0$ and $q^\circ(X) = 0$.

We are interested in the compact Kähler threefolds with only canonical singularities and vanishing first Chern class (e.g., weak Calabi-Yau threefolds). The following theorem, which generalizes [Kaw85, Theorem 8.3] from the projective case, gives a nice structure of the Albanese map (cf. [CGGN20, Theorem 4.1] for a further extension to any dimensional weak Calabi-Yau space):

Theorem 3.2.3 ([Gra18, Theorem 1.10]). *Let X be a normal compact Kähler threefold with only canonical singularities and vanishing first Chern class $c_1(X) = 0$. Let $\text{alb}_X : X \rightarrow A := \text{Alb}(X)$ be the Albanese map. Then there exists a finite étale cover $A_1 \rightarrow A$ such that $X \times_A A_1$ is isomorphic to $F \times A_1$ over A_1 , where F is connected. In particular, alb_X is a surjective analytic fibre bundle with connected fibres.*

Applying Theorem 3.2.3, we classify the normal compact Kähler threefold with canonical singularities and a trivial canonical class. Indeed, one can further follow the idea of [NZ10, Proposition 2.10] to generalize the classification to the case of klt singularities.

Proposition 3.2.4. *Let X be a normal compact Kähler threefold with at worst canonical singularities such that $K_X \sim_{\mathbb{Q}} 0$. Then:*

- (1) $q(X) \leq q^\circ(X) \leq 3$ and $q^\circ(X) \neq 2$. In particular, there is a quasi-étale Galois cover $X' \rightarrow X$ such that $q(X') = q^\circ(X)$.

- (2) If $q^\circ(X) = 3$, then X is a quasi-étale quotient of a complex torus.
- (3) If $q^\circ(X) = 0$, then X is projective, i.e., a weak Calabi-Yau threefold is projective.
- (4) If $q^\circ(X) = 1$, then there exists a quasi-étale cover $E \times F \rightarrow X$, where E is an elliptic curve and F has at worst canonical singularities such that $q^\circ(F) = 0$ and its minimal resolution is a K3 surface.

Proof. We first show (1). For each quasi-étale cover $X' \rightarrow X$, our $K_{X'} \sim_{\mathbb{Q}} 0$ and X' also has only canonical singularities (cf. [KM98, Proposition 5.20]). Then the Albanese map $X' \rightarrow \text{Alb}(X')$ is surjective holomorphic (cf. [KM98, Theorem 5.22] and Theorem 3.2.3), hence $q(X') \leq \dim(X') = 3$. So $q(X) \leq q^\circ(X) \leq 3$ and by the boundedness of $q^\circ(X)$, there exists a quasi-étale cover $X' \rightarrow X$ such that $q(X') = q^\circ(X)$. Taking the Galois closure $X'' \rightarrow X$ of $X' \rightarrow X$ with the induced Galois cover $X'' \rightarrow X'$, we get $q^\circ(X) = q(X') \leq q(X'') \leq q^\circ(X)$. Hence, $X'' \rightarrow X$ is the cover we are looking for.

Suppose that $q^\circ(X) = 2$. Then, there exists a quasi-étale Galois cover $X'' \rightarrow X$ such that $q^\circ(X) = q(X'') = 2$. By Theorem 3.2.3 and the adjunction, the Albanese map $\alpha : X'' \rightarrow \text{Alb}(X'')$ is a fibre bundle with the smooth fibre F being an elliptic curve. After an étale base change $A_1 \rightarrow \text{Alb}(X'')$, there is an étale cover $F \times A_1 \rightarrow X''$ (cf. Theorem 3.2.3) with $q(F \times A_1) = 3$, a contradiction to $q^\circ(X) = 2$. So (1) is proved.

(2) follows from Theorem 3.2.3 and (3) follows from [Gra18, Remark 2.6]. Now, we prove (4). Since $q^\circ(X) = 1$, there exists a quasi-étale Galois cover $X_1 \rightarrow X$ such that $q(X_1) = q^\circ(X) = 1$. Let F be a fibre of $\alpha : X_1 \rightarrow \text{Alb}(X_1)$. By Theorem 3.2.3, there exists an étale cover $T_1 \rightarrow \text{Alb}(X_1)$ such that $T_1 \times F \rightarrow X_1$ is étale. Then $T_1 \times F$ has only canonical singularities (cf. [KM98, Proposition 5.20]). In view of the commutative diagram of the resolutions of $T_1 \times F$ and F , our F has only canonical singularities. Further, $K_{T_1 \times F} \sim_{\mathbb{Q}} 0$ implies that $K_F \sim_{\mathbb{Q}} 0$, hence F is a surface with at worst canonical singularities such that $K_F \sim_{\mathbb{Q}} 0$. Since $q^\circ(X) = 1$, we have $q(F) = q^\circ(F) = 0$. So with F replaced by its global index-one cover (such that $K_F \sim 0$), the minimal resolution of F is a K3 surface. $\square$

In what follows, we discuss the Albanese closure in codimension one of a normal

compact Kähler threefold (cf. [NZ10, Section 2] and [Zho19, Proposition 5.1]).

Definition 3.2.5 (Albanese closure). Let X be a normal compact Kähler threefold with only canonical singularities such that $K_X \sim_{\mathbb{Q}} 0$. We say that the finite morphism $\tau : \tilde{X} \rightarrow X$ is the *Albanese closure in codimension one* of X if the following assertions hold.

- (1) τ is quasi-étale;
- (2) $q^\circ(X) = q(\tilde{X})$;
- (3) τ is Galois; and
- (4) For any other finite morphism $\tau' : X' \rightarrow X$ satisfying the conditions (1) and (2), there exists a quasi-étale morphism $\sigma : X' \rightarrow \tilde{X}$ such that $\tau' = \tau \circ \sigma$.

Now, we prove the existence of Albanese closure in the spirit of [NZ10, Lemma 2.12] (cf. [NZ09, Proposition 4.3]) for readers' convenience.

Proposition 3.2.6. *Let X be a normal compact Kähler threefold with at worst canonical singularities such that $K_X \sim_{\mathbb{Q}} 0$. Then the Albanese closure $\tau : \tilde{X} \rightarrow X$ exists.*

Proof. If $q^\circ(X) = 0$, then X is projective (cf. Proposition 3.2.4 (3)) and our result follows immediately from [NZ10, Lemma 2.12]. So we may assume $q^\circ(X) > 0$. By Proposition 3.2.4, there exists a quasi-étale Galois cover $X_1 \rightarrow X$ with the Galois group G_1 such that $q(X_1) = q^\circ(X)$. Then $K_{X_1} \sim_{\mathbb{Q}} 0$ and X_1 also has at worst canonical singularities. Note that $G_1 = \text{Gal}(X_1/X)$ acts naturally on the Albanese torus $\text{Alb}(X_1)$ (by the universal property of Albanese torus) and hence induces an action on $H_1(\text{Alb}(X_1), \mathbb{Z})$. If we write $\text{Alb}(X_1) := V/\Lambda$ for a vector space V and a lattice Λ , then $H_1(\text{Alb}(X_1), \mathbb{Z}) \cong \Lambda$. Hence, there exists a natural homomorphism $\varphi : G_1 \rightarrow \text{Aut}(\Lambda)$.

Let $H_1 := \ker \varphi$ and $\tilde{X} := X_1/H_1$. Then $\text{Alb}(X_1)/H_1$ is a torus, since H_1 acts on $\text{Alb}(X_1)$ as translations. Then, $\text{Alb}(X_1)/H_1$ is the Albanese torus of $\tilde{X}$ by the universality, hence $q(\tilde{X}) = q(X_1) = q^\circ(X)$. Since H_1 is normal in G_1 , $\tilde{X} \rightarrow X$ is Galois with the induced Galois group G_1/H_1 . Therefore, $\tilde{X} \rightarrow X$ satisfies conditions (1), (2) and (3).

Now, let $X' \rightarrow X$ be an arbitrary quasi-étale cover such that $q(X') = q^\circ(X)$ and take the Galois closure $X_2 \rightarrow X$ of $X_1 \times_X X' \rightarrow X$ with the induced Galois covers $X_2 \rightarrow X'$ and $X_2 \rightarrow X_1$. Let $G_2 := \text{Gal}(X_2/X)$ (resp. $G'_2 := \text{Gal}(X_2/X')$) be the Galois group of $X_2 \rightarrow X$ (resp. $X_2 \rightarrow X'$). Then, with the same argument as above, we denote by $H_2 \subseteq G_2$ the kernel of $G_2 \rightarrow \text{Aut}(H_1(X_2, \mathbb{Z}))$. Note that $\text{Alb}(X_2) \rightarrow \text{Alb}(X_1)$ is an isogeny by choosing a proper origin on $\text{Alb}(X_1)$. Thus, H_2 is the pull-back of $H_1 \subseteq G_1$ under $X_2 \rightarrow X_1$ and then $X_2/H_2 \cong X_1/H_1 = \tilde{X}$. Moreover, G'_2 acts on $H_1(\text{Alb}(X_2), \mathbb{Z})$ trivially by the choice of X' (so that $q(X') = q^\circ(X)$). As a result, $G'_2 \subseteq H_2$ and we have a natural factorization $X' \rightarrow \tilde{X} \rightarrow X$. So (4) is satisfied. $\square$

In the following, we recall the common nef eigenvectors of an automorphism group. Actually, this part will not be used in proving Theorem 3.1.4 and Proposition 3.1.6. However, we still formulate Proposition 3.2.9 to compare with earlier results.

We follow [Zha13, Proof of Theorem 1.2] to find the nef classes ξ which are common eigenvectors of G in our present case. Suppose (X, G) satisfies Hypothesis A. Let $\pi : \tilde{X} \rightarrow X$ be a G -equivariant resolution (cf. [Wlo09, Theorem 2.0.1]). Applying [DS04, Proofs of Theorems 4.3 and 4.7] to the action of G on the pull-back $\pi^*\bar{\mathcal{K}}(X)$ of the nef cone of X , we get nef classes $\pi^*\xi_i$ ($1 \leq i \leq 3$) on $\tilde{X}$ as common eigenvectors of G such that the cup product $\xi_1 \cup \xi_2 \cup \xi_3 \neq 0$, where ξ_i are nef classes on X . Since ξ_i ($1 \leq i \leq 3$) are eigenvectors, for each $g \in G$, we can write $g^*\xi_i = \chi_i(g)\xi_i$ with the characters $\chi_i : G \rightarrow \mathbb{R}_{>0}$. By the projection formula, our $\chi_1\chi_2\chi_3 = 1$. Let

$$\xi := \xi_1 + \xi_2 + \xi_3. \quad (3.1)$$

Then $\xi^3 \geq \xi_1 \cup \xi_2 \cup \xi_3 > 0$, which implies that ξ is a nef and big class on X .

We borrow the following results from [Zha16, Lemma 3.7] and the proofs therein can be adapted into our present version. Readers may refer to [GK20, Section 5] for the information of Chern classes on singular spaces.

Lemma 3.2.7. *Suppose that (X, G) satisfies Hypothesis A. Then, for the ξ in Equation (3.1), the following assertions hold.*

- (1) For every G -periodic (k, k) -class η with $k = 1$ or 2 , $\xi^{3-k} \cdot \eta = 0$; in particular,
 $\xi^2 \cdot c_1(X) = \xi \cdot c_1(X)^2 = 0$.
- (2) Let $Z \subseteq X$ be a G -periodic positive-dimensional proper subvariety of X . Then
 $\xi^{\dim(Z)} \cdot Z = 0$.

The lemma below is a direct consequence of [DS04, Théorème 3.1 and Corollaire 3.4]. It is also known as a generalization of Hodge-Riemann Theorem.

Lemma 3.2.8. *Let X be a normal compact Kähler threefold, $u_i \in H_{\text{BC}}^{1,1}(X)$ ($i = 1, 2$) nef classes and $\eta \in H_{\text{BC}}^{1,1}(X)$ such that $u_1 \cup u_2 \cup \eta = 0$. Then we have:*

- (1) $u_1 \cup \eta^2 \leq 0$.
- (2) Suppose that $u_1 = u_2$ is nef and big, and η is nef. Then $u_1 \cup \eta^2 = 0$ holds if and only if $\eta \equiv 0$, i.e., $\eta \cdot \Gamma = 0$ for each $\Gamma \in \mathbf{N}_1(X)$.

Proof. The first statement is easy to see by pulling back u_i and η to a smooth model $\tilde{X}$ of X (cf. [Zho19, Proposition 2.6] and [DS04, Corollaire 3.4]). For the second statement, note that for every big class u , there exists a suitable resolution $\pi : \tilde{X} \rightarrow X$ and an effective $\mathbb{R}$ -divisor E on $\tilde{X}$ supported in the exceptional locus of π such that $\pi^*u - E$ is a Kähler class on $\tilde{X}$ (cf. [Zho19, Proposition 2.6] and [BDPP13, Proof of Theorem 3.1]). By the projection formula, $(\pi^*u_1)^2 \cup \pi^*\eta = \pi^*u_1 \cup (\pi^*\eta)^2 = 0$. So it is easy to verify that $(\pi^*u_1 - E)^2 \cup \pi^*\eta = 0$ and $(\pi^*u_1 - E) \cup (\pi^*\eta)^2 = 0$, noting that $\pi^*u_1 - E$ and $\pi^*\eta$ are nef and $E \geq 0$. Then, we apply [DS04, Théorème 3.1] for $\pi^*u_1 - E$ and $\pi^*\eta$ to conclude that $\pi^*\eta \equiv 0$. Hence $\eta \equiv 0$ by the projection formula (cf. [HP16, Proposition 3.14]). $\square$

The following proposition gives a sufficient condition for X to be a Q -complex torus under the assumption of **Hypothesis A**. Note that, ξ being Kähler implies that every G -periodic proper subvariety of X is a single point (cf. Lemma 3.2.7).

Proposition 3.2.9. *Suppose (X, G) satisfies **Hypothesis A**. Suppose further that X is minimal (i.e., K_X is nef) and the ξ in Equation (3.1) is a Kähler class on X . Then, $X \cong T/F$, where T is a complex torus and F is a finite group whose action on T is free outside a finite subset of T . Further, the group G (acting on X) lifts to $G_T \subseteq \text{Aut}(T)$ on T .*

Proof. By Lemma 3.2.7, $\xi^2 \cdot c_1(X) = \xi \cdot c_1(X)^2 = 0$. Since ξ is Kähler, $c_1(X) = 0$ (cf. Lemma 3.2.8). Then, it follows from the abundance that $K_X \sim_{\mathbb{Q}} 0$ (cf. [CHP16, Theorem 1.1]). Let m be the minimal integer such that $mK_X \sim 0$. Taking the associated global index-one cover $f : X' := \text{Spec } \bigoplus_{i=0}^{m-1} \mathcal{O}_X(-iK_X) \rightarrow X$, we can lift G to X' by their actions on K_X .

Now, $K_{X'} \sim 0$ and X has only Gorenstein terminal singularities (cf. [KM98, Proposition 5.20]). Let $\sigma : \tilde{X} \rightarrow X'$ be a G -equivariant resolution minimal in codimension two (cf. [GK20, Definition 2.1 and Proposition/Definition 5.3]), and $c_2(X')$ the “Birational” second Chern class, which is defined as an element $c_2(X') \in H^2(X', \mathbb{R})^\vee$ such that

$$c_2(X') \cdot a := \int_{\tilde{X}} c_2(\tilde{X}) \cup \sigma^* A,$$

for any $a \in H^2(X', \mathbb{R})$. Here, $c_2(\tilde{X})$ is the usual second Chern class of $\tilde{X}$.

Since f is finite, $\xi' := f^*\xi$ is a Kähler class on X' (cf. [GK20, Proposition 3.5]). We aim to show that $\xi' \cdot c_2(X') = 0$. Indeed, for each summand $\xi'_i := f^*\xi_i$ of ξ' , we choose $g_i \in G$ such that $g_i^*\xi'_i = \alpha_i\xi'_i$ with $\alpha_i > 1$ (cf. [Zha16, Lemma 3.7]). Further, by [GK20, Lemma 5.6], $c_2(X') \cdot \xi'_i = 0$ and thus $c_2(X') \cdot \xi' = 0$. Applying [GK20, Corollary 1.2], we have $X \cong T/F$, where T is a complex torus and F is a finite group whose action on T is free outside a finite subset of T . Finally, according to [Zha13, §2.14], G lifts to $G_T \subseteq \text{Aut}(T)$. $\square$

Now we come back to the proof of Proposition 3.1.6. First, we recall the following special case of [Zha09, Lemma 2.10], using the maximality of G in Hypothesis A.

Lemma 3.2.10 (cf. [Zha09]). *Suppose that (X, G) satisfies Hypothesis A. Then there does not exist any G -equivariant surjective holomorphic map $\pi : X \rightarrow Y$ such that $\dim(X) > \dim(Y) > 0$ and G descends to a biregular action on Y .*

As an application of Lemma 3.2.10, we are able to show the proposition below.

Proposition 3.2.11. *Suppose that (X, G) satisfies Hypothesis A. Then the Kodaira dimension $\kappa(X) \leq 0$. In particular, either $\kappa(X) = 0$ or X is uniruled.*

Proof. Since $\text{Bim}(X)$ is an infinite group, $\kappa(X) < 3$ (cf. [Uen75, Corollary 14.3]). Suppose that $1 \leq \kappa(X) < 3$. Then, there exists an Iitaka fibration $\varphi : X \dashrightarrow \mathbb{P}^n$ such that $\dim(\text{Im } \varphi) = \kappa(X)$. Taking a G -equivariant resolution $\tilde{X} \rightarrow X$ (cf. [NZ09, Section 1.4] and [Wlo09, Theorem 2.0.1]), we may assume that X is a compact Kähler smooth threefold. Besides, resolving the indeterminacy locus of φ , we can further assume that φ is holomorphic (cf. [NZ09, Theorem A and the remark therein]). Then, we get a non-trivial G -equivariant fibration $X \rightarrow \text{Im } \varphi$ such that $\dim(\text{Im } \varphi) < 3$, a contradiction to Lemma 3.2.10. Therefore, under the assumption of Hypothesis A, $\kappa(X) \leq 0$.

If K_X is pseudo-effective, applying [HP16, Theorem 1.1] and [CHP16, Theorem 1.1], we have $\kappa(X) = 0$. If K_X is not pseudo-effective, then X is uniruled (cf. [BDPP13]). $\square$

Lemma 3.2.12. *Suppose (X, G) satisfies Hypothesis A and $K_X \equiv 0$. Then X is a Q -complex torus, i.e., a quasi-étale quotient of a torus T . Further, the group G lifts to $G_T \subseteq \text{Aut}(T)$.*

Proof. By abundance (cf. [CHP16, Theorem 1.1]), $K_X \sim_{\mathbb{Q}} 0$. Taking the global index-one cover $\tilde{X} \rightarrow X$, we see that G lifts to $\tilde{X}$ by their actions on K_X . We denote by $G|_{\tilde{X}}$ the group to which G lifts, hence there exists a surjective group homomorphism $G|_{\tilde{X}} \rightarrow G$ with the kernel being the Galois group $\text{Gal}(\tilde{X}/X)$ (cf. [Zha13, §2.15]). By [Zha13, Lemma 2.4], there exists a finite index subgroup G_0 of $G|_{\tilde{X}}$ such that $(\tilde{X}, G_0)$ satisfies Hypothesis A. Then, under the Albanese closure map $\tilde{X}' \rightarrow \tilde{X}$ in Proposition 3.2.6, $G|_{\tilde{X}}$ lifts to $\tilde{X}'$ by the uniqueness. By [Zha13, Lemma 2.4 and §2.15] again, there exists a finite index subgroup G_1 of $G|_{\tilde{X}'}$ such that $(\tilde{X}', G_1)$ satisfies Hypothesis A.

From now on, we assume that $q^\circ(X) = q(X)$ and $K_X \sim 0$. If $q(X) = 0$, then by Proposition 3.2.4 (3), X is projective and thus a Q -complex torus (cf. [Zha16, Theorem 1.1]), a contradiction to the equality $q(X) = q^\circ(X)$. So we may assume that $q^\circ(X) = q(X) > 0$. Let $T := \text{Alb}(X)$ and $\alpha : X \rightarrow T$ the Albanese map. Thanks to Theorem 3.2.3, α is surjective holomorphic with connected fibres, and there exists an étale cover $T_1 \rightarrow T$ such that $X \times_T T_1 \cong F \times T_1$ for a connected fibre F of α such that $\kappa(F) = q^\circ(F) = 0$. So $\dim(F) = 0$ or 2 .

If $\dim(F) = 0$, then α is bimeromorphic. So we have $0 \equiv K_X = \alpha^* K_T + E \equiv E$,

where the support of E equals the exceptional locus of α . Since T is smooth, our $E \geq 0$. Hence, the above equation gives us $E = 0$ and thus $X \cong T$.

If $\dim(F) = 2$, then T is an elliptic curve and thus projective. Also, $G \subseteq \text{Aut}(X)$ descends to a well-defined automorphism group G_T of T . Moreover, $T_1 \rightarrow T$ is an isogeny and then with T_1 replaced by a further isogeny $T_2 \rightarrow T_1$, we may assume that $\theta : T_2 \rightarrow T$ is a multiplication map. By [NZ09, Lemma 4.9], G_T lifts to an automorphism group G_{T_2} of T_2 . See the following commutative diagram:

$$\begin{array}{ccc} F \times T_2 & \longrightarrow & T_2 \\ \downarrow & & \downarrow \\ F \times T_1 & \longrightarrow & T_1 \\ \downarrow & & \downarrow \\ X & \longrightarrow & T \end{array}$$

For each $g \in G$ on X , we get an induced automorphism $g|_T \in G_T$ and then $g|_T$ lifts to $g' \in \text{Aut}(T_2)$ (which is not unique) on T_2 . Here, there are $\deg(\theta)$ of choices for the lifting of g_T to $\text{Aut}(T_2)$ (cf. [NZ09, Proof of Lemma 4.9]).

Since $F \times T_2 \cong X \times_T T_2 \subseteq X \times T_2$ with respect to $\alpha : X \rightarrow T$ and $\theta : T_2 \rightarrow T$, we restrict $g \times g' : X \times T_2 \rightarrow X \times T_2$ to $F \times T_2$, for $g \in G$ and $g' \in G_{T_2}$ such that $\alpha \circ g = g_T \circ \alpha$ and $g_T \circ \theta = \theta \circ g'$. Then, we get an induced subgroup $G_{F \times T_2} \subseteq \text{Aut}(F \times T_2)$ (consisting of all liftings of G) and there is a natural surjective group homomorphism $G_{F \times T_2} \rightarrow G$. By [Zha13, Lemma 2.4], there exists a finite index subgroup $G_0 \subseteq G_{F \times T_2}$ such that $(F \times T_2, G_0)$ satisfies **Hypothesis A**. So we get a G_0 -equivariant fibration $F \times T_2 \rightarrow \text{Alb}(F \times T_2) = T_2$, a contradiction to Lemma 3.2.10. Thus, X is a Q -complex torus. $\square$

Proof of Proposition 3.1.6. By Proposition 3.2.11, we have $\kappa(X) = 0$. Hence, $K_X \equiv 0$ (cf. [CHP16, Theorem 1.1]), and our proposition follows from Lemma 3.2.12. $\square$

3.3 Proof of Theorem 3.1.4

In this section, we prove our main result. We begin with the lemma below, which states the splitting property for certain bimeromorphic transformations on product spaces.

Lemma 3.3.1. *Let T be a complex torus and F a (minimal) K3 surface with $q^\circ(F) = 0$. Then every bimeromorphic transformation $g : F \times T \dashrightarrow F \times T$ is an automorphism and splits into the form (σ, τ) such that $\sigma \in \text{Aut}(F)$ and $\tau \in \text{Aut}(T)$.*

Proof. First, we consider the following commutative diagram

$$\begin{array}{ccc} F \times T & \xrightarrow{g} & F \times T \\ \alpha \downarrow & & \downarrow \alpha \\ T & \xrightarrow{\tau} & T \end{array}$$

which is induced by the Albanese map $\alpha : F \times T \rightarrow \text{Alb}(F \times T) = T$ (cf. [NZ09, Corollary 2.11]) with the descended $\tau : T \rightarrow T$ being an automorphism (cf. [Uen75, Lemma 9.11]). By the diagram, for a general point $t \in T$, there exists a bimeromorphic map $\sigma_t : F \dashrightarrow F$. In other words, there exists a meromorphic map $\chi : T \dashrightarrow \text{Bim}(F)$. Since F is a minimal smooth surface, $\text{Bim}(F) = \text{Aut}(F)$. Further, by the property of K3 surfaces, $h^0(F, T_F) = h^0(F, \Omega_F^1) = q(F) = 0$ (where $T_F \cong \Omega_F^1 \otimes (\Omega_F^2)^\vee \cong \Omega_F^1$ is the tangent sheaf). So $\text{Bim}(F) = \text{Aut}(F)$ is discrete. As a result, χ is constant, i.e., the automorphism σ_t is independent of the choice of $t \in T$. Our lemma is proved. $\square$

Proof of Theorem 3.1.4: First, we note that $\kappa(X) \leq 0$ (cf. Proposition 3.2.11).

(1) Since X has at worst terminal singularities and K_X is not pseudo-effective, our X is uniruled (cf. [BDPP13]). After taking a G -equivariant resolution $\tilde{X} \rightarrow X$ (cf. [Wlo09, Theorem 2.0.1]), we may assume that X is smooth. Let $\psi : X \dashrightarrow C(X)$ be the meromorphic map to the cycle space of X which defines the maximal rationally connected fibration of X , sending a general point $x \in X$ to the maximal rationally connected subvariety containing x (as a point in $C(X)$). Since each $g \in G$ is an automorphism and hence proper, it follows from [Bar75, Chapter IV §2 Théorème 6] that g descends to a self-morphism g_c on $C(X)$ via the push-forward operation g_* . Since g is an automorphism, $g_*(M)$ is reduced for every rationally connected submanifold M , hence $\psi \circ g = g_c \circ \psi$.

Let $X \dashrightarrow Z \rightarrow C(X)$ be the Stein factorization of ψ with $Z \rightarrow C(X)$ being a finite morphism (cf. [NZ09, Definition 3.2]). Since $\psi \circ g = g_c \circ \psi$, the (irreducible) graph Γ_{xc} of the meromorphic map $X \dashrightarrow C(X)$ is $(g \times g_c)$ -invariant. Hence, we get an

endomorphism $g_{\Gamma_{xc}} := (g \times g_c)|_{\Gamma_{xc}}$. Let $\widetilde{\Gamma_{xc}}$ be the normalization of Γ_{xc} . Then $g_{\Gamma_{xc}}$ lifts to $g_{\widetilde{\Gamma_{xc}}}$, and $Z \rightarrow C(X)$ is given by the usual Stein factorization of $p_2 : \widetilde{\Gamma_{xc}} \xrightarrow{\phi_1} Z \xrightarrow{\phi_2} C(X)$ (cf. [NZ09, Remark after Definition 3.2]). Since $p_2 \circ g_{\widetilde{\Gamma_{xc}}} = g_c \circ p_2$ and ϕ_2 is finite, for every (connected) fibre F of ϕ_1 , the image $g_{\widetilde{\Gamma_{xc}}}(F)$ is contracted by p_2 , hence $\phi_1(g_{\widetilde{\Gamma_{xc}}}(F))$ is a point. By the rigidity lemma (cf. [BS95, Lemma 4.1.13]), $\phi_1 \circ g_{\widetilde{\Gamma_{xc}}}$ factors through ϕ_1 , i.e., $g_{\widetilde{\Gamma_{xc}}}$ descends to an automorphism g_z on Z . Further, since ϕ_2 is finite but general maximal rationally connected subvariety of X is contracted along ψ , our Z is not uniruled.

Denote by $\Gamma_{xz} \subseteq X \times Z$ the graph of the dominant meromorphic map $X \dashrightarrow Z$. Then we get an endomorphism via the restriction $g_{\Gamma_{xz}} := (g \times g_z)|_{\Gamma_{xz}}$, noting that the graph Γ_{xz} is $(g \times g_z)$ -invariant. Taking the normalization $X' := \widetilde{\Gamma_{xz}}$ of Γ_{xz} with the induced projection $p : X' \rightarrow X$, we see that $g_{\Gamma_{xz}}$ lifts to (holomorphic) g' on X' and $g \circ p = p \circ g'$. As a result, our g lifts to an automorphism g' on X' and there is an induced automorphism subgroup $G_{X'} \subseteq \text{Aut}(X')$ of rank 2. Replacing X' by a G -equivariant resolution (cf. [Wlo09, Theorem 2.0.1]) if necessary, we may assume that X' is smooth. Applying Lemma 3.2.10 to the G -equivariant (holomorphic) fibration $X' \rightarrow Z$ so that Z is a single point, we have X is rationally connected and Theorem 3.1.4 (1) is proved.

(2) Since X has only terminal singularities and K_X is pseudo-effective, by [HP16, Theorem 1.1], there exists a minimal model $X_{\min}$ of X , i.e., there is a bimeromorphic map which is a composite of divisorial contractions and flips: $X \dashrightarrow X_{\min}$ such that $K_{X_{\min}}$ is nef. Then, our G descends to a bimeromorphic transformation subgroup $G|_{X_{\min}} \subseteq \text{Bim}(X_{\min})$ of rank 2. By Proposition 3.2.11, $\kappa(X) = \kappa(X_{\min}) = 0$. Thus, $K_{X_{\min}} \sim_{\mathbb{Q}} 0$ by the abundance (cf. [CHP16, Theorem 1.1]).

In the following, we aim to construct a quasi-étale splitting cover: $\widetilde{X_{\min}} := F \times T \rightarrow X_{\min}$ such that F is a weak Calabi-Yau space (and thus $\dim(F) = 0$ or 2), T is a torus and $G|_{X_{\min}}$ lifts to a subgroup $G_{\widetilde{X_{\min}}} \subseteq \text{Aut}(\widetilde{X_{\min}})$.

By Proposition 3.2.4 and [Zha16, Theorem 1.1], the result holds when $q^\circ(X_{\min}) = 0$. We prepare to exclude the case $q^\circ(X_{\min}) = 1$ (cf. **Step 1-3**) so that the remaining case $q^\circ(X_{\min}) = 3$ is what we desire. Suppose on the contrary that $q^\circ(X_{\min}) = 1$.

Step 1. We reduce to the Albanese closure of $X_{\min}$. Fixing an element $g \in G|_{X_{\min}} \subseteq$

$\text{Bim}(X_{\min})$ and taking the Albanese closure $\tau : X'_{\min} \rightarrow X_{\min}$, we get a meromorphic map $X'_{\min} \rightarrow X_{\min} \dashrightarrow X_{\min}$. After the Stein factorization (cf. [NZ09, Definition 3.2]), we get:

$$\begin{array}{ccccc} X'_{\min} & \xleftarrow{\sigma} & X_1 & \xleftarrow{g'} & X'_{\min} \\ & \searrow \tau & \downarrow \tau' & & \downarrow \tau \\ & & X_{\min} & \xleftarrow{g} & X_{\min} \end{array}$$

where τ' is a finite morphism and g' is bimeromorphic. Since g is isomorphic in codimension one (cf. Theorem 3.2.1), by the property of Stein factorization and the quasi-étaleness of τ , our g' is also isomorphic in codimension one and τ' is quasi-étale. Moreover,

$$q^\circ(X_1) = q^\circ(X_{\min}) = q(X'_{\min}) = q(X_1) = 1.$$

By Definition 3.2.5 (4) and Proposition 3.2.6, τ' factors through τ , i.e., there exists a surjective morphism $\sigma : X_1 \rightarrow X'_{\min}$ such that $\tau' = \tau \circ \sigma$. Since $\deg(\tau) = \deg(\tau')$, our σ is bimeromorphic and we see that $g \in G|_{X_{\min}}$ lifts to a bimeromorphic transformation on its Albanese closure. Note that $G|_{X_{\min}}$ lifts to $G_{X'_{\min}}$ such that its quotient is $G|_{X_{\min}}$.

Step 2. From now on, we may assume that $X_{\min}$ has at worst terminal singularities, $q^\circ(X_{\min}) = q(X_{\min}) = 1$ and $K_{X_{\min}} \sim_{\mathbb{Q}} 0$. Let $\alpha : X_{\min} \rightarrow \text{Alb}(X_{\min}) =: E$ denote the Albanese map. By Theorem 3.2.3, there exists an étale cover $E_1 \rightarrow E$ such that $X_{\min} \times_E E_1 \cong F \times E_1$ for a fibre F of α . By the choice of the Albanese closure, $q^\circ(F) = 0$ and $\dim(F) = 2$. With F further replaced by its global index-one cover, we may assume that $K_F \sim 0$. Also, the subgroup $G|_{X_{\min}} \subseteq \text{Bim}(X_{\min})$ descends to $G_E \subseteq \text{Aut}(E)$ on E (cf. [Uen75, Lemma 9.11]). Since $E_1 \rightarrow E$ is an isogeny, there exists another isogeny $E_2 \rightarrow E_1$ such that the composite map $\theta : E_2 \rightarrow E$ is a multiplication (finite) map. Then, by [NZ09, Lemma 4.9], G_E lifts to G_{E_2} such that $G_{E_2} \rightarrow G_E$ is surjective. Moreover, after the base change $E_2 \rightarrow E_1$, $X_{\min} \times_E E_2 \cong F \times E_2$ with respect to α and θ . Therefore, regarding $F \times E_2$ as a subset of $X_{\min} \times E_2$, we get a natural lifting $G_{F \times E_2} \subseteq \text{Bim}(F \times E_2)$ by restricting $g \times g'$ on $X_{\min} \times E_2$ to $F \times E_2$ for each $g \in G|_{X_{\min}}$ and $g' \in G_{E_2}$ such that $g_E \circ \alpha = \alpha \circ g$ and $\theta \circ g_E = g' \circ \theta$. Here, $g \in G|_{X_{\min}}$ descends to g_E on E and lifts to g' (which is not unique) on G_{E_2} . By the proof in Lemma 3.2.12, there exists a finite index subgroup G_1 of $G_{F \times E_2}$ such that G_1 is free of rank 2 (cf. [Zha13, Lemma 2.4]).

Step 3. Taking a minimal resolution $\tilde{F} \rightarrow F$, we see that $\tilde{F}$ is a minimal K3 surface. Then we get a natural lifting $\tilde{G}_1 \subseteq \text{Bim}(\tilde{F} \times E_2)$ of G_1 on $F \times E_2$. By Lemma 3.3.1,

$$\text{Bim}(\tilde{F} \times E_2) = \text{Aut}(\tilde{F} \times E_2) \subseteq \text{Aut}(\tilde{F}) \times \text{Aut}(E_2).$$

Hence, $(\tilde{F} \times E_2, \tilde{G}_1)$ satisfies **Hypothesis A** (cf. [DNT12, Theorem 1.1]). As a result, there exists a $\tilde{G}_1$ -equivariant non-trivial fibration $\tilde{F} \times E_2 \rightarrow E_2$, contradicting Lemma 3.2.10. Then the case $q^\circ(X) = 1$ cannot happen. So $q^\circ(X_{\min}) = 3$ and $X_{\min}$ is a Q -complex torus.

Step 4. So far we have completed the proof of the first part of Theorem 3.1.4 (2). Since $X_{\min}$ is a Q -complex torus, its Albanese closure $X'_{\min}$ satisfies the following properties: $K_{X'_{\min}} \sim_{\mathbb{Q}} 0$, $q(X'_{\min}) = 3$, $X'_{\min}$ has at worst terminal singularities and the subgroup $G|_{X_{\min}} \subseteq \text{Bim}(X_{\min})$ lifts to $G_{X'_{\min}} \subseteq \text{Bim}(X'_{\min})$ as in **Step 1** such that a finite index subgroup of $G_{X'_{\min}}$ has rank 2 (cf. [Zha13, Lemma 2.4]). With the same proof as in Lemma 3.2.12, $X'_{\min}$ is isomorphic to a complex 3-torus by showing that the Albanese map $X'_{\min} \rightarrow \text{Alb}(X'_{\min})$ is an isomorphism. Thus, $G_{X'_{\min}} \subseteq \text{Bim}(X'_{\min}) = \text{Aut}(X'_{\min})$.

Step 5. In this step, we shall prove that each $g \in G|_{X_{\min}}$ is actually an automorphism. Let $\pi : T_1 := X'_{\min} \rightarrow X_{\min}$ be the Albanese closure of $X_{\min}$ as in **Step 4**. Then, $G|_{X_{\min}}$ lifts to $G_{T_1} \subseteq \text{Aut}(T_1)$ as in **Step 1**. Consider the following commutative diagram,

$$\begin{array}{ccc} T_1 & \xrightarrow{g|_{T_1}} & T_1 \\ \pi \downarrow & & \downarrow \pi \\ X_{\min} & \xrightarrow{g} & X_{\min} \end{array}$$

where $g \in G|_{X_{\min}}$ lifts to $g|_{T_1}$ (which is not unique) on T_1 . Let $H := \text{Gal}(T_1/X_{\min})$. Since each $g|_{T_1}$ normalizes H , i.e., $g|_{T_1} H g|_{T_1}^{-1} \subseteq H$ for any $g|_{T_1} \in G_{T_1}$, the composite map $\pi \circ g|_{T_1}$ is H -invariant. By the universality of the quotient morphism π over the étale loci, the composite $\pi \circ g|_{T_1}$ factors through π . Thus, $g|_{T_1}$ descends to a well-defined morphism $\bar{g} : X_{\min} \rightarrow X_{\min}$, acting biregularly on $X_{\min}$. Since the meromorphic map g is determined by an open dense subset $U \subseteq X_{\min}$ and $g|_U = \bar{g}|_U$, our $g = \bar{g}$ is biholomorphic.

End of Proof of Theorem 3.1.4. Now, each $g \in G|_{X_{\min}}$ is a (biholomorphic) automorphism and $(X_{\min}, G|_{X_{\min}})$ satisfies **Hypothesis A**. Thanks to [Zha13, Lemma 2.4 and

§2.16], $X_{\min} \cong T_1/H$ for a finite group H acting freely outside a finite set of a complex 3-torus T_1 . This completes the proof of Theorem 3.1.4. $\square$

We end up this chapter with the following remark, which extends our main result Theorem 3.1.4 to the solvable group case (cf. [Zha09, Theorem 1.1] and [Zha16, Remark 2.7]). For the organization, we shall only give a sketch for its proof.

Recall that a group G is *virtually unipotent* (resp. *virtually solvable*), if a finite-index subgroup G_1 of G is unipotent (resp. solvable). We say that a group action G on a $\mathbb{C}$ -vector space V is *virtually Z -connected*, if a finite-index subgroup G_1 (acting on V) is Z -connected, i.e., $G_1|_V$ has connected Zariski closure in $\mathrm{GL}(V)$.

Remark 3.3.2 (A further generalization to the solvable group case). Let X be a compact Kähler (smooth) threefold, $G \subseteq \mathrm{Aut}(X)$ a subgroup such that $G|_{H^{1,1}(X, \mathbb{C})}$ is virtually solvable and virtually Z -connected, and $G/N(G) \cong \mathbb{Z}^{\oplus 2}$ (this is the case when the smooth pair (X, G) satisfies **Hypothesis A**). Here, $N(G) \subseteq G$ denotes the set of elements $g \in G$ of null entropy. Then with the same strategy as in Proof of Theorem 3.1.4, we can show that either X is rationally connected, or X is G -equivariantly bimeromorphic to a Q -complex torus. In the following, we give a sketch of the proof for readers' convenience.

First, [Zha09, Lemma 2.10] still works in this case. Second, for any equivariant finite étale Galois cover $\pi : (\tilde{X}, \tilde{G}) \rightarrow (X, G)$ with $G \cong \tilde{G}/\mathrm{Gal}(\tilde{X}/X)$, we have a natural surjection $\tilde{G}|_{H^{1,1}(\tilde{X}, \mathbb{C})} \twoheadrightarrow \tilde{G}|_{\pi^* H^{1,1}(X, \mathbb{C})} \cong G|_{H^{1,1}(X, \mathbb{C})}$. By considering the virtually unipotent kernel (cf. [CWZ14, Theorem 2.2]), if $G|_{H^{1,1}(X, \mathbb{C})}$ is virtually solvable and virtually Z -connected, then so is $\tilde{G}|_{H^{1,1}(\tilde{X}, \mathbb{C})}$ (cf. [Din12, Lemma 5.5]). Moreover, $N(\tilde{G}) = \pi^{-1}(N(G))$ (cf. [NZ09, Appendix A, Lemma A.8]), hence π induces an isomorphism $\tilde{G}/N(\tilde{G}) \cong G/N(G) \cong \mathbb{Z}^{\oplus 2}$.

For terminal threefolds (as the end product of the MMP), the same arguments work after we replace $H^{1,1}(X, \mathbb{C})$ by the complexified Bott-Chern cohomology $H_{\mathrm{BC}}^{1,1}(X) \otimes_{\mathbb{R}} \mathbb{C}$. Indeed, for a G -equivariant generically finite dominant meromorphic map $\pi : V \dashrightarrow W$ between normal compact Kähler spaces with only rational singularities, we can show that, $G|_{H_{\mathrm{BC}}^{1,1}(V) \otimes \mathbb{C}}$ is virtually solvable and virtually Z -connected if and only if so is $G|_{H_{\mathrm{BC}}^{1,1}(W) \otimes \mathbb{C}}$ (cf. [Zho19, Lemma 2.7] and [Hu20, Lemma 4.2 and its proof]).

Dynamical rigidity of rationally connected projective varieties

4.1 Introduction

In this chapter, we work over an algebraically closed field k of characteristic 0. Let X be a smooth projective variety which is *rationally connected*, i.e., any two general points of X can be connected by a chain of rational curves; see [Cam92] and [KMM92]. A natural interest is to characterize such varieties in terms of some dynamic assumptions.

Let $f : X \rightarrow X$ be a surjective endomorphism. Then f^* induces an invertible linear map on $N^1(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ where $\text{NS}(X)$ is the Néron-Severi group of X (cf. Notation 4.2.1). In this chapter, we focus on those f under the condition that all the eigenvalues of $f^*|_{N^1(X)}$ have modulus greater than 1. An endomorphism f having this condition is equivalent to f being *int-amplified*, i.e., $f^*L - L$ is ample for some ample Cartier divisor L ; see [Men20, Theorem 1.1]. It is a natural extension of the *q-polarized* endomorphism, i.e., $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$.

To the best of our knowledge, the int-amplified assumption is necessary for one to get restriction on the rigidity of X . In [Men20, Theorem 1.10], Meng proved the equivariant minimal model program (**MMP** for short) for int-amplified endomorphisms (cf. [MZ20a, Definition 2.1]), generalizing an early result for polarized endomorphisms by Meng and

Zhang (cf. [MZ18a, Theorem 1.8]). We refer to [MZ18a], [CMZ20], [Men20], [MZ20a], [Zho19]=Chapter 2 for details and generalizations about the equivariant MMP. Making use of it, Yoshikawa proved the following result, partially answering a conjecture of Broustet and Gongyo (cf. [BG17, Conjecture 1.2]). It is also the motivation of this project.

Theorem 4.1.1 (cf. [Yos20, Corollary 1.4]). *A rationally connected smooth projective variety X is of Fano type if it admits an int-amplified endomorphism.*

When X is a rationally connected smooth projective surface admitting a non-isomorphic endomorphism, it was proved by Nakayama [Nak02, Theorem 3] that X is then a toric surface, answering affirmatively a conjecture proposed by Sato (cf. [Nak02, Conjecture 2]). Note that every toric variety is of Fano type (cf. Notation 4.2.1); however, there exist many non-rational Fano varieties in higher dimensions. Based on Theorem 4.1.1, we ask the following question which is a higher dimensional analogue of Sato's conjecture; see also [Fak03, Question 4.4] for the polarized case.

Question 4.1.2. Let X be a rationally connected smooth projective variety admitting an int-amplified endomorphism. Is X a toric variety?

The following question proposed by Meng and Zhang aimed to generalize the results for polarized endomorphisms in [HN11, Theorem 2.1] and [MZ19a, Corollary 1.4] to the int-amplified case.

Question 4.1.3 ([MZ19b, Question 10.1]). Let $f : X \rightarrow X$ be an int-amplified endomorphism of a rationally connected smooth projective variety. Suppose there is an f^{-1} -stable reduced divisor D such that $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is étale. Is (X, D) a toric pair?

However, due to an obstacle about the slope semistability, Meng and Zhang can only deal with the case when $f^*|_{N^1(X)}$ has at most two eigenvalues in [MZ19b, Theorem 10.6]. We strongly recommend [MZ19b, Section 10] for a detailed explanation. In this chapter, we will mainly overcome this obstacle and answer Question 4.1.3 affirmatively.

Theorem 4.1.4 ([MZ20b, Theorem 1.5]). *Let X be a rationally connected smooth projective variety with D a reduced divisor. Then (X, D) is a toric pair if and only if X admits an int-amplified endomorphism f such that $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is étale.*

In [Men20, Theorem 1.11], replacing f by a suitable power, $f^*|_{N^1(X)}$ can be viewed as a diagonal matrix $\text{Diag}[\lambda_1, \dots, \lambda_\rho]$, where $\rho := \rho(X) = \dim_{\mathbb{R}} N^1(X)$ and λ_i are (possibly the same) integers greater than 1. Here, ρ can be arbitrarily large in general, even if $n = \dim(X)$ is fixed. Nevertheless, the number r of eigenvalues of $f^*|_{N^1(X)}$ (without counting multiplicities) is bounded by n ; see Proposition 4.4.5. Note that if $r = 1$, then f is the usual polarized endomorphism. For another extremal case when $r = n$, we show the strongest splitting rigidity of X . The key idea is to show that the equivariant MMP of X only involves (conic bundle type) Fano contractions of smooth Fano varieties, and then we can apply the adjunction formula in the most comfortable way and prove by induction on the dimension of X .

Theorem 4.1.5 ([MZ20b, Theorem 1.6]). *Let X be a rationally connected smooth projective variety of dimension n . Then*

$$X \cong (\mathbb{P}^1)^{\times n}$$

if and only if X admits a surjective endomorphism f such that the eigenvalues of $f^|_{N^1(X)}$ (without counting multiplicities) are n distinct real numbers greater than 1.*

Remark 4.1.6. In the proof of Theorem 4.1.5, we get a finite surjective morphism

$$\psi : X \rightarrow (\mathbb{P}^1)^{\times n}$$

with $\rho(X) = n$ and all the effective divisors of X being nef. The remaining main difficulty is to show that ψ is an isomorphism or simply $X \cong (\mathbb{P}^1)^{\times n}$. When $n = \rho(X) = 2$, we have $X \cong \mathbb{P}^1 \times \mathbb{P}^1$ by an easy observation of smooth Fano surfaces. However, for the higher dimensional cases, this is in general not true without dynamical constraint. For example, when $n = \rho(X) = 3$, there exists a double cover ψ whose branch locus is a divisor of tridegree $(2, 2, 2)$ (cf. [MM82, Table 3]). Such X admits three different Fano contractions to $\mathbb{P}^1 \times \mathbb{P}^1$ which are singular conic bundles with non-empty discriminant loci. In particular, we have $X \not\cong \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$.

We propose the following splitting question for the general $1 \leq r \leq n$ which reduces Question 4.1.2 to the polarized case. Question 4.1.7 is not true in the general settings of (not necessarily rationally connected) smooth projective varieties; see examples

in [Men20, Section 10].

Question 4.1.7. Let $f : X \rightarrow X$ be a surjective endomorphism of a rationally connected smooth projective variety such that the eigenvalues of $f^*|_{N^1(X)}$ (without counting multiplicities) are distinct integers $\lambda_1, \dots, \lambda_r$ greater than 1. Will we have $X \cong X_1 \times \dots \times X_r$ where each X_i is a smooth projective variety of Fano type and f splits as a product of $f|_{X_i}$ which is λ_i -polarized?

This chapter is based on the joint work with Sheng Meng [MZ20b].

4.2 Preliminaries

Notation 4.2.1. Let X be a projective variety. We use the following notation throughout this chapter.

$\text{Pic}(X)$	the group of Cartier divisors of X modulo linear equivalence $\sim$
$\text{Pic}^\circ(X)$	the neutral connected component of $\text{Pic}(X)$
$\text{NS}(X)$	$\text{Pic}(X)/\text{Pic}^\circ(X)$, the Néron-Severi group of X
$N^1(X)$	$\text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, the space of $\mathbb{R}$ -Cartier divisors modulo numerical equivalence $\equiv$
$N_r(X)$	the space of r -cycles modulo weak numerical equivalence $\equiv_w$ (cf. [MZ18a, Definition 2.2])
$\rho(X)$	$\dim_{\mathbb{R}} N^1(X)$, the Picard number of X
$\kappa(X, D)$	the Iitaka dimension of a $\mathbb{Q}$ -Cartier divisor D
$\text{Nef}(X)$	the cone of nef classes in $N^1(X)$
$\text{PE}^1(X)$	the cone of pseudo-effective classes in $N^1(X)$
$\overline{\text{NE}}(X)$	the cone of pseudo-effective classes in $N_1(X)$

- The above cones are $(f^*)^{\pm 1}$ -invariant for any surjective endomorphism $f : X \rightarrow X$.
- A surjective endomorphism $f : X \rightarrow X$ is *q-polarized* if $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$, or equivalently if $f^*B \equiv qB$ for some

big $\mathbb{R}$ -Cartier divisor B and integer $q > 1$ (cf. [MZ18a, Proposition 3.6]).

- A surjective endomorphism $f : X \rightarrow X$ is *int-amplified* if $f^*L - L = H$ for some ample Cartier divisors L and H , or equivalently if $f^*L - L = H$ for some big $\mathbb{R}$ -Cartier divisors L and H (cf. [Men20, Theorem 1.1]).
- We say that a normal projective variety X is of *Fano type*, if there exists an effective Weil $\mathbb{Q}$ -divisor Δ on X such that the pair (X, Δ) has at worst klt singularities and $-(K_X + \Delta)$ is ample (cf. [PS09, Lemma-Definition 2.6]). If $\Delta = 0$, we say that X is a *Fano variety*.
- We say that a normal variety X is a *toric variety* if X contains an algebraic torus $T = (\mathbb{A}^1)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on X . In this case, let $D := X \setminus T$, which is a divisor; the pair (X, D) is said to be a *toric pair* with B the *toric boundary*.
- A finite surjective morphism is *quasi-étale* if it is étale in codimension one.

The following result is well-known due to the cone theorem.

Lemma 4.2.2 (cf. [KM98, Theorem 3.7], [BCHM10, Corollary 1.3.2]). *Let X be a normal projective variety of Fano type. Then $\text{Nef}(X)$ is generated by finitely many base point free (extremal) Cartier divisors.*

Lemma 4.2.3. *Let X be a projective variety with $\text{Nef}(X) = \text{PE}^1(X)$. Then any generically finite surjective morphism $\pi : X \rightarrow Y$ to a projective variety Y is finite.*

Proof. Let $\pi : X \rightarrow Y$ be a generically finite surjective morphism. Fixing an ample Cartier divisor H on Y , we have that π^*H is big and thus ample by assumption. So π does not contract any curve and hence π is finite. $\square$

Lemma 4.2.4. *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal projective variety such that all the eigenvalues of $f^*|_{\text{N}^1(X)}$ are positive real numbers. Suppose that $(f^m)^*D \equiv \lambda^m D$ for some $\lambda > 0$ and integer $m \geq 1$. Then $f^*D \equiv \lambda D$.*

Proof. We may assume $D \neq 0$ and $m \geq 2$. Let $\varphi := f^*|_{N^1(X)}$ and we regard φ as a matrix. Note that $D \in \ker(\varphi^m - \lambda^m)$ and

$$\varphi^m - \lambda^m = \prod_{i=0}^{m-1} (\varphi - \lambda \cdot \xi_m^i),$$

where ξ_m is a primitive m -th root of unity. Suppose that $D \notin \ker(\varphi - \lambda)$. Then there exists $1 \leq j \leq m-1$ such that

$$\tilde{D} := \left(\prod_{i=0}^{j-1} (\varphi - \lambda \cdot \xi_m^i) \right) (D) \neq 0 \text{ and } (\varphi - \lambda \cdot \xi_m^j)(\tilde{D}) \equiv 0.$$

Since $\lambda \cdot \xi_m^j$ is not a real number but an eigenvalue of φ , we get a contradiction. $\square$

Remark 4.2.5. Let $X := C_1 \times \cdots \times C_n$ with each $C_i \cong \mathbb{P}^1$. Let $f : X \rightarrow X$ be a surjective endomorphism. Then replacing f by a power, we have

$$f = f_1 \times \cdots \times f_n$$

where each $f_i : C_i \rightarrow C_i$ is a surjective endomorphism. Moreover, if all the eigenvalues of $f^*|_{N^1(X)}$ are already positive real numbers, then we don't need to replace f by a power by Lemma 4.2.4. Note that this kind of splitting result holds true when each C_i is a normal projective variety with $H^1(C_i, \mathcal{O}_{C_i}) = 0$; see [San20, Theorem 4.6].

4.3 Proof of Theorem 4.1.4

In this section, we shall answer Question 4.1.3 affirmatively. Let X be a normal projective variety and D a reduced divisor on X . Let $j : U \hookrightarrow X$ be a smooth open subset of X with $\text{codim}(X \setminus U) \geq 2$ and $D \cap U$ being a normal crossing divisor. Denote by

$$\hat{\Omega}_X^1(\log D) := j_* \Omega_U^1(\log D \cap U)$$

where $\Omega_U^1(\log D \cap U)$ is the locally free sheaf of germs of logarithmic 1-forms over U with poles only along $D \cap U$. Note that $\hat{\Omega}_X^1(\log D)$ is a reflexive coherent sheaf on X which is independent of the choice of U .

Proposition 4.3.1 (cf. [MZ19b, Proposition 10.3]). *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X which is of dimension $n \geq 2$ and smooth in codimension 2. Let H be an ample Cartier divisor and $D \subseteq X$ a reduced divisor. Suppose that $f^{-1}(D) = D$ and $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is quasi-étale. Then*

$$c_1(\hat{\Omega}_X^1(\log D)) \cdot H^{n-1} = c_1(\hat{\Omega}_X^1(\log D))^2 \cdot H^{n-2} = c_2(\hat{\Omega}_X^1(\log D)) \cdot H^{n-2} = 0.$$

We follow the idea of [MZ19b, Proposition 10.5] to prove the next proposition.

Proposition 4.3.2. *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety, which is of dimension n , of Fano type and smooth in codimension 2. Let H be an ample Cartier divisor and $D \subseteq X$ a reduced divisor. Suppose that $f^{-1}(D) = D$ and $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is quasi-étale. Then $\hat{\Omega}_X^1(\log D)$ is H -slope semistable.*

Proof. By Lemma 4.2.2, $\text{Nef}(X)$ is generated by finitely many nef divisors. Replacing f by a power, we may assume f^* fixes each extremal ray of $\text{Nef}(X)$. Then, there exist nef divisors $D_1, \dots, D_k$ on X such that $f^*D_i \equiv \lambda_i D_i$ and $H \equiv \sum_{i=1}^k a_i D_i$ with each $a_i > 0$.

Suppose on the contrary that $\hat{\Omega}_X^1(\log D)$ is not H -slope semistable. Let $\mathcal{F} \subseteq \hat{\Omega}_X^1(\log D)$ be the maximal destabilizing subsheaf with respect to H such that

$$\mu_H(\mathcal{F}) := \frac{c_1(\mathcal{F}) \cdot H^{n-1}}{\text{rank } \mathcal{F}} > \mu_H(\hat{\Omega}_X^1(\log D)) = 0.$$

Note that the last equality is due to Proposition 4.3.1. Since $H \equiv \sum a_i D_i$ with each $a_i > 0$, there exists a summand $D_{i_1} \cdots D_{i_{n-1}}$ of H^{n-1} such that $c_1(\mathcal{F}) \cdot D_{i_1} \cdots D_{i_{n-1}} > 0$. Then $D_{i_1} \cdots D_{i_{n-1}} \not\equiv_w 0$ and there exists a nef Cartier divisor D_l such that

$$f^*D_l = \lambda_l D_l \text{ and } D_{i_1} \cdots D_{i_{n-1}} \cdot D_l > 0.$$

So $\deg(f) = \lambda_{i_1} \cdots \lambda_{i_{n-1}} \cdot \lambda_l$ by the projection formula. Since all the D_{i_j} are nef, we have

$$s = \sup \{c_1(\mathcal{F}) \cdot D_{i_1} \cdots D_{i_{n-1}} \mid \mathcal{F} \subseteq \hat{\Omega}_X^1(\log D)\} < \infty.$$

Note that $\lambda_l > 1$ (cf. [Men20, Theorem 1.1]). Then for some $k \gg 1$ and $g := f^k$, we get the following inequality by the projection formula

$$c_1(g^*\mathcal{F}) \cdot D_{i_1} \cdots D_{i_{n-1}} = \lambda_l^k \cdot c_1(\mathcal{F}) \cdot D_{i_1} \cdots D_{i_{n-1}} > s.$$

Let U be a smooth open subset in X such that $\text{codim}(X \setminus U) \geq 3$ and $D \cap U$ is a normal crossing divisor (cf. [MZ19b, Proposition 10.2]). Let $j : g^{-1}(U) \hookrightarrow X$ be the inclusion map and $\mathcal{G} := j_*((g^*\mathcal{F})|_{g^{-1}(U)})$. Then we have

$$c_1(\mathcal{G}) \cdot D_{i_1} \cdots D_{i_{n-1}} = c_1(g^*\mathcal{F}) \cdot D_{i_1} \cdots D_{i_{n-1}} > s.$$

Note that $(g^*\mathcal{F})|_{g^{-1}(U)} \subseteq (g^*\hat{\Omega}_X^1(\log D))|_{g^{-1}(U)} \cong \hat{\Omega}_X^1(\log D)|_{g^{-1}(U)}$, the latter of which is a locally free sheaf. Since $\text{codim}(X \setminus g^{-1}(U)) \geq 2$ and j_* is left exact, $\mathcal{G}$ is a coherent subsheaf of $\hat{\Omega}_X^1(\log D)$. So we get a contradiction. $\square$

Proof of Theorem 4.1.4. Let $n = \dim(X)$. Suppose (X, D) is a toric pair and denote by $T := X \setminus D \cong (k^*)^n$ the big torus. Then the power map

$$T \rightarrow T \text{ via } (x_1, \dots, x_n) \mapsto (x_1^q, \dots, x_n^q)$$

extends to a surjective endomorphism $f : X \rightarrow X$; see [Nak02, Lemma 4]. Note that it also extends to a q -polarized endomorphism $g : \mathbb{P}^n \rightarrow \mathbb{P}^n$. Then f is q -polarized by [MZ18a, Corollary 3.12].

For another direction, X first is of Fano type by Theorem 4.1.1. If $\dim(X) = 1$, then $X \cong \mathbb{P}^1$ and D is a divisor of two distinct points. Assume that $\dim(X) \geq 2$. By Propositions 4.3.1, 4.3.2 and [GKP16, Theorem 1.20], $\hat{\Omega}_X^1(\log D)$ is locally free of rank n . So our theorem follows from [MZ19a, Theorem 4.5] and [BMSZ18, Theorem 1.2]. $\square$

4.4 Dynamics with Hodge index theorem

We begin with the following type of Hodge index theorem which is known to experts.

Lemma 4.4.1 (cf. [DS04, Corollarie 3.2] or [Zha16, Lemma 3.2]). *Let X be a normal projective variety of dimension n . Let $D_1 \not\equiv 0$ and $D_2 \not\equiv 0$ be two nef $\mathbb{R}$ -Cartier divisors such that $D_1 \cdot D_2 \equiv_w 0$. Then $D_1 \equiv aD_2$ for some $a > 0$.*

Proof. We may assume $n \geq 2$. Let H be a very ample Cartier divisor on X . Let S be a general surface on X such that $H^{n-2} \equiv_w S$. Then $D_1|_S \cdot D_2|_S = D_1 \cdot D_2 \cdot H^{n-2} = 0$. By

the Hodge index theorem on S , we have $D_1|_S \equiv aD_2|_S$ for some $a > 0$. Therefore,

$$(D_1 - aD_2) \cdot H^{n-1} = (D_1 - aD_2)^2 \cdot H^{n-2} = 0.$$

By [Zha16, Lemma 3.2], $D_1 \equiv aD_2$. □

We slightly generalize [Zha16, Claim 3.3] (also cf. [DS04, Théorème 3.3]) to the following, which states the semi-negativity of the generalized Hodge index theorem.

Lemma 4.4.2. *Let X be a normal projective variety of dimension n and M some $\mathbb{R}$ -Cartier divisor. Let $D_1, \dots, D_{n-1}$ be nef $\mathbb{R}$ -Cartier divisors such that $D_1 \cdots D_{n-1} \not\equiv_w 0$ and $M \cdot D_1 \cdots D_{n-1} = 0$. Then $M^2 \cdot D_1 \cdots D_{n-2} \leq 0$.*

Proof. We may assume $n \geq 2$. Write $D_i = \lim_{m \rightarrow \infty} D_{i,m}$ with $D_{i,m}$ ample $\mathbb{R}$ -Cartier divisors for each i . Fix an ample Cartier divisor H on X . Since $H \cdot D_{1,m} \cdots D_{n-1,m} > 0$, we have

$$(M + r(m)H) \cdot D_{1,m} \cdots D_{n-1,m} = 0$$

for some unique real number $r(m)$. Therefore,

$$(M + r(m)H)^2 \cdot D_{1,m} \cdots D_{n-2,m} \leq 0$$

by the negativity in [Zha16, Claim 3.3]. By the assumption $D_1 \cdots D_{n-1} \not\equiv_w 0$, we have $H \cdot D_{1,m} \cdots D_{n-1,m} > 0$ and hence $\lim_{m \rightarrow \infty} r(m) = 0$. Therefore, $M^2 \cdot D_1 \cdots D_{n-2} \leq 0$ by letting $m \rightarrow \infty$. □

The following lemma is known in [DS04, Corollaire 3.5] for the case of compact Kähler manifolds. We follow the idea there and reprove it in the algebraic context.

Lemma 4.4.3. *Let X be a normal projective variety of dimension n . Let $D, D', D_1, \dots, D_k$ ($k \leq n - 2$) be nef $\mathbb{R}$ -Cartier divisors such that $D \cdot D' \cdot D_1 \cdots D_k \equiv_w 0$. Then $(aD + a'D') \cdot D_1 \cdots D_k \equiv_w 0$ for some real numbers $(a, a') \neq (0, 0)$. Furthermore, if $D \cdot D_1 \cdots D_k \not\equiv_w 0$, then $a' \neq 0$ and (a, a') is unique up to a scalar.*

Proof. We may assume $n \geq 2$. If $D \cdot D_1 \cdots D_k \equiv_w 0$, we simply take $(a, a') = (1, 0)$. Now we assume that $D \cdot D_1 \cdots D_k \not\equiv_w 0$ (and hence $D_1 \cdots D_k \not\equiv_w 0$).

Fix ample Cartier divisors $A_1, \dots, A_{n-k-1}$ on X . Denote by $V := \mathbb{R}D + \mathbb{R}D'$ and

$$W := \{x \in N^1(X) \mid x \cdot D_1 \cdots D_k \cdot A_1 \cdots A_{n-k-1} = 0\}$$

subspaces of $N^1(X)$. Note that

$$D \notin W \text{ and } \tilde{D} := aD + a'D' \in V \cap W$$

where $a := D' \cdot D_1 \cdots D_k \cdot A_1 \cdots A_{n-k-1}$ and $a' := -D \cdot D_1 \cdots D_k \cdot A_1 \cdots A_{n-k-1} \neq 0$.

Then $\dim_{\mathbb{R}} V \cap W \leq 1$ and the uniqueness follows.

For each $1 \leq i \leq n - k - 1$, consider the following bilinear form on $N^1(X)$:

$$q_i(x, y) := x \cdot y \cdot D_1 \cdots D_k \cdot A_1 \cdots A_{i-1} \cdot A_{i+1} \cdots A_{n-k-1}.$$

Then it follows from Lemma 4.4.2 and $D \cdot D' \cdot D_1 \cdots D_k \equiv_w 0$ that q_i is semi-negative on W but semi-positive on V . Hence $q_i(\tilde{D}, \tilde{D}) = 0$.

For any $w \in W$ and $\lambda \in \mathbb{R}$, we have $q_i(\lambda\tilde{D} - w, \lambda\tilde{D} - w) \leq 0$. Then the inequality

$$q_i(w, w) - 2\lambda q_i(\tilde{D}, w) \leq 0$$

holds for any $\lambda \in \mathbb{R}$ and $w \in W$. This happens only when $q_i(\tilde{D}, w) = 0$ for any $w \in W$.

Note that W and A_i span $N^1(X)$ because W is a hyperplane of $N^1(X)$ and $A_i \notin W$. Note also that $q_i(\tilde{D}, A_i) = 0$. Then

$$q_i(\tilde{D}, x) = \tilde{D} \cdot D_1 \cdots D_k \cdot A_1 \cdots A_{i-1} \cdot x \cdot A_{i+1} \cdots A_{n-k-1} = 0$$

for any $x \in N^1(X)$. This implies that $V \cap W$ is independent of the choice of each A_i . So

$$\tilde{D} \cdot D_1 \cdots D_k \cdot x_1 \cdots x_{n-k-1} = 0$$

for any divisors $x_1, \dots, x_{n-k-1} \in N^1(X)$, which means $\tilde{D} \cdot D_1 \cdots D_k \equiv_w 0$. $\square$

Proposition 4.4.4 (cf. [DS04, Lemme 4.4]). *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal projective variety X of dimension n . Let $D, D', D_1, \dots, D_k$ ($k \leq n - 2$) be nef $\mathbb{R}$ -Cartier divisors such that $D \cdot D_1 \cdots D_k \not\equiv_w 0$ and $D' \cdot D_1 \cdots D_k \not\equiv_w 0$. Suppose $f^*(D \cdot D_1 \cdots D_k) \equiv_w \lambda D \cdot D_1 \cdots D_k$ and $f^*(D' \cdot D_1 \cdots D_k) \equiv_w \lambda' D' \cdot D_1 \cdots D_k$ for two real numbers $\lambda \neq \lambda'$. Then $D \cdot D' \cdot D_1 \cdots D_k \not\equiv_w 0$.*

Proof. Note that $n \geq 2$ since $\rho(X) \geq 2$ by the assumption. Suppose on the contrary that $D \cdot D' \cdot D_1 \cdots D_k \equiv_w 0$. By Lemma 4.4.3, $(aD + a'D') \cdot D_1 \cdots D_k \equiv_w 0$ for some $(a, a') \in \mathbb{R}^* \times \mathbb{R}^*$ unique up to a scalar. So for any $\mathbb{R}$ -Cartier divisors $H_1, \dots, H_{n-k-2}$, we have

$$(aD + a'D') \cdot D_1 \cdots D_k \cdot H_1 \cdots H_{n-k-2} = 0.$$

Hence, by the projection formula, we have

$$(a\lambda D + a'\lambda'D') \cdot D_1 \cdots D_k \cdot f^*H_1 \cdots f^*H_{n-k-2} = 0.$$

Note that $f^*|_{N^1(X)}$ is invertible. Then $(a\lambda D + a'\lambda'D') \cdot D_1 \cdots D_k \equiv_w 0$, a contradiction with the uniqueness of (a, a') up to a scalar and $\lambda \neq \lambda'$. $\square$

Now we state the main proposition in this section about the dynamical rigidity at the first glance, which will be crucially used in Section 4.5.

Proposition 4.4.5. *Let $f : X \rightarrow X$ be a surjective endomorphism of an n -dimensional $\mathbb{Q}$ -Gorenstein normal projective variety X of Fano type. Suppose that the eigenvalues of $f^*|_{N^1(X)}$ (without counting multiplicities) are distinct real positive numbers $\{\lambda_1, \dots, \lambda_r\}$ with $r \geq n$. Then the following assertions hold.*

- (1) $\rho(X) = n = r$.
- (2) The nef cone $\text{Nef}(X)$ is generated by base-point-free divisors $D_1, \dots, D_n$ such that $f^*D_i \sim \lambda_i D_i$.
- (3) For each i , λ_i is a positive integer, $D_i^2 \equiv_w 0$ and $\kappa(X, D_i) = 1$.
- (4) $D_1 \cdots D_n > 0$ and $\deg(f) = \lambda_1 \cdots \lambda_n$.
- (5) $\text{Nef}(X) = \text{PE}^1(X)$. In particular, X is a Fano variety.

Proof. It is trivial if $n = \dim(X) = 1$. Assume that $n \geq 2$.

By Lemma 4.2.2, $\text{Nef}(X)$ is generated by base-point-free divisors $D_1, \dots, D_m$. Note that $m \geq \rho(X) \geq r \geq n$. Since $f^*|_{N^1(X)}$ is linearly invertible and $\text{Nef}(X)$ is $f^*|_{N^1(X)}$ -invariant, $(f^s)^*$ fixes all the extremal rays $R_{D_1}, \dots, R_{D_m}$ of $\text{Nef}(X)$ for some $s > 0$. By

Lemma 4.2.4, we may assume that $f^*D_i \equiv \lambda_i D_i$ for $1 \leq i \leq n$ and $f^*D_i \equiv \mu_i D_i$ for $n+1 \leq i \leq m$ with $\mu_i \in \{\lambda_1, \dots, \lambda_r\}$. Since D_i is integral, λ_i is an integer.

We apply Proposition 4.4.4 several times. Since $\lambda_1 \neq \lambda_2$ and $\lambda_1 \neq \lambda_3$, we have $D_1 \cdot D_2 \not\equiv_w 0$ and $D_1 \cdot D_3 \not\equiv_w 0$. Since $\lambda_1 \lambda_2 \neq \lambda_1 \lambda_3$, we further have $D_1 \cdot D_2 \cdot D_3 \not\equiv_w 0$. Repeating the same argument, we have $D_1 \cdot D_2 \cdots D_{n-1} \cdot D_n \neq 0$ and $\deg(f) = \lambda_1 \cdots \lambda_n$ by the projection formula. So (4) is proved.

Suppose $m > n$. Note that D_m and D_n are nef and linearly independent in $N^1(X)$. Then $D_m \cdot D_n \not\equiv_w 0$ by Lemma 4.4.1. Similarly, $D_m \cdot D_{n-2} \not\equiv_w 0$. Hence $D_m \cdot D_n \cdot D_{n-2} \not\equiv_w 0$ by Proposition 4.4.4. Repeatedly, we have $D_m \cdot D_n \cdot D_{n-2} \cdots D_1 \neq 0$. Applying the projection formula, we have $\deg(f) = \mu_m \lambda_1 \cdots \lambda_{n-2} \lambda_n$, which implies $\mu_m = \lambda_{n-1}$. By the same argument after replacing D_n by D_{n-1} , our $D_m \cdot D_{n-1} \cdot D_{n-2} \cdots D_1 \neq 0$ and thus $\mu_m = \lambda_n = \lambda_{n-1}$, a contradiction. In particular, $m = \rho(X) = r = n$.

Replacing D_m by D_i in the above argument, we have $D_i^2 \equiv_w 0$ for each i . Since D_i is base point free, $\kappa(X, D_i) = 1$. Note that $\text{Pic}_{\mathbb{Q}}(X) \cong \text{NS}_{\mathbb{Q}}(X)$. Then we have $f^*D_i \sim \lambda_i D_i$ after replacing D_i by a suitable multiple. So (1)~(3) are proved.

Let $D \in \partial \text{Nef}(X)$. Then without loss of generality, by (2), we may write $D = \sum_{i=1}^{n-1} a_i D_i$ with $a_i \geq 0$. By (3), $D^n = 0$ and thus D is not big. So $\text{Nef}(X) = \text{PE}^1(X)$. Note that $-K_X$ is a $\mathbb{Q}$ -Cartier big divisor. Then $-K_X$ is further ample. So (5) is proved. $\square$

4.5 Proof of Theorem 4.1.5

In this section, we prove Theorem 4.1.5 and use Notation 4.5.1 throughout this section.

Notation 4.5.1. Let X be a smooth projective variety which is rationally connected of dimension n . Let $f : X \rightarrow X$ be a surjective endomorphism such that the eigenvalues of $f^*|_{N^1(X)}$ (without counting multiplicities) are n distinct real numbers $\Lambda := \{\lambda_1, \dots, \lambda_n\}$ which are greater than 1.

Proposition 4.5.2. *There exist f -periodic prime divisors $D_1, \dots, D_n$ such that:*

- (1) *each D_i is a smooth Fano projective variety and $f^*D_i \sim \lambda_i D_i$;*

(2) *there exist unique (up to isomorphism) f -equivariant fibrations*

$$\phi_i : X \rightarrow Y_i \cong \mathbb{P}^1$$

with $f|_{Y_i}$ being λ_i -polarized. Each D_i is a smooth fibre of ϕ_i ; and

(3) *replacing f by a positive power, $(f|_{D_i})^*|_{N^1(D_i)}$ has eigenvalues $\Lambda \setminus \{\lambda_i\}$.*

Proof. Since f is int-amplified (cf. [Men20, Theorem 1.1]), X is of Fano type by Theorem 4.1.1. By Proposition 4.4.5, X is Fano, $\rho(X) = n$, and there are n base-point-free effective divisors $D_i \neq 0$ such that $f^*D_i \sim \lambda_i D_i$ and $\text{Nef}(X)$ is generated by $D_1, \dots, D_n$.

Denote by $\phi_i : X \rightarrow Y_i$ the Iitaka fibration of D_i . Then $D_i = \phi_i^* H_i$ for some ample $\mathbb{R}$ -Cartier divisor H_i on Y_i . By Proposition 4.4.5, $\dim(Y_i) = \kappa(X, D_i) = 1$. Since X is rationally connected, $Y_i \cong \mathbb{P}^1$. For any curve C with $\phi_i(C)$ being a point, by the projection formula, we have

$$D_i \cdot f_* C = f^* D_i \cdot C = \lambda_i D_i \cdot C = \lambda_i H_i \cdot (\phi_i)_* C = 0.$$

Then $f(C)$ is also contracted by ϕ_i . By the rigidity lemma (cf. [Deb01, Lemma 1.15]), ϕ_i is f -equivariant and denote by $g_i := f|_{Y_i}$. Note that g_i is then λ_i -polarized.

Let $p_i : X \rightarrow Z_i \cong \mathbb{P}^1$ be another f -equivariant fibration such that $f|_{Z_i}$ is λ_i -polarized. Let F_i be a general fibre of p_i . Then p_i is the Iitaka fibration of F_i and $f^* F_i \sim \lambda_i F_i$. Then F_i lies in the extremal ray R_{D_i} ; hence p_i and ϕ_i are the same up to an isomorphism. We may replace D_i by a general fibre of ϕ_i . Then D_i is smooth. So (2) is satisfied.

By the adjunction formula,

$$-K_{D_i} = -(K_X + D_i)|_{D_i} \sim -K_X|_{D_i}$$

is ample. So D_i is Fano and (1) is satisfied.

By [Fak03, Theorem 5.1], g_i has Zariski dense periodic points. So we may further replace D_i by an f -periodic one. By Proposition 4.4.5, $D_1 \cdots D_n > 0$. So $D_j|_{D_i} \neq 0$ for $j \neq i$ and after a suitable iteration of f , we have

$$(f|_{D_i})^*(D_j|_{D_i}) \sim \lambda_j D_j|_{D_i}$$

for $j \neq i$. So $(f|_{D_i})^*|_{N^1(D_i)}$ has at least $n-1$ distinct real eigenvalues $\{\lambda_1, \dots, \widehat{\lambda_i}, \dots, \lambda_n\}$. Further all the eigenvalues of $(f|_{D_i})^*|_{N^1(D_i)}$ are positive integers after replacing f by a power, since $\text{Nef}(D_i)$ is a rational polyhedron. So (3) is satisfied by applying Proposition 4.4.5 for D_i . $\square$

Proposition 4.5.3. *There are f -equivariant Fano contractions*

$$\pi_i : X \rightarrow X_i, \quad 1 \leq i \leq n$$

of K_X -negative extremal rays, such that:

- (1) *The eigenvalues of $(f|_{X_i})^*|_{N^1(X_i)}$ are $\Lambda \setminus \{\lambda_i\}$.*
- (2) *π_i is a conic bundle and X_i is a smooth Fano projective variety.*

Proof. We apply Proposition 4.5.2 and use the same notation there. First note that $\sum_{i=1}^n D_i$ is ample and

$$\left(\sum_{i=1}^n D_i\right)^n = (n!)D_1 \cdots D_n > 0.$$

So we have

$$D_1 \cap \cdots \cap \widehat{D_i} \cap \cdots \cap D_n \neq \emptyset.$$

Let C_i be an irreducible curve of $D_1 \cap \cdots \cap \widehat{D_i} \cap \cdots \cap D_n$. By Proposition 4.4.5, X is Fano. Then we have

$$K_X \cdot C_i < 0 \text{ and } D_j \cdot C_i = 0$$

for $j \neq i$ (recall that $D_i^2 \equiv_w 0$). Since the space spanned by $\text{nef } D_1, \dots, \widehat{D_i}, \dots, D_n$ is an f^* -invariant hyperplane of $N^1(X)$, its dual space is f_* -invariant 1-dimensional and contains R_{C_i} as an extremal ray in $\overline{\text{NE}}(X)$. Therefore, R_{C_i} induces an f -equivariant contraction (cf. [KM98, Theorem 3.7]) $\pi_i : X \rightarrow X_i$ for $1 \leq i \leq n$. By Proposition 4.4.5 (5), $\text{Nef}(X) = \text{PE}^1(X)$. So it follows from Lemma 4.2.3 that π_i is a Fano contraction, i.e., $\dim(X_i) < \dim(X)$. By the cone theorem (cf. [KM98, Theorem 3.7]), for any $j \neq i$, $D_j = \pi_i^* L_j$ for some Cartier divisor L_j on X_i . Then $(f|_{X_i})^* L_j \equiv \lambda_j L_j$. By Proposition 4.4.5, $\rho(X) = n$ and hence

$$\rho(X_i) = \rho(X) - 1 = n - 1.$$

So the set of eigenvalues of $(f|_{X_i})^*|_{N^1(X_i)}$ contains $\Lambda \setminus \{\lambda_i\}$ and hence coincides with it (cf. Proposition 4.4.5). This implies (1).

Consider the morphism

$$\psi : X \rightarrow X_i \times Y_i$$

induced from $\pi_i : X \rightarrow X_i$ and $\phi_i : X \rightarrow Y_i$ (cf. Proposition 4.5.2). Denote by $p_1 : X_i \times Y_i \rightarrow X_i$ and $p_2 : X_i \times Y_i \rightarrow Y_i$ the two projections. Let

$$H := p_1^*\left(\sum_{j \neq i} L_j\right) + p_2^*(\phi_i(D_i))$$

be a Cartier divisor on $X_i \times Y_i$. Then $\psi^*H = \sum_{i=1}^n D_i$ is ample and hence ψ contracts no curve of X . Note that

$$\dim(X_i \times Y_i) = \dim(X_i) + 1 \leq \dim(X).$$

So ψ is a finite surjective morphism and we have $\dim(X_i) = \dim(X) - 1$. Now both ψ and p_1 are equi-dimensional. Then $\pi_i = p_1 \circ \psi$ is also equi-dimensional. By [And85, Theorem 3.1], π_i is a conic bundle and X_i is smooth. Note that X_i is rationally connected. Applying Proposition 4.4.5 together with (1), we see that X_i is Fano. So (2) is proved. $\square$

Proof of Theorem 4.1.5. One direction is simple, for example we may take $f = f_1 \times \cdots \times f_n$, where $f_i([a : b]) = [a^{i+1} : b^{i+1}]$.

Now we consider another direction and show by induction on $n = \dim(X)$. It is trivial if $n = 1$. If $n = 2$, by Proposition 4.4.5, X is a smooth Fano surface of Picard number 2 with $\text{Nef}(X) = \text{PE}^1(X)$. Then $X \cong \mathbb{P}^1 \times \mathbb{P}^1$. From now on, suppose Theorem 4.1.5 holds for $n - 1$ with $n \geq 3$.

We use the same notation as in Propositions 4.5.2 and 4.5.3. By induction, we have $X_i \cong D_i \cong (\mathbb{P}^1)^{\times(n-1)}$. Note that $f|_{X_i}$ splits (cf. Remark 4.2.5) and for $j \neq i$, there are f -equivariant natural projections

$$p_j : X_i \cong (\mathbb{P}^1)^{\times(n-1)} \rightarrow Z_j \cong \mathbb{P}^1$$

such that $f|_{Z_j}$ is λ_j -polarized. We may assume $Z_j = Y_j$ and $p_j \circ \pi_i = \phi_j$ by the uniqueness property in Proposition 4.5.2. Then $D_1 \cap \cdots \cap \widehat{D_i} \cap \cdots \cap D_n$ intersects transversally and

is a general fibre ℓ_i of π_i . In particular, we have

$$d := D_1 \cdots D_n = D_i \cdot \ell_i$$

and $D_i \cdot \ell_j = 0$ for $i \neq j$ since $D_i^2 \equiv_w 0$. Note that $K_X \cdot \ell_i = -2$. Since $D_1, \dots, D_n$ is a basis for $N^1(X)$, we may write

$$K_X \equiv \sum_{i=1}^n a_i D_i$$

for some rational numbers a_i . Intersecting the above numerical equivalence with ℓ_i , we have $a_i = -2/d$ for each i and thus

$$-dK_X \equiv 2 \sum_{i=1}^n D_i.$$

Therefore,

$$(-dK_X)^{n-1} \cdot D_1 = 2^{n-1}(n-1)!d.$$

From another aspect, we apply the adjunction formula

$$K_{D_1} = (K_X + D_1)|_{D_1} = K_X|_{D_1}$$

and note that $(-K_{D_1})^{n-1} = 2^{n-1}(n-1)!$ since $D_1 \cong (\mathbb{P}^1)^{\times(n-1)}$. Then

$$(-dK_X)^{n-1} \cdot D_1 = (-dK_{D_1})^{n-1} = 2^{n-1}(n-1)!d^{n-1}$$

Therefore, $d^{n-1} = d$ and hence $d = 1$ since we assumed $n \geq 3$. Consider the morphism

$$\psi : X \rightarrow X_1 \times Y_1 \cong (\mathbb{P}^1)^{\times n}$$

induced from $\pi_1 : X \rightarrow X_1$ and $\phi_1 : X \rightarrow Y_1$. Note that

$$\deg(\psi) = D_1 \cdot \ell_1 = d = 1.$$

So the theorem is proved. □

We end up this chapter with the following remark.

Remark 4.5.4. It seems also to be an interesting (and doable) topic if we turn to the Kähler setting. In other words, one may generalize our Theorem 4.1.5 and Question 4.1.7 with $N^1(X)$ replaced by the Bott-Chern cohomology group $H_{\text{BC}}^{1,1}(X)$, and then consider the eigenvalues. At least for compact Kähler threefold case, we have developed the tool of the equivariant minimal model program (see [Zho19]=Chapter 2).

Non-isomorphic endomorphisms of Fano threefolds

5.1 Introduction

In this chapter, we work over an algebraically closed field k of characteristic 0. We continue to study [MZ20b, Question 1.2]=Question 4.1.2. Let $f : X \rightarrow X$ be a surjective endomorphism of a projective variety X . Recall that f is *int-amplified*, if $f^*L - L$ is ample for some ample Cartier divisor L on X , or equivalently, if every eigenvalue of $f^*|_{\mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{C}}$ has modulus > 1 (cf. Lemma 5.2.5). We say that f is *polarized* if $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$ (cf. Lemma 5.2.4). Note that every polarized endomorphism is int-amplified.

Question 4.1.2 itself generalizes the following long-standing Conjecture 5.1.1 of the 1980's. It has been proved for homogeneous spaces (cf. [PS89]), for Fano threefolds of Picard number one (cf. [ARVdV99], [HM03]), and for hypersurfaces of the projective space (cf. [PS89, Proposition 8], [Bea01]). Recall that a normal projective variety X is *Fano* if the anti-canonical divisor $-K_X$ is an ample $\mathbb{Q}$ -Cartier divisor.

Conjecture 5.1.1. Let X be a smooth Fano variety of Picard number one. Suppose X admits a non-isomorphic surjective endomorphism. Then X is a projective space.

In this chapter, we shall give a positive answer to Question 4.1.2 for smooth Fano threefolds (cf. Theorem 5.1.4). More generally, in Theorem 5.1.2 below, we give the criterion for the existence of a non-isomorphic surjective endomorphism.

Now we state our main results of this chapter.

Theorem 5.1.2 ([MZZ21, Main Theorem]). *A smooth Fano threefold is either toric or a product of $\mathbb{P}^1$ and a del Pezzo surface if and only if it admits a non-isomorphic surjective endomorphism.*

Theorem 5.1.2 follows immediately from Theorems 5.1.3 and 5.1.4 below.

Recall that a surjective endomorphism $f : X \rightarrow X$ is *amplified* if $f^*L - L$ is ample for some (not necessarily ample) Cartier divisor L ([KR17]). Note that “int-amplified” $\Rightarrow$ “amplified” and the converse direction holds true if X is Fano (cf. Lemma 5.2.6).

Theorem 5.1.3 ([MZZ21, Theorem 1.3]). *Let X be a smooth Fano threefold. Then the following assertions are equivalent.*

- (1) *X is a product of $\mathbb{P}^1$ and a del Pezzo surface.*
- (2) *X admits a non-isomorphic surjective endomorphism which is not polarized even after iteration.*
- (3) *X admits a non-isomorphic surjective endomorphism which is non-amplified (or equivalently, non-int-amplified) (Lemma 5.2.6).*

Theorem 5.1.4 ([MZZ21, Theorem 1.4]). *Let X be a smooth Fano threefold. Then the following assertions are equivalent.*

- (1) *X is a toric variety.*
- (2) *X admits a polarized endomorphism.*
- (3) *X admits an amplified (or equivalently, int-amplified) endomorphism (Lemma 5.2.6).*

Question 4.1.2 is known to be true under further assumption that f has totally invariant ramification (cf. [MZ20b, Theorem 1.5]=Theorem 4.1.4, [MZ19a, Corollary 1.4] and

[HN11, Theorem 1.2]). However, even when X is toric, f^{-1} may not fix its big torus, e.g., take a general endomorphism of $\mathbb{P}^n$. This actually makes the proving more difficult.

Fano threefolds do not admit any “primitive” endomorphisms unless they are toric:

Corollary 5.1.5 ([MZZ21, Corollary 1.5]). *Let X be a smooth Fano threefold, and $f : X \rightarrow X$ a non-isomorphic surjective endomorphism. Then either X is toric; or $X = \mathbb{P}^1 \times S$ where S is a del Pezzo surface and (after iteration) $f = f_P \times f_S$ with $f_P : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ and $f_S \in \text{Aut}(S)$.*

Del Pezzo surfaces being rational, our Theorem 5.1.2 recovers [Zha12, Theorem 1.2]:

Corollary 5.1.6 ([MZZ21, Corollary 1.6]). *A smooth Fano threefold admitting a non-isomorphic surjective endomorphism is rational.*

Remark 5.1.7 (Classification-free approach). In the first trial of proving our theorems, we utilized the classification of smooth Fano varieties which is available in dimension ≤ 3 but is not plausible in higher dimensions; it needs tedious and patient case by case checking and it would actually be 10-page longer than the present (second) classification-free approach; this second approach adopted here might be useful in solving Questions and Conjectures: 4.1.2, 5.1.1, 5.1.8 and 5.1.9 in higher dimensions.

Besides the enlightening structural propositions in [MM83], our key ingredient of proving Theorems 5.1.3 and 5.1.4 (both for “(3) $\Rightarrow$ (1)”), without resorting to the detailed classification of Fano threefolds, is to show that a fibre-dimension one Fano contraction (if exists) $\tau : X \rightarrow Y$ is eventually a splitting $\mathbb{P}^1$ -bundle over a toric surface by the following steps, where the second and last implications require the dynamical assumption:

$$\tau \Rightarrow \text{conic bundle} \Rightarrow \mathbb{P}^1\text{-bundle} \Rightarrow \text{algebraic } \mathbb{P}^1\text{-bundle} \Rightarrow \text{splitting } \mathbb{P}^1\text{-bundle}.$$

The first three implications are proved in Theorem 5.4.1. The last implication is discussed in Theorems 5.2.13 and 5.6.4 even for not necessarily Fano X , which is crucial in Step B below. Note that a splitting $\mathbb{P}^1$ -bundle over a toric variety is toric (cf. Proposition 5.2.9).

In the situation of Theorem 5.1.4 (for “(3) $\Rightarrow$ (1)”), we further show:

(Step A) If $\rho(X) \leq 2$, then X is either $\mathbb{P}^3$, a (toric) splitting $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$, or a (toric) blow-up of $\mathbb{P}^3$ along a line (cf. Theorem 5.5.1).

(Step B) If $\rho(X) \geq 3$, then X is a (toric) blow-up of a (not necessarily Fano) splitting $\mathbb{P}^1$ -bundle X' over a toric surface along disjoint curves which are intersections of some $(f|_{X'})^{-1}$ -invariant (after iteration) prime divisors (cf. Proof of Theorem 5.8.1).

The proof of **Step B** heavily relies on the study of the following Question 5.1.8 on f^{-1} -periodic subvarieties which generalizes another long-standing Conjecture 5.1.9 below.

Question 5.1.8. Let f be an int-amplified endomorphism of a rationally connected (or rational, or toric) smooth projective variety X . Let Σ be the union of f^{-1} -periodic (closed) subvarieties. Is X toric, and if so, is there a toric pair (X, Δ) such that $\Sigma \subseteq \Delta$?

We will study Question 5.1.8 for the case when Σ is a divisor, with results Theorems 5.3.2 and 5.3.3, confirming the surface case and the case of splitting $\mathbb{P}^1$ -bundles over rational surfaces.

For the surface case, Nakayama confirmed Sato's conjecture that a smooth projective rational surface Y with a non-isomorphic surjective endomorphism is toric (cf. [Nak02]). For our purpose of studying a conic bundle X over a toric surface Y , i.e., for **Step B** above, this is not enough; we need the info on the $(f|_Y)^{-1}$ -periodic curves as in Question 5.1.8. Such info is completed, in view of Lemmas 5.2.10, 5.3.1 and Theorem 5.3.2.

Conjecture 5.1.9 below appears in the study of dynamical systems. We refer to Theorem 5.2.16 for the known cases of Conjecture 5.1.9 by the works [PS89], [CLN00], [Bea01], [Gur03], [NZ10, Section 5] and [Hör17].

Conjecture 5.1.9. Let f be an endomorphism of $X := \mathbb{P}^n$ with $\deg(f) \geq 2$. Then any f^{-1} -invariant (or f^{-1} -periodic) closed subvariety Y is a linear subspace of X .

This chapter is based on the joint work with Sheng Meng and De-Qi Zhang [MZZ21].

5.2 Preliminaries

We use the following notation throughout this chapter.

Notation 5.2.1. Let X be a projective variety.

- The symbols $\sim$ (resp. $\sim_{\mathbb{Q}}, \equiv$) denote the *linear equivalence* (resp. $\mathbb{Q}$ -*linear equivalence*, *numerical equivalence*) on $\mathbb{Q}$ - (or $\mathbb{R}$ -) Cartier divisors. We also use $\equiv$ to denote the *numerical equivalence* of 1-cycles on X .
- Denote by $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^{\circ}(X)$ the *Néron-Severi group* of X . Let $\text{N}^1(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ the space of $\mathbb{R}$ -Cartier divisors modulo numerical equivalence and $\rho(X) := \dim_{\mathbb{R}} \text{N}^1(X)$ the *Picard number* of X . Let $\text{N}_1(X)$ be the dual space of $\text{N}^1(X)$ consisting of 1-cycles. Denote by $\text{Nef}(X)$ the cone of *nef divisors* in $\text{N}^1(X)$ and $\overline{\text{NE}}(X)$ the dual cone consisting of *pseudo-effective 1-cycles* in $\text{N}_1(X)$.
- Let $f : X \rightarrow X$ be a surjective endomorphism. A subset $Y \subseteq X$ is f^{-1} -*invariant* (resp. f^{-1} -*periodic*) if $f^{-1}(Y) = Y$ (resp. $f^{-s}(Y) = Y$ for some $s > 0$).
- A surjective endomorphism $f : X \rightarrow X$ is q -*polarized* if $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$; see Lemma 5.2.4.
- A surjective endomorphism $f : X \rightarrow X$ is *amplified* if $f^*L - L$ is ample for some Cartier divisor L . We say that f is *int-amplified* if $f^*L - L$ is ample for some *ample* Cartier divisor L ; see Lemma 5.2.5.
Clearly, “polarized” $\Rightarrow$ “int-amplified” $\Rightarrow$ “amplified”.
- A smooth projective variety X is *rationally connected* if any two general points of X can be connected by a chain of rational curves.
- A normal projective variety X is of *Fano type*, if there is an effective Weil $\mathbb{Q}$ -divisor Δ on X such that the pair (X, Δ) has at worst klt singularities and $-(K_X + \Delta)$ is ample and $\mathbb{Q}$ -Cartier. If $\Delta = 0$, we say that X is a (*klt*) *Fano variety*. A smooth Fano surface is usually called a *del Pezzo surface*.
- Let X be a smooth Fano threefold. We say that X is *imprimitive* if it is isomorphic to the blow-up of a smooth Fano threefold Y along a smooth irreducible curve. We say that X is *primitive* if it is not imprimitive (cf. [MM83, Definition 1.3]).

- A normal projective variety X of dimension n is a *toric variety* if X contains a *big torus* $T = (\mathbb{A}^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on X . In this case, $B := X \setminus T$ is a divisor; the pair (X, B) is said to be a *toric pair* with B the *toric boundary*.

When (X, B) is a toric pair, it is known that the reflexive sheaf of logarithmic 1-form $\widehat{\Omega}_X^1(\log B) \cong \mathcal{O}_X^{\oplus n}$ (cf. [Ful93, Section 4.3, page 87] and [MZ19a, Remark 4.6]) and $K_X + B \sim 0$. Hence, if X is further smooth projective, then B has exactly $\rho(X) + \dim(X)$ irreducible components (see e.g., the inequality and cohomology exact sequence in [MZ19a, Theorem 4.5] and its proof).

- Let $\tau : X \rightarrow Y$ be the blow-up of a smooth toric variety Y along a smooth closed subvariety C . We say that τ is a *toric blow-up* if there exists some big torus T acting on Y with $T(C) = C$. In this case, X is still toric.
- A fibration $\tau : X \rightarrow Y$ of smooth projective varieties is a *conic bundle* if every fibre is isomorphic to a conic, i.e., a scheme of zeros of a nonzero homogeneous form of degree 2 on $\mathbb{P}^2$.

Denote by $\Delta_\tau := \{y \in Y \mid \tau \text{ is not smooth over } y\}$ the *discriminant* of τ . When Y is a surface, either $\Delta_\tau = \emptyset$ or Δ_τ is a (not necessarily irreducible) curve with only ordinary double points (cf. [MM83, Section 6]). If X is further assumed to be Fano, then we say τ is a *Fano conic bundle*.

- Let $\tau : X \rightarrow Y$ be a fibration of smooth projective varieties. A subvariety $S \subseteq X$ is said to be a *(cross-)section* of τ if the restriction $\tau|_S : S \cong Y$ is an isomorphism onto Y . We say that τ is a (smooth) $\mathbb{P}^1$ -*bundle*, if τ is smooth and every fibre is isomorphic to $\mathbb{P}^1$. We say that τ is an *algebraic* $\mathbb{P}^1$ -*bundle*, if τ is a $\mathbb{P}^1$ -bundle and $X \cong \mathbb{P}_Y(\mathcal{E})$ for some locally free rank-two sheaf $\mathcal{E}$ on Y . An algebraic $\mathbb{P}^1$ -bundle $X := \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ is said to be a *splitting* $\mathbb{P}^1$ -*bundle* if $\mathcal{E}$ is a direct sum of two invertible sheaves. Note that a splitting $\mathbb{P}^1$ -bundle over a toric normal projective variety is toric (cf. Proposition 5.2.9).

For the convenience of readers, we now recall several results to be used in the subsequent sections. Lemmas 5.2.2 ~ 5.2.6 supplement Notation 5.2.1 on endomorphisms.

Lemma 5.2.2. *Let f be a surjective endomorphism of a normal projective variety X . Then any finite sequence of minimal model program starting from X , is f -equivariant (after iterating f), if one of the following conditions is satisfied.*

- (1) *The (closed) Mori cone $\overline{\text{NE}}(X)$ of pseudo-effective 1-cycles has only finitely many extremal rays (this holds when X is of Fano type) (cf. [Zha10, Lemma 2.11], [KM98, Theorem 3.7]).*
- (2) *$\deg(f) \geq 2$ and $\dim(X) = 2$ (then X is known to have only lc singularities) (cf. [Nak02, Proposition 11], [MZ19b, Theorem 4.7]).*
- (3) *X admits an int-amplified endomorphism (cf. [MZ20a, Theorem 1.1]).*

The following generalizes [Zha10, Lemma 2.9] to the int-amplified case.

Lemma 5.2.3 (cf. [Zha10, Lemma 2.9]). *Let $f : X \rightarrow X$ be a surjective endomorphism of a $\mathbb{Q}$ -factorial normal projective variety of dimension n . Let $S(X)$ be the set of prime divisors D on X such that $D|_D$ is not pseudo-effective. Then $f^{-1}(S(X)) = S(X)$. If f is further assumed to be int-amplified, then $S(X)$ is a finite set; hence after iteration, $f^{-1}(D) = D$ for every prime divisor $D \in S(X)$.*

Proof. $f^{-1}(S(X)) = S(X)$ is from Step 4 of [Zha10, Proof of Lemma 2.9] which holds true for any surjective endomorphism. Take any $D \in S(X)$. Then $f^{-i}f^i(D) = D$ for any $i > 0$. Hence, D is f^{-1} -periodic by [MZ20a, Lemma 3.5]. Thus $S(X)$ consists of f^{-1} -periodic prime divisors and is hence a finite set by [MZ20a, Proposition 3.6]. $\square$

Lemma 5.2.4 (cf. [MZ18a, Propositions 1.1, 2.9 and Lemma 3.4]). *Let f be a surjective endomorphism of a projective variety X and $q > 0$. Then the following are equivalent.*

- (1) *$f^*|_{N^1(X)}$ is diagonalizable with all the eigenvalues being of modulus q .*
- (2) *$f^*B \equiv qB$ for some big $\mathbb{R}$ -Cartier divisor B .*

If further $q > 1$ is an integer, then the above are equivalent to

- (3) f is q -polarized, i.e., $f^*H \sim qH$ for some ample Cartier divisor H (and $q > 1$).

In particular, $\deg(f) = q^{\dim(X)}$ if one of the above conditions holds.

Lemma 5.2.5 (cf. [Men20, Theorem 1.1], [MZ19b, Propositions 3.4 and 3.7]). *Let f be a surjective endomorphism of a projective variety X . Then the following are equivalent.*

- (1) f is int-amplified, i.e., $f^*L - L$ is ample for some ample Cartier divisor L .
- (2) All the eigenvalues of $\varphi := f^*|_{N^1(X)}$ are of modulus greater than 1.
- (3) All the eigenvalues of $f^*|_{\text{Pic}(X) \otimes_{\mathbb{Z}} \mathbb{Q}}$ are of modulus greater than 1.
- (4) If C is a φ -invariant convex cone in $N^1(X)$, then $\emptyset \neq (\varphi - \text{id}_{N^1(X)})^{-1}(C) \subseteq C$.

In our Fano setting, “int-amplified” is equivalent to “amplified”.

Lemma 5.2.6. *Let f be a surjective endomorphism of a normal projective variety X with the nef cone $\text{Nef}(X)$ being a rational polyhedron (this is the case when X is Fano). Then f is amplified if and only if it is int-amplified.*

Proof. One direction is clear by the definition. Suppose $\dim(X) > 0$ and $A = f^*L - L$ is ample for some Cartier divisor L . Since $\text{Nef}(X)$ is a rational polyhedron, f^* fixes each extremal ray of $\text{Nef}(X)$ after iteration. Note that $\text{Nef}(X)$ spans $N^1(X)$. Then $N^1(X)$ admits a nef integral basis $\{D_1, \dots, D_m\}$ such that $f^*D_i \equiv \lambda_i D_i$ for positive integers λ_i .

Write $L \equiv \sum_{i=1}^m a_i D_i$ and $L' := \sum_{i=1}^m D_i$. Then $A = f^*L - L \equiv \sum a_i(\lambda_i - 1)D_i$ and $A' := f^*L' - L' \equiv \sum(\lambda_i - 1)D_i$. For $c \gg 1$, our $cA' \equiv A + \sum(c - a_i)(\lambda_i - 1)D_i$ and $cL' \equiv A + \sum(c - a_i(\lambda_i - 1))D_i$ are both sums of the ample divisor A and some nef divisors. Hence, A' and L' are ample. So f is int-amplified. $\square$

We recall the well-known result ([Mor82, Theorem 3.5]) on the Fano contractions of smooth projective threefolds. For our purpose, we focus on the rationally connected case.

Lemma 5.2.7. *Let X be a smooth projective threefold and $X \rightarrow Y$ the Fano contraction of a K_X -negative extremal ray to a normal projective variety Y . Suppose X is rationally connected (this is the case when X is Fano or of Fano type). Then the following hold.*

- (1) *If $\dim(Y) = 1$, then $Y \cong \mathbb{P}^1$.*
- (2) *Suppose that $\dim(Y) = 2$. Then Y is a smooth rational surface; if further Y has Picard number one, then $Y \cong \mathbb{P}^2$.*

Proof. Since X is rationally connected, so is Y ; hence Y is rational when $\dim(Y) \leq 2$. Then (1) follows. For (2), just note that Y is smooth by [Mor82, Theorem 3.5]. $\square$

Given a smooth projective toric variety, its toric boundary is always nice.

Lemma 5.2.8. *Let (X, B) be a toric pair with X being smooth projective. Then B is of simple normal crossing, i.e., the pair (X, B) is log smooth, and $K_X + B \sim 0$. If X is further a surface, then B is a simple loop of smooth rational curves.*

Proof. The first part is from [CLS11, p. 360 and Theorem 8.2.3] while the second follows from the first and the adjunction (cf. [CLS11, Proposition 3.2.7] and Notation 5.2.1). $\square$

The result below should be well-known; we give the proof for readers' convenience.

Proposition 5.2.9. *Let $X := \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ be a splitting $\mathbb{P}^1$ -bundle over a normal projective variety Y . Suppose (Y, Δ_Y) is a toric pair. Then X is toric and $(X, S_0 + S_1 + \tau^{-1}(\Delta_Y))$ is a toric pair for any two disjoint (cross) sections S_0, S_1 of τ .*

Proof. Let $T(Y) = Y \setminus \Delta_Y$ and fix a toric action $T(Y)$ on Y . Write $\mathcal{E} = \mathcal{L}_0 \oplus \mathcal{L}_1$ as a direct sum of two line bundles on Y . Denote by $\tau_i : \mathcal{L}_i \rightarrow Y$ the induced fibre bundles. For each $x \in X$, we write $x = (y, [v_0 : v_1])$ with $y = \tau_i(x)$ and $v_i \in (\mathcal{L}_i)_y := \tau_i^{-1}(y)$. Note that $\mathcal{L}_i \cong \mathcal{O}_Y(D_i)$ for some Cartier divisors D_i with support contained in Δ_Y which are $T(Y)$ -invariant. Then $\mathcal{L}_i$ are $T(Y)$ -equivariant line bundles, i.e., there are $T(Y)$ -actions on $\mathcal{L}_i$ such that $\tau_i : \mathcal{L}_i \rightarrow Y$ are $T(Y)$ -equivariant and the $T(Y)$ -actions on the

fibres of τ_i are linear (cf. [Oda88, §2]). Thus we can construct a faithful action of a torus $T := T(Y) \times k^* = \{(g, t)\}$ on X via:

$$(g, t) \cdot (y, [v_0 : v_1]) = (g(y), [g(v_0) : t \cdot g(v_1)])$$

where the $T(Y)$ action on X is determined by $\mathcal{L}_i$. So X is toric.

Let S_0 and S_1 be two disjoint sections of τ . They are uniquely determined by sub line bundles $\mathcal{H}_0, \mathcal{H}_1$ of $\mathcal{E}$. Since $S_0 \cap S_1 = \emptyset$, we have $(\mathcal{H}_0)_y \neq (\mathcal{H}_1)_y$ for any $y \in Y$. So $\mathcal{E} = \mathcal{H}_0 \oplus \mathcal{H}_1$. Updating the above T -action by using the new decomposition, we see that $\tau^{-1}(\Delta_Y), S_0, S_1$ are T -invariant and T acts transitively on $X \setminus (S_0 \cup S_1 \cup \tau^{-1}(\Delta_Y))$. $\square$

The following lemma allows us to focus on polarized endomorphisms whenever given non-isomorphic surface endomorphisms.

Lemma 5.2.10. *Let S be a smooth projective rational surface admitting a non-isomorphic surjective endomorphism f which is not polarized even after iteration. Then $S \cong \mathbb{P}^1 \times \mathbb{P}^1$.*

Proof. By [MZ19b, Theorem 5.4], $\rho(S) = 2$ and hence S has an f -equivariant (after iterating f) ruling $\pi : S \rightarrow C \cong \mathbb{P}^1$. Suppose on the contrary that $S \not\cong \mathbb{P}^1 \times \mathbb{P}^1$. Then the ruling admits a negative section C_0 which is an f^{-1} -invariant curve after iterating f (cf. [Nak02, Lemma 9 and Proposition 11] or [MZ19b, Lemma 4.3]). Write $f^*C_0 = qC_0$ with $q \geq 1$. By the projection formula, $q^2 = \deg(f) > 1$ and hence $q > 1$. Then $f|_{C_0}$ and hence $f|_C$ are q -polarized. Let F be a fibre of π . Then $f^*F \sim qF$. Since $C_0 + F$ is big and $f^*(C_0 + F) \sim q(C_0 + F)$, our f is q -polarized (cf. Lemma 5.2.4), a contradiction! $\square$

In Theorem 5.2.11 below, we apply [BH14, Theorem 1.4] and [MZ19b, Theorem 6.2] to give some restrictions on totally invariant prime divisors. Locally, for example when X is smooth, the log canonical pair (X, Δ) implies that Δ has at most two components containing a common codimensional 2 subvariety of X (cf. [KM98, Lemma 2.29]). Globally, for example when $X = \mathbb{P}^n$, the effectivity of $-(K_X + \Delta)$ implies that $\deg(\Delta) \leq n + 1$ and hence Δ has at most $n + 1$ components.

Theorem 5.2.11. *Let f be an int-amplified endomorphism of a normal projective variety X . Let Δ be an f^{-1} -invariant reduced divisor such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Then*

(1) (X, Δ) has at worst log canonical singularities, and

(2) $-(K_X + \Delta)$ is effective.

Proof. (1) follows immediately from Lemma 5.2.5 and [BH14, Theorem 1.4]. To prove (2), we consider the log ramification divisor formula $K_X + \Delta = f^*(K_X + \Delta) + R'$ where R' is a $\mathbb{Q}$ -Cartier effective divisor. Let $D := -(K_X + \Delta)$. Then $f^*D - D = R'$. By the same proof of [MZ19b, Theorem 6.2], D is effective; see also [MZ19b, Propositions 3.4, 3.7] and [Men20, Proposition 3.2]. $\square$

Below is a criterion for a $\mathbb{P}^1$ -bundle to be algebraic.

Lemma 5.2.12 ([Ele82]). *Let $\tau : X \rightarrow Y$ be a $\mathbb{P}^1$ -bundle over a smooth projective rational variety Y . Then $X = \mathbb{P}_Y(\mathcal{E})$ for some locally free rank-two sheaf $\mathcal{E}$ over Y .*

Theorem 5.2.13 and Proposition 5.2.14 below are important in showing the splitting-ness of algebraic $\mathbb{P}^1$ -bundles admitting non-isomorphic endomorphisms. For certain X as in Theorem 5.6.4, we can dispose the semistability assumption in Theorem 5.2.13.

Theorem 5.2.13 ([Ame03, Theorem 2, Proposition 2.4]). *Let $X = \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ be an algebraic $\mathbb{P}^1$ -bundle over a smooth projective rational variety Y . If X admits an int-amplified endomorphism f and $\mathcal{E}$ is H -semistable for some ample divisor H , then τ is a splitting $\mathbb{P}^1$ -bundle.*

Proof. Since f is int-amplified, we see that $\deg(f) > \deg(f|_Y)$ (cf. Lemma 5.2.5). Note that rational varieties are simply connected. Hence, the H -semistable sheaf $\mathcal{E}$ splits by [Ame03, Theorem 2 and Proposition 2.4]. $\square$

Proposition 5.2.14 ([Ame03, Proposition 3]). *Suppose $B = \mathbb{P}^n$ and $X = \mathbb{P}_B(\mathcal{E})$ with $\mathcal{E}$ a vector bundle of rank two. Then there exists an endomorphism of X of degree bigger than one if and only if $\mathcal{E}$ is a direct sum of two line bundles.*

In what follows, we recall the known results of Conjectures 5.1.1 and 5.1.9.

Theorem 5.2.15 ([ARVdV99], [HM03]). *Let X be a smooth Fano threefold of Picard number one admitting a non-isomorphic surjective endomorphism. Then $X \cong \mathbb{P}^3$.*

Theorem 5.2.16. *Conjecture 5.1.9 holds if one of the following cases occurs.*

- (1) $X = \mathbb{P}^2$ (cf. [Gur03]).
- (2) $X = \mathbb{P}^3$; Y is an f^{-1} -invariant prime divisor (cf. [Hör17], [NZ10, Theorem 1.5]).
- (3) Y is a smooth hypersurface of X (cf. [CLN00], [Bea01], [Gur03]).

To end the section, we recall the following well-known fact which will be heavily used in Section 5.4.

Proposition 5.2.17 (cf. [KM98, Theorem 3.7 (4)]). *Let $\pi : X \rightarrow Y$ be a Fano contraction of a K_X -negative extremal ray R between projective varieties. Suppose that X is smooth. Then for every prime divisor Q on Y , its pull-back π^*Q is reduced and irreducible.*

Proof. Suppose on the contrary that $\pi^*Q = P_1 + P_2$ where P_1 is a prime divisor and P_2 is a non-zero integral divisor on X . Then for a general curve ℓ contracted by π (and hence a generator of R) we have $P_i \cdot \ell = 0$ for each i . Since X is smooth, both P_i are effective Cartier divisors. By the cone theorem (cf. [KM98, Theorem 3.7 (4)]), there exist non-zero effective Cartier divisors Q_i on Y such that $P_i = \pi^*Q_i$. Therefore, $Q = Q_1 + Q_2$ being non-reduced or reducible gives the contradiction. $\square$

5.3 Totally periodic prime divisors

This section describes totally periodic prime divisors in surfaces and splitting $\mathbb{P}^1$ -bundles.

Lemma 5.3.1. *Let f be a non-isomorphic surjective endomorphism of $X = Y \times Z \cong \mathbb{P}^1 \times \mathbb{P}^1$. Let C be an f^{-1} -periodic curve in X . Then C is either $\{\cdot\} \times \mathbb{P}^1$ or $\mathbb{P}^1 \times \{\cdot\}$.*

Proof. Denote by $p_Y : X \rightarrow Y$ and $p_Z : X \rightarrow Z$ the natural projections. After iteration, we may assume C is f^{-1} -invariant and f splits into the form $g \times h$ with g and h being surjective endomorphisms.

Suppose on the contrary that C dominates both Y and Z , via p_Y and p_Z . Then $q := \deg(g) = \deg(f|_C) = \deg(h)$. Since f is non-isomorphic, we have $q > 1$. Hence, f is q -polarized. So $f^*C = qC$ (cf. Lemma 5.2.4) and thus C is a component of the ramification

divisor R_f of f . Since $f = g \times h$, we have $R_f = p_Y^* R_g + p_Z^* R_h$ where R_g and R_h are the ramification divisors of g and h , respectively. But then, C is of the form $\{\cdot\} \times \mathbb{P}^1$ or $\mathbb{P}^1 \times \{\cdot\}$, a contradiction to our assumption. This proves the lemma. $\square$

Now we consider general smooth projective rational surfaces.

Theorem 5.3.2. *Let X be a smooth projective rational surface admitting an int-amplified endomorphism f . Then X is toric and there is a toric pair (X, Δ) such that Δ contains the union Σ of all the f^{-1} -periodic prime divisors.*

In particular, if $\rho(X) = 2$, then Δ is a union of two disjoint (cross) sections of a fixed ruling of X and two distinct fibres.

Proof. Since there are only finitely many f^{-1} -periodic subvarieties by [MZ20a, Corollary 3.8], we may assume that they are all f^{-1} -invariant, after iterating f . Further, (X, Σ) is lc and $-(K_X + \Sigma)$ is effective by Theorem 5.2.11.

Case $\rho(X) = 1$. Then $X \cong \mathbb{P}^2$ and Σ is a union of at most three lines by Theorem 5.2.16; noting that $\deg(\Sigma) \leq 3$. Since (X, Σ) is lc, Σ is contained in some Δ , a union of three lines with no common intersection (looking like $\begin{smallmatrix} \diagup & & \diagdown \\ & \times & \\ \diagdown & & \diagup \end{smallmatrix}$). Clearly, (X, Δ) is a toric pair.

Case $\rho(X) = 2$. Suppose $X \cong \mathbb{P}^1 \times \mathbb{P}^1$. By Lemma 5.3.1, Σ is contained in some Δ , a union of two fibres each from two projections of X (looking like $\begin{smallmatrix} \top & & \bot \\ \hline \end{smallmatrix}$). Clearly, (X, Δ) is a toric pair.

Suppose $X \not\cong \mathbb{P}^1 \times \mathbb{P}^1$. Consider the f -equivariant (after iteration) ruling $\tau : X \rightarrow Z \cong \mathbb{P}^1$ (which is also a splitting $\mathbb{P}^1$ -bundle). Write $\Sigma = \Sigma_h \cup \Sigma_v$ where Σ_h (resp. Σ_v) is the union of components of Σ dominating (resp. not dominating) Z . Since $f|_Z$ is also int-amplified on $Z \cong \mathbb{P}^1$ (cf. Lemma 5.2.5), there are at most two $(f|_Z)^{-1}$ -periodic points. So Σ_v is a union of at most two fibres of τ (cf. [CMZ20, Lemma 7.5]). In particular, there exists a toric pair (Z, Δ_Z) such that $\Sigma_v \subseteq \tau^{-1}(\Delta_Z)$.

Let $F \cong \mathbb{P}^1$ be a fibre of τ which is nef on X . Then $-(K_X + \Sigma) \geq 0$ implies that

$$\Sigma_h \cdot F = \Sigma \cdot F \leq -K_X \cdot F = -\deg(K_F) = 2$$

by the adjunction formula. Note that Σ_h contains the unique negative section C_0 of τ . Therefore, either $\Sigma_h = C_0$ or Σ_h is a union of C_0 and another section C of τ . In the first

case, we are done by applying Proposition 5.2.9 for $\Delta = C_1 \cup C_0 \cup \tau^{-1}(\Delta_Z)$ where C_1 is a section of τ disjoint with C_0 (cf. [Har77, Chapter V, Theorem 2.17]).

Similarly, in the second case, it suffices for us to show that $C \cap C_0 = \emptyset$. Suppose on the contrary and we take $x \in C \cap C_0$ which is f^{-1} -invariant after iteration. Note that $\tau^{-1}(\tau(x))$ is f^{-1} -invariant (cf. [CMZ20, Lemma 7.5]). Then Σ contains three curves $C, C_0, \tau^{-1}(\tau(x))$ with a common intersection point x . In particular, (X, Σ) is not lc at x (cf. [KM98, Lemma 2.29]), a contradiction to Theorem 5.2.11. So $C \cap C_0 = \emptyset$ and the second case is done by applying Proposition 5.2.9 again for $\Delta = C \cup C_0 \cup \tau^{-1}(\Delta_Z)$.

Case $\rho(X) \geq 3$. By Lemma 5.2.2, running an f -equivariant (after iterating f) MMP, we have a composition $\pi : X \rightarrow Y$ of smooth blow-downs of (f^{-1} -invariant successively) (-1) -curves with the target Y being a Hirzebruch surface and the ruling $\tau_Y : Y \rightarrow Z \cong \mathbb{P}^1$. Set $\tau := \tau_Y \circ \pi$, and $f_Y := f|_Y$. Since $\rho(X) \geq 3$, there is an f_Y^{-1} -invariant point y_0 in Y .

We claim that $\tau_Y : Y \rightarrow Z$ admits at least one f_Y^{-1} -invariant (cross) section C_Y . Indeed, if $Y \cong \mathbb{P}^1 \times \mathbb{P}^1$, then the horizontal (fibre) section C_Y of τ_Y containing y_0 is f_Y^{-1} -invariant (cf. Lemma 5.3.1). If $Y \not\cong \mathbb{P}^1 \times \mathbb{P}^1$, then the unique negative section C_Y of τ_Y is f_Y^{-1} -invariant. So the claim is proved. Let $C_X \subseteq X$ be the strict transform of C_Y . Then $C_X \subseteq \Sigma$.

Since $\rho(X) \geq 3$, τ admits at least one reducible fibre $F = \bigcup_{i=1}^n F_i$ with $F_i \cong \mathbb{P}^1$ for all i and $n \geq 2$. By the generic smoothness, τ has only finitely many reducible fibres. So F is f^{-1} -periodic by [CMZ20, Lemma 7.4] and hence $C_X \cup F \subseteq \Sigma$. After iteration, we may assume $f^{-1}(F_i) = F_i$ for each i .

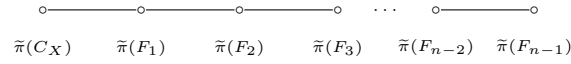
We claim that $F \cup C_X$ is a linear chain of smooth rational curves with dual graph

$$\begin{array}{ccccccc} \circ & \text{---} & \circ & \text{---} & \circ & \text{---} & \circ & \dots & \circ & \text{---} & \circ \\ C_X & & F_1 & & F_2 & & F_3 & & F_{n-1} & & F_n \end{array}$$

Denote by $F_0 := C_X$ for convenience. Note that $f|_{F_i}$ (like f) is int-amplified for each $0 \leq i \leq n$. So each F_i admits at most two $(f|_{F_i})^{-1}$ -periodic points. Since $F_i \cap F_j$ is $(f|_{F_i})^{-1}$ -invariant for any j , we have $\sharp F_i \cap (F_0 \cup F \setminus F_i) \leq 2$ for each $0 \leq i \leq n$. Recall that $\pi(F)$ intersects with $C_Y = \pi(C_X) = \pi(F_0)$ transversally at only one point. So C_X

intersects F transversally at only one point and $C_X \cup F$ is a divisor of simple normal crossing. Finally, note that $C_X \cup F$ is connected. So the claim follows.

We claim further that one can find some $F_{i_0}^2 = -1$ with $0 < i_0 < n$. We show by induction on n . If $n = 2$, then F_1 and F_2 are both (-1) -curves, since $\pi(F)^2 = 0$. In general, note that at least one component of F is a (-1) -curve because π is a composition of blow-ups from Y . Suppose F_n is the only (-1) -curve contained in F . Let $\tilde{\pi} : X \rightarrow \tilde{X}$ be the (f -equivariant) contraction of F_n . Then we have the following dual graph of $\tilde{\pi}(C_X \cup F)$:



Note that $\tilde{\pi}(F_i)^2 = F_i^2 \neq -1$ for $0 < i < n - 1$, a contradiction by induction on n . So the claim is proved.

By the above claim, let $\pi' : X \rightarrow X'$ be the (f -equivariant) contraction of the (-1) -curve F_{i_0} over Z (which may no longer be over Y). Then X' is still smooth and $\rho(X') = \rho(X) - 1$. Let Σ' be the union of $(f|_{X'})^{-1}$ -periodic curves in X' . Then $\Sigma' = \pi'(\Sigma)$ (cf. [CMZ20, Lemma 7.5]). By induction on the Picard number, there is a toric pair (X', Δ') such that $\Sigma' \subseteq \Delta'$. Denote by $F_0 := C_X$. Since $0 < i_0 < n$, we see that $\pi'(F_{i_0}) = \pi'(F_{i_0-1}) \cap \pi'(F_{i_0+1})$ is an intersection of two components of Σ' (and also Δ'). In particular, π' is a toric blow-up and (X, Δ) is a toric pair with Δ being the union of F_{i_0} and the strict transform of Δ' . Clearly, $\Sigma \subseteq \Delta$. This proves the theorem. $\square$

Theorem 5.3.3. *Let $\tau : X \rightarrow Y$ be a splitting $\mathbb{P}^1$ -bundle over a smooth projective rational surface. Suppose X admits an int-amplified endomorphism f . Then X is toric and there is a toric pair (X, Δ) such that Δ contains the union of all the f^{-1} -periodic prime divisors.*

Proof. Let Σ be the union of the f^{-1} -periodic prime divisors. Since there are only finitely many f^{-1} -periodic subvarieties by [MZ20a, Corollary 3.8], we may assume that components of Σ are all f^{-1} -invariant after iteration. Write $\Sigma = \Sigma_h \cup \Sigma_v$ where Σ_h (resp. Σ_v) is the union of components of Σ dominating (resp. not dominating) Y . Note that τ is a Fano contraction of some K_X -negative extremal ray. After iteration, τ is f -equivariant and $f|_Y$ (like f) is also int-amplified (cf. Lemmas 5.2.2 and 5.2.5). Note that $\Sigma_v = \tau^{-1}(\Sigma_Y)$

where Σ_Y is the union of $(f|_Y)^{-1}$ -periodic curves in Y (cf. [CMZ20, Lemma 7.4]). By Theorem 5.3.2, there exists a toric pair (Y, Δ_Y) such that $\Sigma_v \subseteq \tau^{-1}(\Delta_Y)$. By Proposition 5.2.9, it suffices for us to show that Σ_h is contained in two disjoint sections of τ .

Assume first τ is a trivial bundle so that $X = Y \times Z \cong Y \times \mathbb{P}^1$ and we may assume that $f = g \times h$, after iteration (cf. Lemma 5.2.2). By Lemma 5.2.5 and the same proof of Lemma 5.3.1, $\Sigma_h \subseteq \text{Supp } R_f = (\text{Supp } R_g \times Z) \cup (Y \times \text{Supp } R_h)$. Hence, $\Sigma_h \subseteq Y \times \text{Supp } R_h$. Since h on $Z \cong \mathbb{P}^1$ (like f) is also int-amplified, $\text{Supp } R_h$ has at most two h^{-1} -periodic points. So Σ_h is contained in two disjoint fibres of $X \rightarrow Z$ which are two disjoint sections of τ as desired.

Assume next τ is non-trivial. Write $X = \mathbb{P}_Y(\mathcal{E})$ with $\mathcal{E} = \mathcal{O}_Y \oplus \mathcal{L}$ such that $\mathcal{L}$ is not pseudo-effective. Let S_0 be the cross-section of τ determined by the projection $\mathcal{O}_Y \oplus \mathcal{L} \rightarrow \mathcal{L}$. Note that $\mathcal{O}_X(S_0)|_{S_0} \cong (\tau|_{S_0})^* \mathcal{L}$ is not pseudo-effective. By Lemma 5.2.3, S_0 is f^{-1} -periodic and hence $S_0 \subseteq \Sigma_h$.

Let $X_y := \tau^{-1}(y) \cong \mathbb{P}^1$ be a fibre of τ . By Theorem 5.2.11, $-(K_X + \Sigma)$ is effective. Since X_y is movable, we have $-(K_X + \Sigma) \cdot X_y \geq 0$ and hence

$$\Sigma_h \cdot X_y = \Sigma \cdot X_y \leq -K_X \cdot X_y = -\deg(K_{X_y}) = 2$$

by the adjunction formula. Note that Σ_h contains S_0 . Therefore, either $\Sigma_h = S_0$ or Σ_h is a union of S_0 and another prime divisor S with $S \cdot X_y = 1$ (which is not known to be a section of τ currently). In the first case, we are done by applying Proposition 5.2.9 for $\Delta = S_1 \cup S_0 \cup \tau^{-1}(\Delta_Y)$ where S_1 is a section of τ disjoint with S_0 by the splitting structure of $\mathcal{E}$ (cf. Proof of Proposition 5.2.9). Similarly, in the second case, it suffices for us to show that S is a section of τ and $S \cap S_0 = \emptyset$.

We first claim that $S \cap S_0 = \emptyset$ if Σ_h contains another prime divisor $S \neq S_0$. Suppose on the contrary and let C be an irreducible curve in $S \cap S_0$ which is f^{-1} -invariant after iteration. Then $F := \tau^{-1}(\tau(C))$ is an f^{-1} -invariant divisor (cf. [CMZ20, Lemma 7.5]). Note that $C \subseteq F \cap S \cap S_0$. Then the pair $(X, S + S_0 + F)$ is not lc at C (cf. [KM98, Lemma 2.29]). However, this contradicts Theorem 5.2.11. The claim is proved.

The above claim further implies that S (if exists) does not contain any fibre of τ . Then $S \cdot X_y = 1$ implies that $S \cap X_y$ is a single point for any $y \in Y$. Therefore, S is a section

disjoint with S_0 as desired. This proves the theorem. $\square$

Corollary 5.3.4. *Let f be a non-isomorphic surjective endomorphism of a smooth projective rational surface X . Let C be an f^{-1} -periodic curve. Then C is a smooth rational curve. In particular, if X is further assumed to be Fano, then $C^2 \leq 1$.*

Proof. If f is not int-amplified, then $X \cong \mathbb{P}^1 \times \mathbb{P}^1$ by Lemma 5.2.10, and our corollary follows from Lemma 5.3.1. If f is int-amplified, then there exists a toric pair (X, Δ) such that Δ contains C (cf. Theorem 5.3.2); thus $C \cong \mathbb{P}^1$ by Lemma 5.2.8.

Suppose further that X is Fano. By Lemma 5.2.2, there is an f -equivariant (after iteration) composition $\pi : X \rightarrow Y$ of smooth blow-downs such that $Y \cong \mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$ since X and hence Y are still Fano. Note also that either C is an exceptional curve of π or $\pi(C)$ is an $(f|_Y)^{-1}$ -periodic curve as described in Theorem 5.2.16 or Lemma 5.3.1 (cf. [CMZ20, Lemma 7.5]). Thus $C^2 = -1$ or $C^2 \leq \pi(C)^2 \leq 1$. $\square$

5.4 Fano contractions to surfaces

The whole section is devoted to the proof of the following theorem.

Theorem 5.4.1. *Let $\tau : X \rightarrow Y$ be a Fano contraction of an extremal ray of a smooth Fano threefold X with $\dim(Y) = 2$. Suppose X admits a non-isomorphic surjective endomorphism f . Then X is an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$ over Y .*

Now we begin to prove Theorem 5.4.1. Recall that Y is smooth rational and τ is a conic bundle with the discriminant locus denoted as Δ_τ (cf. [Mor82, Theorem 3.5] and Lemma 5.2.7). Besides, we can descend $f : X \rightarrow X$ to $g := f|_Y$ after iteration by Lemma 5.2.2. By Lemma 5.2.12, to prove Theorem 5.4.1, it suffices for us to show $\Delta_\tau = \emptyset$.

Notation 5.4.2. We will use the following notation throughout this section.

- (1) X is a smooth Fano threefold, and Y is a smooth projective rational surface; the conic bundle $\tau : X \rightarrow Y$ is flat (cf. [Bea77, Proposition 1.2]); $\rho(X) = \rho(Y) + 1$.
- (2) $f : X \rightarrow X$ is a non-isomorphic surjective endomorphism and $g := f|_Y$.

- (3) Δ_τ is the discriminant locus of τ , which is a union of irreducible curves C_i .
- (4) $S_i := \tau^{-1}C_i = \tau^*C_i$ is reduced and irreducible (cf. [KM98, Theorem 3.7]).
- (5) The fibres X_y with $y \in \Delta_\tau$ satisfy the following: If y is a smooth point of Δ_τ , then X_y is a reducible conic in $\mathbb{P}^2$. If y is a double point of Δ_τ , then X_y is isomorphic to a double line in $\mathbb{P}^2$ (cf. [Bea77, Proposition 1.2]).
- (6) If f is polarized (or int-amplified), then so is g (cf. Lemmas 5.2.4 and 5.2.5).

Proposition 5.4.3 (cf. [Zha12, Claim 3.3]). *Suppose $\Delta_\tau \neq \emptyset$. Then $\deg(g) > 1$.*

Lemma 5.4.4. *Suppose that $\Delta_\tau \neq \emptyset$. Then Δ_τ is g^{-1} -invariant and a union of (at least two) g^{-1} -periodic smooth rational curves C_i with $C_i \cdot (\Delta_\tau - C_i) \geq 2$.*

Proof. By Proposition 5.4.3, $\deg(g) > 1$. By [CMZ20, Lemma 7.4], $g^{-1}(\Delta_\tau) = \Delta_\tau$. Thus all C_i are g^{-1} -periodic. Hence, C_i are all smooth rational curves by Corollary 5.3.4. Note that each C_i meets at least two points of other irreducible components of Δ_τ (cf. [Miy83, Remark 4.2]). So the lemma follows. $\square$

Lemma 5.4.5. *Suppose that $\Delta_\tau \neq \emptyset$ and g is int-amplified. Then $\text{Supp } R_g = \Delta_\tau$ and (Y, Δ_τ) is a toric pair. In particular, Δ_τ is a simple loop of smooth rational curves, and $K_Y + \Delta_\tau \sim 0$ (cf. Lemma 5.2.8).*

Proof. By Lemma 5.4.4 and Theorem 5.3.2, there is a toric pair (Y, Δ) such that $\Delta_\tau \subseteq \Delta$. Note that Δ is a simple loop of smooth rational curves, i.e., each component of Δ intersects (transversally) exactly two other components of Δ at two distinct points (cf. Lemma 5.2.8). So Lemma 5.4.4 further implies that $\Delta_\tau = \Delta$, which is g^{-1} -invariant. Since $K_Y + \Delta_\tau \sim 0$, we obtain the log ramification divisor formula $K_Y + \Delta_\tau = g^*(K_Y + \Delta_\tau)$. Therefore, $\text{Supp } R_g = \Delta_\tau$. So the lemma is proved. $\square$

Lemma 5.4.6. *Suppose that $\Delta_\tau \neq \emptyset$. Then $Y \cong \mathbb{P}^2$ or $Y \cong \mathbb{P}^1 \times \mathbb{P}^1$.*

Proof. Suppose on the contrary that the lemma is false. After iteration, g is q -polarized by Lemma 5.2.10 and Y admits a g^{-1} -invariant negative curve E (cf. [Nak02] or [MZ19b,

Lemma 4.3]). Then $g^*E = qE$ and $E \subseteq \text{Supp } R_g$. By Lemma 5.4.5, E is a smooth rational curve contained in Δ_τ . Hence, E is a rational connected component of Δ_τ by [MM83, Corollary 6.7]. This contradicts Lemma 5.4.4. So the lemma holds true. $\square$

Proposition 5.4.7. *Suppose that $\Delta_\tau \neq \emptyset$ and f is int-amplified. Then X has no divisorial contraction.*

We now prove Proposition 5.4.7. It will last till Claim 5.4.11. Suppose on the contrary that X has a divisorial contraction π with E the exceptional divisor. After iteration, we may assume $f^{-1}(E) = E$ (cf. Lemma 5.2.2). By [Mor82, Theorem 3.3], E is normal and $\mathbb{Q}$ -factorial. Note that g and $f|_E$ (like f) are also int-amplified. Let $T := Y \setminus \Delta_\tau$ and $T_E := E \setminus \tau^{-1}(\Delta_\tau)$. By Lemma 5.4.4, $f^{-1}(T_E) = T_E$ after iteration.

Claim 5.4.8. $\tau|_E : E \rightarrow Y$ is a generically finite surjective morphism of degree 2.

Proof. Consider the case $\tau(E) \neq Y$. We have $E = \tau^*\tau(E)$ by the cone theorem (cf. [KM98, Theorem 3.7]). By Lemma 5.4.6, Y has no negative curves. Then $\tau(E)$ and hence E are nef, a contradiction to $-E$ being π -ample. Thus $\tau(E) = Y$. Note that τ is a contraction of an extremal ray. So E is τ -ample.

We still need to show that $E \cdot \ell = 2$ for a general fibre ℓ of τ . Note that $-(K_X + E)$ is effective by Theorem 5.2.11 and ℓ is movable. So $-(K_X + E) \cdot \ell \geq 0$ and hence $E \cdot \ell \leq -K_X \cdot \ell = 2$ by the adjunction formula. On the other hand, since τ is a (flat) conic bundle, $\ell \equiv \ell' + \ell''$ where $\ell' \cup \ell''$ is a (reduced) reducible fibre lying over a smooth point of Δ_τ . So we have

$$E \cdot \ell = E \cdot (\ell' + \ell'') \geq 1 + 1$$

where the second inequality is because E is integral and τ -ample. Hence, $E \cdot \ell = 2$ and the claim is proved. $\square$

Claim 5.4.9. The restriction $f|_{T_E}$ is an étale morphism.

Proof. Consider the following commutative diagram:

$$\begin{array}{ccccc}
 & & f|_{T_E} & & \\
 & \nearrow & & \searrow & \\
 T_E & \xrightarrow{\phi} & T' & \xrightarrow{p_1} & T_E \\
 & \searrow \tau|_{T_E} & \downarrow p_2 & & \downarrow \tau|_{T_E} \\
 & & T & \xrightarrow{g|_T} & T
 \end{array}$$

where T' is the fibre product of $g|_T$ and $\tau|_{T_E}$. Since $g|_T$ is finite étale by Lemma 5.4.5, so is p_1 , hence T' is normal. Note that $\tau|_{T_E}$ is generically finite. Then $\deg(f|_{T_E}) = \deg(g|_T) = \deg(p_1)$, and hence $\deg(\phi) = 1$, i.e., ϕ is birational. Since ϕ (like $f|_{T_E}$) is also finite, and both T_E and T' are normal, ϕ is an isomorphism. So the claim is proved. $\square$

Claim 5.4.10. $\tau^{-1}(\Delta_\tau)|_E = \tau^{-1}(\Delta_\tau) \cap E$, a reduced divisor, and $K_E + \tau^{-1}(\Delta_\tau)|_E \sim_{\mathbb{Q}} 0$.

Proof. By Claim 5.4.9, the ramification divisor of $f|_E$ has support contained in the $(f|_E)^{-1}$ -invariant divisor $\tau^{-1}(\Delta_\tau) \cap E$. Thus, by the log ramification divisor formula,

$$K_E + \tau^{-1}(\Delta_\tau) \cap E = (f|_E)^*(K_E + \tau^{-1}(\Delta_\tau) \cap E). \quad (\dagger)$$

This and Lemma 5.2.5 imply $K_E + \tau^{-1}(\Delta_\tau) \cap E \equiv 0$, since $f|_E$ is int-amplified.

On the other hand, by the log ramification divisor formula on X , we have

$$K_X + \tau^{-1}(\Delta_\tau) + E = f^*(K_X + \tau^{-1}(\Delta_\tau) + E) + R'_f,$$

where R'_f is an effective divisor with no common component with $\tau^{-1}(\Delta_\tau) + E$. Restricting the above equation to E for both sides, we have

$$(K_X + \tau^{-1}(\Delta_\tau) + E)|_E = (f|_E)^*(K_X + \tau^{-1}(\Delta_\tau) + E)|_E + R'_f|_E,$$

where $R'_f|_E$ is effective since $E \not\subseteq \text{Supp } R'_f$. By the adjunction formula, we have

$$K_E + \tau^{-1}(\Delta_\tau)|_E = (f|_E)^*(K_E + \tau^{-1}(\Delta_\tau)|_E) + R'_f|_E,$$

which implies that $-(K_E + \tau^{-1}(\Delta_\tau)|_E)$ is pseudo-effective by Lemma 5.2.5. Note that

$$-(K_E + \tau^{-1}(\Delta_\tau)|_E) \leq -(K_E + \tau^{-1}(\Delta_\tau) \cap E) \equiv 0.$$

So the inequality above is in fact an equality. This proves the first half of the claim, while the second half follows from the first half, Equation $(\dagger)$ above and Lemma 5.2.5 (noting that E is normal and $\mathbb{Q}$ -factorial). $\square$

Claim 5.4.11. $K_X + \tau^{-1}(\Delta_\tau) + E \not\equiv 0$.

Proof. Suppose on the contrary that $K_X + \tau^{-1}(\Delta_\tau) + E \equiv 0$. Then we have

$$K_X + \tau^{-1}(\Delta_\tau) + E = f^*(K_X + \tau^{-1}(\Delta_\tau) + E)$$

which implies that the restriction map

$$f|_{X \setminus (\tau^{-1}(\Delta_\tau) \cup E)} : X \setminus (\tau^{-1}(\Delta_\tau) \cup E) \rightarrow X \setminus (\tau^{-1}(\Delta_\tau) \cup E)$$

is étale by the purity of branch loci. By [MZ20b, Theorem 1.4], $(X, \tau^{-1}(\Delta_\tau) + E)$ is a toric pair. In particular, $\tau^{-1}(\Delta_\tau) + E$ has $\dim(X) + \rho(X)$ components.

On the other hand, both Δ_τ and $\tau^{-1}(\Delta_\tau)$ have $\dim(Y) + \rho(Y)$ components each by Lemma 5.4.5 (cf. Notation 5.4.2). So $\tau^{-1}(\Delta_\tau) + E$ is a divisor with only $\dim(Y) + \rho(Y) + 1 = \dim(X) + \rho(X) - 1$ components. We have reached a contradiction. $\square$

End of Proof of Proposition 5.4.7. By the adjunction formula, $K_X \cdot \ell = -2$ for a general fibre ℓ of $\tau : X \rightarrow Y$. This and Claim 5.4.8 imply that $K_X + \tau^{-1}(\Delta_\tau) + E$ is τ -trivial. By the cone theorem (cf. [KM98, Theorem 3.7]) and Claim 5.4.11,

$$K_X + \tau^{-1}(\Delta_\tau) + E \sim \tau^*D$$

for some Cartier divisor $D \not\equiv 0$ on Y . Therefore, restricting the above equation to E for both sides and applying the adjunction formula, we have

$$K_E + \tau^{-1}(\Delta_\tau)|_E = (K_X + \tau^{-1}(\Delta_\tau) + E)|_E \sim \tau^*D|_E = (\tau|_E)^*D \not\equiv 0,$$

where the last inequality follows from Claim 5.4.8. This contradicts Claim 5.4.10. The proof of Proposition 5.4.7 is completed. $\square$

Lemma 5.4.12. *Suppose that $\Delta_\tau \neq \emptyset$. Suppose further that there exists an extremal contraction $\pi : X \rightarrow Z \cong \mathbb{P}^1$ of a K_X -negative extremal face, with reduced and irreducible fibres such that the induced map $\psi : X \rightarrow Y \times Z$ is finite surjective. Denote by $h := f|_Z$ after iteration. Then the following assertions hold:*

- (I) $\deg(\psi) > 1$ and ψ has branch loci $B_\psi \neq \emptyset$.

(2) Each prime divisor of B_ψ dominates both Y and Z .

(3) $(g \times h)^* B_\psi = (g \times h)^{-1}(B_\psi)$.

(4) B_ψ is $(g \times h)$ -invariant.

Proof. Let $p_Y : Y \times Z \rightarrow Y$ and $p_Z : Y \times Z \rightarrow Z$ be the projections. If $\deg(\psi) = 1$, then $X \cong Y \times Z$ and τ is a trivial $\mathbb{P}^1$ -bundle, contradicting that $\Delta_\tau \neq \emptyset$. So $\deg(\psi) > 1$. Since $Y \times Z$ is simply connected, ψ has branch loci $B_\psi \neq \emptyset$. This proves (1).

By the purity of branch loci, write $B_\psi = \bigcup_{i=1}^n P_i$ with P_i being prime divisors. Let Q be a prime divisor of $Y \times Z$ with $p_Z(Q) \neq Z$. Then we have $Q = p_Z^* p_Z(Q)$ by the coordinate projection and hence $\psi^* Q = \pi^* p_Z(Q)$. Since $p_Z(Q)$ is a single point, by our assumption, $\psi^* Q = \pi^* p_Z(Q)$ is reduced and irreducible. Similar arguments show that $\psi^* Q$ is also a reduced and irreducible prime divisor if Q is a prime divisor of $Y \times Z$ with $p_Y(Q) \neq Y$, noting that τ is a Fano contraction of a K_X -negative extremal ray (cf. Proposition 5.2.17). As a result, each prime divisor $P_i \subseteq B_\psi$ dominates both Y and Z . So (2) is proved.

Note that the branch locus of $g \times h$: $B_{g \times h} = p_Y^{-1}(B_g) + p_Z^{-1}(B_h)$. By (2), no prime divisor of B_ψ lies in the branch locus of $g \times h$. So $(g \times h)^* P_i = (g \times h)^{-1}(P_i) = \sum_k P_i(k)$ with $P_i(k)$ being prime divisors. In particular, (3) is proved.

By (3) and comparing the coefficients on the both sides of $\psi^*(g \times h)^* P_i = f^* \psi^* P_i$, we may assume $P_i(1) \subseteq B_\psi$ for each i . Note that $P_i(1) \neq P_j(1)$ for any $i \neq j$. Then $B_\psi = \bigcup_{i=1}^n P_i(1)$ and hence $(g \times h)(B_\psi) = B_\psi$. So (4) is proved. $\square$

Lemma 5.4.13. *Suppose that $\Delta_\tau \neq \emptyset$. Suppose further that f is polarized. Then it is impossible that X has an extremal contraction $\pi : X \rightarrow Z \cong \mathbb{P}^1$ of a K_X -negative extremal face, with reduced and irreducible fibres such that the induced map $\psi : X \rightarrow Y \times Z$ is finite surjective.*

Proof. Suppose on the contrary that such π exists. Denote by $h := f|_Z$ after iteration (cf. Lemma 5.2.2) and $p_Y : Y \times Z \rightarrow Y$, $p_Z : Y \times Z \rightarrow Z$ the two projections.

We claim that $\deg(\psi) = 2$. Note that h (like f) is also polarized. Let F be an f -periodic smooth general fibre of $\pi : X \rightarrow Z$ (cf. [Fak03, Theorem 5.1]). After iteration,

we may assume $f(F) = F$. Consider the following commutative diagram

$$\begin{array}{ccc} F & \xrightarrow{f|_F} & F \\ \tau|_F \downarrow & & \downarrow \tau|_F \\ Y & \xrightarrow{g} & Y \end{array}$$

where $\tau|_F$ is finite surjective by the finiteness of ψ . Since F is general, we may assume $\tau^{-1}(\Delta_\tau)|_F = \tau^{-1}(\Delta_\tau) \cap F$ is a reduced divisor on F . By the adjunction formula, the log ramification divisor formula on $\tau|_F$ and Lemma 5.4.5, we have

$$(K_X + \tau^{-1}(\Delta_\tau))|_F = K_F + \tau^{-1}(\Delta_\tau)|_F = (\tau|_F)^*(K_Y + \Delta_\tau) + R \sim R \quad (*)$$

with R an effective divisor on F . Consider the log ramification divisor formula on $f|_F$:

$$K_F + \tau^{-1}(\Delta_\tau)|_F = (f|_F)^*(K_F + \tau^{-1}(\Delta_\tau)|_F) + R';$$

here R' is effective on F . Hence, $-(K_F + \tau^{-1}(\Delta_\tau)|_F)$ is pseudo-effective by Lemma 5.2.5, since $f|_F$ is still polarized. This and the $(*)$ above imply $(K_X + \tau^{-1}(\Delta_\tau))|_F \equiv 0$. Hence, $K_X + \tau^{-1}(\Delta_\tau) = \pi^*H$ for some divisor H on Z , by the cone theorem (cf. [KM98, Theorem 3.7]). Write $\pi^*H \sim nF$ for some $n \in \mathbb{Z}$. Let ℓ be a general fibre of τ and $\ell' \cup \ell''$ a (reduced) reducible fibre. Since τ is a (flat) conic bundle, $\ell \equiv \ell' + \ell''$. Then we have

$$nF \cdot \ell = \pi^*H \cdot \ell = (K_X + \tau^{-1}(\Delta_\tau)) \cdot \ell = K_X \cdot \ell = -2$$

by the adjunction formula. Since F is integral and τ -ample, $F \cdot \ell = F \cdot (\ell' + \ell'') \geq 1 + 1$. So $n = -1$ and $F \cdot \ell = 2$. In particular, $\deg(\psi) = 2$ and the claim is proved.

Note that ψ being a double cover implies $\psi^*B_\psi = 2\psi^{-1}(B_\psi)$. By Lemma 5.4.12 (3), $(g \times h)^*B_\psi = (g \times h)^{-1}(B_\psi)$. Then we have:

$$2f^*\psi^{-1}(B_\psi) = f^*\psi^*B_\psi = \psi^*(g \times h)^*B_\psi = \psi^*((g \times h)^{-1}(B_\psi));$$

hence $(g \times h)^{-1}(B_\psi) \subseteq B_\psi$. Since $g \times h$ is finite surjective, we have $(g \times h)^{-1}(B_\psi) = B_\psi$. Then $B_\psi \subseteq B_{g \times h} = p_Y^{-1}(B_g) + p_Z^{-1}(B_h)$ since $g \times h$ (like its lifting f) is also polarized (cf. Lemma 5.2.4). This contradicts Lemma 5.4.12 (2). $\square$

Lemma 5.4.14. *Suppose $\Delta_\tau \neq \emptyset$ and $Y \cong \mathbb{P}^2$. Then f is polarized after iteration.*

Proof. Note that $\rho(X) = \rho(Y) + 1 = 2$ and $-K_X$ is ample. Let $\pi : X \rightarrow Z$ be the second contraction of a K_X -negative extremal ray. After iteration, we may assume π is f -equivariant (cf. Lemma 5.2.2). Note that $\rho(Z) = 1$. If $\dim(Z) = \dim(X)$, then $f|_Z$ is non-isomorphic and hence polarized; thus f is polarized (cf. Lemma 5.2.4). So we may assume π is a Fano contraction. Then $Z \cong \mathbb{P}^2$ or $\mathbb{P}^1$ (cf. Lemma 5.2.7). Set $h := f|_Z$. Then $h^*\mathcal{O}_Z(1) = \mathcal{O}_Z(p)$ for some $p \geq 1$. On the other hand, since $g = f|_Y$ has $\deg(g) > 1$ by Proposition 5.4.3 and $Y \cong \mathbb{P}^2$, we have $g^*\mathcal{O}_Y(1) = \mathcal{O}_Y(q)$ for some $q > 1$. Now $f^*|_{N^1(X)} = \text{Diag}[q, p]$. It then suffices to show that $p = q$ (cf. Lemma 5.2.4).

Consider the case $Z \cong \mathbb{P}^2$. Let $D_1 = \tau^*\mathcal{O}_Y(1)$ and $D_2 = \pi^*\mathcal{O}_Z(1)$. Then $f^*D_1 \sim qD_1$ and $f^*D_2 \sim pD_2$. Since $\tau \neq \pi$, we have $D_1^2 \cdot D_2 > 0$ and $D_2^2 \cdot D_1 > 0$. By the projection formula, $\deg(f) = q^2p = p^2q$. So $q = p$ and f is polarized.

Consider the case $Z \cong \mathbb{P}^1$. Then our π here is a Fano contraction of a K_X -negative extremal ray (and hence has reduced and irreducible fibres; cf. Proposition 5.2.17). Then the induced map $\psi : X \rightarrow Y \times Z$ is finite surjective since $\tau \neq \pi$ and $\rho(X) = \rho(Y \times Z)$. By Lemma 5.4.12, $\deg(\psi) > 1$, the branch locus $B_\psi \neq \emptyset$ and we may assume (after iteration) $(g \times h)(P_1) = P_1$ for some prime divisor $P_1 \subseteq B_\psi$. Denote by $p_Y : Y \times Z \rightarrow Y$ and $p_Z : Y \times Z \rightarrow Z$ the two projections. Since $p_Y|_{P_1} : P_1 \rightarrow Y$ is generically finite and surjective by Lemma 5.4.12 (2), $(g \times h)|_{P_1}$ (like its descending g on Y) is q -polarized by Lemma 5.2.4. Since $p_Z|_{P_1} : P_1 \rightarrow Z$ is surjective, h on Z (like its lifting $(g \times h)|_{P_1}$) is q -polarized by Lemma 5.2.4. So $p = q$ and f is polarized. $\square$

Lemma 5.4.15. *Suppose $\Delta_\tau \neq \emptyset$ and $Y \cong \mathbb{P}^1 \times \mathbb{P}^1$. Then f is polarized after iteration.*

Proof. First, note that $\rho(X) = \rho(Y) + 1 = 3$. After iteration, we may write $f^*|_{N^1(X)} = \text{Diag}[\lambda_1, \lambda_2, \lambda_3]$ and $g^*|_{N^1(Y)} = \text{Diag}[\lambda_2, \lambda_3]$ (cf. [MZ20a, Lemma 5.4] or [Men20, Lemma 9.2]). We may assume $\lambda_2 \leq \lambda_3$. By Proposition 5.4.3, $\deg(g) > 1$, hence $\lambda_3 > 1$. Denote by $p_i : Y \rightarrow Y_i \cong \mathbb{P}^1$ ($i = 2, 3$) the two projections such that $\deg(g|_{Y_i}) = \lambda_i$. By Lemma 5.4.4, $g^{-1}(\Delta_\tau) = \Delta_\tau$. Hence, Δ_τ contains at least (and indeed exactly) two fibres each of p_2 and p_3 by Lemmas 5.3.1 and 5.4.5, so it is ample.

We first claim that either $\lambda_2 > 1$ or $\lambda_1 = \lambda_3 > 1$. Suppose $\lambda_2 = 1$. Let L be a general fibre of p_2 such that $L \not\subseteq \Delta_\tau$. Since $g^{-1}(\Delta_\tau) = \Delta_\tau$, we have $g(L) \not\subseteq \Delta_\tau$. Since

$g|_{Y_2}$ is an automorphism, $\#L \cap \Delta_\tau = L \cdot \Delta_\tau = g(L) \cdot \Delta_\tau = \#g(L) \cap \Delta_\tau \geq 1$. Let $F := \tau^{-1}(L)$. Then F (resp. $f(F)$) is a smooth ruled surface with a ruling $\tau|_F : F \rightarrow L$ (resp. $\tau|_{f(F)} : f(F) \rightarrow g(L)$) having $\#L \cap \Delta_\tau (> 0)$ singular fibres being intersecting (-1) -curves. Let ℓ be an irreducible component of a singular fibre of $\tau|_F$ lying over $P := \tau(\ell) \in L$. Since $P \in L \cap \Delta_\tau$, we have $g^{-1}(g(P)) = P$ and $(f|_F)^*(f(\ell)) = \lambda_3 \ell$ by noting that $\deg(g|_L) = \lambda_3$. By the projection formula,

$$-\deg(f|_F) = \deg(f|_F) \cdot \ell^2 = ((f|_F)^*(f(\ell)))^2 = \lambda_3^2 \ell^2 = -\lambda_3^2.$$

Note that $\deg(f|_F) = \lambda_1 \lambda_3$. Then we have $\lambda_1 = \lambda_3$ and the claim is proved.

Let L' be a g -periodic general fibre of p_3 such that $L' \not\subseteq \Delta_\tau$ (cf. [Fak03, Theorem 5.1]). After iteration, we may assume $g(L') = L'$ with $\deg(g|_{L'}) = \lambda_2$. Denote by $F' := \tau^{-1}(L')$. Then $(f|_{F'})^*|_{N^1(F')}$ has two positive integral eigenvalues λ_1, λ_2 with either $\lambda_1 > 1$ or $\lambda_2 > 1$ by the claim above. In particular, $f|_{F'}$ is non-isomorphic. Our F' is a smooth ruled surface over L' with $\#L' \cap \Delta_\tau (> 0)$ singular fibres. By Lemma 5.2.10, $f|_{F'}$ is polarized after iteration. So $\lambda_1 = \lambda_2$. If $\lambda_1 = \lambda_3$, then f is polarized.

Suppose $\lambda_1 \neq \lambda_3$. By the claim above, $\lambda_2 > 1$. Taking a g -periodic fibre of p_2 , and arguing as above, we get $\lambda_1 = \lambda_3$, a contradiction. So f is polarized as required. $\square$

Proposition 5.4.16. *Suppose $\Delta_\tau \neq \emptyset$. Then $Y \not\cong \mathbb{P}^2$.*

Proof. Suppose on the contrary that $Y \cong \mathbb{P}^2$. Then $\rho(X) = \rho(Y) + 1 = 2$. By Lemma 5.4.14, f is polarized (after iteration). Let $\pi : X \rightarrow Z$ be the second contraction of a K_X -negative extremal ray. After iteration, we may assume π is f -equivariant (cf. Lemma 5.2.2). Then $h := f|_Z$ (like f) is also polarized. By Proposition 5.4.7, π has to be a Fano contraction. Since $\rho(Z) = \rho(X) - 1 = 1$, our $Z \cong \mathbb{P}^2$ or $\mathbb{P}^1$ by Lemma 5.2.7. Suppose $Z \cong \mathbb{P}^1$. Then the induced map $\psi : X \rightarrow Y \times Z$ is finite surjective since $\tau \neq \pi$ and $\rho(X) = \rho(Y \times Z) = 2$. However, this contradicts Lemma 5.4.13 (cf. Proposition 5.2.17). So we may assume $Z \cong \mathbb{P}^2$.

After iterating f , we may assume that every component C_i of Δ_τ is g^{-1} -invariant (cf. Lemma 5.4.4). Hence, $S_i = \tau^{-1}(C_i)$ and $S_i \cap S_j$ are f^{-1} -invariant. Consider the irreducible non-reduced (double line) fibre with reduced structure $\ell_d = S_1 \cap S_2 =$

$\tau^{-1}(C_1 \cap C_2)$, which is non-empty (cf. [Mor82, Theorem 3.5] and Lemma 5.4.4) Since ℓ_d is not contracted by π , the image $\pi(\ell_d)$ is an h^{-1} -invariant irreducible curve by [CMZ20, Lemma 7.5]. Denote by $D := \pi^{-1}(\pi(\ell_d))$ which is a prime divisor on X by the cone theorem (cf. [KM98, Theorem 3.7]). Then $f^{-1}(D) = D$ and D is different from S_1, S_2 . Hence, the pair $(X, S_1 + S_2 + D)$ is not lc at ℓ_d since all the three prime divisors S_1, S_2, D contain ℓ_d (cf. [KM98, Lemma 2.29]). This contradicts Theorem 5.2.11. $\square$

Proposition 5.4.17. *Suppose $\Delta_\tau \neq \emptyset$. Then $Y \not\cong \mathbb{P}^1 \times \mathbb{P}^1$.*

Proof. Suppose on the contrary that $Y \cong \mathbb{P}^1 \times \mathbb{P}^1$. Then $\rho(X) = \rho(Y) + 1 = 3$. By Lemma 5.4.15, f is polarized (after iteration). Let $\pi_1 : X \rightarrow Z_1$ be the second contraction of a K_X -negative extremal ray. After iteration, we may assume that π_1 is f -equivariant (cf. Lemma 5.2.2). By Proposition 5.4.7, π_1 has to be a Fano contraction and Z_1 is a smooth rational (ruled) surface with $\rho(Z_1) = \rho(X) - 1 = 2$ (cf. Lemma 5.2.7).

We claim that $Z_1 \cong \mathbb{P}^1 \times \mathbb{P}^1$. Suppose on the contrary. Then Z_1 contains some negative curve C . Let $E := \pi_1^{-1}(C)$ ($= \pi_1^*(C)$ by the cone theorem). By the projection formula, $E \cdot \tilde{C} < 0$ for any curve $\tilde{C} \subseteq X$ dominating C . Since $-K_X$ is ample, the Mori cone $\overline{\text{NE}}(X)$ is generated by finitely many extremal (rational) curves, one of which, still denoted as $\tilde{C}$, is E -negative and is of course K_X -negative too. Then the contraction of the “non-nef” extremal ray $\mathbb{R}_{\geq 0}[\tilde{C}]$ is birational and hence divisorial by [Mor82, Theorem 3.3]. However, this contradicts Proposition 5.4.7. So the claim is proved.

Let ℓ be a general fibre of τ . Then ℓ is not contracted by π_1 . Let $p : Z_1 \rightarrow Z \cong \mathbb{P}^1$ be one of the natural projections such that $\pi_1(\ell)$ is not contracted by p . Denote by $\pi : X \xrightarrow{p \circ \pi_1} Z$ the composition. Then the induced map $\psi : X \rightarrow Y \times Z$ is finite surjective since $\tau \neq \pi$ and $\rho(X) = \rho(Y \times Z) = 3$. For any $t \in Z$, the fibre $p^*(t)$ is a prime divisor in $Z_1 \cong \mathbb{P}^1 \times \mathbb{P}^1$. By Proposition 5.2.17, π has reduced and irreducible fibres.

Let H be an ample divisor of Z and denote by

$$F := \{C \in \overline{\text{NE}}(X) \mid \pi^*H \cdot C = 0\}.$$

Since π^*H is nef, F is a (K_X -negative) extremal face of $\overline{\text{NE}}(X)$. By the projection formula, any curve C is contracted by π if and only if $\pi^*H \cdot C = 0$. So π is the contraction

of the face F . However, this contradicts Lemma 5.4.13. $\square$

End of Proof of Theorem 5.4.1. Suppose $\Delta_\tau \neq \emptyset$. By Lemma 5.4.6, $Y \cong \mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$. However, this contradicts Propositions 5.4.16 and 5.4.17. Thus, τ is a (smooth) $\mathbb{P}^1$ -bundle over Y . By Lemma 5.2.12, X is an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$ over Y . We are done. $\square$

5.5 Fano threefolds of Picard number two

In this section, we deal with the case where the Picard number $\rho(X) = 2$ and prove:

Theorem 5.5.1. *Let X be a smooth Fano threefold of Picard number $\rho(X) \leq 2$. Suppose that X admits a non-isomorphic surjective endomorphism f . Then X is toric. To be precise, X is either $\mathbb{P}^3$, or a splitting $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$, or a blow-up of $\mathbb{P}^3$ along a line.*

We prepare two lemmas before proving Theorem 5.5.1.

Lemma 5.5.2. *Let $\pi : X \rightarrow Y \cong \mathbb{P}^3$ be a blow-up of Y along a smooth curve C . Suppose that X is Fano and admits a non-isomorphic surjective endomorphism f . Then the second extremal contraction $\tau : X \rightarrow Z$ is a Fano contraction.*

Proof. Note that $\rho(X) = \rho(Y) + 1 = 2$. Suppose on the contrary that $\tau : X \rightarrow Z$ is birational. Then it is a divisorial contraction by [Mor82, Theorems 3.3]. Let E and F be the exceptional loci of π and τ , respectively, which are known to be prime divisors. After iteration, we may assume π and τ are f -equivariant and denote by $g := f|_Y$ and $h := f|_Z$ (cf. Lemma 5.2.2). Note that g and hence f and h are all polarized (cf. Lemma 5.2.4).

We claim that $E \neq F$. Suppose on the contrary that $E = F$. Then $\tau(F)$ is not a point; otherwise π and τ will contract some common curve, a contradiction. Hence, τ is the blow-up of a smooth threefold Z along a smooth curve by [Mor82, Theorem 3.3]. By [KM98, Lemma 2.29], $K_X = \pi^*K_Y + E = \tau^*K_Z + F$ and hence $K_X - E$ is both π -trivial and τ -trivial. Since $\rho(X) = 2$, we have $K_X - E \equiv 0$, a contradiction to X being Fano. The claim is proved.

By this claim, $\pi(F)$ and $\tau(E)$ are both surfaces. If $E \cap F = \emptyset$, then by the cone theorem (cf. [KM98, Theorem 3.7]), $F = \pi^*\pi(F)$ is nef, a contradiction to $-F$ being

τ -ample. So $E \cap F \neq \emptyset$. Let C_0 be an f^{-1} -invariant (after iteration) irreducible curve contained in $E \cap F$. Since $\pi \neq \tau$, this C_0 is not contracted by π and τ at the same time.

Suppose $\tau(C_0)$ is not a point. Then $\tau(C_0) = \tau(F) \subseteq \tau(E)$, and $\tau : X \rightarrow Z$ is the blow-up along the (smooth) curve $\tau(F)$ by [Mor82, Theorem 3.3]. Hence, the rationally connected smooth threefold Z of Picard number one has $Z \cong \mathbb{P}^3$ (cf. Theorem 5.2.15). Note that $\tau(E)$ is a surface in Z . By [CMZ20, Lemma 7.5], $h^{-1}(\tau(E)) = \tau(E)$. So the h^{-1} -periodic $\tau(E)$ and hence $\tau(C_0) = \tau(F)$ are both linear by Theorem 5.2.16. Thus $\tau : X \rightarrow Z \cong \mathbb{P}^3$ is a toric blow-up along the line $\tau(F)$, and X is toric.

Suppose $\tau(C_0)$ is a point. Then $\pi(C_0)$ is not a point and $C = \pi(C_0) = \pi(E) \subseteq \pi(F) \subseteq Y \cong \mathbb{P}^3$. By Theorem 5.2.16, the g^{-1} -periodic $\pi(F)$ and hence C are both linear. So $\pi : X \rightarrow Y \cong \mathbb{P}^3$ is the toric blow-up along the line C and is toric.

Thus, in both cases, X is a toric blow-up of $\mathbb{P}^3$ along a line $\ell (= C \text{ or } \tau(F))$. Then there is a free pencil $\varphi : X \rightarrow B := \mathbb{P}^1$, parameterizing hyperplanes in $\mathbb{P}^3$ passing through ℓ . Since $\rho(X) = \rho(B) + 1$ and $-K_X$ is ample, $\varphi (\neq \pi)$ is a Fano contraction of a K_X -negative extremal ray and hence $\varphi = \tau$, a contradiction. So our lemma holds. $\square$

Lemma 5.5.3. *Let $\pi : X \rightarrow Y \cong \mathbb{P}^3$ be a blow-up of Y along a smooth curve C . Suppose that X is Fano and admits a non-isomorphic surjective endomorphism f . Suppose further that X admits a Fano contraction $\tau : X \rightarrow Z$ to a curve. Then C is a line, π is a toric blow-up and X is toric.*

Proof. By Lemma 5.2.7, $Z \cong \mathbb{P}^1$. We may assume π and τ are f -equivariant after iteration and denote by $g := f|_Y$ and $h := f|_Z$ (cf. Lemma 5.2.2). Note that g and hence f and h are all polarized by Lemma 5.2.4. Let E be the (f^{-1} -invariant) exceptional divisor of π . Note that $\tau|_E : E \rightarrow Z$ is surjective because π and τ contract no common curves.

Let F be a general f -periodic fibre of τ (cf. [Fak03, Theorem 5.1]). After iteration, we assume that $f(F) = F$. Then $f|_F$ (like f) is also polarized. Hence, F is a smooth toric Fano surface (cf. Theorem 5.3.2). Since E is τ -ample, $E|_F$ is an ample divisor on F ; hence the support $E \cap F$ is connected (cf. [Har77, Chapter III, Corollary 7.9]). By the generic smoothness of $\tau|_E : E \rightarrow Z$ (cf. [Har77, Chapter III, Corollary 10.7]), we see that $C_0 := E \cap F$ is smooth and hence irreducible. Since $f^{-1}(E) = E$, we have

$(f|_F)^{-1}(C_0) = C_0$ and thus $C_0 \cong \mathbb{P}^1$ (cf. Theorem 5.3.2 and Lemma 5.2.8). Now that $\pi \neq \tau$ and C_0 dominates C via $\pi|_E : E \rightarrow C$, we have $C \cong \mathbb{P}^1$. As a result, the exceptional divisor E , as a smooth rational surface of Picard number two admitting two rulings, has to be $\mathbb{P}^1 \times \mathbb{P}^1$ with C_0 being the ruling of $\tau|_E$.

Further, if ℓ is any fibre of $\pi|_E : \mathbb{P}^1 \times \mathbb{P}^1 \cong E \rightarrow C$, we have $(C_0 \cdot \ell)_E = 1$. Note that $C \subseteq \pi(F)$. Write $\pi^*\pi(F) = F + eE$ for some integer $e > 0$. Then we have

$$F \cdot \ell = (F|_E \cdot \ell)_E = (C_0 \cdot \ell)_E = 1.$$

Since $\pi^*\pi(F)$ is π -trivial, we have

$$1 = F \cdot \ell = (F + eE) \cdot \ell - eE \cdot \ell = -eE \cdot \ell$$

which implies $e = 1$ and $E \cdot \ell = -1$. Note that

$$1 \leq \pi(F)^3 = (\pi^*\pi(F))^2 \cdot F = (F + E)^2 \cdot F = E^2 \cdot F = (C_0^2)_F,$$

Note also that $C_0 = E|_F$ is $(f|_F)^{-1}$ -invariant. By Corollary 5.3.4, $(C_0^2)_F \leq 1$. So $\pi(F)^3 = 1$ and hence $\pi(F) \cong \mathbb{P}^2$ is a hyperplane in $Y \cong \mathbb{P}^3$. Since $g|_{\pi(F)}$ (like g) is polarized, the $(g|_{\pi(F)})^{-1}$ -invariant curve $C = \pi(C_0)$ is a line in $\pi(F)$ (and also in Y) by Theorem 5.2.16. Thus, π is a toric blow-up along a line $C \subseteq Y \cong \mathbb{P}^3$ and X is toric. $\square$

Proof of Theorem 5.5.1. If $\rho(X) = 1$, then $X \cong \mathbb{P}^3$ by Theorem 5.2.15.

Assume $\rho(X) = 2$. If there is a Fano contraction $\tau : X \rightarrow Y$ to a surface (this is the case when X is primitive by [MM83, Theorem 1.6]), then $\rho(Y) = \rho(X) - 1 = 1$ implies $Y \cong \mathbb{P}^2$ (cf. Lemma 5.2.7). By Theorem 5.4.1, $X := \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ is an algebraic $\mathbb{P}^1$ -bundle with $\mathcal{E}$ a locally free sheaf of rank 2 over $Y \cong \mathbb{P}^2$. This $\mathcal{E}$ splits, and hence X is toric (cf. Propositions 5.2.14 and 5.2.9).

Suppose X has no Fano contraction to a surface. Then X is imprimitive, so there is a divisorial contraction $\pi : X \rightarrow X_1$ to a smooth Fano threefold with $\rho(X_1) = \rho(X) - 1 = 1$, and π is the blow-up along a smooth curve $C \subseteq X_1$. We may assume that f descends to a (non-isomorphic) surjective endomorphism $f|_{X_1}$ after iteration (cf. Lemma 5.2.2). By Theorem 5.2.15, $X_1 \cong \mathbb{P}^3$. Now our theorem follows from Lemmas 5.5.2 and 5.5.3. $\square$

5.6 Conic bundle and minimal model program

In this section, we deal with general Fano threefolds. First, we observe:

Theorem 5.6.1. *Let X be a smooth Fano threefold with $\rho(X) \geq 3$. Suppose that X admits a non-isomorphic surjective endomorphism f . Then X admits a conic bundle.*

Proof. By [MM83, Section 9], either X admits a conic bundle $\tau : X \rightarrow Y$ or X is isomorphic to the blow-up of $\mathbb{P}^3$ along a disjoint union of a line and a conic (and hence of Picard number 3). For the latter case, let X' be the blow-up of $\mathbb{P}^3$ along the conic curve which is still Fano by the ramification divisor formula or by [MM83, Corollary 4.6]. One can verify that X' is not toric. By Theorem 5.5.1, X' and hence X admit no non-isomorphic surjective endomorphism, a contradiction. So X admits a conic bundle. $\square$

Encouraged by the above result, next we aim to reduce a general (Fano) conic bundle to a Fano contraction $X \rightarrow Y$ to a surface Y as in Theorem 5.4.1, so that $\rho(X) = \rho(Y) + 1$. So we apply the relative minimal model program.

Theorem 5.6.2. *Let X be a smooth Fano threefold with a conic bundle $\tau : X \rightarrow Y$. Suppose that X admits a non-isomorphic surjective endomorphism f . Then there exists an f -equivariant (after iteration) minimal model program*

$$X = X_1 \rightarrow \cdots \rightarrow X_i \rightarrow \cdots \rightarrow X_{r+1} \rightarrow Y$$

such that the following assertions hold.

- (1) $r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth Fano threefold.
- (2) $X_{r+1} \xrightarrow{\tau_{r+1}} Y$ is a Fano contraction and an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$ over Y .
- (3) The composite $\tau_i : X_i \rightarrow Y$ is a conic bundle with $(f|_Y)^{-1}$ -invariant discriminant $\Delta_{\tau_i} = C_i \cup \cdots \cup C_r$ a disjoint union of $r + 1 - i$ smooth curves on Y .
- (4) The composite $\pi : X \rightarrow X_{r+1}$ is the blow-up of X_{r+1} along r disjoint union of $(f|_{X_{r+1}})^{-1}$ -invariant smooth curves $\bigcup_{i=1}^r \overline{C_i}$ with $\tau_{r+1}(\overline{C_i}) = C_i$.

Proof. By [MM83, Propositions 6.2, 6.3, Corollary 6.4 and Propositions 6.5, 6.8], we may run a relative minimal model program $X = X_1 \rightarrow \cdots \rightarrow X_i \rightarrow \cdots \rightarrow X_{r+1}$ of X over Y which is f -equivariant after iteration (cf. Lemma 5.2.2), such that:

- (i) $r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth Fano threefold; and
- (ii) The composition $\tau_i : X_i \rightarrow Y$ is a conic bundle with $\Delta_{\tau_i} = \Delta_{\tau_{r+1}} \cup C_i \cup \cdots \cup C_r$ where $C_i, \dots, C_r$ are connected (and irreducible) components of Δ_{τ_i} .

So (1) is proved. Note that $\rho(X_{r+1}) = \rho(Y) + 1$. So the conic bundle $\tau_{r+1} : X_{r+1} \rightarrow Y$ is automatically a Fano contraction of a $K_{X_{r+1}}$ -negative extremal ray. After iteration, f descends to a non-isomorphic surjective endomorphism of X_{r+1} . By Theorem 5.4.1, τ_{r+1} is an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$ over Y . So (2) is proved. In particular, $\Delta_{\tau_{r+1}} = \emptyset$ and (3) is proved. Finally, (4) follows easily from (3) by noting that each divisorial contraction is the blow-up along some smooth curve which is either a smooth fibre of the conic bundle $X_i \rightarrow Y$ or a subsection over Y (cf. [MM83, Proposition 6.8]). $\square$

In order to prove Theorem 5.8.1 and hence 5.1.4, we need to study other possible minimal model program which is a sequence of smooth (but not necessarily Fano) threefolds.

Lemma 5.6.3. *Let X be a smooth Fano threefold with a conic bundle $\tau : X \rightarrow Y$ over a Hirzebruch surface $Y \xrightarrow{p} Z \cong \mathbb{P}^1$. Suppose that τ factors through $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ with τ_0 being a (smooth) $\mathbb{P}^1$ -bundle. Suppose further that X admits an int-amplified endomorphism f . Then $p \circ \tau_0 : W \rightarrow Z$ is a fibre bundle such that either all fibres are $\mathbb{F}_0$ or all fibres are $\mathbb{F}_1$, where $\mathbb{F}_d$ denotes the Hirzebruch surface of degree d .*

Proof. For $z \in Z$, let $Y_z := p^{-1}(z) \cong \mathbb{P}^1$, $X_z := (p \circ \tau)^{-1}(z)$ and $W_z := (p \circ \tau_0)^{-1}(z)$. Then W_z is a smooth (rational) ruled surface of Picard number 2. Hence, $W_z \cong \mathbb{F}_d$ for some $d \geq 0$. The lemma is equivalent to the claim that $W_z \cong \mathbb{F}_d$ has anti-ample canonical divisor, i.e., $d = 0, 1$. Indeed, once this claim is proved, the lemma then follows from the fact that $\mathbb{F}_0$ and $\mathbb{F}_1$ are not deformed to each other.

We prove this claim by induction on $r := \rho(X) - \rho(Y) - 1$. If $r = 0$, then π is isomorphic and $W \cong X$ is Fano. By the adjunction formula, $K_{W_z} = K_W|_{W_z}$ is anti-ample, as claimed.

Suppose $r > 0$. By [MM83, Proposition 6.3 and Corollary 6.4], since τ_0 is a (smooth) $\mathbb{P}^1$ -bundle, Δ_τ is a disjoint union of r smooth curves C_i , and π is the blow-up of W along the disjoint union $\coprod_{i=1}^r \overline{C_i}$ of r many $(f|_W)^{-1}$ -invariant smooth curves $\overline{C_i}$ with $\tau_0(\overline{C_i}) = C_i$. Write $\tau^{-1}(C_i) = \tau^*C_i = E_i + F_i$ where E_i is the π -exceptional divisor with center $\overline{C_i}$ and F_i is the strict transform of $\tau_0^{-1}(C_i)$. Denote by $L_i := E_i \cap F_i$ the transversal intersection. Note that $F_i \cong \tau_0^{-1}(C_i)$ and $L_i \cong \overline{C_i}$. Let $z \in Z$.

If X_z is irreducible, then X_z is the blow-up of W_z along points $\coprod_{i=1}^r \overline{C_i} \cap W_z$. In particular, X_z is smooth. By the adjunction formula, $K_{X_z} = K_X|_{X_z}$ is anti-ample, so is K_{W_z} on the image W_z of X_z , as claimed.

If $X_z (= \tau^{-1}(Y_z))$ is reducible, then Y_z equals some C_i (cf. [MM83, Proposition 6.3]) and $F_i \cong W_z$ via $\pi : X \rightarrow W$. Let $X \rightarrow X'$ be the contraction of E_i over W . If $E_i \not\cong \mathbb{F}_0$, then X' is Fano (and smooth) by [MM83, Proposition 6.5]. Replacing X by X' , we are done by induction. So we may assume $E_i \cong \mathbb{F}_0$. Note that E_i, F_i and hence L_i are f^{-1} -invariant after iteration (cf. Lemmas 5.2.2 and 5.4.4). By Lemma 5.3.1, we have

$$F_i \cdot L_i = (F_i|_{E_i} \cdot L_i)_{E_i} = (L_i^2)_{E_i} = 0$$

since $f|_{E_i}$ (like f) is int-amplified. This implies (working over F_i):

$$(L_i^2)_{F_i} = E_i \cdot L_i = (\tau^*C_i - F_i) \cdot L_i = C_i^2 = Y_z^2 = 0,$$

so $W_z \cong F_i \cong \mathbb{F}_0$, because $L_i \subseteq F_i$ is horizontal to (indeed a cross-section of) the ruling $F_i \rightarrow C_i$. This proves the claim and also the lemma. $\square$

Theorem 5.6.4 below replaces the semi-stability condition in Theorem 5.2.13 by the birational dominance of a Fano threefold. It is used crucially in proving Theorem 5.8.1.

Theorem 5.6.4. *Let X be a smooth Fano threefold admitting an int-amplified endomorphism f . Suppose X admits a conic bundle $\tau : X \rightarrow Y$ (with Y smooth), which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ with τ_0 being a (smooth) $\mathbb{P}^1$ -bundle. Then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

Proof. After iteration, we may assume that τ is f -equivariant by Lemma 5.2.2. Let $g := f|_W$ and $h := f|_Y$, which are both int-amplified (cf. Lemma 5.2.5). Note that Y is a

smooth rational surface (cf. Proof of Lemma 5.2.7). By Lemma 5.2.12, we may write $W = \mathbb{P}_Y(\mathcal{E})$ with $\mathcal{E}$ a rank-two vector bundle on Y .

If $\rho(Y) = 1$, then $Y \cong \mathbb{P}^2$, and the theorem follows from Proposition 5.2.14. If $\rho(Y) \geq 3$, then $\tau = \tau_0$ is a trivial bundle by [MM83, Proposition 9.10].

Thus we may assume $\rho(Y) = 2$. Then $Y \cong \mathbb{F}_d$ (Hirzebruch surface). By [MM83, Corollary 6.7], $d = 0, 1$. Fix a ruling $p : Y \rightarrow Z \cong \mathbb{P}^1$. For $z \in Z$, let $Y_z := p^{-1}(z)$, $X_z := (p \circ \tau)^{-1}(z)$ and $W_z := (p \circ \tau_0)^{-1}(z)$. Let ℓ be the fibre class of p and C_0 the section class of p which is extremal in $\overline{\text{NE}}(Y)$ and $C_0^2 = -d$. Twisting with a suitable line bundle of Y , we may assume that $c_1(\mathcal{E}) = aC_0 + b\ell$ with $-1 \leq a, b \leq 0$. Then $\mathcal{O}_{Y_z}(c_1(\mathcal{E}|_{Y_z})) \cong \mathcal{O}_{Y_z}(a)$ for any $z \in Z$. This, $a \in \{0, -1\}$, and $\mathbb{P}(\mathcal{E}|_{Y_z}) = W_z \cong \mathbb{F}_c$ ($c \leq 1$, cf. Lemma 5.6.3) by base change, imply $\mathcal{E}|_{Y_z} \cong \mathcal{O}_{Y_z} \oplus \mathcal{O}_{Y_z}(a)$ for any $z \in Z$, and $c = -a$.

Note that our a here is independent of the choice of z ; hence the function $z \mapsto h^0(Y_z, \mathcal{E}|_{Y_z})$ is a constant function. So the natural morphism $\varphi : p^*p_*\mathcal{E} \rightarrow \mathcal{E}$ has domain a locally free sheaf, which is an evaluation map on every fibre (cf. [Har77, Ch III, Corollary 12.9]). Note that the global sections of $\mathcal{E}|_{Y_z}$ are constant, since $a \leq 0$. Then we have an exact sequence

$$(*) \quad 0 \rightarrow p^*p_*\mathcal{E} \xrightarrow{\varphi} \mathcal{E} \rightarrow \mathcal{Q} \rightarrow 0$$

where $\mathcal{Q}$ is still a vector bundle. If $a = 0$, then $p_*\mathcal{E}$ has rank two (over $\mathbb{P}^1$) and splits always, and this implies $\mathcal{E} \cong p^*p_*\mathcal{E}$ splits.

We still have to consider the case $a = -1$. Then $p_*\mathcal{E}$ is a line bundle, say $\mathcal{O}_Z(e)$ for some $e \in \mathbb{Z}$. Then $p^*p_*\mathcal{E} \cong \mathcal{O}_Y(e\ell)$ and $\mathcal{Q} \cong \mathcal{O}_Y(-C_0 + (b - e)\ell)$. Note that

$$\text{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E}) = H^1(Y, C_0 + (2e - b)\ell) = H^1(Y, K_Y + (C_0 - K_Y) + (2e - b)\ell)$$

vanishes if $2e - b \geq 0$ by the Kodaira vanishing theorem and the nef and bigness of $(C_0 - K_Y) + (2e - b)\ell$ when $Y = \mathbb{F}_d$ ($d = 0, 1$). So $2e - b \geq 0$ implies that $(*)$ and hence $\mathcal{E}$ split. Therefore, we may assume that $2e - b < 0$. Denote by $\xi := 2c_1(p^*p_*\mathcal{E}) - c_1(\mathcal{E}) \sim C_0 + (2e - b)\ell$. Then neither ξ nor $-\xi$ is pseudo-effective. Hence, ξ defines a non-empty wall $W := \{A \in \text{Amp}(Y) \mid A \cdot \xi = 0\}$ (cf. [Qin93, Chapter I, Definition 2.1.5 and

Chapter II, Definition 1.2.1]) where $\text{Amp}(Y)$ is the ample cone in $\text{NS}_{\mathbb{R}}(Y)$. By [Qin93, Chapter II, Theorem 1.2.3, Page 406], either $\mathcal{E}$ splits or $\mathcal{E}$ is H -stable for (any) ample divisor $H \in \mathcal{C}$ where $\mathcal{C}$ is a chamber (surrounded by walls) near W (i.e., $\overline{\mathcal{C}} \cap W \neq \emptyset$) and $A \cdot \xi < 0$ for any $A \in \mathcal{C}$. If $\mathcal{E}$ is H -stable, then by our dynamical assumption and Theorem 5.2.13, $\mathcal{E}$ still splits, a contradiction. This proves the theorem. $\square$

5.7 Proof of Theorem 5.1.3

In this section, we prove Theorem 5.1.3. The key step is the following:

Theorem 5.7.1. *Let f be a non-isomorphic surjective endomorphism of a smooth Fano threefold X . Then either f is polarized after iteration or $X \cong Y \times \mathbb{P}^1$ for a del Pezzo surface Y .*

Lemma 5.7.2. *Theorem 5.7.1 holds for the case $\rho(X) = 3$.*

Proof. Note that $\text{Nef}(X)$ is a rational polyhedron with r extremal rays. Since $\text{Nef}(X)$ spans $N^1(X)$, we have $r \geq \rho(X) = 3$. After iteration, we may assume $f^*|_{N^1(X)} = \text{Diag}[\lambda_1, \lambda_2, \lambda_3]$ is diagonal with λ_i being positive integers. We may assume that f is not polarized even after iteration. Then $\#\{\lambda_1, \lambda_2, \lambda_3\} = 2, 3$ (cf. Lemma 5.2.4).

We show that $r = 3$. Suppose $r \geq 4$. Note that any three of the extremal rays are linearly independent. If $f^*|_{N^1(X)}$ has three distinct eigenvalues, then there are exactly three linearly independent one-dimensional eigenspaces. However, one of them contains two extremal rays, a contradiction. If $f^*|_{N^1(X)}$ has two distinct eigenvalues, then there is a two-dimensional eigenspace containing three extremal rays, a contradiction. So $r = 3$ and $\text{Nef}(X)$ (and hence $N^1(X)$) is generated by three extremal nef divisors D_1, D_2, D_3 .

We may assume $f^*D_i = \lambda_i D_i$. Since f is non-isomorphic and non-polarized, D_i is not big by Lemma 5.2.4, so $D_i^3 = 0$ for all i . Since $-K_X$ is ample, there are three extremal ray contractions $\pi_i : X \rightarrow Y_i$ such that $\pi_i^* \text{Nef}(Y_i)$ is generated by $\{D_1, D_2, D_3\} \setminus \{D_i\}$ and $\rho(Y_i) = 2$. Denote by $f_i := f|_{Y_i}$.

Consider the case where every π_i is a divisorial contraction. Then each exceptional locus of π_i consists of a single prime divisor E_i by [Mor82, Theorem 3.3]. Note that

each E_i is π_i -anti ample and hence a non-nef (effective) eigenvector of $f^*|_{N^1(X)}$ different from D_1, D_2, D_3 . Thus $\#\{\lambda_1, \lambda_2, \lambda_3\} = 2$. We may assume $\lambda_1 = \lambda_2 \neq \lambda_3$. Note that f_3 (like its birational lifting f) is non-isomorphic. Then f_3 is λ_1 -polarized and hence so is f (cf. Lemma 5.2.4), a contradiction.

Thus we may assume that $\pi_1 : X \rightarrow Y_1$ is a Fano contraction. Since $\rho(Y_1) = 2$, we have $Y_1 \cong \mathbb{F}_d$, a Hirzebruch surface (cf. Lemma 5.2.7). By Theorem 5.4.1, π_1 is further an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_{Y_1}(\mathcal{E})$. By Lemma 5.2.10, $\lambda_2 = \lambda_3$ if $Y_1 \not\cong \mathbb{P}^1 \times \mathbb{P}^1$. Since f is not polarized, either $\lambda_1 \neq \lambda_2$ or $\lambda_1 \neq \lambda_3$. Without loss of generality, we may assume $\lambda_1 \neq \lambda_3$. Since $Y_1 \cong \mathbb{F}_d$, we can choose a suitable ruling $p : Y_1 \rightarrow C \cong \mathbb{P}^1$ such that $\deg(f_1|_C) = \lambda_2$. By Lefschetz fixed point formula on C , there exists an f_1 -invariant fibre $L \cong \mathbb{P}^1$ of p . Denote by $F := \pi_1^{-1}(L)$ which is f -invariant. Note that $(f|_F)^*|_{N^1(F)}$ has two distinct eigenvalues λ_1, λ_3 by our assumption. Then $F \cong \mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ by Lemma 5.2.10. Since π_1 is a (smooth) $\mathbb{P}^1$ -bundle and X is Fano, each fibre of $p \circ \pi_1$ is isomorphic to $\mathbb{F}_0$ (cf. e.g., [MM83, Lemma 9.4]). By [MM83, Lemma 9.3], $X \cong Z \times_{\mathbb{P}^1} Y_1$ for a $\mathbb{P}^1$ -bundle Z over $\mathbb{P}^1$. In particular, the induced map $X \rightarrow Z$ is another Fano contraction which is either π_2 or π_3 , i.e., $Z = Y_j$ for $j = 2$ or 3 . If $Y_1 \cong \mathbb{P}^1 \times \mathbb{P}^1$, then X is a trivial $\mathbb{P}^1$ -bundle over Z by the base change. If $Y_1 \not\cong \mathbb{P}^1 \times \mathbb{P}^1$, then $\lambda_2 = \lambda_3 \neq \lambda_1$ implies that $f_2^*|_{N^1(Y_2)}$ and $f_3^*|_{N^1(Y_3)}$ have two distinct positive eigenvalues. Then $Z = Y_j \cong \mathbb{P}^1 \times \mathbb{P}^1$ by Lemma 5.2.10 and hence X is a trivial $\mathbb{P}^1$ -bundle over Y_1 by the base change. This proves the lemma. $\square$

Proof of Theorem 5.7.1. We divide the proof into three cases.

Case $\rho(X) = 1$. In this case, f is polarized.

Case $\rho(X) = 2$. If X has a divisorial contraction to a projective variety X_1 which is f -equivariant after iteration (cf. Lemma 5.2.2), then $\rho(X_1) = \rho(X) - 1 = 1$, and hence $f|_{X_1}$ is polarized; thus so is f (cf. Lemma 5.2.4). So we may assume X has only Fano contractions. Thus X is primitive. By [MM83, Theorem 1.6], there is an (f -equivariant, after iterating f) Fano contraction $\tau : X \rightarrow Y$ to a surface Y . Since $\rho(Y) = \rho(X) - 1 = 1$, we have $Y \cong \mathbb{P}^2$ by Lemma 5.2.7. Set $g := f|_Y$. By Theorem 5.4.1 and Proposition

5.2.14, τ is a splitting $\mathbb{P}^1$ -bundle.

We may assume τ is not trivial; otherwise, X splits. We aim to prove f is polarized after iteration. After suitable twisting, we may write $X \cong \mathbb{P}_Y(\mathcal{O}_Y \oplus \mathcal{L})$ with the line bundle $\mathcal{L}$ being not pseudo-effective on Y . Then there exists a section S of τ such that $\mathcal{O}_X(S)|_S \cong (\tau|_S)^*\mathcal{L}$ is not pseudo-effective. By Lemma 5.2.3, $f^{-1}(S) = S$ after iteration. Write $f^*S = aS$ for some integer $a \geq 1$ and $g^*|_{N^1(Y)} = q \text{ id}$ for some integer $q \geq 1$. Since $N^1(X)$ is spanned by S and $\tau^*N^1(Y) \cong \mathbb{R}$, we have $f^*|_{N^1(X)} = \text{Diag}[a, q]$. Note that $(f|_S)^*(S|_S) = aS|_S$ and $S|_S \not\equiv 0$. Then a is an eigenvalue of $(f|_S)^*|_{N^1(S)}$, which implies $a = q$ is an eigenvalue of $g^*|_{N^1(Y)}$ since S is f -equivariantly isomorphic to Y . Since f is non-isomorphic, our $a = q > 1$ and hence f is polarized (cf. Lemma 5.2.4).

Case $\rho(X) \geq 3$. By Theorems 5.6.1 and 5.6.2, there is an f -equivariant (after iteration) conic bundle $\tau : X \rightarrow Y$ and denote by $g := f|_Y$ (cf. Lemma 5.2.2). In the following, we apply Theorem 5.6.2 and use the notations there.

If $\rho(X) = 3$, then Theorem 5.7.1 follows from Lemma 5.7.2. Thus we may assume $\rho(X) \geq 4$. If $\tau : X \rightarrow Y$ has the base $\rho(Y) \geq 3$, then it is a trivial conic bundle by [MM83, Proposition 9.10], and we are done. So we assume $\rho(Y) \leq 2$. Hence, $Y = \mathbb{P}^2$ or $Y = \mathbb{F}_d$ (cf. Lemma 5.2.7) with $d \leq 1$ (cf. [MM83, Corollary 6.7]). Since the g^{-1} -invariant divisor Δ_τ is a disjoint union of $r = \rho(X) - \rho(Y) - 1$ of $\mathbb{P}^1$'s (cf. Theorem 5.6.2), we may assume $Y \cong \mathbb{F}_1$ (cf. [MM83, (9.5), (9.6)], Lemma 5.3.1).

Thus Δ_τ is a subset of the disjoint union of the (-1) -curve C_0 on $Y \cong \mathbb{F}_1$ and another section C_1 disjoint from C_0 (cf. [MM83, Corollary 6.7]). Let $L \cong \mathbb{P}^1$ be a g -invariant fibre along the ruling $\sigma : Y \rightarrow Z \cong \mathbb{P}^1$ by the Lefschetz fixed point formula for $f|_Z$. Let $F := \tau^{-1}(L)$. Then F is an f -invariant smooth projective rational surface, which has some reducible fibres lying over $L \cap \Delta_\tau$. In particular, $F \not\cong \mathbb{P}^1 \times \mathbb{P}^1$. If $f|_F$ is isomorphic, then $f|_Z$ and hence $g = f|_Y$ are non-isomorphic, so g is polarized after iteration (cf. Lemma 5.2.10); thus $g|_L$ is polarized and non-isomorphic, which implies $f|_F$ is also non-isomorphic, a contradiction. Hence, $f|_F$ is non-isomorphic and thus q -polarized for some $q > 1$, since $F \not\cong \mathbb{P}^1 \times \mathbb{P}^1$ (cf. Lemma 5.2.10). As a result, $g|_L$ (and hence g) are both q -polarized. Moreover, we have $f^*F \sim qF$.

Finally, we prove f is q -polarized after iteration. By Lemma 5.2.4, it suffices to show f_{r+1} is q -polarized (cf. the notations in Theorem 5.6.2). Note that $\rho(X_{r+1}) = \rho(Y) + 1 = 3$ and τ_{r+1} is an algebraic $\mathbb{P}^1$ -bundle over Y . Write $g^*|_{N^1(Y)} = \text{Diag}[q, q]$ with $q > 1$ and $f_{r+1}^*|_{N^1(X_{r+1})/\tau_{r+1}^* N^1(Y)} = a \text{ id}$ for some integer $a \geq 1$. Then $f_{r+1}^*|_{N^1(X_{r+1})} = \text{Diag}[a, q, q]$ and it suffices to show that $a = q$ (cf. Lemma 5.2.4). By the projection formula, $\deg(f_{r+1}) = aq^2$. On the other hand, $\deg(f|_F) \cdot F = f_* F = (\deg(f)/q)F$ by the projection formula. Since $f|_F$ is q -polarized, we have $\deg(f_{r+1}) = \deg(f) = q^3$ and hence $a = q$ as desired. This completes the proof of our theorem. $\square$

Proof of Theorem 5.1.3. Clearly, (3) implies (2). By Theorem 5.7.1, (2) implies (1). If $X \cong \mathbb{P}^1 \times S$ for a del Pezzo surface S , then taking $f_1 : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ to be the square map $[x : y] \mapsto [x^2 : y^2]$ and $f := f_1 \times \text{id}_S$ the product map, we have f is non-isomorphic and non-int-amplified. So (1) implies (3). $\square$

5.8 Proofs of Theorem 5.1.4 and Corollary 5.1.5

To prove Theorem 5.1.4, we begin with the following, using results in Section 5.6.

Theorem 5.8.1. *Let X be a smooth Fano threefold with $\rho(X) \geq 3$. Suppose that X admits an int-amplified endomorphism f . Then X is toric.*

Proof. We apply Theorems 5.6.1 and 5.6.2 and employ the same notations there. Especially, $X_{r+1} = \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau_{r+1}} Y$ is a splitting $\mathbb{P}^1$ -bundle (cf. Theorem 5.6.4). Since there are only finitely many f^{-1} -periodic subvarieties by [MZ20a, Corollary 3.8], we may assume that they are all f^{-1} -invariant (after iterating f). If $r = 0$, then $\tau : X \rightarrow Y$ is a splitting $\mathbb{P}^1$ -bundle; hence X is toric by Theorem 5.3.3. We now assume $1 \leq r (= \rho(X/Y) - 1)$.

Then, by Theorem 5.6.2, τ is a conic bundle and $\Delta_\tau = \coprod_{i=1}^r C_i$ with $C_i \cong \mathbb{P}^1$. By [MM83, Proposition 6.3], $\tau^{-1}(C_i) = E_i \cup F_i$, and E_i and F_i are (smooth) $\mathbb{P}^1$ -bundles over C_i , with the f^{-1} -invariant curve $E_i \cap F_i$ dominating C_i (and indeed, a cross-section).

We claim that there is an f^{-1} -invariant section S of τ dominating Y . Recall the birational contraction $\pi : X \rightarrow X_{r+1}$ and the extremal Fano contraction $\tau_{r+1} : X_{r+1} \rightarrow$

Y . Note that $\pi(E_1 \cap F_1)$ is an f_{r+1}^{-1} -invariant subsection over Y (not contracted by τ_{r+1}) and $\pi(E_1 \cup F_1) = \tau_{r+1}^{-1}(C_1)$ is a $\mathbb{P}^1$ -bundle over C_1 . If τ_{r+1} is a trivial bundle so that $X_{r+1} = Y \times Z \cong Y \times \mathbb{P}^1$, then its second projection induces a second ruling on $\pi(E_1 \cup F_1)$; thus $\pi(E_1 \cup F_1) \cong \mathbb{P}^1 \times \mathbb{P}^1$; now $\pi(E_1 \cap F_1)$ is contained in an f_{r+1}^{-1} -invariant horizontal section S_{r+1} of τ_{r+1} (cf. [CMZ20, Lemma 7.5] and Lemma 5.3.1 applied to $\pi(E_1 \cup F_1)$). If τ_{r+1} is not a trivial bundle, as in Proof of Theorem 5.3.3, some section S_{r+1} of τ_{r+1} has $S_{r+1}|_{S_{r+1}}$ not being pseudo-effective and is hence f_{r+1}^{-1} -invariant after iteration by Lemma 5.2.3. In both cases, take $S \subseteq X$ to be the proper transform of S_{r+1} . Note also that $\pi|_S : S \cong S_{r+1}$ by the observation of π as in Theorem 5.6.2. So the claim is proved.

Since $\pi(S) = Y$, we have $(E_i \cup F_i) \cap S \neq \emptyset$. So we may assume $S \cap E_i \neq \emptyset$ for each i . By [MM83, Proposition 6.3] and Lemma 5.2.2, after iteration, there is an f -equivariant birational morphism $\pi' : X \rightarrow X'$ over Y contracting all E_i with $f' := f|_{X'}$, such that:

- (i) the induced morphism $\tau' : X' \rightarrow Y$ is a conic bundle with $\Delta_{\tau'} = \emptyset$; and
- (ii) π' is the blow-up of X' along r smooth curves $C'_i := F'_i \cap S'$, where $F'_i := \tau'^{-1}(C_i)$ and $S' := \pi'(S)$ are f'^{-1} -invariant prime divisors.

Our τ' is a $\mathbb{P}^1$ -bundle, but X' may not be Fano. By Theorem 5.6.4, τ' is a splitting $\mathbb{P}^1$ -bundle. Hence, by Theorem 5.3.3, there is a toric pair (X', Δ') such that Δ' contains all f'^{-1} -invariant prime divisors (including F'_i, S'). By the construction, π' is the composition of toric blow-ups of the intersection of two prime divisors in the log smooth toric boundary starting from the (log smooth) toric pair (X', Δ') . Thus X is toric. $\square$

Proof of Theorem 5.1.4. Clearly, (2) implies (3). By Theorems 5.5.1 and 5.8.1, (3) implies (1). Note that every projective toric variety has a polarized endomorphism; see [Nak02, Lemma 4] and [MZ20b, Proof of Theorem 1.4]. So (1) implies (2). $\square$

Proof of Corollary 5.1.5. By Theorem 5.1.2, we may assume $X = \mathbb{P}^1 \times S$ with S a del Pezzo surface. Iterating f , we may assume f fixes all (finitely many and automatically K_X -negative) extremal rays of $\overline{\text{NE}}(X)$. So f descends to f_P on $\mathbb{P}^1$ and f_S on S via the two projections which are contractions of extremal faces of $\overline{\text{NE}}(X)$, and $f = f_P \times f_S$. If $\deg(f_S) \geq 2$, then S and hence X are toric by [Nak02, Theorem 3]. We are done. $\square$

Chapter 6

Amplified endomorphisms of Fano fourfolds

6.1 Introduction

In this chapter, we work over an algebraically closed field k of characteristic 0. We continue to study Question 4.1.2 and extend Theorem 5.1.4 to the fourfold case.

Recall that f is q -polarized if $f^*H \sim qH$ for some ample Cartier divisor H on X and integer $q > 1$, *int-amplified* if $f^*L - L$ is ample for some ample Cartier divisor L on X , and *amplified* if $f^*L - L$ is ample for some (not necessarily ample) Cartier divisor L on X (cf. [KR17]); see [MZ18a, Propositions 1.1, 2.9] and [Men20, Theorem 1.1] for the equivalent definitions. Clearly,

$$\text{“polarized”} \Rightarrow \text{“int-amplified”} \Rightarrow \text{“amplified”}.$$

We reformulate the following theorem which was proved in [MZZ21, Theorem 1.4](=Theorem 5.1.4) for readers’ convenience.

Theorem 6.1.1 ([MZZ21, Theorem 1.4]=Theorem 5.1.4). *Let Y be a smooth Fano threefold admitting an (int-)amplified endomorphism g . Then Y is toric. Further, after iteration, either one of the following holds.*

- (A) *If $\rho(Y) \leq 2$, then Y is either $\mathbb{P}^3$, a splitting $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$, or a (toric) blow-up*

of $Y' := \mathbb{P}^3$ along a $(g|_{Y'})^{-1}$ -invariant line (cf. [MZZ21, Theorem 5.1]).

(B) If $\rho(Y) \geq 3$, then Y admits a conic bundle over a toric surface Z which factors as $Y \xrightarrow{\varphi} Y' \xrightarrow{p_0} Z$ where φ is a (toric) blow-up of a (not necessarily Fano) splitting $\mathbb{P}^1$ -bundle Y' over Z along disjoint curves which are intersections of $(g|_{Y'})^{-1}$ -invariant prime divisors (cf. [MZZ21, Theorems 6.1 and 8.1]).

Remark 6.1.2. Note that, when X is a smooth Fano variety, the surjective endomorphism f on X being amplified is equivalent to being int-amplified (cf. [MZZ21, Lemma 2.6]).

In this chapter, we shall give a positive answer to Question 4.1.2 for smooth Fano fourfolds admitting conic bundles (cf. Notation 6.2.1 (10)).

Theorem 6.1.3 and Corollary 6.1.4 below are our main results of this chapter.

Theorem 6.1.3 ([JZ20, Theorem 1.4]). *Let X be a smooth Fano fourfold admitting a conic bundle. Then the following assertions are equivalent.*

- (1) X is a toric variety.
- (2) X admits a polarized endomorphism.
- (3) X admits an (int-)amplified endomorphism.

In particular, X is rational if one of the above equivalent conditions holds.

Corollary 6.1.4 below is a generalization of [MZZ21, Theorem 6.4], and it follows from Theorem 6.1.3. In contrast, [MZZ21, Theorem 6.4] is the main ingredient in the proof of Theorem 6.1.1 (cf. [MZZ21, Proof of Theorem 8.1]).

Corollary 6.1.4 ([JZ20, Corollary 1.5]). *Let $f: X \rightarrow X$ be an (int-)amplified endomorphism of a smooth Fano fourfold. Suppose that X admits a conic bundle $\tau: X \rightarrow Y$ which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ where π is a composition of blow-ups along disjoint smooth projective surfaces and τ_0 is an elementary conic bundle (cf. Notation 2.1 (11 c)). Then W is a splitting $\mathbb{P}^1$ -bundle over Y (cf. Notation 6.2.1 (11)).*

We briefly explain the strategy for the proof of Theorem 6.1.3 (3) $\Rightarrow$ (1). Let $\tau: X \rightarrow Y$ be a conic bundle. By Theorem 6.5.1, τ factors through an elementary conic bundle $\tau_0: W \rightarrow Y$. If one of the following holds: (1) $\rho(X) - \rho(Y) \neq 2$; (2) $Y \cong \mathbb{P}^3$ or the blow-up of a line on $\mathbb{P}^3$; or (3) Y is a $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$, $\mathbb{F}_0$ or $\mathbb{F}_1$ (cf. Notation 6.2.1 (6)), then W is a splitting $\mathbb{P}^1$ -bundle. Otherwise, after replacing τ by a new conic bundle $\hat{\tau}$ if necessary, our new $\hat{\tau}_0$ is a splitting $\mathbb{P}^1$ -bundle (cf. Theorem 6.8.1). Choosing a suitable equivariant minimal model program for τ , one can verify that X is a toric blow-up of a splitting $\mathbb{P}^1$ -bundle over Y ; hence X is toric (cf. [MZZ21, Proposition 2.9] and Theorem 6.9.1).

This chapter is based on the joint work with Jia Jia [JZ20].

6.2 Preliminaries

Notation 6.2.1. Let X be a projective variety. We use the following notation throughout this chapter (cf. Notation 5.2.1).

- 1) The symbols $\sim$ (resp. $\equiv$) denote the *linear equivalence* (resp. *numerical equivalence*) on $\mathbb{Q}$ - (or $\mathbb{R}$ -) Cartier divisors.
- 2) Let $N^1(X)$ be the space of $\mathbb{R}$ -Cartier divisors modulo numerical equivalence $\equiv$, and $\rho(X) := \dim_{\mathbb{R}} N^1(X)$ the *Picard number* of X . Let $N_1(X)$ be the dual space of $N^1(X)$ consisting of 1-cycles, $\text{Amp}(X)$ the cone of *ample divisors* in $N^1(X)$ and $\overline{NE}(X)$ the *Mori cone* of pseudo-effective 1-cycles in $N_1(X)$.
- 3) Let $f: X \rightarrow X$ be a surjective endomorphism. A subset $D \subseteq X$ is *f^{-1} -invariant* (resp. *f^{-1} -periodic*) if $f^{-1}(D) = D$ (resp. $f^{-s}(D) = D$ for some $s \geq 1$).
- 4) A normal projective variety X is of *Fano type*, if there is an effective Weil $\mathbb{Q}$ -divisor Δ on X such that the pair (X, Δ) has at worst klt singularities and $-(K_X + \Delta)$ is an ample $\mathbb{Q}$ -Cartier divisor. If $\Delta = 0$, we say that X is a (*klt*) *Fano variety*.
- 5) A smooth Fano surface is usually called a *del Pezzo surface*. The *degree* of a del Pezzo surface is defined as the self-intersection number of its canonical divisor.

- 6) Denote by $\mathbb{F}_d := \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(-d))$ the Hirzebruch surface of degree d with $d \in \mathbb{Z}_{\geq 0}$.
- 7) A smooth Fano threefold is *imprimitive* if it is isomorphic to the blow-up of another smooth Fano threefold along a smooth irreducible curve (cf. [MM83, Definition 1.3]).
- 8) A normal projective variety X of dimension n is a *toric variety* if X contains a *big torus* $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on X .
- 9) Let $\pi: X \rightarrow W$ be the blow-up of a smooth toric variety W along a smooth closed subvariety S . We say that π is a *toric blow-up* if there exists some big torus T acting on W with $T(S) = S$. In this case, X is still toric.
- 10) A fibration $\tau: X \rightarrow Y$ of smooth projective varieties is a *(regular) conic bundle* if every fibre of τ is a conic, i.e., it is isomorphic as a scheme to the zeros of a nontrivial section $\mathcal{O}_{\mathbb{P}^2}(2)$. If X is further assumed to be Fano, then τ is a *Fano conic bundle*.
 - (a) Denote by $\Delta_\tau := \{y \in Y \mid \tau \text{ is not smooth over } y\}$ the *discriminant* of τ , which is a reduced divisor on Y (cf. [Sar83, Proposition 1.8 and Corollary 1.9]).
 - (b) There is a double cover $\sigma: \widetilde{\Delta}_\tau \rightarrow \Delta_\tau$, which is étale over the regular locus of Δ_τ (cf. [Bea77, Proposition 1.5] or [Sar83, §1.17]).
 - (c) A conic bundle $\tau: X \rightarrow Y$ is said to be *elementary*, if $\rho(X) = \rho(Y) + 1$; in this case, the double cover σ in (b) is nontrivial over each irreducible component $D_i \subseteq \Delta_\tau$; hence $\sigma^{-1}(D_i)$'s are all connected (cf. [Sar83, §1.17]).
 - (d) Let $\mathcal{E}$ be a locally free sheaf of rank 3 on a smooth projective variety Y , and $\pi: \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ the standard projection. An irreducible reduced (possibly singular) divisor V such that the generic fibre of $\pi|_V$ is a smooth rational curve is called an *embedded conic* over Y (cf. [Sar83, Definition 1.1 and §1.5]).
- 11) A fibration $\tau: X \rightarrow Y$ is a (smooth) $\mathbb{P}^1$ -*bundle*, if τ is a conic bundle and $\Delta_\tau = \emptyset$. A fibration $\tau: X \rightarrow Y$ is an *algebraic $\mathbb{P}^1$ -bundle*, if $X \cong \mathbb{P}_Y(\mathcal{E})$ for some locally

free sheaf $\mathcal{E}$ of rank 2 on Y . An algebraic $\mathbb{P}^1$ -bundle $X := \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ is a *splitting $\mathbb{P}^1$ -bundle* if $\mathcal{E}$ is a direct sum of two invertible sheaves.

- 12) For a prime divisor D on X , the inclusion $i: D \hookrightarrow X$ induces a push-forward of 1-cycles $i_*: N_1(D) \rightarrow N_1(X)$ with the image a linear subspace. The *Lefschetz defect* δ_X is defined as $\delta_X := \max\{\text{codim}_{N_1(X)} i_*(N_1(D)) \mid D \text{ a prime divisor}\}$ (cf. [Cas12]).
- 13) Fix a locally free sheaf $\mathcal{E}$ of rank 2 on a smooth Fano threefold Y . For every saturated subsheaf $\mathcal{F} \subseteq \mathcal{E}$, i.e., $\mathcal{E}/\mathcal{F}$ is torsion free (so $\mathcal{F}$ is reflexive and thus locally free), denote by $\xi_{\mathcal{F}} := 2c_1(\mathcal{F}) - c_1(\mathcal{E})$ and denote by

$$W_{\mathcal{E}}(\mathcal{F}) := \{H^2 \mid H \in \text{Amp}(Y), H^2 \cdot \xi_{\mathcal{F}} = 0\} \subseteq P(Y) := \{H^2 \mid H \in \text{Amp}(Y)\}.$$

the *generalized wall* of $P(Y) \subseteq N_1(Y)$ with respect to $\mathcal{E}$ (cf. [GT17, Section 6]).

In the rest of this section, we gather several results to be used frequently in the subsequent sections. We begin with Lemmas 6.2.2 ~ 6.2.5, which are on endomorphisms.

Lemma 6.2.2. *Let $f: X \rightarrow X$ be a surjective endomorphism on a normal projective variety. Then any finite sequence of a minimal model program starting from X , is f -equivariant (after iteration), if either of the following conditions is satisfied.*

- (1) *The (closed) Mori cone $\overline{\text{NE}}(X)$ has only finitely many extremal rays (this holds when X is of Fano type) (cf. [KM98, Theorem 3.7], [Zha10, Lemma 2.11]).*
- (2) *X admits an int-amplified endomorphism (cf. [MZ20a, Theorem 1.1]).*

Lemma 6.2.3 (cf. [Men20, Theorem 1.1]). *Let f be a surjective endomorphism of a projective variety X . Then the following assertions are equivalent.*

- (1) *f is int-amplified, i.e., $f^*L - L$ is ample for some ample Cartier divisor L .*
- (2) *All the eigenvalues of $f^*|_{N^1(X)}$ are of modulus greater than 1.*

Lemma 6.2.4. *Let $f: X \rightarrow X$ be an int-amplified endomorphism on the product space $X := X_1 \times X_2$ such that f splits into $f_1 \times f_2$ after iteration (this is the case when both X_i are of Fano type). Then every f^{-1} -periodic prime divisor D is of the form $D_1 \times X_2$ or $X_1 \times D_2$ where D_i is some prime divisor on X_i for $i = 1, 2$.*

Proof. Let R_f and R_{f_i} be the ramification divisors of f and f_i , respectively. After iteration, we assume $f^*D = qD$ for some $q > 1$ (cf. Lemma 6.2.3); thus D is a component of R_f . Since f splits, we have $R_f = p_1^*R_{f_1} + p_2^*R_{f_2}$ with p_i being the natural projections. Hence, D is of the form $D_1 \times X_2$ or $X_1 \times D_2$ where D_i is some prime divisor on X_i . $\square$

Lemma 6.2.5 (cf. [BH14, Theorem 1.4], [MZZ21, Theorem 2.11]=Theorem 5.2.11). *Let $f: X \rightarrow X$ be an int-amplified endomorphism on a normal projective variety. Let Δ be an f^{-1} -invariant reduced divisor such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Then*

- (1) (X, Δ) has at worst log canonical singularities, and
- (2) $-(K_X + \Delta)$ is effective.

The following Lemma 6.2.6 was first proved in [And85] and reformulated in [Wis91, Theorem 1.2] (cf. [Rom19b, Theorem 1.2 and Remark 2.4]).

Lemma 6.2.6 ([And85, Theorems 2.3 and 3.1]). *Let $\chi: X \rightarrow X'$ be a contraction of a K_X -negative extremal ray of a smooth projective variety X . If every fibre of χ is of dimension ≤ 1 , then X' is smooth and either*

- (1) χ is a conic bundle, or
- (2) χ is a blow-up of X' along a smooth subvariety of codimension 2.

Lemma 6.2.7 (cf. [Cas12, Theorem 1.1]). *Let X be a smooth Fano fourfold and δ_X its Lefschetz defect. If $\delta_X \geq 4$, then $X \cong S_1 \times S_2$ with S_i being del Pezzo surfaces. If $\delta_X = 3$, then there exists a conic bundle $X \rightarrow Y$ such that $\rho(X) - \rho(Y) = 3$.*

Proof. The first assertion is a direct consequence from [Cas12, Theorem 1.1]. For the second assertion, see [Cas12, Step 3.3.15 of Proof of Proposition 3.3.1]. $\square$

Lemma 6.2.8 (cf. [Wis91, pp. 156, Corollary]). *Let $\tau: X \rightarrow Y$ be a conic bundle. If X is smooth Fano and either $\dim(Y) \leq 3$ or $\rho(Y) = 1$, then Y is also smooth Fano.*

Lemma 6.2.9 below characterizes the behaviour of the non-elementary Fano conic bundles. We refer readers to Section 6.5 for descriptions with a dynamical assumption.

Lemma 6.2.9 (cf. [Rom19a, Theorem 4.2]). *Let X be a smooth Fano fourfold and $\tau: X \rightarrow Y$ a conic bundle which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ where π is a composition of blow-ups along disjoint smooth projective surfaces and τ_0 is an elementary conic bundle.*

- (1) *If $\rho(X) - \rho(Y) \geq 4$, then $X \cong S_1 \times S_2$, where S_i are del Pezzo surfaces.*
- (2) *If $\rho(X) - \rho(Y) = 3$, then τ_0 is a smooth $\mathbb{P}^1$ -bundle, and there exists a smooth $\mathbb{P}^1$ -bundle $p: Y \rightarrow Z$ with Z being a del Pezzo surface.*
- (3) *If $\rho(X) - \rho(Y) = 2$, and τ_0 is singular, then there exists a smooth $\mathbb{P}^1$ -bundle $p: Y \rightarrow Z$ with Z being a del Pezzo surface.*
- (4) *If $X \cong S_1 \times S_2$ with S_i being del Pezzo surfaces (this is the case when (1) occurs), then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

Proof. All the assertions follow from [Rom19a, Theorem 4.2] and note that (4) is a consequence of [Rom19a, Proof of Theorem 4.2 (1)]. $\square$

Lemma 6.2.10 below is well known and we rewrite it here for readers' convenience.

Lemma 6.2.10. *Let $W \rightarrow Z$ be a flat morphism of algebraic varieties. Let $C \subseteq Z$ be a closed subscheme and D the inverse image of C in W . Let W' be the blow-up of W along D and Z' the blow-up of Z along C . Then $W' \cong W \times_Z Z'$.*

In what follows, we prove two lemmas that naturally complement Notation 6.2.1 (10). Lemma 6.2.11 was first proved in [Sar83, Proposition 1.16]. It shows, at each singular point of the discriminant of a conic bundle, there are precisely two components intersecting transversally. We give an alternative proof here for readers' convenience.

Lemma 6.2.11. *Suppose that the discriminant locus Δ_τ of a conic bundle $\tau: X \rightarrow Y$ contains three distinct irreducible components $D_i (i = 1, 2, 3)$. Then $D_1 \cap D_2 \cap D_3 = \emptyset$.*

Proof. Suppose on the contrary that there exists $v \in D_1 \cap D_2 \cap D_3 \subseteq \text{Sing } \Delta_\tau$. Choose an affine open neighbourhood $U \subseteq Y$ of v with local coordinates y_i such that one can write the local equation of $X_U := \tau^{-1}(U) \subseteq U \times \mathbb{P}_{x_0, x_1, x_2}^2$ in the form $Q_U :=$

$\sum_{0 \leq i, j \leq 2} a_{ij} x_i x_j = 0$, where x_i are the coordinates in $\mathbb{P}^2$ and $a_{ij} \in \mathcal{O}_Y(U)$. Locally, Δ_τ is given by the determinant equation $\det Q_U = 0$. Since Δ_τ has at least three components near v , we have $\det Q_U \in \mathfrak{m}_v^3$. It follows that $\text{rk}(\partial^2 \det Q_U / \partial y_i \partial y_j)_v = 0$, contradicting [Sar83, Proposition 1.8 5.c]. $\square$

Lemma 6.2.12. *Let $\tau: X \rightarrow Y$ be a conic bundle, and $\pi: Y' \rightarrow Y$ a morphism such that $\pi(Y') \not\subseteq \Delta_\tau$. Then the morphism $\tau': X' \rightarrow Y'$ by the base change is an embedded conic. In particular, if $Y' \subseteq Y$ is a smooth closed subvariety not contained in Δ_τ , and $\Delta_{\tau'} := \Delta_\tau|_{Y'}$ is a reduced divisor with simple normal crossings on Y' , then τ' is also a conic bundle.*

Proof. Since τ is a conic bundle, $\mathcal{E} := \tau_* \mathcal{O}(-K_X)$ is locally free of rank 3 on Y (cf. [Har77, Chapter III, Corollary 12.9]). Let $X' := X \times_Y Y'$, $\tilde{X} := \mathbb{P}_Y(\mathcal{E})$ and $\tilde{X}' := \mathbb{P}_{Y'}(\pi^* \mathcal{E})$.

$$\begin{array}{ccc}
 \tilde{X}' & \xrightarrow{q} & \tilde{X} \\
 \downarrow i' & & \downarrow i \\
 X' & \xrightarrow{p} & X \\
 \downarrow \tau' & & \downarrow \tau \\
 Y' & \xrightarrow{\pi} & Y
 \end{array}
 \quad \begin{array}{c}
 \varphi' \curvearrowright \\
 \varphi \curvearrowright
 \end{array}$$

Clearly, the generic fibre of τ' is an irreducible rational curve by the base change. By [Sar83, §1.5], X embeds into $\tilde{X}$ and $\tau = \varphi \circ i$. Then $\tilde{X}' \cong \tilde{X} \times_Y Y'$ with φ' and q being the natural projections. Since $\pi \circ \tau' = \varphi \circ i \circ p$, our τ' factors through φ' by the universal property and we get the morphism i' . Then i' is an embedding since $X' \cong X \times_{\tilde{X}} \tilde{X}'$; hence τ' is an embedded conic. The second part follows from the first part, τ being a (regular) conic bundle and [Sar83, Corollary 1.11]. $\square$

At the end of this section, we recall the following lemma, which works in the proof of Lemma 6.6.6 under dynamical assumptions. Note that the system of walls given by [GT17, Theorem 6.6] yields an obvious stratification of $P(X)$ into connected chambers.

Lemma 6.2.13 (cf. [GT17, Proposition 6.5 and Theorem 6.6]). *There is a homeomorphism from $\text{Amp}(Y)$ to $P(Y)$ (cf. Notation 6.2.1 (13)). The set of walls $\{W_{\mathcal{E}}(\mathcal{F})\}_{\mathcal{F}}$ is*

locally finite in $P(Y)$, i.e., there are only finitely many walls $W_{\mathcal{E}}(\mathcal{F})$ in each compact set $K \subseteq P(Y)$.

6.3 Totally periodic subvarieties on Fano threefolds

In this section, we shall study [MZZ21, Question 1.8] for Fano threefolds. The main results are Theorem 6.3.1 (confirming the divisor case) and Proposition 6.3.4.

Theorem 6.3.1. *Let Y be a smooth Fano threefold admitting an int-amplified endomorphism g . Then Y is toric and there is a toric pair (Y, Δ) such that Δ contains the union Σ of all the g^{-1} -periodic prime divisors. Further, every g^{-1} -periodic prime divisor is a smooth rational surface.*

Proof. By [MZ20a, Corollary 3.8], there are only finitely many g^{-1} -periodic subvarieties; hence we may assume that they are all g^{-1} -invariant, after iteration. Further, (Y, Σ) is lc and $-(K_Y + \Sigma)$ is effective (cf. Lemma 6.2.5). By Theorem 6.1.1, Y is toric. We shall treat all cases in Theorem 6.1.1 and replace g by a power if necessary (cf. Lemma 6.2.2).

Case (1): $Y \cong \mathbb{P}^3$. Then Σ is a union of at most four planes (cf. [Hör17, Corollary 1.2], [NZ10, Theorem 1.5 (5)] and Lemma 6.2.5). Since (Y, Σ) is lc, the boundary Σ is a simple normal crossing divisor (cf. [KM98, Lemma 2.29]); thus Σ is contained in a tetrahedron Δ (looking like $\diamond$) in $\mathbb{P}^3$. Clearly, (Y, Δ) is a toric pair.

Case (2): $\pi: Y \rightarrow Z \cong \mathbb{P}^3$ is a (toric) blow-up along an $(h := g|_Z)^{-1}$ -invariant line L after iteration. Then, Σ consists of the π -exceptional divisor E and the π -strict transform of the h^{-1} -invariant divisor $\Sigma_Z := \pi_*\Sigma$ on Z (cf. [CMZ20, Lemma 7.5]).

Claim 6.3.2. There exists a reduced divisor Δ_Z containing Σ_Z such that (Z, Δ_Z) is a toric pair and L is contained in the intersection of toric boundary components of (Z, Δ_Z) .

Proof of Claim 6.3.2. If Σ_Z has at most one component, then we can choose a suitable Δ_Z satisfying the condition of our claim. Assume $D_1, D_2, \dots$ are irreducible components of Σ_Z and denote $\ell_{ij} := D_i \cap D_j$. If $L = \ell_{ij}$, we are done (cf. **Case (1)**); hence we assume this is not the case. If $L \not\subset D_i$, $L \not\subset D_j$ but $L \cap \ell_{ij} \neq \emptyset$, then the three g^{-1} -invariant

prime divisors E, π^*D_i and π^*D_j intersect along the curve $\pi^{-1}(L \cap \ell_{ij})$, contradicting Lemma 6.2.5. Hence, for each ℓ_{ij} , either $L \cap \ell_{ij} = \emptyset$, or L is contained in D_i or D_j .

If Σ_Z consists of two components, then we can choose a suitable Δ_Z satisfying the condition. Note that there are at most two $(g|_L)^{-1}$ -invariant points in L by Lemma 6.2.5.

If Σ_Z has at least three components, then L must lie in one of them; otherwise, L intersects them at three distinct points by assumption, which are $(g|_L)^{-1}$ -invariant after iteration, a contradiction. Assume $L \subseteq D_1$. Then $L \cap \ell_{1i} \cap \ell_{1j} = \emptyset$ by Lemma 6.2.5 since they are $(g|_{D_1})^{-1}$ -invariant. Hence if Σ_Z has three components, we may choose another plane $D \supseteq L$ such that $\Delta_Z = \Sigma_Z + D$. Finally, if Σ_Z has four components, then L intersects ℓ_{1i} ($i = 2, 3, 4$) at three distinct points, a contradiction to Lemma 6.2.5 again.

We come back to the proof of Case (2). By Claim 6.3.2, $(Y, \Delta := \pi_*^{-1}\Delta_Z + E)$ is a toric pair. For $E \neq D \subseteq \Delta$, if $L \subseteq \pi(D)$, then $D = \pi_*^{-1}(\pi(D)) \cong \mathbb{P}^2$ by **Case (1)**; if $L \not\subseteq \pi(D)$, then $D = \pi_*^{-1}(\pi(D)) \cong \mathbb{F}_1$. On the other hand, Y admits another Fano contraction to $L' \cong \mathbb{P}^1$ along which, E dominates L' . So E admits two (distinct) rulings and hence $E \cong \mathbb{F}_0$. As a result, each component of Σ_Z is smooth rational.

Case (3): $\pi: Y \rightarrow Y'$ is a (toric) blow-up along disjoint curves C_i which are intersections of $(g|_{Y'})^{-1}$ -invariant prime divisors, and $Y' \rightarrow Z$ is a splitting $\mathbb{P}^1$ -bundle over a smooth toric surface. By [MZZ21, Theorem 3.3], there is a reduced divisor $\Delta_{Y'}$ containing all the $(g|_{Y'})^{-1}$ -periodic prime divisors such that $(Y', \Delta_{Y'})$ is a (smooth) toric pair. Let $\sum E_i$ be the sum of π -exceptional divisors. Then $(Y, \Delta_Y := \sum E_i + \pi_*^{-1}\Delta_{Y'})$ is a toric pair. Note that Y is a conic bundle over Z (cf. Theorem 6.1.1). So every $(g^{-1}$ -invariant) π -exceptional divisor E_i is a $\mathbb{P}^1$ -bundle over $C_i \cong \mathbb{P}^1$ (cf. [MM83, Proposition 6.3], [CMZ20, Lemma 7.5] and [MZZ21, Corollary 3.4]). Then every g^{-1} -invariant prime divisor ($\neq E_i$) is a smooth blow-up of a component of $\Delta_{Y'}$, and thus smooth rational. $\square$

As an application of Theorem 6.3.1, the following corollary generalizes [MZZ21, Theorem 3.3] to higher dimensional cases. The proof is the same as [MZZ21, Theorem 3.3] after replacing [MZZ21, Theorem 3.2] by Theorem 6.3.1.

Corollary 6.3.3. *Let $\tau: X \rightarrow Y$ be a splitting $\mathbb{P}^1$ -bundle over a smooth Fano threefold*

Y. Suppose that X admits an int-amplified endomorphism f . Then X is toric and there is a toric pair (X, Δ) such that Δ contains the union of all the f^{-1} -periodic prime divisors.

At the end of this section, we show the following result, which takes the first step towards the understanding of the totally periodic curves of an int-amplified endomorphism on Fano threefolds. So far, we are only able to deal with the case when Y admits a conic bundle. It is conjectured that every totally periodic curve on $\mathbb{P}^3$ is linear (cf. e.g., [MZZ21, Conjecture 1.9]).

Proposition 6.3.4. *Let Y be a smooth Fano threefold admitting a conic bundle and an int-amplified endomorphism g . Then every g^{-1} -periodic curve is smooth rational.*

Proof. Let C be a g^{-1} -periodic curve. Let $Y \rightarrow Z$ be the conic bundle, which factors as $Y \xrightarrow{\varphi} Y' \xrightarrow{p_0} Z$ such that φ is the blow-up of W along a disjoint union of some $(g|_{Y'})^{-1}$ -periodic smooth curves, and p_0 is an algebraic $\mathbb{P}^1$ -bundle over a smooth rational surface Z (cf. [MZZ21, Theorem 6.2]). By Lemma 6.2.2, we may assume that both φ and p_0 are g -equivariant and C is g^{-1} -invariant after iteration.

Suppose that C is contained in some $(g^{-1}$ -invariant) φ -exceptional prime divisor E . Then C is a $(g|_E)^{-1}$ -invariant curve on the smooth rational surface E (cf. Theorem 6.3.1); thus C is a smooth rational curve by [MZZ21, Corollary 3.4].

Suppose that C is not contained in the exceptional locus of φ . Then C is birational to its image $C' := \varphi(C)$ on Y' , which is $(g|_{Y'})^{-1}$ -invariant (after iteration) by [CMZ20, Lemma 7.5]. If p_0 contracts C' , then $C' \cong \mathbb{P}^1$. If p_0 does not contract C' , then $p_0(C')$ is $(g|_Z)^{-1}$ -periodic and hence a smooth rational curve (cf. [CMZ20, Lemma 7.5] and [MZZ21, Corollary 3.4]). After iteration, C' is contained in the g^{-1} -invariant smooth rational surface $F' := p_0^{-1}(p_0(C'))$. Then C' , as a $(g|_{F'})^{-1}$ -invariant curve on F' , is also smooth rational by [MZZ21, Corollary 3.4]. In both cases, $\varphi|_C$ is an isomorphism. Indeed, taking the normalization $\tilde{C} \rightarrow C \rightarrow C'$, the composition is birational and thus an isomorphism, which forces $\tilde{C} \cong C$. So C is smooth rational as desired. $\square$

6.4 Elementary Fano conic bundles

The whole section is devoted to proving the smoothness of an elementary Fano conic bundle with the dynamical assumption. To be more precise, we shall prove the following:

Theorem 6.4.1. *Let $\tau: X \rightarrow Y$ be an elementary Fano conic bundle from a smooth fourfold X . Suppose that X admits an int-amplified endomorphism f . Then $\Delta_\tau = \emptyset$, i.e., τ is a smooth (and hence an algebraic) $\mathbb{P}^1$ -bundle.*

Notation 6.4.2. We will assume and use the following notation throughout this section.

- (1) X is a smooth Fano fourfold, and $\tau: X \rightarrow Y$ is an elementary conic bundle; hence Y is a smooth Fano threefold (cf. Lemma 6.2.8).
- (2) $f: X \rightarrow X$ is int-amplified and it descends to an int-amplified endomorphism $g := f|_Y$ on Y after iteration (cf. Lemmas 6.2.2 and 6.2.3).
- (3) We use $E_\bullet$ to denote the exceptional locus $\text{Exc}(\bullet)$ for simplicity.

Before proving Theorem 6.4.1, we prepare two lemmas (cf. Lemmas 6.4.3 and 6.4.4).

Lemma 6.4.3 (cf. [Cas08, Theorem 3.14]). *Let $p: Y \rightarrow Z$ be a $\mathbb{P}^1$ -bundle over a smooth projective surface Z and $\phi: Z \rightarrow Z'$ the blow-up of a point Q on a smooth projective surface Z' . Then Z' is Fano. Moreover, one has the following commutative diagram*

$$\begin{array}{ccc} Y & \xrightarrow{\varphi} & Y' \\ p \downarrow & & \downarrow p' \\ Z & \xrightarrow{\phi} & Z' \end{array}$$

where φ is the blow-up of a smooth Fano threefold Y' along a smooth rational curve, the exceptional divisor $E_\varphi = p^*E_\phi$, and p' is a Fano contraction and further a $\mathbb{P}^1$ -bundle.

Proof. First, Z is Fano (cf. Lemma 6.2.8) and thus Z' is also Fano by the ramification divisor formula of ϕ . Let R_ϕ (resp. R_p) be the extremal ray of $\overline{\text{NE}}(Z)$ (resp. $\overline{\text{NE}}(Y)$) contracted by ϕ (resp. p). Given that Y is Fano and thus $\overline{\text{NE}}(Y)$ is polyhedral, there is an extremal ray R_φ of $\overline{\text{NE}}(Y)$ such that $p_*R_\varphi = R_\phi$ and $R_\varphi \cap R_p = \{0\}$. Denote by

$\varphi: Y \rightarrow Y'$ the (K_Y -negative) contraction of R_φ . By the rigidity lemma (cf. [Deb01, Lemma 1.15]), $\phi \circ p$ factors through φ and we get the above commutative diagram.

By [Cas08, Theorem 3.14 (v)], φ is the blow-up of a smooth Fano threefold Y' along a smooth rational curve with $E_\varphi = p^*E_\phi$. By Theorem 6.3.1, E_φ being g^{-1} -invariant is smooth rational. Since the fibre of p' over Q is $\varphi(E_\varphi) \cong \mathbb{P}^1$ and every other fibre of p' (which is isomorphic to the corresponding fibre of p) is $\mathbb{P}^1$, our p' is a $\mathbb{P}^1$ -bundle. $\square$

Lemma 6.4.4. *Suppose Y is imprimitive. Then one has the commutative diagram*

$$\begin{array}{ccc} X & \xrightarrow{\chi} & X' \\ \tau \downarrow & & \downarrow \tau' \\ Y & \xrightarrow{\varphi} & Y' \end{array}$$

such that the following assertions hold.

- (1) φ is the blow-up of a smooth Fano threefold Y' along a smooth curve C .
- (2) χ is the blow-up of a smooth Fano fourfold X' along a smooth projective surface.
- (3) The exceptional divisor $E_\chi = \tau^*E_\varphi$.
- (4) τ' is an elementary Fano conic bundle and $X \cong X' \times_{Y'} Y$. In particular, if τ' is a smooth $\mathbb{P}^1$ -bundle, then so is τ .
- (5) The above commutative diagram is f -equivariant after replacing f by a power.

Proof. (1) follows from the imprimitivity of Y . Let R_φ be the K_Y -negative extremal ray contracted by φ . Since $\overline{\text{NE}}(X)$ is rational polyhedral and τ is a K_X -negative contraction of an extremal ray R_τ , there is an extremal ray R_χ of $\overline{\text{NE}}(X)$ such that $\tau_*R_\chi = R_\varphi$ and $R_\chi \cap R_\tau = \{0\}$, noting that faces of $\overline{\text{NE}}(Y)$ are in bijection with the faces of $\overline{\text{NE}}(X)$ containing R_τ . Since X is Fano, there exists a K_X -negative contraction $\chi: X \rightarrow X'$ of R_χ . By the rigidity lemma, $\varphi \circ \tau$ factors through χ . So (5) follows from Lemma 6.2.2.

We claim that all the fibres of χ have dimension ≤ 1 . Indeed, if there exists a fibre component F_0 of χ with $\dim(F_0) \geq 2$, then $\varphi(\tau(F_0))$ is a point. Since $R_\chi \cap R_\tau = \{0\}$, the restriction $\tau|_{F_0}$ is finite; thus $\tau(F_0)$ cannot be contracted to a point along φ by (1).

Hence our claim holds. Since $\tau_* R_\chi = R_\varphi$, we have $\tau(E_\chi) \subseteq E_\varphi$; then χ is birational with $E_\chi \subseteq \tau^* E_\varphi$. By Lemma 6.2.6, χ is the blow-up along a smooth projective surface. Since $\tau^* E_\varphi$ is irreducible (cf. [KM98, Theorem 3.7]), (3) is proved.

Next we show that X' is Fano. Suppose on the contrary. Then it follows from [Wis91, Proposition 3.4] that there exists an extremal ray $R_1 \neq R_\chi$ of $\overline{NE}(X)$ such that $E_\chi \cdot R_1 < 0$; thus the locus of R_1 is contained in E_χ . By (3), R_1 is not contracted by τ . Denote by $\chi_1: X \rightarrow X_1$ the $(K_X$ -negative) contraction of R_1 . **We claim that all fibres of χ_1 have dimension ≤ 1 .** Indeed, if there exists a fibre component F_0 of χ_1 such that $\dim(F_0) \geq 2$, then $E_\chi \cong \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ (cf. [Wis91, Proof of Proposition 3.6 (ii)]). As a result, $\chi_1|_{E_\chi}$ is a Fano contraction onto $\mathbb{F}_0$, a contradiction to $\dim(F_0) \geq 2$. So our claim holds. Then, it follows from Lemma 6.2.6 that χ_1 is the blow-up of smooth X_1 along a smooth surface with $E_{\chi_1} = E_\chi$ and $E_\chi \rightarrow \chi_1(E_\chi)$ is a smooth $\mathbb{P}^1$ -bundle. Since the ruling $\chi_1|_{E_\chi}$ gives rise to a ruling on E_φ different from $\varphi|_{E_\varphi}$, we see that $E_\varphi \cong \mathbb{F}_0$. Moreover, for a fibre ℓ_1 of χ_1 , by the projection formula, we have $E_\varphi \cdot \tau(\ell_1) = E_\chi \cdot \ell_1 = -1$. Therefore, $E_\varphi|_{E_\varphi} \cong \mathcal{O}_{E_\varphi}(-1, -1)$ and thus Y' is not Fano (cf. [MM83, Lemma 4.4]), a contradiction to (1). So our assumption is absurd and (2) is proved.

Note that τ' is an elementary Fano contraction. For any point $y' \in Y' \setminus C$, we have $X'_{y'} \cong X_{\varphi^{-1}(y')}$ and hence $\dim(X'_{y'}) = 1$. On the other hand, $\chi(E_\chi) = \tau'^{-1}(C) \rightarrow C$ is flat (onto a smooth curve) and hence also has fibre dimension one. By Lemma 6.2.6, τ' is an elementary (flat) conic bundle. So (4) follows from Lemma 6.2.10. $\square$

Now we begin to prove Theorem 6.4.1. In view of Lemmas 6.4.3, 6.4.4 and Theorem 6.1.1, to prove Theorem 6.4.1, it suffices to focus on the following three cases: (1) $Y \cong \mathbb{P}^3$; (2) $p: Y \rightarrow Z \cong \mathbb{P}^2$ is a splitting $\mathbb{P}^1$ -bundle; and (3) $p: Y \rightarrow Z \cong \mathbb{F}_0$ is a splitting $\mathbb{P}^1$ -bundle.

Suppose on the contrary that $\Delta_\tau \neq \emptyset$. By [CMZ20, Lemma 7.4], $g^{-1}(\Delta_\tau) = \Delta_\tau$. After iteration, we may assume that each component D_i of Δ_τ is g^{-1} -invariant. By Theorem 6.3.1, Δ_τ is contained in some toric boundary of Y and thus Δ_τ has simple normal crossings.

Theorem 6.4.5. *Suppose that $Y \cong \mathbb{P}^3$. Then $\Delta_\tau = \emptyset$.*

Proof. By [Hör17, Corollary 1.2], each component $D_i \cong \mathbb{P}^2$. By Lemma 6.2.11 and the ampleness of each D_i , we see that Δ_τ has at most two components. However, both $D_i \cong \mathbb{P}^2$ and $D_i \setminus (D_i \cap D_j) \cong \mathbb{P}^2 \setminus \mathbb{P}^1$ are simply connected; thus there is no nontrivial étale cover over them. This contradicts our τ being elementary (cf. Notation 6.2.1 (10)). $\square$

Theorem 6.4.6. *Suppose that $p: Y \rightarrow Z \cong \mathbb{P}^2$ is a splitting $\mathbb{P}^1$ -bundle. Then $\Delta_\tau = \emptyset$.*

Proof. Up to a twist, we may write $Y = \mathbb{P}_Z(\mathcal{F})$ where $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(a)$ with $-2 \leq a \leq 0$ by an easy calculation (cf. [SW90]). There are two possibilities:

Case 1: Δ_τ contains no section of p . Then each component D_i of Δ_τ , being the pull-back of some line on $Z \cong \mathbb{P}^2$, is a g^{-1} -invariant Hirzebruch surface by Theorem 6.3.1. Since $D_i \cong \mathbb{F}_d$ and $D_i \setminus (D_i \cap D_j) \cong \mathbb{A}^1 \times \mathbb{P}^1$ are both simply connected, the double cover $\sigma: \widetilde{\Delta_\tau} \rightarrow \Delta_\tau$ being nontrivial implies that Δ_τ is the pull-back of a union of three $(g|_Z)^{-1}$ -periodic lines with no common intersection (looking like $\bigwedge$ on $Z \cong \mathbb{P}^2$ (cf. Lemma 6.2.5)).

If p is a trivial bundle with $Y \cong Z \times T \cong Z \times \mathbb{P}^1$, then after iteration, $g = g|_Z \times g|_T$ with $g|_T$ being polarized on T (cf. Lemma 6.2.3). So we can pick a $g|_T$ -periodic point $t \in T$ (cf. [Fak03, Theorem 5.1]) and define $S := Z \times \{t\} \cong \mathbb{P}^2$, a g -periodic section of p . If p is nontrivial, then we take S to be the “negative section” of this splitting $\mathbb{P}^1$ -bundle such that $\mathcal{O}_Y(S)|_S = \mathcal{O}_S(a)$, which is g^{-1} -periodic (cf. [MZZ21, Lemma 2.3]). In both cases, $g|_S$ is int-amplified after iteration. Then the base change $\tau_S: X_S := X \times_Y S \rightarrow S$ is proper and flat. Moreover, $S \not\subset \Delta_\tau$ and $\Delta_{\tau_S} = \Delta_\tau|_S$ is a loop consisting of three rational curves on S with simple normal crossings. By Lemma 6.2.12, X_S is smooth and τ_S is a conic bundle. Since $S \cong \mathbb{P}^2$ and $\Delta_{\tau_S} \neq \emptyset$, one has $\rho(X_S/S) = 1$ (cf. [MM83, Corollary 6.4]). On the other hand, X_S is Fano. Indeed, for any curve $C \subseteq X_S$, we have

$$K_{X_S} \cdot C = (K_X + X_S)|_{X_S} \cdot C = K_X \cdot C + X_S \cdot C = K_X \cdot C + (S \cdot \tau_* C)_Y < 0,$$

since $\mathcal{O}_Y(S)|_S \cong \mathcal{O}(a)$ with $a \leq 0$. Since $\overline{\text{NE}}(X_S)$ has only two extremal rays, our $-K_{X_S}$ is ample. So we apply [MZZ21, Theorem 4.1] to get $\Delta_{\tau_S} = \emptyset$, a contradiction.

Case 2: Δ_τ contains at least one section S of p . Then $S \cong \mathbb{P}^2$ and $S \setminus (S \cap D_i) \cong \mathbb{P}^2 \setminus \mathbb{P}^1$ are both simply connected where D_i is a vertical component (if exists). So there exist

at least two components D_1, D_2 of Δ_τ , which are the pull-back of some lines ℓ_1, ℓ_2 on $Z \cong \mathbb{P}^2$. Given that $\ell_1 \cap \ell_2$ is nonempty in $\mathbb{P}^2$, so is $S \cap D_1 \cap D_2$. But this violates Lemma 6.2.11.

So we finish the proof of our theorem. $\square$

Theorem 6.4.7. *Suppose that $p: Y \rightarrow Z \cong \mathbb{F}_0$ is a splitting $\mathbb{P}^1$ -bundle. Then $\Delta_\tau = \emptyset$.*

Proof. According to [SW90], up to a twist, we may write $Y = \mathbb{P}_Z(\mathcal{F})$ where $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(a, b)$ with $(a, b) = (0, 0), (0, -1), (-1, -1)$ or $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(1, -1)$.

Case 1: Δ_τ contains no section of p . Then, similar to Proof of Theorem 6.4.6, Δ_τ is the pull-back of a loop (looking like $\begin{smallmatrix} \uparrow\downarrow \\ \uparrow\downarrow \end{smallmatrix}$ on Z). If $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(1, -1)$, then for a general fibre ℓ of the first projection $Z \rightarrow \mathbb{P}^1$, one has $\mathcal{F}|_\ell \cong \mathcal{O}_\ell \oplus \mathcal{O}_\ell(-1)$. Hence $H := p^*\ell \cong \mathbb{P}_\ell(\mathcal{F}|_\ell) \cong \mathbb{F}_1$, and $\tau_H: \tau^*H \rightarrow H$ is a Fano conic bundle with $\Delta_{\tau_H} = \Delta_\tau|_H$ being two fibres (cf. Lemma 6.2.12), a contradiction to [MM83, Corollary 6.7]. In the remaining cases, $S|_S$ is anti-nef, and then with the same proof as in Theorem 6.4.6, we can deduce a contradiction to [MZZ21, Theorem 4.1].

Case 2: Δ_τ contains at least one section $S \cong \mathbb{F}_0$ of p . After iteration, we may assume that $Y \rightarrow Z := Z_1 \times Z_2 \rightarrow Z_i \cong \mathbb{P}^1$ is g -equivariant (cf. Lemma 6.2.2). Take a general point $z \in Z$ and let $Y_z := p^{-1}(z)$ be its (movable) fibre. By Lemma 6.2.5, we have $\Delta_\tau \cdot Y_z \leq -K_Y \cdot Y_z = 2$; thus Δ_τ contains at most two sections of p .

Since S is simply connected, there exists a component $D_1 = p^*\ell_1 \subseteq \Delta_\tau$, with some curve ℓ_1 on Z (cf. Notation 2.1 (11 c)). By [CMZ20, Lemma 7.5], ℓ_1 is $(g|_Z)^{-1}$ -periodic. So we may assume ℓ_1 is $\{\cdot\} \times \mathbb{P}^1$ (cf. [MZZ21, Lemma 3.1]). Clearly, $D_1 \cong \mathbb{F}_d$ for some $d \geq 0$, and both $S \setminus (S \cap D_1) \cong \mathbb{A}^1 \times \mathbb{P}^1$ and $D_1 \setminus (S \cap D_1) \cong \mathbb{F}_d \setminus (\text{a section of } \mathbb{F}_d)$ are simply connected. On the other hand, by Lemma 6.2.11, any three components of Δ_τ have no common intersection. So the double cover $\sigma: \widetilde{\Delta}_\tau \rightarrow \Delta_\tau$ being nontrivial implies that Δ_τ consists of exact two sections of p , and two disjoint components D_1, D_2 which are pull-backs of some curves ℓ_1, ℓ_2 on Z , respectively. In particular, ℓ_2 is of the form $\{\cdot\} \times \mathbb{P}^1$.

Take a $g|_{Z_2}$ -periodic point $\{z_2\} \in Z_2$ (cf. Lemma 6.2.3 and [Fak03, Theorem 5.1])

and define $L_{z_2} := Z_1 \times \{z_2\}$. Then $Y_{z_2} := p^*L_{z_2}$ is a Hirzebruch surface and $X_{z_2} := (p \circ \tau)^*L_{z_2}$ is a Fano threefold admitting an int-amplified endomorphism after iteration. Since $Y_{z_2} \not\subset \Delta_\tau$ and $\Delta_{\tau_{z_2}} = \Delta_\tau|_{Y_{z_2}}$ is a loop of four rational curves (looking like $\boxplus$ on Y_{z_2} with simple normal crossings, by Lemma 6.2.12, X_{z_2} is smooth and $\tau_{z_2}: X_{z_2} \rightarrow Y_{z_2}$ is a conic bundle. Since $\Delta_{\tau_{z_2}}$ is connected and ample on Y_{z_2} , we have $\rho(X_{z_2}/Y_{z_2}) = 1$ (cf. [MM83, Corollary 6.4]). By [MZZ21, Theorem 4.1], $\Delta_{\tau_{z_2}} = \emptyset$, a contradiction. $\square$

End of Proof of Theorem 6.4.1. It follows from Theorems 6.4.5 ~ 6.4.7 and [MZZ21, Lemma 2.12], noting that Y is rational (cf. [Zha12, Theorem 1.2]). $\square$

6.5 Main reduction for non-elementary conic bundles

In this section, we shall study non-elementary (singular) conic bundles. The main results of this section are Theorem 6.5.1 and Lemma 6.5.2.

Theorem 6.5.1 (Equivariant minimal model for conic bundles). *Let X be a smooth Fano fourfold and $\tau: X \rightarrow Y$ a conic bundle. Suppose X admits an int-amplified endomorphism f . Then, after iteration, there exists an f -equivariant minimal model program*

$$X = X_0 \xrightarrow{\pi_1} X_1 \xrightarrow{\pi_2} X_2 \xrightarrow{\pi_2} \cdots \xrightarrow{\pi_r} X_r =: W \xrightarrow{\tau_0} Y$$

such that the following assertions hold.

- (1) $r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth manifold.
- (2) $\tau_0: W = \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ is an algebraic $\mathbb{P}^1$ -bundle over a smooth Fano threefold.
- (3) There are r disjoint prime divisors $D_1, \dots, D_r$ on Y and r pairs of prime divisors $E_i, \widetilde{E}_i$ on X such that $\tau^*D_i = E_i + \widetilde{E}_i$ and $D_i \cap D_j = \emptyset$ for $i \neq j$.
- (4) τ_0 is smooth and $\Delta_\tau = \bigsqcup_{i=1}^r D_i$ with each D_i a $(g := f|_Y)^{-1}$ -invariant smooth rational connected component of Δ_τ . Further, τ has reduced fibres over $\bigcup_{i=1}^r D_i$.
- (5) The composite $X_i \rightarrow Y$ is a conic bundle with the discriminant $D_{i+1} \cup \cdots \cup D_r$.

(6) The composite $\pi = \pi_r \circ \cdots \circ \pi_1: X \rightarrow W$ is the blow-up of W along r disjoint union of $(f|_W)^{-1}$ -invariant smooth rational surfaces $\bigcup_{i=1}^r \overline{D}_i$ with $\overline{D}_i \cong D_i$.

Proof. First, by Lemma 6.2.8, Y is smooth Fano. By [Rom19a, Propositions 3.4, 3.5], we may run a relative minimal model program $X \rightarrow \cdots \rightarrow X_i \rightarrow \cdots \rightarrow X_r$ of X over Y which is f -equivariant after iteration (cf. Lemma 6.2.2), such that (1) and (3) hold.

We will show the smoothness of τ_0 in the following two paragraphs. By Lemma 6.2.9 and Theorem 6.4.1, we only need to consider the case $r = 1$, i.e., $\pi = \pi_1$ is a single blow-up along a smooth projective surface. Suppose on the contrary that τ_0 is singular, i.e., $\Delta_{\tau_0} \neq \emptyset$. By Lemma 2.9 (3), there exists a (smooth) $\mathbb{P}^1$ -bundle $p: Y \rightarrow Z$ to a smooth rational surface. After iteration, we may assume f descends to an int-amplified endomorphism $f|_Z$ on Z and Δ_τ is g^{-1} -invariant on Y (cf. Lemmas 6.2.2, 6.2.3 and [CMZZ20, Lemma 7.4]). By [MZZ21, Theorem 3.3], each component of Δ_τ is either a section of p or the pull-back of some $(f|_Z)^{-1}$ -periodic (rational) curve on Z (cf. [CMZZ20, Lemma 7.5]).

Suppose that Δ_{τ_0} contains a section S of p . Since $S \cong Z$ is simply connected, the nontrivial double cover $\widetilde{\Delta_{\tau_0}} \rightarrow \Delta_{\tau_0}$ (cf. Notation 6.2.1 (10)) implies that there is another irreducible (vertical) component F of Δ_{τ_0} intersecting S . Then, F is the pull-back of some $(f|_Z)^{-1}$ -periodic (rational) curve on Z . But now there is no D_1 disjoint from Δ_{τ_0} , a contradiction. Hence each component of Δ_τ is a pull-back of some curve C_i on Z . Since C_i 's are $(f|_Z)^{-1}$ -periodic, there is a toric pair (Z, Δ_Z) such that each $C_i \subseteq \Delta_Z$ (cf. [MZZ21, Theorem 3.2]). Clearly, Δ_Z is a simple loop of smooth rational curves. Since D_1 is disjoint from the connected Δ_{τ_0} , there is some C_i such that $p^{-1}(C_i) \subseteq \Delta_{\tau_0}$ and C_i intersects exact one of other C_j 's. However, $p^{-1}(C_i) \setminus (p^{-1}(C_i \cap C_j)) \cong \mathbb{A}^1 \times \mathbb{P}^1$ being simply connected contradicts the existence of the nontrivial double cover (cf. Notation 6.2.1 (10)). So τ_0 is a smooth $\mathbb{P}^1$ -bundle.

Since τ_0 is smooth, by [MZZ21, Lemma 2.12], $W = \mathbb{P}_Y(\mathcal{E})$ for some locally free sheaf $\mathcal{E}$ of rank 2, since Y is rational (cf. [Zha12, Theorem 1.2]). So (2) is proved.

Let $f_i := f|_{X_i}$. Since the exceptional divisor $\text{Exc}(\pi_i)$ is f_{i-1}^{-1} -invariant and the surface $D'_i \subseteq X_i$ blown up by π_i is f_i^{-1} -invariant, its image $\overline{D}_i$ on W is f_r^{-1} -invariant, and its

image D_i on Y is g^{-1} -periodic (cf. [CMZ20, Lemma 7.5]). Hence, Theorem 6.3.1 gives that D_i is rational for all i . Together with (3), (4) $\sim$ (6) are proved. $\square$

From now on till the end of this section, we assume Notation 6.4.2 except that our τ here may not be elementary. We shall prove Lemma 6.5.2 below, which generalizes Lemma 6.4.4.

Lemma 6.5.2. *Let $\tau: X \rightarrow Y$ be a Fano conic bundle, which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ where π is the blow-up along disjoint surfaces and $W = \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ is an algebraic $\mathbb{P}^1$ -bundle. Suppose that Y is imprimitive. Then we have the following commutative diagram*

$$\begin{array}{ccc}
 X & \xrightarrow{\chi} & X' \\
 \downarrow \pi & \searrow \eta & \downarrow \pi' \\
 W & \xrightarrow{\psi} & W' \\
 \downarrow \tau_0 & \searrow \tau'_0 & \downarrow \\
 Y & \xrightarrow{\varphi} & Y'
 \end{array}
 \begin{array}{c}
 \tau \swarrow \quad \searrow \tau' \\
 \end{array}$$

such that the following hold.

- (1) φ is the blow-up of a smooth Fano threefold Y' along a smooth curve C .
- (2) χ is the blow-up of a smooth fourfold X' along a smooth projective surface.
- (3) The above commutative diagram is f -equivariant after replacing f by a power.

Moreover, if the exceptional divisor $E_\varphi \not\subset \text{Supp } \Delta_\tau$, then the following assertions hold.

- (i) ψ is a K_W -negative contraction and is a blow-up along a smooth projective surface.
- (ii) $E_\psi = \tau_0^* E_\varphi$ and $E_\chi = \pi^* E_\psi = \tau^* E_\varphi$.
- (iii) $\tau'_0: W' = \mathbb{P}_{Y'}(\mathcal{E}') \rightarrow Y'$ is an algebraic $\mathbb{P}^1$ -bundle such that $\mathcal{E} = \varphi^* \mathcal{E}'$.
- (iv) X' is Fano, and τ' is a Fano conic bundle.
- (v) $W \cong W' \times_{Y'} Y$ and $X \cong X' \times_{Y'} Y \cong X' \times_{W'} W$.

Proof. (1) follows from the imprimitivity of Y . Let R_φ be the K_Y -negative extremal ray contracted by φ . Since τ (resp. τ_0) is a K_X (resp. K_W)-negative contraction of an extremal face F_τ of $\overline{\text{NE}}(X)$ (resp. F_{τ_0} of $\overline{\text{NE}}(W)$), there are extremal rays R_ψ and R_χ of $\overline{\text{NE}}(W)$ and $\overline{\text{NE}}(X)$, respectively such that $\tau_*R_\chi = (\tau_0)_*R_\psi = R_\varphi$ and $R_\chi \cap F_\tau = R_\psi \cap F_{\tau_0} = \{0\}$ (cf. Proof of Lemma 6.4.4). Since X is Fano, there exists a K_X -negative contraction $\chi: X \rightarrow X'$ of R_χ . By the rigidity lemma (cf. [Deb01, Lemma 1.15]), $\varphi \circ \tau$ factors through χ . Then, the same proof of Lemma 4.4 (2) shows (2) (here, X' may not be Fano).

Similarly, there exists a contraction $\eta: X \rightarrow W'$ of the K_X -negative extremal face F_η containing the π -contracted extremal face F_π such that $\pi_*F_\eta = R_\chi$. By the rigidity lemma several times, we see that η factors through π , η factors through χ , and τ' factors through π' . So we get the commutative diagram and (3) follows from Lemma 6.2.2.

From now on, we further assume that $E_\varphi \not\subseteq \text{Supp } \Delta_\tau$. Then τ^*E_φ is irreducible. Since $\tau_*E_\chi \subseteq E_\varphi$, it follows that $\tau^*E_\varphi = E_\chi$ (cf. [KM98, Theorem 3.7]). With the same proof of Lemma 4.4 (2), our X' is Fano.

Note that different components of E_π are disjoint from each other; hence $E_\pi \cdot \ell_\pi < 0$ for every fibre ℓ_π of π . Since $E_\chi \cdot \ell_\pi = 0$ by the projection formula, our E_χ **is not a component of E_π** . Let $\ell_\psi \in R_\psi$ be a curve on W , and $\ell_\chi \in R_\chi$ a curve on X . Then $\pi_*\ell_\chi = a\ell_\psi$ for some $a \in \mathbb{Z}_{>0}$, and we have

$$K_W \cdot (a\ell_\psi) = \pi^*K_W \cdot \ell_\chi = (K_X - E_\pi) \cdot \ell_\chi < 0.$$

Here, $E_\pi \cdot \ell_\chi \geq 0$; otherwise, E_χ will coincide with some component of E_π by (2), a contradiction. So ψ is a K_W -negative contraction. Applying [Cas08, Theorem 3.14] for the diagram $\tau'_0 \circ \psi = \varphi \circ \tau_0$, (i) and (ii) are proved.

With the same proof of Lemma 4.4 (4), τ'_0 is an elementary conic bundle. Note that, outside the curve $C \subseteq Y'$ blown up by φ , every fibre of τ'_0 is a smooth conic since so is τ_0 . Thus the divisor $\Delta_{\tau'_0} \subseteq C$, which is absurd (cf. Notation 2.1 (11 a)). Hence, τ'_0 is a smooth $\mathbb{P}^1$ -bundle. By (3) and [Zha12, Theorem 1.2], Y' is rational; thus $W' = \mathbb{P}_{Y'}(\mathcal{E}')$ for some locally free sheaf $\mathcal{E}'$ of rank 2 on Y' (cf. [MZZ21, Lemma 2.12]). By Lemma 6.2.10, $W \cong W' \times_{Y'} Y$; hence up to a twist, $\mathcal{E} = \varphi^*\mathcal{E}'$, which implies (iii).

We claim that τ' is equidimensional. Consider the behaviour of π' and note that

$E_{\pi'} \subseteq \chi(E_\pi)$ is a disjoint union of prime divisors. On the one hand, outside $\psi(E_\psi)$, the fibres of π' have dimension ≤ 1 since so are fibres of π over $W \setminus E_\psi$. On the other hand, $(\pi')^{-1}(\psi(E_\psi)) = \chi(E_\chi)$ is a smooth surface by (1); thus the fibres of π' over $\psi(E_\psi)$ have dimension ≤ 1 . So, all fibres of π' have dimension ≤ 1 and it follows from X' being Fano, Lemma 6.2.6 and the induction on $\rho(X'/W')$ that π' is the blow-up along disjoint surfaces S' on W' . Moreover, S' is the image under ψ of the surfaces blown up by π by the above diagram; hence S' is the union of subsections of τ'_0 . This further implies the fibres of τ' over C are of dimension 1 and our claim holds.

Since X' is Cohen-Macaulay and Y' is smooth, our τ' is flat. By (iii) and Lemma 6.2.10, (iv) and (v) are proved. So we complete the proof of Lemma 6.5.2. $\square$

Remark 6.5.3. Removing the condition “ $E_\varphi \not\subset \text{Supp } \Delta_\tau$ ”, we still have the commutative diagram in Lemma 6.5.2. However, in this case, ψ is possibly not a K_W -negative contraction; hence Lemma 6.2.6 cannot be applied and W' may not even be $\mathbb{Q}$ -factorial!

6.6 Conic bundles onto $\mathbb{P}^1$ -bundles over rational surfaces

In this section, we consider the conic bundles with the base isomorphic to a $\mathbb{P}^1$ -bundle over a rational surface. Theorem 6.6.2 is our main result in this section.

Notation 6.6.1. We follow the notations below throughout this section.

- (1) $\tau: X \rightarrow Y$ is a conic bundle from a smooth Fano fourfold X , which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ where π is a composition of blow-ups along disjoint smooth projective surfaces and τ_0 is an elementary conic bundle.
- (2) By Theorem 6.5.1, Y is a smooth Fano threefold and $W = \mathbb{P}_Y(\mathcal{E})$ with $\mathcal{E}$ being a locally free sheaf of rank 2 on Y .
- (3) $f: X \rightarrow X$ is an int-amplified endomorphism. After iteration, f descends to $g := f|_Y$ and $h := f|_W$, which are both int-amplified (cf. Lemmas 6.2.2 and 6.2.3).

- (4) $p: Y \rightarrow Z$ is a $\mathbb{P}^1$ -bundle over a smooth rational surface Z . By [MZZ21, Theorem 6.4], $Y = \mathbb{P}_Z(\mathcal{F})$ with $\mathcal{F}$ being a splitting locally free sheaf of rank 2 on Z .

$$f \circ X \xrightarrow[\pi]{\tau} h \circ W \xrightarrow[\tau_0]{\tau} g \circ Y \xrightarrow{p} Z$$

- (5) For each $z \in Z$, let $Y_z := p^{-1}(z)$, $X_z := (p \circ \tau)^{-1}(z)$ and $W_z := (p \circ \tau_0)^{-1}(z)$.

- (6) We use $E_\bullet$ to denote the exceptional locus $\text{Exc}(\bullet)$ for simplicity.

Theorem 6.6.2. *In the setting of Notation 6.6.1, suppose that Z is isomorphic to $\mathbb{P}^2$, $\mathbb{F}_0$ or $\mathbb{F}_1$. Then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

In what follows, we extend [MZZ21, Lemma 6.3] to the following four dimensional case. The proof of Lemma 6.6.3 will last till the paragraph before Lemma 6.6.6.

Lemma 6.6.3. *The composite $p \circ \tau_0: W \rightarrow Z$ is a fibre bundle such that either all the fibres are $\mathbb{F}_0$, or all the fibres are $\mathbb{F}_1$.*

Proof. First, note that each fibre $W_z \cong \mathbb{F}_d$ for some $d \geq 0$. So the lemma is equivalent to showing that $d \leq 1$, i.e., W_z is Fano, since $\mathbb{F}_0$ and $\mathbb{F}_1$ cannot deform to each other.

Let $r := \rho(X) - \rho(W)$. If $r = 0$, then $W = X$ is Fano; hence our lemma follows from the adjunction formula. So we may assume $r > 0$. By Theorem 6.5.1, Δ_τ is a disjoint union of r smooth g^{-1} -invariant surfaces D_i , and π is the blow-up of W along the disjoint union $\bigsqcup_{i=1}^r \overline{D_i}$ of h^{-1} -invariant surfaces $\overline{D_i}$.

Write $\tau^{-1}(D_i) = \tau^*D_i = E_i + \widetilde{E_i}$ where E_i is the π -exceptional divisor with center $\overline{D_i}$ and $\widetilde{E_i}$ is the π -strict transform of $\tau_0^{-1}(D_i)$. Let $S_i := E_i \cap \widetilde{E_i}$, which is f^{-1} -invariant. Then $\widetilde{E_i} \cong \tau_0^{-1}(D_i)$ and $S_i \cong \overline{D_i}$. By Theorem 6.3.1, Δ_τ is contained in some toric boundary of Y . We shall discuss case by case in terms of Δ_τ .

Case 1: Δ_τ contains at least one section of p . Then the g^{-1} -invariant Δ_τ consists of either one or two disjoint sections of p (cf. Theorem 6.5.1 and [MZZ21, Theorem 3.3]). Hence, for every $z \in Z$, the fibre $Y_z \not\subset \Delta_\tau$. Then every surface $\overline{D_i}$ blown up by π is either disjoint from W_z or intersects with W_z at some points; otherwise, $\tau_0(W_z \cap \overline{D_i})$ being a curve contradicts $Y_z \not\subset \Delta_\tau$. So X_z is the blow-up of W_z along several points

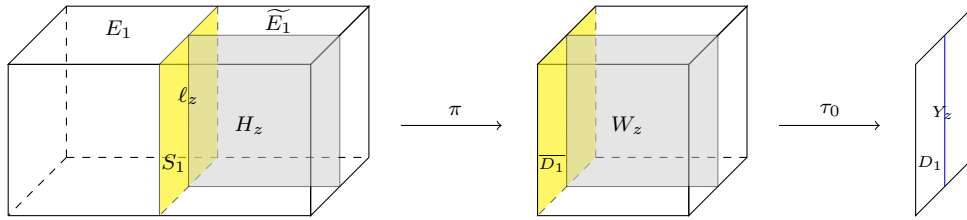
$\bigsqcup_{i=1}^r \overline{D_i} \cap W_z$ and hence X_z is smooth and irreducible. Since X is Fano and $p \circ \tau$ is flat, each X_z is a del Pezzo surface by the adjunction formula. Then W_z is also Fano by the ramification divisor formula. So our lemma holds in this case.

Case 2: Δ_τ contains no section of p . Then Δ_τ consists of prime divisors D_i , each of which is the pull-back of a smooth rational curve on Z along p (cf. [MZZ21, Corollary 3.4]). We may assume $Y_z \subseteq D_i$ for some i ; otherwise, $W_z \cong X_z$ being Fano follows from the adjunction. Without loss of generality, we may further assume $Y_z \subseteq D_1$ and π factors as $X \xrightarrow{\pi_1} X_1 \rightarrow W$ where π_1 is the blow-up with $E_{\pi_1} = E_1 = \pi^{-1}(\overline{D_1})$.

If X_1 is Fano, then we are done by induction on r . Thus we may assume that X_1 is not Fano. Let $H_z := (\pi|_{\widetilde{E_1}})^{-1}(W_z) \cong W_z$ and

$$\ell_z := S_1 \cap H_z = (E_1 \cap \widetilde{E_1}) \cap H_z = (\pi|_{\widetilde{E_1}})^{-1}(W_z \cap \overline{D_1}),$$

which is a cross-section of the ruling $H_z \cong W_z \rightarrow Y_z$; see the following picture.



Claim 6.6.4. $(\ell_z \cdot S_1)_{\widetilde{E_1}} = 0$.

Suppose Claim 6.6.4 for the time being. Then we have

$$(\ell_z^2)_{H_z} = (\ell_z \cdot S_1|_{H_z})_{H_z} = (\ell_z \cdot S_1)_{\widetilde{E_1}} = 0.$$

Since ℓ_z is an (irreducible) cross-section of the ruling $H_z \rightarrow Y_z$, our $H_z \cong \mathbb{F}_0$ (cf. [Har77, Chapter V, Proposition 2.20]). As a result, $W_z \cong H_z \cong \mathbb{F}_0$ and hence our lemma follows. So we are only left to show Claim 6.6.4.

Proof of Claim 6.6.4. Since X_1 is not Fano and our $S_1 \cong \overline{D_1}$ has Picard number 2, by [Wis91, Proposition 3.6], either one of the following cases occurs:

- (1) $S_1 \cong \overline{D_1} \cong \mathbb{F}_0$ and $E_1 \cong S_1 \times \mathbb{P}^1$ is a trivial bundle;

- (2) X admits another blow-down $\pi'_1: X \rightarrow X'_1$ onto a smooth fourfold X'_1 which contracts the divisor E_1 to a smooth surface $S' \subseteq X'_1$.

If Case (1) holds, then we have $E_1 \cong \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ and hence

$$(\ell_z \cdot S_1)_{\widetilde{E_1}} = (\ell_z \cdot E_1|_{\widetilde{E_1}})_{\widetilde{E_1}} = \ell_z \cdot E_1 = -\ell_z \cdot \widetilde{E_1} = -(\ell_z \cdot \widetilde{E_1}|_{E_1})_{E_1} = -(\ell_z \cdot S_1)_{E_1} = 0,$$

noting that $(E_1 + \widetilde{E_1}) \cdot \ell_z = (D_1 \cdot Y_z)_Y = 0$ by the projection formula. So Claim 6.6.4 holds for Case (1). From now on, we assume Case (2).

Claim 6.6.5. Under the condition of Case (2), S_1 is contracted by π'_1 .

Proof. Suppose on the contrary that $\pi'_1(S_1) = S'$. Note that $\rho(S') = \rho(E_1) - 1 = \rho(\overline{D_1}) = 2$, hence $S' \cong S_1$ is a (rational) ruled surface. Note also that the ruling of $\pi|_{E_1}$ induces a natural ruling on S' since π'_1 does not contract any fibre of π . By Lemma 6.2.2, π'_1 is f -equivariant after iteration. Take a general $f|_{S'}$ -periodic fibre of $S' \rightarrow \mathbb{P}^1$ (cf. [Fak03]*Theorem 5.1) and denote its π'_1 -inverse image in $E_1 \subseteq X$ by T , which is an f -periodic Hirzebruch surface. After iteration, we may assume $f(T) = T$.

On the one hand, T contains a fibre of π since every fibre of $S' \rightarrow \mathbb{P}^1$ is dominated by a fibre of $E_1 \rightarrow \overline{D_1}$. Hence, considering $\rho(T) = \rho(\overline{D_1}) = 2$, $\pi(T)$ is a curve. Since $S_1 \cong \pi(S_1) = \overline{D_1}$, we have $\pi(T) = \pi(c := S_1 \cap T) \cong \mathbb{P}^1$ on $\overline{D_1}$. So $\pi|_T$ gives another ruling of T different from $\pi'_1|_T$; thus $T \cong \mathbb{F}_0$. On the other hand, since $\pi'_1(S_1) = S'$, we see that c is an $(f|_T)^{-1}$ -invariant curve. By [MZZ21]*Lemma 3.1, $c = \{\cdot\} \times \mathbb{P}^1$ or $\mathbb{P}^1 \times \{\cdot\}$. Nevertheless, this is impossible since c is neither a fibre of π nor π'_1 . $\square$

End of Proof of Claim 6.6.4 (and Lemma 6.6.3). By Claim 6.6.5, S_1 is contracted to a curve, since the dimension of each fibre of π'_1 is no more than one. If ℓ_z is contracted, then we have $E_1 \cdot \ell_z < 0$ since E_1 is π'_1 -anti-ample. Then

$$(K_{E_1} \cdot \ell_z)_{E_1} = (K_X + E_1) \cdot \ell_z < 0.$$

So $\pi'_1|_{E_1}$ is a Fano contraction. In particular, $(\ell_z \cdot S_1)_{E_1} = 0$ by the cone theorem.

If ℓ_z is not contracted by π'_1 , our $\overline{D_1} \cong S_1 \cong \mathbb{F}_0$ since the minimal surface S_1 admits another ruling (induced by $\pi'_1|_{S_1}$). In this case, for c' being a fibre of $\pi'_1|_{S_1}$, we have

$$(K_{\overline{D_1}} \cdot \pi(c'))_{\overline{D_1}} = (K_{S_1} \cdot c')_{S_1} = ((K_{E_1} + S_1) \cdot c')_{E_1} = (K_{E_1} \cdot c')_{E_1},$$

since $(S_1 \cdot c')_{E_1} = 0$ by the cone theorem and $S_1 \cong \overline{D_1}$. So $E_1 \cong \overline{D_1} \times \mathbb{P}^1$ (cf. [Wis91, Remark following Lemma 3.3]). With the same argument as in Case (1), $(\ell_z \cdot S_1)_{E_1} = 0$. Now Claim 6.6.4 is proved and we have finished the proof of Lemma 6.6.3. $\square$

The following lemma contributes to showing the splitting-ness of $\mathcal{E}$. We recall Notation 6.2.1 (13) and Lemma 6.2.13 for the related notations and properties.

Lemma 6.6.6. *Suppose Notation 6.6.1 and the existence of the following exact sequence*

$$0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{E} \rightarrow \mathcal{Q} \rightarrow 0, \quad (\dagger)$$

with $\mathcal{F}_1$ and $\mathcal{Q}$ being invertible sheaves such that the wall $W_{\mathcal{E}}(\mathcal{F}_1) \neq \emptyset$. Then $\mathcal{E}$ splits.

Proof. Since $W_{\mathcal{E}}(\mathcal{F}_1) \neq \emptyset$, there exists one $A_0 \in \text{Amp}(Y)$ such that $\xi_{\mathcal{F}_1} \cdot A_0^2 = 0$. By Lemma 6.2.13, we can take a sufficiently small convex compact neighbourhood $\widehat{K}$ of $A_0 \in \text{Amp}(Y)$ and let $K \subseteq P(Y)$ be its homeomorphic image such that all the walls in K pass through A_0^2 . Since $\xi_{\mathcal{F}_1} \cdot H^2 = 0$ for each $H \in W_{\mathcal{E}}(\mathcal{F}_1)$, there is a chamber $\mathcal{C}$ in K such that $\xi_{\mathcal{F}_1} \cdot A^2 < 0$ for any ample ($\mathbb{R}$ -Cartier) divisor A with $A^2 \in \mathcal{C}$. Fix one such A_1 .

If $\mathcal{E}$ is A_1 -semistable, then by our dynamical assumption and [Ame03, Proposition 2.4], $\mathcal{E}$ splits and our lemma holds.

Suppose that $\mathcal{E}$ is not A_1 -semistable. Let $\mathcal{F}_2$ be a maximal destabilizing (saturated invertible) sheaf associated to A_1 . Then one has $\xi_{\mathcal{F}_2} \cdot A_1^2 > 0$ by definition; thus $\mathcal{F}_2 \not\subset \mathcal{F}_1$. Consider the natural restriction of the exact sequence $(\dagger)$ to $\mathcal{F}_2$. Since $\mathcal{F}_2 \not\subset \mathcal{F}_1$ and $\mathcal{Q}$ is locally free, the map $\mathcal{F}_2 \rightarrow \mathcal{Q}$ is an injection. If $c_1(\mathcal{F}_2) = c_1(\mathcal{Q})$, then it is easy to verify $\mathcal{E} = \mathcal{F}_1 \oplus \mathcal{F}_2$, which shows the splitting-ness of our $\mathcal{E}$. If $c_1(\mathcal{F}_2) < c_1(\mathcal{Q})$, then

$$\xi_{\mathcal{F}_2} \cdot A_0^2 < \xi_{\mathcal{Q}} \cdot A_0^2 = -\xi_{\mathcal{F}_1} \cdot A_0^2 = 0 < \xi_{\mathcal{F}_2} \cdot A_1^2.$$

This in turn implies that $(x_0 A_0 + (1 - x_0) A_1)^2$ lies in the wall $W_{\mathcal{E}}(\mathcal{F}_2)$ for some $0 < x_0 < 1$. However, $x_0 A_0 + (1 - x_0) A_1 \in \widehat{K}$ by the convexity; hence $(x_0 A_0 + (1 - x_0) A_1)^2 \in K$, a contradiction to the choice of K , noting that $W_{\mathcal{E}}(\mathcal{F}_2)$ does not pass through A_0^2 . $\square$

Proof of Theorem 6.6.2. We divide the proof into three cases in terms of Z .

Case A: $Z \cong \mathbb{P}^2$. We may write $Y = \mathbb{P}_Z(\mathcal{F})$ where $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(-k)$ with $k = 0, 1$ or 2 (cf. [SW90]). Let $F := p^*L$ where L is any line on $Z \cong \mathbb{P}^2$ and S the section class of Y such that $\mathcal{O}_Y(S)|_S \cong \mathcal{O}_S(-k)$ (with respect to the surjection $\mathcal{F} \rightarrow \mathcal{O}_Z(-k)$). If $k = 1$ or 2 , then $S|_S$ is not pseudo-effective and hence S is f^{-1} -periodic (cf. [MZZ21, Lemma 2.3]); if $k = 0$, let S be an f -periodic section (cf. [Fak03, Theorem 5.1]). In both cases, we may assume S is f -invariant after iteration. Up to a twist, we assume $c_1(\mathcal{E}) = aS + bF$ with $-1 \leq a, b \leq 0$. Then $\mathcal{O}_{Y_z}(c_1(\mathcal{E}|_{Y_z})) \cong \mathcal{O}_{Y_z}(a)$ for any $z \in Z$. Applying $\mathbb{P}_{Y_z}(\mathcal{E}|_{Y_z}) = W_z \cong \mathbb{F}_c$ with $c \leq 1$ by Lemma 6.6.3, we have $\mathcal{E}|_{Y_z} \cong \mathcal{O}_{Y_z} \oplus \mathcal{O}_{Y_z}(a)$ for any $z \in Z$. Thus the function $z \mapsto h^0(Y_z, \mathcal{E}|_{Y_z})$ is constant and the natural morphism $p^*p_*\mathcal{E} \rightarrow \mathcal{E}$ has domain a locally free sheaf, which is an evaluation map on every fibre (cf. [Har77, Chapter III, Corollary 12.9]). Since $a \leq 0$, the global sections of $\mathcal{E}|_{Y_z}$ are constant. Then we have an exact sequence with $\mathcal{Q}$ being a vector bundle.

$$0 \rightarrow p^*p_*\mathcal{E} \rightarrow \mathcal{E} \rightarrow \mathcal{Q} \rightarrow 0 \quad (*)$$

Suppose $a = 0$. Then $p_*\mathcal{E}$ is locally free of rank 2; thus $p^*p_*\mathcal{E} \cong \mathcal{E}$ and $W \cong Y' \times_Z Y$ with $Y' = \mathbb{P}_Z(p_*\mathcal{E})$. Since $Z \cong \mathbb{P}^2$ and $h|_{Y'}$ is int-amplified (cf. Lemma 6.2.3), our $p_*\mathcal{E}$ and hence $\mathcal{E}$ split (cf. [Ame03, Proposition 3]).

Suppose $a = -1$. Then $p_*\mathcal{E}$ is a line bundle, say $\mathcal{O}_Z(e)$ for some $e \in \mathbb{Z}$. Hence $\mathcal{F}_1 := p^*p_*\mathcal{E} \cong \mathcal{O}_Y(eF)$ and $\mathcal{Q} \cong \mathcal{O}_Y(-S + (b - e)F)$. Note that

$$\mathrm{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E}) = H^1(Y, S + (2e - b)F) = H^1(Y, K_Y + (S - K_Y) + (2e - b)F).$$

Using the relative canonical bundle formula, we have

$$S - K_Y \equiv S - (-2S + p^*(K_Z + \det \mathcal{F})) = \frac{5}{2}S + (3 + k)F + \frac{1}{2}S$$

where $\mathcal{O}_Y(S)|_S \cong \mathcal{O}_S(-k)$ with $k = 0, 1, 2$. Then one can easily verify that $\frac{5}{2}S + (3 + k)F$ is nef and big for any $k = 0, 1, 2$. Since $\frac{1}{2}S$ has the support with only normal crossings, by the Kawamata–Viehweg vanishing theorem (cf. e.g., [KM98, Theorem 2.64]), $\mathrm{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E})$ vanishes if $t := 2e - b \geq 0$, noting that F is nef on Y . So $t \geq 0$ implies that $(*)$ and hence $\mathcal{E}$ split. Therefore, we may assume that $t < 0$. Let $\xi_{\mathcal{F}_1} :=$

$2c_1(p^*p_*\mathcal{E}) - c_1(\mathcal{E}) \sim S + tF$. Then our theorem for the case $Z \cong \mathbb{P}^2$ follows from Claim 6.6.7 and Lemma 6.6.6.

Claim 6.6.7. $W_{\mathcal{E}}(\mathcal{F}_1) \neq \emptyset$ when $t < 0$.

Proof of Claim 6.6.7. Note that up to a multiple, any ample divisor on Y can be written as $D = D(u) := S + uF$ with $u > k$. So we have the following:

$$\xi_{\mathcal{F}_1} \cdot D(u)^2 = (S + tF) \cdot (S + uF)^2 = (u - (k - t))^2 - t^2 + kt.$$

Since $k - t > k$ and $-t^2 + kt = t(k - t) < 0$, there exists $u' > k$ such that $\xi_{\mathcal{F}_1} \cdot D(u')^2 = 0$ by the continuity, which completes the proof of our claim.

Case B: $Z \cong \mathbb{F}_0$. First, we may write $Y = \mathbb{P}_Z(\mathcal{F})$ where $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(-k_1, -k_2)$ with $0 \leq k_1, k_2 \leq 1$ (resp. $\mathcal{O}_Z \oplus \mathcal{O}_Z(1, -1)$) (cf. [SW90]). Note that these Fano threefolds Y with $\rho(Y) = 3$ has exactly 3 extremal rays in $\overline{\text{NE}}(Y)$. Let $F_i := p^*L_i$ be the “fibre” class of p where $L_1 \cong \mathcal{O}_Z(1, 0)$ and $L_2 \cong \mathcal{O}_Z(0, 1)$, and S the section class of p such that $\mathcal{O}_Y(S)|_S \cong \mathcal{O}_S(-k_1, -k_2)$ (resp. $\mathcal{O}_S(1, -1)$). Similar to **Case A**, we may assume our S is f -invariant after iteration. Up to a twist, we assume $c_1(\mathcal{E}) = aS + b_1F_1 + b_2F_2$ with $-1 \leq a, b_i \leq 0$. Then we have $\mathcal{E}|_{Y_z} \cong \mathcal{O}_{Y_z} \oplus \mathcal{O}_{Y_z}(a)$ for any $z \in Z$ and we get the exact sequence (*) again, noting that the global sections of $\mathcal{E}|_{Y_z}$ are constant.

Assume first $a = -1$. Then $p_*\mathcal{E}$ is a line bundle, say $\mathcal{O}_Z(e_1, e_2)$ for some $e_i \in \mathbb{Z}$. Hence $\mathcal{F}_1 := p^*p_*\mathcal{E} \cong \mathcal{O}_Y(e_1F_1 + e_2F_2)$ and $\mathcal{Q} \cong \mathcal{O}_Y(-S + (b_1 - e_1)F_1 + (b_2 - e_2)F_2)$. Then

$$\text{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E}) = H^1(Y, K_Y + (S - K_Y) + (2e_1 - b_1)F_1 + (2e_2 - b_2)F_2).$$

Applying the relative canonical bundle formula, we have

$$S - K_Y \equiv S - (-2S + p^*(K_Z + \det \mathcal{F})) = 3S + (2 + k_1)F_1 + (2 + k_2)F_2 \text{ (resp. } 3S + F_1 + 3F_2).$$

Using the three extremal rays of $\overline{\text{NE}}(Y)$, we can verify that $S - K_Y$ (resp. $S - K_Y - F_1$) is nef and big, noting that the bigness follows from the nefness and the positive self-intersection, and when $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(1, -1)$, the two horizontal extremal curves lie in distinct sections of p . By the Kawamata–Viehweg vanishing theorem (cf. e.g., [KM98, Theorem 2.64]), $\text{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E})$ vanishes if $t_i := 2e_i - b_i \geq 0$ for $i = 1, 2$ (resp. $t_1 \geq -1$

and $t_2 \geq 0$). Therefore, $t_i \geq 0$ ($i = 1, 2$) (resp. $t_1 \geq -1$ and $t_2 \geq 0$) implies that $(*)$ and hence $\mathcal{E}$ split. So we may assume $t_i < 0$ for some i (resp. $t_1 < -1$ or $t_2 < 0$). Let $\xi_{\mathcal{F}_1} := 2c_1(p^*p_*\mathcal{E}) - c_1(\mathcal{E}) \sim S + t_1F_1 + t_2F_2$. Then our theorem for the case $Z \cong \mathbb{F}_0$ and $a = -1$ follows from Claim 6.6.8 and Lemma 6.6.6.

Claim 6.6.8. $W_{\mathcal{E}}(\mathcal{F}_1) \neq \emptyset$ when $t_i < 0$ for some i (resp. $t_1 < -1$ or $t_2 < 0$).

Proof of Claim 6.6.8. Recall that $Y = \mathbb{P}_Z(\mathcal{F})$ with $\det \mathcal{F} = \mathcal{O}(-k_1, -k_2)$ with $k_i = 0$ or 1 (resp. $\mathcal{O}(1, -1)$). Then up to a multiple, any ample divisor D on Y can be written as $D = D(u_1, u_2) := S + u_1F_1 + u_2F_2$ with $u_i > k_i$ (resp. $u_1 > 0$ and $u_2 > 1$). Hence,

$$\begin{aligned} \xi_{\mathcal{F}_1} \cdot D(u_1, u_2)^2 &= (S + t_1F_1 + t_2F_2) \cdot (S + u_1F_1 + u_2F_2)^2 \\ &= 2(u_1 - (k_1 - t_1))(u_2 - (k_2 - t_2)) - 2t_1t_2 + t_1k_2 + t_2k_1 \\ &\text{(resp. } = 2(u_1 + (t_1 + 1))(u_2 + (t_2 - 1)) - 2t_1t_2 + t_1 - t_2). \end{aligned}$$

Since $u_i > k_i$ for each i (resp. $u_1 > 0$ and $u_2 > 1$) and $t_j < 0$ for some j (resp. $t_1 < -1$ or $t_2 < 0$), there exist $u'_i > k_i$ (resp. $u'_1 > 0$ and $u'_2 > 1$) such that $\xi_{\mathcal{F}_1} \cdot D(u'_1, u'_2)^2 = 0$ by the continuity. So our claim holds.

Assume second $a = 0$. Now $p_*\mathcal{E}$ is locally free of rank 2 and one gets the following commutative diagram such that $p^*p_*\mathcal{E} \cong \mathcal{E}$ and $W \cong Y \times_Z Y'$.

$$\begin{array}{ccccc} X & \xrightarrow{\quad} & X_{r-1} & \xrightarrow{q_{r-1}} & T \\ & \searrow \pi & \downarrow \pi_r & & \downarrow \phi \\ & & W = \mathbb{P}_Y(\mathcal{E}) & \xrightarrow{q} & Y' := \mathbb{P}_Z(p_*\mathcal{E}) \\ & \searrow \tau & \downarrow \tau_0 & & \downarrow p' \\ & & Y & \xrightarrow{p} & Z \end{array}$$

Clearly, $q: W \rightarrow Y'$ is also a (smooth) $\mathbb{P}^1$ -bundle. Let $r := \rho(X) - \rho(W)$ and $\overline{D} := \bigsqcup_{i=1}^r \overline{D}_i \subseteq W$ be the blown up centres of π , each component of which is of dimension 2.

1°: If $\dim(q(\overline{D}_i)) = 2$ for all $1 \leq i \leq r$, then each $\overline{D}_i$, as an h^{-1} -invariant divisor on $q^{-1}(q(\overline{D}_i))$, is a subsection of q ; hence $q \circ \pi$ is a conic bundle and hence Y' is a smooth Fano threefold (cf. Lemma 6.2.8).

2°: Otherwise, after rearranging the blow-ups, we may assume that $q(\overline{D}_r) =: C$ is a curve in Y' , where $\overline{D}_r$ is the blow-up centre of π_r . **We claim that $\overline{D}_r = q^{-1}(C)$ in**

this case. Note that $\tau_0(\overline{D_r})$ is a divisor on Y and thus $p'(C) = p \circ \tau_0(\overline{D_r})$ cannot be a point. Let $F = p'^{-1}(p'(C))$, a Hirzebruch surface. Note also that $\overline{D_r} \subseteq q^{-1}(F)$ is a (prime) divisor, and $q|_{q^{-1}(F)}: q^{-1}(F) \rightarrow F$ is a $\mathbb{P}^1$ -bundle by the base change. Then $q^{-1}(C) = (q|_{q^{-1}(F)})^{-1}(C) \supseteq \overline{D_r}$ is irreducible; hence they coincide and our claim holds.

Let T be the blow-up of Y' along the curve C . By Lemma 6.2.10, $X_{r-1} \cong T \times_{Y'} W$ and we denote by $q_{r-1}: X_{r-1} \rightarrow T$ the natural projection. Then q_{r-1} is a $\mathbb{P}^1$ -bundle. Note that $p' \circ \phi: T \rightarrow Z$ is a conic bundle and $\rho(X) - \rho(X_{r-1}) = r - 1$. By induction, Y' is dominated by a smooth Fano threefold.

No matter 1° or 2° occurs, the above commutative diagram is f -equivariant after iteration (cf. Lemma 6.2.2). Hence, $p_*\mathcal{E}$ splits by [MZZ21, Theorem 6.4] and then $\mathcal{E} = p^*p_*\mathcal{E}$ splits. This completes the proof for the case $Z \cong \mathbb{F}_0$.

Case C: $Z \cong \mathbb{F}_1$. Denote by c and ℓ the negative section and a fibre of the Hirzebruch surface Z , respectively. We may write $Y = \mathbb{P}_Z(\mathcal{F})$ where $\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(-k(C+\ell))$ with $k = 0$ or 1 (cf. [SW90]). Let $C := p^*c$, $L := p^*\ell$ be two “fibre” classes, and S the section class of p such that $\mathcal{O}_Y(S)|_S \cong \mathcal{O}_Z(-k(C+\ell))$. Similar to **Case A**, we may assume our S is f -invariant after iteration. Up to a twist, we may assume $c_1(\mathcal{E}) = aS + b_1C + b_2L$ with $-1 \leq a, b_i \leq 0$. Then, $\mathcal{O}_{Y_z}(c_1(\mathcal{E}|_{Y_z})) \cong \mathcal{O}_{Y_z}(a)$ for any $z \in Z$, and we get the exact sequence (*) again, noting that the global sections of $\mathcal{E}|_{Y_z}$ are constant.

Assume that $a = -1$. Then $p_*\mathcal{E}$ is a line bundle, say $\mathcal{O}_Z(e_1c + e_2\ell)$ for some $e_i \in \mathbb{Z}$. So $\mathcal{F}_1 := p^*p_*\mathcal{E} \cong \mathcal{O}_Y(e_1C + e_2L)$ and $\mathcal{Q} \cong \mathcal{O}_Y(-S + (b_1 - e_1)C + (b_2 - e_2)L)$. Similar to **Case B**, one can verify that

$$S - K_Y \equiv S - (-2S + p^*(K_Z + \det \mathcal{F})) = 3S + (2+k)C + (3+k)L$$

is nef and big for $k = 0, 1$ and then $\text{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E})$ vanishes if $t_i := 2e_i - b_i \geq 0$ for $i = 1, 2$. So $t_i \geq 0$ ($i = 1, 2$) implies that (*) and hence $\mathcal{E}$ split. Therefore, we assume $t_i < 0$ for some i . Let $\xi_{\mathcal{F}_1} := 2c_1(p^*p_*\mathcal{E}) - c_1(\mathcal{E}) \sim S + t_1C + t_2L$. Then our theorem for the case $Z \cong \mathbb{F}_1$ and $a = -1$ follows from Claim 6.6.9 and Lemma 6.6.6.

Claim 6.6.9. $W_{\mathcal{E}}(\mathcal{F}_1) \neq \emptyset$ when $t_i < 0$ for some i .

Proof of Claim 6.6.9. First, any ample divisor D on Y can be written as $D = D(u_1, u_2) :=$

$S + u_1C + u_2L$ with $u_2 > u_1 > k$ after replacing D by a multiple. So

$$\xi_{\mathcal{F}_1} \cdot D(u_1, u_2)^2 = u_1^2 + 2(u_1 - (k - t_1))((u_2 - u_1) - (k - t_2)) - (k - 2t_1)(k - t_2).$$

If $t_1 < 0$, taking $k < u_1 < k - t_1$ and $u_2 - u_1 \gg 1$, we have $\xi_{\mathcal{F}_1} \cdot D(u_1, u_2) < 0$. If $t_2 < 0$, taking $u_1 = k - t_2$ and $0 < u_2 - u_1 \ll k - t_2$, we have $\xi_{\mathcal{F}_1} \cdot D(u_1, u_2)^2 \simeq t_2(k - t_2) < 0$. In both cases, there exist $u'_2 > u'_1 > k$ such that $\xi_{\mathcal{F}_1} \cdot D(u'_1, u'_2)^2 = 0$. Our claim holds.

The same proof as in **Case B** works for the case $a = 0$ in **Case C**. We have completed the proof of Theorem 6.6.2. $\square$

6.7 Conic bundles over (the blow-up of) $\mathbb{P}^3$

In this section, we shall study conic bundles over $\mathbb{P}^3$ or the blow-up of $\mathbb{P}^3$ along a line. Our main results of this section are Theorems 6.7.1 and 6.7.2 below.

Theorem 6.7.1. *Let X be a smooth Fano fourfold admitting an int-amplified endomorphism f . Suppose that X admits a conic bundle $\tau: X \rightarrow Y \cong \mathbb{P}^3$, which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ as in Notation 6.6.1 (1). Then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

Proof. By Theorem 5.1 (2), $W = \mathbb{P}_Y(\mathcal{E})$ for some locally free rank 2 sheaf $\mathcal{E}$ on Y . So our theorem follows from Lemmas 6.2.2, 6.2.3 and [Ame03, Proposition 3]. $\square$

Theorem 6.7.2. *Let X be a smooth Fano fourfold admitting an int-amplified endomorphism f . Suppose that X admits a conic bundle $\tau: X \rightarrow Y$ with $\varphi: Y \rightarrow Y' \cong \mathbb{P}^3$ being the blow-up along a line. Suppose further that τ factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ as in Notation 6.6.1 (1). Then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

Lemma 6.7.3. *Let $q: Y \rightarrow Z \cong \mathbb{P}^2$ be a conic bundle from a smooth Fano threefold Y . Suppose that q factors as $Y \xrightarrow{\varphi} Y' \xrightarrow{q_0} Z$, where φ is the blow-up of a smooth threefold Y' along a smooth curve C and $q_0: Y' \rightarrow Z$ is a splitting $\mathbb{P}^1$ -bundle. Then Y' is Fano.*

Proof. Suppose on the contrary that Y' is not Fano. Then $K_{Y'} \cdot C = 0$ (cf. [MM83, Lemma 4.4 and Proposition 4.5]). After twisting, we may assume $Y' = \mathbb{P}_Z(\mathcal{F})$ with

$\mathcal{F} \cong \mathcal{O}_Z \oplus \mathcal{O}_Z(a)$ and $a \leq 0$. Then by the relative canonical bundle formula,

$$K_{Y'} = -2\xi + q_0^*(K_Z + \det \mathcal{F}) \sim -2\xi + (a - 3)q_0^*H,$$

where $H \subseteq Z$ is a line and $\xi \cong \mathcal{O}_{Y'}(1)$ is the tautological divisor which is a section of q_0 (with respect to the surjection $\mathcal{F} \rightarrow \mathcal{O}_Z(a)$; thus $\mathcal{O}_{Y'}(\xi)|_\xi \cong \mathcal{O}_\xi(a)$). Then

$$0 = K_{Y'} \cdot C = -2(\xi \cdot C) + (a - 3)H \cdot (q_0)_*C < -2(\xi \cdot C).$$

This implies $\xi \cdot C < 0$ and hence $C \subseteq \xi$. Take a curve $\ell \subseteq \varphi^{-1}(\xi) \subseteq Y$ such that $\varphi_*\ell \equiv tC$ on $\xi \cong \mathbb{P}^2$ for some $t \in \mathbb{Z}_{>0}$. Then $E_\varphi \cdot \ell \geq 0$ and

$$K_Y \cdot \ell = \varphi^*K_{Y'} \cdot \ell + E_\varphi \cdot \ell \geq K_{Y'} \cdot \varphi_*\ell = (K_{Y'}|_\xi \cdot tC)_\xi = tK_{Y'} \cdot C = 0,$$

This is absurd and our lemma is proved. $\square$

In the rest of this section, we will focus on the proof of Theorem 6.7.2. During the proof, we stick to Notation 6.6.1 except that our p here is a $\mathbb{P}^2$ -bundle.

Proof of Theorem 6.7.2. By Lemma 6.5.2 and Theorem 6.7.1, we assume $E_\varphi \subseteq \text{Supp } \Delta_\tau$. Further, we have the following f -equivariant commutative diagram after iteration.

$$\begin{array}{ccccc} X & \xrightarrow{\pi} & W = \mathbb{P}_Y(\mathcal{E}) & \xrightarrow{\psi} & W' = \mathbb{P}_{Y'}(\mathcal{E}') \\ & \searrow \tau & \downarrow \tau_0 & & \downarrow \tau'_0 \\ & & Y & \xrightarrow{\varphi} & Y' \cong \mathbb{P}^3 \\ & & \downarrow p & & \\ & & Z \cong \mathbb{P}^1 & & \end{array}$$

Note that there is a (smooth) $\mathbb{P}^2$ -bundle $p: Y \rightarrow Z \cong \mathbb{P}^1$ and $\Delta_\tau \subseteq Y$ is a disjoint union of r components with $r = \rho(X) - \rho(W)$. Since every $(f|_{Y'})^{-1}$ -invariant prime divisor on Y' is an (ample) hyperplane (cf. [Hör17, Corollary 1.2]), we have $r = 1$ (i.e., $\Delta_\tau = E_\varphi$), and π is a single blow-up along a surface $\bar{S} \subseteq W$ (cf. [CMZ20, Lemma 7.5]).

For each $z \in Z$, the fibre X_z is a (smooth) Fano threefold and $\pi|_{X_z}$ is the blow-up of $W_z = \mathbb{P}_{Y_z}(\mathcal{E}|_{Y_z})$ along a (smooth) rational curve $C_z := W_z|_{\bar{S}}$. Take an $f|_Z$ -periodic point $z_0 \in Z$ (cf. [Fak03, Theorem 5.1]). After iteration, f restricts to an int-amplified

endomorphism on $W_{z_0} \cong \mathbb{P}_{Y_{z_0}}(\mathcal{E}|_{Y_{z_0}})$. Hence, it follows from [Ame03, Proposition 3] that $\mathcal{E}|_{Y_{z_0}}$ splits, noting that $Y_{z_0} \cong \mathbb{P}^2$. By Lemma 6.7.3, W_{z_0} is Fano. Note that $(p \circ \tau_0)|_{\bar{S}}$ is flat onto a smooth curve Z . Then the C_z 's are numerically equivalent on $\bar{S}$ and hence on W . Now for every $z \in Z$, applying the adjunction, we have

$$(K_{W_z} \cdot C_z)_{W_z} = (K_W|_{W_z} \cdot C_z)_{W_z} = K_W \cdot C_z = K_W \cdot C_{z_0} = (K_{W_{z_0}} \cdot C_{z_0})_{W_{z_0}} < 0.$$

Therefore, every W_z is Fano (cf. [MM83, Lemma 4.4 and Proposition 4.5]). On the other hand, Fano $\mathbb{P}^1$ -bundles over $\mathbb{P}^2$ cannot deform to each other (cf. e.g., [MM83]). So for all $z \in Z$, $W_z \cong \mathbb{P}_{Y_z}(\mathcal{E}|_{Y_z}) \cong \mathbb{P}_{Y_z}(\mathcal{O} \oplus \mathcal{O}(-k))$ with $k = 0, 1$ or 2 (cf. [SW90]).

Let $F \cong \mathbb{P}^2$ and $S \cong \mathbb{F}_0$ be the fibre class of p and the exceptional divisor of φ , respectively. Up to a twist, we can write $c_1(\mathcal{E}) = aS + bF$ with $-1 \leq a, b \leq 0$. Then $\mathcal{O}_{Y_z}(c_1(\mathcal{E}|_{Y_z})) \cong \mathcal{O}_{Y_z}(a)$. By assumption, $S = E_\varphi = \Delta_\tau$; hence $\tau^*S = E + \tilde{E}$ with $E = E_\pi$ being π -exceptional. Now we refer to Lemma 6.5.2 and use the notations therein. By the projection formula, $(E + \tilde{E}) \cdot \ell_\chi < 0$, where $\ell_\chi \cong \mathbb{P}^1$ is a fibre of χ lying in R_χ . Since χ is a divisorial contraction, the locus of R_χ either equals E or equals $\tilde{E}$. In both cases, $E \cap \tilde{E}$ is contracted to a curve (cf. Claim 6.6.5). So we may assume $\ell_\chi \subseteq E \cap \tilde{E}$.

If the locus of R_χ equals $\tilde{E}$, then $\tilde{E} \cdot \ell_\chi = -1$ since χ is a blow-up. Since $\tau|_{\tilde{E} \cap E}$ is an isomorphism, for a fibre ℓ_φ of φ , we have

$$(E + \tilde{E}) \cdot \ell_\chi = \tau^*S \cdot \ell_\chi = S \cdot \ell_\varphi = -1.$$

So $E \cdot \ell_\chi = 0$ and thus $K_W \cdot \ell_\psi = K_X \cdot \ell_\chi < 0$. Then, ψ is a K_W -negative contraction. By Lemma 5.2 (iii) (or [Cas08, Theorem 3.14] for $\tau'_0 \circ \psi = \varphi \circ \tau_0$), we have $\mathcal{E} = \varphi^*\mathcal{E}'$. Since $f|_{W'}$ is int-amplified, $\mathcal{E}'$ and hence $\mathcal{E}$ split (cf. [Ame03, Proposition 3]).

Thus, we may assume that the locus of R_χ equals E , $E \cdot \ell_\chi = -1$, and then $\tilde{E} \cdot \ell_\chi = 0$. Then $\chi|_{\tilde{E}}$ induces a divisorial contraction contracting $E \cap \tilde{E}$, noting that $K_{\tilde{E}} \cdot \ell_\chi = K_X \cdot \ell_\chi < 0$. Let $H := \pi_*^{-1}(\tau_0^{-1}\ell_\varphi) \subseteq \tilde{E}$ be the $\mathbb{P}^1$ -bundle over $\ell_\varphi \cong \mathbb{P}^1$. Since $H \cap (E \cap \tilde{E})$ is contracted by χ , our $\chi|_H$ is a (smooth) blow-down on H ; hence $\mathbb{P}_{\ell_\varphi}(\mathcal{E}|_{\ell_\varphi}) = \tau_0^{-1}(\ell_\varphi) \cong H \cong \mathbb{F}_1$. Now $c_1(\mathcal{E})|_{\ell_\varphi} = (aS + bF)|_{\ell_\varphi} = b - a$ and $-1 \leq a, b \leq 0$ imply $b - a = \pm 1$. Since $W_z = \mathbb{P}_{Y_z}(\mathcal{E}|_{Y_z}) \cong \mathbb{P}_{Y_z}(\mathcal{O} \oplus \mathcal{O}(-k))$ with $k = 0, 1$ or 2 , there are three cases:

- (1) $a = -1, b = 0, c_1(\mathcal{E}) = -S$ and $\mathcal{E}|_{Y_z} = \mathcal{O} \oplus \mathcal{O}(-1)$ for all $z \in Z$;
- (2) $a = 0, b = -1, c_1(\mathcal{E}) = -F$ and $\mathcal{E}|_{Y_z} = \mathcal{O} \oplus \mathcal{O}$ for all $z \in Z$;
- (3) $a = 0, b = -1, c_1(\mathcal{E}) = -F$ and $\mathcal{E}|_{Y_z} = \mathcal{O}(1) \oplus \mathcal{O}(-1)$ for all $z \in Z$.

In Case (3), replacing $\mathcal{E}$ by $\mathcal{E} \otimes \mathcal{O}_Y(-S)$, we may assume

- (3') $a = -2, b = -1, c_1(\mathcal{E}) = -2S - F$ and $\mathcal{E}|_{Y_z} = \mathcal{O} \oplus \mathcal{O}(-2)$ for all $z \in Z$.

In all the cases, the function $z \mapsto h^0(Y_z, \mathcal{E}|_{Y_z})$ is constant on Z . With the same argument as in Proof of Theorem 6.6.2, global sections of $\mathcal{E}|_{Y_z}$ are constant in the cases: (1), (2) and (3'). Then we get the same exact sequence (*).

If Case (2) occurs, then $\mathcal{E}|_{Y_z} \cong \mathcal{O} \oplus \mathcal{O}$ for all z ; hence $p_*\mathcal{E}$ is locally free of rank 2 (on $\mathbb{P}^1$) and splits always, which implies that $\mathcal{E} \cong p^*p_*\mathcal{E}$ splits.

If Cases (1) or (3') occur, then $p_*\mathcal{E}$ is a line bundle, say $\mathcal{O}_Z(e)$ for some $e \in \mathbb{Z}$. Then $\mathcal{F}_1 := p^*p_*\mathcal{E} \cong \mathcal{O}_Y(eF)$ and $\mathcal{Q} \cong \mathcal{O}_Y(aS + (b - e)F)$. Note that

$$\text{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E}) = H^1(Y, -aS + (2e - b)F) = H^1(Y, K_Y + (-aS - K_Y) + (2e - b)F).$$

Since $(-aS - K_Y) + (2e - b)F \equiv (3 - a)S + (4 - b)F + 2eF$, it is nef and big (for both cases (1) and (3')) when $e \geq 0$. By the Kawamata–Viehweg vanishing theorem (cf. e.g., [KM98, Theorem 2.64]), $\text{Ext}^1(\mathcal{Q}, p^*p_*\mathcal{E})$ vanishes if $e \geq 0$. So $e \geq 0$ implies that (*) and hence $\mathcal{E}$ split. Therefore, we may assume $e < 0$. Then $\xi_{\mathcal{F}_1} := 2c_1(p^*p_*\mathcal{F}_1) - c_1(\mathcal{E}) \sim -aS + (2e - b)F$. Since $e < 0$, no matter (1) or (3') occurs, $2e - b < 0$. So our theorem follows from Claim 6.7.4 and Lemma 6.6.6.

Claim 6.7.4. $W_{\mathcal{E}}(\mathcal{F}_1) \neq \emptyset$ when $t := 2e - b < 0$.

Proof of Claim 6.7.4. Note that any ample divisor D on Y can be written as $D = D(u) := S + uF$ with $u > 1$ after replacing D by a multiple. So we have:

$$\xi_{\mathcal{F}_1} \cdot D(u)^2 = -aS^3 + (-2au + t)S^2 \cdot F = -2a(u - 1) + t.$$

The last equality is due to $\mathcal{O}_Y(S)|_S \cong \mathcal{O}_S(-1, 1)$, noting that for a fibre C_0 of $p|_S$ (which is the second ruling of S different from $\varphi|_S$), we have $S \cdot C_0 = 1$ by the ramification divisor formula of φ and the adjunction. Since $-2a > 0$ and $t < 0$, there exists a real number $u_0 > 1$ such that $\xi_{\mathcal{F}_1} \cdot D(u_0)^2 = 0$. So our claim holds. $\square$

6.8 Splitting-ness of algebraic $\mathbb{P}^1$ -bundles

Combining with the results in Sections 6.6 and 6.7, our main focus in this section is to show Theorem 6.8.1, which is a key to proving Theorem 6.1.3.

Theorem 6.8.1. *Let X be a smooth Fano fourfold admitting an int-amplified endomorphism f . Let $\tau: X \rightarrow Y$ be a conic bundle which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ as in Notation 6.6.1 (1). If one of the following holds, then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

- (1) $\rho(X) - \rho(W) \neq 1$, i.e., either $\tau = \tau_0$ is elementary or $\rho(X) - \rho(W) \geq 2$;
- (2) $Y \cong \mathbb{P}^3$ or the blow-up of a line on $\mathbb{P}^3$;
- (3) Y is a $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$, $\mathbb{F}_0$ or $\mathbb{F}_1$.

Furthermore, if τ_0 is not a splitting $\mathbb{P}^1$ -bundle, then $(\ddagger)$ there exists another conic bundle $\widehat{\tau}: X \rightarrow \widehat{W} \rightarrow \widehat{Y}$ with $\widehat{\tau}_0: \widehat{W} \rightarrow \widehat{Y}$ being elementary such that $\rho(X) - \rho(\widehat{W}) = 2$. In particular, $\widehat{W}$ is a splitting $\mathbb{P}^1$ -bundle over Y .

Remark 6.8.2. (1) We shall assume Notation 6.6.1 throughout this section. The first part of Theorem 6.8.1 follows from Theorems 6.6.2, 6.7.1, 6.7.2, 6.8.3, and 6.8.4.

- (2) We are unable to show the splitting-ness of τ_0 when π is a single blow-up, since Lemma 6.5.2 cannot be applied if $E_\varphi \subseteq \Delta_\tau$; see Remark 6.5.3.

Theorem 6.8.3. *Suppose that $X = W$ (which is Fano). Then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

Proof. By Theorem 6.1.1 and Lemmas 6.4.3, 6.4.4, we only need to consider when Y is $\mathbb{P}^3$ or a $\mathbb{P}^1$ -bundle over $\mathbb{P}^2$ or $\mathbb{F}_0$. This is clear by Theorems 6.7.1 and 6.6.2. $\square$

Theorem 6.8.4. *Suppose that $\rho(X) - \rho(W) \geq 2$. Then τ_0 is a splitting $\mathbb{P}^1$ -bundle.*

Proof. If $\rho(X/W) \geq 3$, then X is a product of del Pezzo surfaces and thus our theorem follows (cf. Lemma 6.2.9). So we may assume that $\rho(X/W) = 2$ and X does not split. By Lemma 2.9 (2), Y is a smooth $\mathbb{P}^1$ -bundle over a del Pezzo surface Z . By Lemmas 6.4.3

and 6.5.2, we have the following f -equivariant commutative diagram after iteration,

$$\begin{array}{ccc}
 X & \xrightarrow{\chi} & X' \\
 \downarrow \pi & \searrow \eta & \downarrow \pi' \\
 W & \xrightarrow{\psi} & W' \\
 \downarrow \tau_0 & & \downarrow \tau'_0 \\
 Y & \xrightarrow{\varphi} & Y' \\
 \downarrow p & & \downarrow p' \\
 Z & \xrightarrow{\phi} & Z'
 \end{array}
 \quad (6.1)$$

where ϕ is the blow-up of a point on a Fano surface Z' and φ is the blow-up of a smooth curve on a smooth Fano threefold Y' with the exceptional $E_\varphi = p^*E_\phi$. By Theorem 6.6.2, we may further assume Z is a toric Fano surface of degree 6 or 7 ([Nak02, Theorem 3]).

We claim that E_φ and Δ_τ have no common component. Suppose on the contrary that there exists a surface S as a component of both E_φ and Δ_τ . Note that S is a $\mathbb{P}^1$ -bundle over $C := \varphi(S)$ and hence $\rho(S) = 2$. Consider the proper transform $D := \pi_*^{-1}(\tau_0^*S)$, which is a prime divisor of Picard number 3 on X . Note also that $\rho(X) = \rho(Z) + 4 \geq 7$. Then the Lefschetz defect $\delta_X \geq 4$ and hence $X \cong S_1 \times S_2$ with S_i being del Pezzo surfaces (cf. Lemma 6.2.7), contradicting our assumption at the beginning of the proof.

By Lemma 6.5.2, τ' is a Fano conic bundle, $W' = \mathbb{P}_{Y'}(\mathcal{E}')$ and $\mathcal{E} = \varphi^*\mathcal{E}'$; thus our theorem follows from Theorem 6.6.2 **Case A**: $Z \cong \mathbb{P}^2$ (and induction on $\rho(Z)$). $\square$

Proof of Theorem 6.8.1. By Remark 6.8.2, we only need to show the second part. Since τ_0 is not a splitting $\mathbb{P}^1$ -bundle, we have $\rho(X) - \rho(W) = 1$ (and hence Δ_τ is irreducible) by the first part. In the light of Theorems 6.1.1, 6.6.2, 6.7.1, 6.7.2 and Lemma 6.2.8, we may further assume either one of the following.

- (a) Y admits a non-elementary conic bundle $Y \rightarrow Z$ to a smooth Fano surface Z ;
- (b) Y is a $\mathbb{P}^1$ -bundle over Z where Z is a del Pezzo surface with degree 6 or 7.

Note that, in both cases, Z is a toric Fano surface (cf. [Nak02, Theorem 3]).

Suppose (a) occurs. We will use [MM83, Proposition 9.10]. Then $\rho(Z) \leq 2$ and we have two small cases: (i) Δ_τ is not φ -exceptional, or (ii) Δ_τ is φ -exceptional.

If (i) occurs, Lemma 6.5.2 and Theorem 6.6.2 would imply $\mathcal{E}$ splits, a contradiction.

If (ii) occurs, then we have the following commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{\tau} & Y & \xrightarrow{p} & Z \\ \pi \searrow & & \tau_0 \searrow & & \varphi \searrow \\ & W & \xrightarrow{\varphi} & Y' & \xrightarrow{p_0} Z \end{array} \quad (6.2)$$

where Y' is smooth Fano, $Z \cong \mathbb{P}^2$, $\mathbb{F}_0$ or $\mathbb{F}_1$, τ and p are singular conic bundles, and τ_0 and p_0 are smooth $\mathbb{P}^1$ -bundles (cf. Theorem 6.5.1 and [MZZ21, Theorem 6.2]).

We claim that the Lefschetz defect $\delta_X \geq 3$. Let $D_{Y'}$ be a section of p_0 and disjoint from $\varphi(\Delta_\tau)$ (cf. Theorem 6.1.1 (B)). Then the irreducible divisor $\tau^*\varphi_*^{-1}D_{Y'}$ has Picard number $\rho(Z) + 1$. On the other hand, the inequality $\rho(X) \geq \rho(Z) + 4$ implies that $\delta_X \geq \rho(X) - \rho(\tau^*\varphi_*^{-1}D_{Y'}) \geq 3$; hence our claim holds.

By Lemma 6.2.7, there exists another Fano conic bundle $X \rightarrow \hat{Y}$ such that $\rho(X) - \rho(\hat{Y}) = 3$. Together with Theorems 6.5.1 and 6.8.4, $(\ddagger)$ holds.

Suppose (b) occurs. Then we also have two small cases: (i') Δ_τ is a section of $p: Y \rightarrow Z$, or (ii') Δ_τ is the pull-back of an $(f|_Z)^{-1}$ -invariant curve on Z (cf. Corollary 6.3.3).

If (i') happens, then we have the commutative diagram (6.1). Since Δ_τ is the section of p , it is not φ -exceptional. Hence $X' \rightarrow Y'$ is a Fano conic bundle, $W \cong W' \times_{Y'} Y$ and $\varphi(\Delta_\tau)$ is a section of $Y' \rightarrow Z'$ (cf. Lemma 6.5.2). After one more reduction if necessary, we may assume $Z' \cong \mathbb{P}^2$. So $\mathcal{E}'$ and hence $\mathcal{E}$ split by Theorem 6.6.2, a contradiction.

If (ii') happens, then choosing a rational curve C disjoint from $p(\Delta_\tau)$ in Z (this is doable by considering Z as blow-ups from $\mathbb{P}^2$), we have $\rho(X) - \rho(\tau^*p^*C) = \rho(Z) \geq 3$. Applying Lemma 6.2.7 again, there exists another Fano conic bundle $X \rightarrow \hat{Y}$ such that $\rho(X) - \rho(\hat{Y}) = 3$. By Theorem 6.8.4, we have $(\ddagger)$. $\square$

6.9 Proofs of Theorem 6.1.3 and Corollary 6.1.4

To prove Theorem 6.1.3, we begin with the theorem below. For its proof, we follow the idea of [MZZ21, Proof of Theorem 8.1].

Theorem 6.9.1. *Let X be a smooth Fano fourfold admitting a conic bundle $\tau: X \rightarrow Y$. Suppose that X admits an int-amplified endomorphism f . Then X is toric.*

Proof. We follow the notations in Theorem 6.5.1. If $\rho(X) - \rho(Y) = 1$, then X is a splitting $\mathbb{P}^1$ -bundle over the smooth toric Fano threefold Y (cf. Lemma 6.2.8, Theorems 6.8.3 and 6.1.1); thus X is toric by [MZZ21, Proposition 2.9]. If $\rho(X) - \rho(Y) \geq 4$, then X is a product of del Pezzo surfaces (cf. Lemma 2.9(1)); hence $f = f_1 \times f_2$ after iteration, noting that $\overline{\text{NE}}(X)$ is rational polyhedral. So it follows from Lemma 6.2.3 and [Nak02, Theorem 3] that both S_i are toric. As a result, X is also toric. In the following, we may assume $r := \rho(X) - \rho(Y) - 1 = 1$ or 2 . By Theorem 6.5.1, $\tau^{-1}(D_i) = E_i \cup \widetilde{E}_i$, and $E_i, \widetilde{E}_i$ are (smooth) $\mathbb{P}^1$ -bundles over D_i , with $E_i \cap \widetilde{E}_i \cong D_i$ being an f^{-1} -invariant surface.

Case: $r = 2$. By Theorem 6.8.4, $X_r = \mathbb{P}_Y(\mathcal{E})$ is a splitting $\mathbb{P}^1$ -bundle over Y ; hence after a suitable twisting, we may assume $\mathcal{E} = \mathcal{O}_Y \oplus \mathcal{L}$ with $\mathcal{L}$ being trivial or not pseudo-effective. We claim that there is an f^{-1} -invariant section S of τ dominating Y . Note that $\pi(E_1 \cap \widetilde{E}_1)$ is an f_r^{-1} -invariant subsection over Y (not contracted by τ_0) and $\pi(E_1 \cup \widetilde{E}_1) = \tau_0^{-1}(D_1)$ is a $\mathbb{P}^1$ -bundle over D_1 . If τ_0 is a trivial bundle so that $X_r = Y \times Z \cong Y \times \mathbb{P}^1$, then $\pi(E_1 \cup \widetilde{E}_1) \cong D_1 \times \mathbb{P}^1$. Now $\pi(E_1 \cap \widetilde{E}_1)$ is contained in an f_r^{-1} -invariant horizontal section S_r of τ_0 (cf. [CMZ20, Lemma 7.5] and Lemma 6.2.4 applied to $\pi(E_1 \cup \widetilde{E}_1)$). If τ_0 is not a trivial bundle, then some section S_r has $S_r|_{S_r} \cong \det \mathcal{E} = \mathcal{L}$ being not pseudo-effective and hence is f_r^{-1} -invariant after iteration by [MZZ21, Lemma 2.3].

In both cases, let $S \subseteq X$ be the proper transform of S_r . Note that the surfaces in W blown up by π are either contained in S_r or disjoint from S_r . Indeed, if there is some $\overline{D}_i$ intersects S_r along a curve C , which is also f_r^{-1} -invariant, then the three f_r^{-1} -invariant prime divisors $\overline{D}_i, \tau_0^{-1}(\tau_0(C))$ and $\tau_0^{-1}(\tau_0(\overline{D}_i)) \cap S_r$ in the f_r^{-1} -invariant smooth projective threefold $\tau_0^{-1}(\tau_0(\overline{D}_i))$ will have a common intersection curve C , a contradiction to Lemma 6.2.5. So we get the f^{-1} -invariant section $S \cong S_r$ of τ . Since $\pi(S) = Y$, we have $(E_i \cup \widetilde{E}_i) \cap S \neq \emptyset$. Hence we may assume $S \cap E_i \neq \emptyset$ for each i . By Theorem 6.5.1 (cf. [Rom19a, Remark 3.6]), after iteration, there is an f -equivariant birational morphism $\pi': X \rightarrow X'$ over Y contracting all the E_i with $f' := f|_{X'}$, such that:

- (a) the induced morphism $\tau'_0: X' \rightarrow Y$ is an algebraic $\mathbb{P}^1$ -bundle;
- (b) π' is the blow-up of X' along 2 smooth projective surfaces $D'_i := F'_i \cap S'$, where

$F'_i := (\tau'_0)^{-1}(D_i)$ and $S' := \tau'_0(S)$ are $(f')^{-1}$ -invariant prime divisors.

Since $\rho(X) - \rho(X') = 2$, it follows from Theorem 6.8.4 that X' is a splitting $\mathbb{P}^1$ -bundle over Y ; thus X' is toric (cf. [MZZ21, Proposition 2.9]). By Corollary 6.3.3, there is a toric pair (X', Δ') such that Δ' contains all the $(f')^{-1}$ -invariant prime divisors (including F'_i, S'). By the construction, π' is the composition of toric blow-ups of the intersection of prime divisors in the toric boundary starting from (X', Δ') . Thus X is toric.

Case: $r = 1$. By Theorem 6.5.1, τ factors as $\tau_0: X_1 \rightarrow Y$ (resp. $\tilde{\tau}_0: \tilde{X}_1 \rightarrow Y$). If both τ_0 and $\tilde{\tau}_0$ are splitting $\mathbb{P}^1$ -bundles, then we are done with the same argument as above. Otherwise, by Theorem 6.8.1, there exists another conic bundle $X \rightarrow \hat{Y}$ such that $\rho(X) - \rho(\hat{Y}) = 3$. So we reduce to **Case:** $r = 2$ and our theorem holds. $\square$

Proof of Theorem 6.1.3. Clearly, (2) implies (3). By Theorem 6.9.1, (3) implies (1). Since every projective toric variety has a polarized endomorphism (cf. [Nak02, Lemma 4] and [MZ20b, Proof of Theorem 1.4]), (1) implies (2). $\square$

Proof of Corollary 6.1.4. By Theorem 6.5.1, $W = \mathbb{P}_Y(\mathcal{E})$ for some locally free rank two sheaf $\mathcal{E}$ on Y . Then our X is toric (cf. Theorem 6.1.3). Fix a toric action G on X such that it descends to a unique action G_Y on Y which is τ -equivariant (cf. [Bri11, Proposition 2.1]). Since X is toric and $X \rightarrow W$ has connected fibres, by [Bri11, Proposition 2.1] again, we see that W is toric with the (uniquely descended) toric action G_W on W . By [Dru99, Lemma 1], $\mathcal{E}$ splits. $\square$

We end up this chapter with the following general remark on Question 4.1.2.

Remark 6.9.2. To give a positive answer to our Question 4.1.2, the first step is to consider the smooth projective variety of Fano type admitting a conic bundle structure (cf. [Yos20, Corollary 1.4]). However, even in dimension three, there exists a gap where the birational contraction may possibly be a small contraction (cf. the discussion in Section 5.4).

Totally invariant divisors of int-amplified endomorphisms

7.1 Introduction

In this chapter, we work over the field $\mathbb{C}$ of complex numbers. We shall extend the early result [Zha14, Theorem 1.3] for the case of polarized endomorphisms to the case of int-amplified endomorphisms. The main method we use is to run the minimal model program (MMP for short) equivariantly so that we can reduce the dimension of varieties and then use the induction. It is accessible under some good conditions for the variety X (cf. [Men20, Theorem 1.10]).

Let X be a projective variety and $f : X \rightarrow X$ a surjective endomorphism. Recall that the endomorphism f is said to be *int-amplified*, if there exist ample Cartier divisors H and L such that $f^*L - L = H$. A prime divisor V on X is said to be f^{-1} -invariant (or *f-totally invariant*), if $f^{-1}(V) = V$ set-theoretically. To simplify the expression, we first introduce a notation for the set of f -totally invariant prime divisors for the int-amplified endomorphism $f : X \rightarrow X$:

$$\text{TI}_f(X) := \{V \mid V \text{ is a prime divisor on } X \text{ such that } f^{-1}(V) = V\}.$$

In the following, we will bound the cardinality $c := \#\text{TI}_f(X)$. It turns out that there

exists an optimal upper bound which is determined by the Picard number and dimension of X . Also, we want to give a sufficient condition for $f^*|_{\text{Pic}(X)}$ to be diagonalizable over $\mathbb{Q}$ by applying some early results. Moreover, under some extra conditions, we show that the variety X is rationally connected and simply connected with respect to complex topology.

Now we state our main result of this chapter.

Theorem 7.1.1 ([Zho21c, Theorem 1.1]). *Let X be a projective variety of dimension n with only $\mathbb{Q}$ -factorial Kawamata log terminal singularities, and $f : X \rightarrow X$ an int-amplified endomorphism. Let V_j ($1 \leq j \leq c$) be all the prime divisors in $\text{TI}_f(X)$. Then we have (with $\rho := \rho(X)$):*

- (1) $c \leq n + \rho$. Furthermore, if $c \geq 1$, then the pair $(X, \sum V_j)$ is log canonical and X is uniruled.
- (2) Suppose $c \geq n + \rho - 2$. Then either X is rationally connected and simply connected with respect to complex topology, or there is a fibration $X \rightarrow E$ onto an elliptic curve E such that every fibre is normal, irreducible, equi-dimensional and rationally connected. Furthermore, in the latter case, for some integer $t \geq 1$, f^t descends to an int-amplified endomorphism $g_E : E \rightarrow E$. In both cases, $(f^t)^*|_{\text{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{Q}$.
- (3) Suppose $c \geq n + \rho - 1$. Then X is rationally connected and simply connected with respect to complex topology. Furthermore, for some integer $t \geq 1$, $(f^t)^*|_{\text{Pic}(X)}$ is diagonalizable over $\mathbb{Q}$.
- (4) Suppose $c \geq n + \rho$. Then $c = n + \rho$, $K_X + \sum_{j=1}^{n+\rho} V_j \sim_{\mathbb{Q}} 0$ and f is étale outside $(\bigcup V_j) \cup f^{-1}(\text{Sing}(X))$, where $\text{Sing}(X)$ denotes the singular locus of X .

Note that, the upper bound $\rho(X) + \dim(X)$ of $c = \#\text{TI}_f(X)$ given in Theorem 7.1.1 is optimal. For example, one can easily consider the coordinate power map of $\mathbb{P}^1$.

The following corollary follows immediately from our main theorem and [BMSZ18, Theorem 1.2], in which case, the complexity (cf. [BMSZ18, Definition 1.1]) is zero. So $(X, \sum V_j)$ is a toric pair.

Corollary 7.1.2 ([Zho21c, Corollary 1.2]). *Let X be a normal projective variety with only $\mathbb{Q}$ -factorial Kawamata log terminal singularities, and $f : X \rightarrow X$ an int-amplified endomorphism of X . If the cardinality $c = \sharp \text{TI}_f(X)$ satisfies $c = \rho(X) + \dim(X)$ (achieving the upper bound) as in Theorem 7.1.1 (4), then $(X, \sum V_j)$ is a toric pair.*

Let $N^1(X) = \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$. Comparing with the case of polarized endomorphisms, we remove the hypothesis (cf. [Zha14, Theorem 1.3]): either $f^*|_{N^1(X)} = q \text{id}_{N^1(X)}$, or $n \leq 3$. In general, $f^*|_{N^1(X)}$ may not be a scalar matrix even in the case of polarized endomorphisms (cf. [MZ18a, Example 7.1]). In our case of int-amplified endomorphisms, we show that the diagonalizable result holds even when $c = \rho(X) + \dim(X) - 1$, which greatly strengthens [Zha14, Theorem 1.3]. Moreover, we add more details to [Zha14, Proofs of Theorem 1.3 and Proposition 2.12].

For the organization, we begin with several preliminaries in Section 7.2. Then we give a detailed proof for our Theorem 7.1.1 in Section 7.3.

This chapter is based on my work [Zho21c].

7.2 Preliminaries

7.2.1 Notation and terminology

Let X be a projective variety of dimension n over $\mathbb{C}$. Let D be a Cartier divisor on X , and $\mathcal{O}_X(D)$ its corresponding invertible sheaf. We often identify D with $\mathcal{O}_X(D)$. If X is $\mathbb{Q}$ -factorial, then any Weil divisor B on X is $\mathbb{Q}$ -Cartier, i.e., there exists some integer m , such that mB is Cartier. We use K_X to denote a canonical divisor of the variety X .

In correspondence with notations in [Har77], denote by $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^\circ(X)$ the Néron-Severi group of X . Let $\text{NS}_{\mathbb{Q}}(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$ and $N^1(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$. Denote by $q(X) := \dim H^1(X, \mathcal{O}_X)$ the irregularity.

Let $N_r(X)$ denote the quotient space of r -cycles with real coefficients modulo the weak numerical equivalence $\equiv_w$ (cf. [MZ18a, Definition 2.2]). When X is normal, we can regard $N^1(X)$ as a subspace of $N_{n-1}(X)$ (cf. [Zha16, Lemma 3.2]).

Let $f : X \rightarrow X$ be a (finite) surjective endomorphism of a projective variety X . We define the pull-back of r -cycles for f , such that f^* induces an automorphism of $N_r(X)$ and $f_* f^* = (\deg(f)) \text{id}$. More precisely, the pull-back $f^* : N_r(X) \rightarrow N_r(X)$ is defined by $f^* = (\deg(f)) (f_*)^{-1}$. When X is normal, one can also define the pull-back of a Weil divisor H , as the closure of $(f|_U)^*(H|_U)$ where $U \subseteq X$ is a smooth open locus on X and $X \setminus U$ is a codimension ≥ 2 closed subset. Furthermore, if H is $\mathbb{Q}$ -Cartier, the pull-back we discussed above coincides with the usual pull-back of a $\mathbb{Q}$ -Cartier divisor.

We recall the following $(f^*)^\pm$ -invariant cones (cf. [MZ18a, Definition 2.4]).

- **Amp**(X): the set of classes of ample $\mathbb{R}$ -Cartier divisors in $N^1(X)$;
- **Nef**(X): the set of classes of nef $\mathbb{R}$ -Cartier divisors in $N^1(X)$;
- **PE_r**(X): the closure of the set of classes of effective r -cycles with $\mathbb{R}$ -coefficients in $N_r(X)$.

The Theorem of the Base of Néron-Severi asserts that the real space **N₁**(X) of 1-cycles with real coefficients modulo numerical equivalence is a finite dimensional $\mathbb{R}$ -vector space. There is a natural perfect pairing:

$$N_1(X) \times N^1(X) \rightarrow \mathbb{R}.$$

Then $\text{rank}_{\mathbb{R}} N^1(X) = \text{rank}_{\mathbb{R}} N_1(X) =: \rho(X)$, the *Picard number* of X .

We use the symbol $\sim$ (resp. $\sim_{\mathbb{Q}}$, $\equiv$ or $\equiv_w$) to denote the linear (resp. $\mathbb{Q}$ -linear, numerical or weak numerical) equivalence relation. By [BG+71, CH. XIII, Theorem 4.6]), each numerically equivalent divisor is algebraically equivalent up to a positive multiple.

7.2.2 Int-amplified endomorphisms

In this subsection, we first recall the definitions of polarized, amplified and int-amplified endomorphisms. Then we refer to [Men20] for the general properties of int-amplified endomorphisms.

Definition 7.2.1 ([Men20]). Let $f : X \rightarrow X$ be a surjective endomorphism of a projective variety X . We say that

- (1) f is *polarized* if $f^*D \sim qD$ for some ample Cartier divisor D and integer $q > 1$;
- (2) f is *amplified* if $f^*D - D = H$ for some Cartier divisor D and ample Cartier divisor H ; and
- (3) f is *int-amplified* if $f^*D - D = H$ for some ample Cartier divisors D and H .

It follows from the definition that $(1) \Rightarrow (3) \Rightarrow (2)$. One can also check directly that if f is an int-amplified endomorphism, then any power of f is also int-amplified.

In what follows, we will recall the theorem below, which ensures that we can run equivariant MMP and then do the induction on the dimension of X . Recall that a normal projective variety X is said to be *$\mathbb{Q}$ -abelian* if there exists a finite surjective morphism $A \rightarrow X$ étale in codimension one (or *quasi-étale* in short) with A an abelian variety.

Theorem 7.2.2 ([Men20, Theorem 1.10]; cf. [MZ18a, Theorem 1.8]). *Let $f : X \rightarrow X$ be an int-amplified endomorphism of a $\mathbb{Q}$ -factorial klt projective variety X . Then replacing f by a positive power, there exist a $\mathbb{Q}$ -abelian variety Y , a morphism $X \rightarrow Y$, and an f -equivariant relative minimal model program over Y*

$$X = X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_l \dashrightarrow X_{l+1} = Y$$

which means a positive power of f descends to an endomorphism g_i on each X_i , for $1 \leq i \leq l+1$, with every $u_i : X_i \dashrightarrow X_{i+1}$ a divisorial contraction, a flip or a Fano contraction over Y , of a K_{X_i} -negative extremal ray. Further, we have:

- (1) If K_X is pseudo-effective, then $X = Y$ and it is $\mathbb{Q}$ -abelian.
- (2) If K_X is not pseudo-effective, then for each i , $X_i \rightarrow Y$ is equi-dimensional and holomorphic with every fibre (irreducible) rationally connected and g_i is int-amplified. The last rational map $X_l \dashrightarrow X_{l+1} = Y$ is a Fano contraction (a morphism).
- (3) $f^*|_{N^1(X)}$ is diagonalizable over $\mathbb{C}$ if and only if so is $g_{l+1}^*|_{N^1(Y)}$.

To make our symbols symmetric, we may denote by g_0 some power f^t , so that g_0 descends to the int-amplified endomorphism g_i of X_i for each i .

Remark 7.2.3. With the same symbols given above, we shall prove later that, for a divisorial contraction or a flip $X_i \dashrightarrow X_{i+1}$, we have the equality $\sharp\mathrm{TI}_{g_{i+1}}(X_{i+1}) = \sharp\mathrm{TI}_{g_i}(X_i) - (\rho(X_i) - \rho(X_{i+1}))$ (cf. Lemma 7.3.4), while for a Mori fibre contraction $X_r \rightarrow X_{r+1}$, the inequality $\sharp\mathrm{TI}_{g_{r+1}}(X_{r+1}) \geq \sharp\mathrm{TI}_{g_r}(X_r) - (\dim(X_r) - \dim(X_{r+1})) - 1$ holds (cf. Lemma 7.3.7). Here, $\sharp\mathrm{TI}_{g_i}(X_i)$ denotes the cardinality of the set $\mathrm{TI}_{g_i}(X_i)$ of totally invariant divisors for the int-amplified endomorphism $g_i : X_i \rightarrow X_i$.

When proving Theorem 7.2.2, the following statements for the case of int-amplified endomorphisms are useful.

Lemma 7.2.4 ([Men20, Lemma 3.5]). *Let $\pi : X \rightarrow Y$ be a generically finite and surjective morphism of projective varieties. Suppose $f : X \rightarrow X$ and $g : Y \rightarrow Y$ are two surjective endomorphisms such that $g \circ \pi = \pi \circ f$. Then f is int-amplified if and only if so is g .*

Lemma 7.2.5 ([Men20, Theorem 3.3]). *Let $f : X \rightarrow X$ be an int-amplified surjective endomorphism of a projective variety X . Then all the eigenvalues of $f^*|_{N^1(X)}$ are of modulus greater than 1.*

If X admits an int-amplified endomorphism f , then for any $D \in \mathrm{TI}_f(X)$, $f^*D = sD$ for some integer $s > 1$. Further, we have $\deg(f) > 1$ (cf. [Men20, Lemma 3.7]).

An amplified morphism was first defined by Krieger and Reschke (cf. [KR17]), and Fakhruddin showed the following very motivating result. Recall that a subset $Z \subseteq X$ is said to be f -periodic if $f^s(Z) = Z$ for some $s > 0$.

Lemma 7.2.6 ([Fak03, Theorem 5.1]). *Let X be a projective variety over an algebraically closed field k . Suppose $f : X \rightarrow X$ is a dominant morphism and $\mathcal{L}$ a line bundle on X such that $f^*\mathcal{L} \otimes \mathcal{L}^{-1}$ is ample. Then the subset $X(k)$ consisting of periodic points of f is Zariski dense in X .*

7.2.3 Rational connectedness of varieties

We refer to [Kol96, CH, IV, Definitions 1.1 and 3.2] for *uniruled varieties* and *rationally connected varieties*. Note that the rational connectedness implies the uniruledness, and

both of them are birational properties. Recall that a projective variety X of Fano type (i.e., there exists an effective $\mathbb{Q}$ -divisor D such that the pair (X, D) is klt and $-(K_X + D)$ is ample $\mathbb{Q}$ -Cartier) is rationally connected (cf. [Zha06a]).

Remark 7.2.7. If X is a normal projective variety with the canonical divisor K_X not pseudo-effective, then X is uniruled (cf. [BDPP13, 0.3 Corollary]). The converse direction also holds when X has only canonical singularities.

Recall that a path connected topological space X is *simply connected* if and only if its fundamental group $\pi_1(X)$ is trivial. Besides, *algebraic fundamental group* $\pi_1^{\text{alg}}(X)$ is a profinite completion of the topological fundamental group $\pi_1(X)$. A connected variety X over a separably closed field is said to be *algebraically simply connected* if it has no nontrivial connected finite étale cover, which is equivalent to $\pi_1^{\text{alg}}(X) = \{1\}$. The following lemma is known by the Universal Coefficient Theorem.

Lemma 7.2.8. *Suppose X is a smooth variety over $\mathbb{C}$ with trivial fundamental group. Then the irregularity $q(X) := h^1(X, \mathcal{O}_X) = \dim H^1(X, \mathcal{O}_X) = 0$.*

Further, the fundamental group of a klt variety is stable when we take the resolution.

Lemma 7.2.9 ([Tak03, Theorem 1.1]). *Let X be a normal variety and $f : Y \rightarrow X$ a resolution of singularities. Then the induced homomorphism $f_* : \pi_1(Y) \rightarrow \pi_1(X)$ is an isomorphism if the pair (X, Δ) is klt for some Δ .*

We end up this subsection with the following well-known theorem.

Theorem 7.2.10 ([Cam91, Theorem 3.5]). *A smooth proper rationally connected variety is simply connected with respect to complex topology.*

7.2.4 Properties for polarized cases

At the end of this preliminary, we consider a polarized endomorphism f on a normal variety X . First, we recall the following result.

Lemma 7.2.11 ([Zha14, Proposition 2.1]). *Let X be a normal variety, $f : X \rightarrow X$ a surjective endomorphism of $\deg(f) > 1$ and D a nonzero reduced divisor with $f^{-1}(D) = D$. Assume:*

- (1) X is log canonical around D ;
- (2) D is $\mathbb{Q}$ -Cartier; and
- (3) f is ramified around D .

Then the pair (X, D) is log canonical around D . In particular, the reduced D is normal crossing outside the union of $\text{Sing}(X)$ and a codimension three subset of X .

For a linear map $\phi : V \rightarrow V$ of a finite dimensional real normed vector space V , denote by $\|\phi\|$ the norm of ϕ . The following proposition gives us a criterion for ϕ to be diagonalizable.

Proposition 7.2.12 ([CMZ20, Proposition 3.1]; cf. [MZ18a, Proposition 2.9]). *Let $\phi : V \rightarrow V$ be an invertible linear map of a positive dimensional real normed vector space V . Assume $\phi(C) = C$ for a convex cone $C \subseteq V$ such that C spans V and its closure $\overline{C}$ contains no line. Let q be a positive number. Then the conditions (i) and (ii) below are equivalent.*

- (i) $\phi(u) = qu$ for some $u \in C^\circ$ (the interior part of C).
- (ii) There exists a constant $N > 0$, such that $\frac{\|\phi^i\|}{q^i} < N$ for all $i \in \mathbb{Z}$.

Assume further the equivalent conditions (i) and (ii). Then we have the following.

- (1) ϕ is a diagonalizable linear map with all eigenvalues of modulus q .
- (2) Suppose $q > 1$. Then for any $v \in V$ such that $\phi(v) - v \in C$, we have $v \in C$.

Corollary 7.2.13. *Let $f : X \rightarrow X$ be a polarized endomorphism on a normal projective variety X such that $f^*H \equiv qH$ for some $q > 0$ and an ample divisor H on X . Then the linear operation $f^*|_{N_{n-1}(X)}$ is diagonalizable with all eigenvalues of modulus q . In particular, if D is an effective reduced Weil divisor such that $f^{-1}D = D$, then with f replaced by its power, $f^*D_i = qD_i$ for each irreducible component D_i of D .*

Proof. We may regard $N^1(X)$ as a subspace of $N_{n-1}(X)$ (cf. [Zha16, Lemma 3.2]) and consider the invertible linear map $\phi = f^*|_{N_{n-1}(X)}$. Let $V := N_{n-1}(X)$ and $C := \text{PE}(X)_{n-1} \subseteq V$ as we defined in Subsection 7.2.1. Since H is ample, the volume $\text{vol}(H) > 0$ and thus H lies in the interior part of $\text{PE}_{n-1}(X)$ if we regard H as a big Weil $\mathbb{R}$ -divisor (cf. [FKL16, Theorem 3.5 (ii), (iii)]). In addition, $\text{PE}_{n-1}(X)$ spans the whole $N_{n-1}(X)$ and then by Proposition 7.2.12, ϕ is a diagonalizable linear map with all eigenvalues of modulus q .

Since D is f^{-1} -invariant, after replacing f by its power, we may assume $f^{-1}D_i = D_i$ for each irreducible component D_i of D . Therefore, we can see from the above discussion that $f^*D_i = \phi(D_i) = qD_i$ for each component D_i . $\square$

Remark 7.2.14. Indeed, the second part of Corollary 7.2.13 also follows easily from the projection formula. Besides, the above corollary will fail if we remove the condition that f is polarized. For example, consider the product $X = \mathbb{P}^1 \times \mathbb{P}^1$ and $f = f_1 \times f_2$, where f_i is the power map of $\mathbb{P}^1$, mapping $[x, y]$ to $[x^{q_i}, y^{q_i}]$ with $q_1 \neq q_2$. Let $D_1 = \{a_1\} \times \mathbb{P}^1$, $D_2 = \mathbb{P}^1 \times \{a_2\}$ and $D = D_1 + D_2$, where a_i is one of the two f_i^{-1} -invariant (coordinate) points. Then D is an ample Cartier divisor. However, by projection formula, $f^*D_i = q_i D_i$. Moreover, in this case, we can get four f^{-1} -invariant prime divisors.

The following proposition extends [Zha14, Proposition 2.12] to the case when V_j are reduced divisors. We first state this result and then do some preparations for its proof.

Proposition 7.2.15. *Let X be a normal projective variety of dimension $n \geq 2$, V_j ($1 \leq j \leq s$) reduced divisors (may have more than one component), and $f : X \rightarrow X$ a polarized endomorphism with $\deg(f) = q^n$ such that K_X is $\mathbb{Q}$ -Cartier and*

- (1) X has only log canonical singularities around $\bigcup V_j$;
- (2) every V_j is $\mathbb{Q}$ -Cartier and ample;
- (3) $f^{-1}(V_j) = V_j$ for all j ; and
- (4) V_i and V_j ($i \neq j$) have no common irreducible components.

Then $s \leq n + 1$; and $s = n + 1$ only if: f is étale outside $(\bigcup V_j) \cup f^{-1}(\text{Sing}(X))$ and $K_X + \sum_{j=1}^{n+1} V_j \equiv 0$.

Remark 7.2.16. With the same assumption in Proposition 7.2.15, the remaining necessary condition for $s = n + 1$ in [Zha14, Proposition 2.12] still holds: each V_j is irreducible and $\bigcap_{i=1}^t V_{b_i} \subseteq X$ is a normal irreducible subvariety for every subset $\{b_1, \dots, b_t\} \subseteq \{1, \dots, n + 1\}$ with $1 \leq t \leq n - 2$ (cf. [Zha14, Claim 2.11], Proof of Proposition 7.2.15 and Remark 7.2.19).

Before proving Proposition 7.2.15, we prepare several lemmas.

Lemma 7.2.17. *Let $f : X \rightarrow X$ be a polarized endomorphism on a normal projective variety X of dimension n and $\deg(f) = q^n > 1$. Then any Weil divisor M (not necessarily effective) on X such that $f^*M \equiv_w M$, is weakly numerically trivial. In particular, suppose further that M is $\mathbb{Q}$ -Cartier. Then $M \equiv 0$.*

Proof. Since f is polarized by an ample divisor H on X , the linear operation $f^*|_{N_{n-1}(X)}$ is diagonalizable with all eigenvalues of modulus $q > 1$ (cf. Corollary 7.2.13). Suppose $M \not\equiv_w 0$. Then 1 is an eigenvalue of the linear operation $f^*|_{N_{n-1}(X)}$, a contradiction. So $M \equiv_w 0$. Further, if M is $\mathbb{Q}$ -Cartier, then $M \equiv 0$ (cf. [Zha16, Lemma 3.2]). $\square$

Lemma 7.2.18 (cf. [Zha14, Lemma 2.7]). *Let $f : X \rightarrow X$ be a finite surjective endomorphism on a normal projective variety X of dimension $n \geq 1$. Suppose that M is a pseudo-effective Weil divisor and E is an effective $\mathbb{Q}$ -divisor such that the following weakly numerical equivalence $(*)$ holds:*

$$(*) \quad M \equiv_w f^*M + E.$$

Then the effective $\mathbb{Q}$ -divisor $E = 0$.

Proof. Suppose $E > 0$. Replacing $(*)$ by a positive multiple, we may assume E is integral. Substituting the above expression of M to the RHS $(k - 1)$ -times, we get

$$M \equiv_w (f^k)^*M + \sum_{i=0}^{k-1} (f^i)^*E.$$

Taking a fixed ample Cartier divisor H on X and then using Nakai–Moishezon criterion (cf. [KM98, Theorem 1.37]), we have the following

$$(M \cdot H^{n-1}) \geq \sum_{i=0}^{k-1} ((f^i)^* E \cdot H^{n-1}).$$

Since E is integral, the right-hand side tends to infinity if we let $k \rightarrow \infty$, a contradiction. Hence, $E = 0$. $\square$

Now, we begin to prove Proposition 7.2.15 (cf. [Zha14, Proof of Proposition 2.12]). Besides, readers may refer to [Zha14, Lemma 2.8] for a further proof of Remark 7.2.16.

Proof of Proposition 7.2.15. Suppose $s \geq n + 1$. If $n = 2$, then we may go to the end product X_{n-2} with $n = 2$. Therefore, we may further assume that initially $n \geq 3$ and $f^*H \sim qH$ for an ample divisor H on X . Since the number of V_j and the irreducible components of V_j are finite, with f replaced by its power, we may assume for each irreducible component B_{ij} of V_j , $f^{-1}(B_{ij}) = B_{ij}$. By Corollary 7.2.13, $f^*B_{ij} = qB_{ij}$. Note that Proposition 7.2.15 still holds with f replaced by its power since f is étale away from $\cup V_j$ if so is its power.

We reduce the dimension of X by continuously taking normalization of divisors. Then we prove Proposition 7.2.15 by an early result for the surface case. Let $D_0 = \sum V_j$ and consider the log ramification divisor formula for the pair (X, D_0) (cf. [Lit82, Theorem 11.5]):

$$K_X + D_0 = f^*(K_X + D_0) + \Delta_f, \quad (7.1)$$

where Δ_f is an effective (integral) divisor, having no common components with D_0 .

Suppose further $\Delta_f > 0$. Since V_1 is ample, $(V_1^{n-1} \cdot \Delta_f) > 0$. Fix a component B_0 of V_1 intersecting Δ_f and take the normalization of B_0 followed by the inclusion:

$$\sigma_1 : X_1 = \widetilde{B_0} \rightarrow B_0 \hookrightarrow X_0 = X.$$

Then one can get a diagram (by the universal property of normalization):

$$f_1^*(\sigma_1^*H) = \sigma_1^*(f^*H) \sim q\sigma_1^*H,$$

which means that this lifting f_1 is polarized by σ_1^*H and $\deg(f_1) = q^{n-1}$. Pulling back Equation (7.1) along the map σ_1 , we have the following:

$$K_{X_1} + D_1 := \sigma_1^*(K_X + D_0) = \sigma_1^*(f^*(K_X + D_0) + \Delta_f) = f_1^*(K_{X_1} + D_1) + \sigma_1^*\Delta_f.$$

Here, $D_1 = \text{Diff}(D_0)$ contains the reduced Weil divisor $\sum_{j \geq 2} \text{Supp } \sigma_1^*V_j$ (cf. [Sho92, Corollary 3.11], [Kol92, Corollary 16.7] and Lemma 7.2.11). Besides, (X_1, D_1) is log canonical by [Kaw07, Theorem 1]. Note that $V_j|_{B_0}$ ($j > 1$) is ample on B_0 and the normalization is a finite surjective morphism. Hence, each $\sigma_1^*V_j$ ($j > 1$) is nonzero and still ample on X_1 (cf. [KM98, Theorem 1.37]). Moreover, $\text{Supp } \sigma_1^*V_j$ ($j > 1$) is connected since $\dim(X_1) = n - 1 \geq 2$ (cf. [Har77, Corollary 7.9]).

By the choice of B_0 , $\Delta_f|_{B_0}$ is a nonzero effective $\mathbb{Q}$ -divisor on B_0 . Since the normalization is finite, $\sigma_1^*\Delta_f$ is a nonzero effective $\mathbb{Q}$ -divisor on X_1 . Repeatedly, we fix an (integral) irreducible component B_1 of $\text{Supp } \sigma_1^*V_2$ ($\leq D_1$) intersecting $\sigma_1^*\Delta_f$. This is possible since $\sigma_1^*V_2$ is ample on X_1 . With f replaced by its power, we may assume $f_1^{-1}(B_1) = B_1$. Since f_1 is polarized, by Corollary 7.2.13, we have $f_1^*B_1 = qB_1$. Taking the normalization of B_1 followed by the inclusion map, we get $\sigma_2 : X_2 \rightarrow X_1$ with $\dim(X_2) = n - 2$. Similarly, $f_1|_{B_1}$ lifts to a polarized endomorphism f_2 of X_2 with $\deg(f_2) = q^{n-2}$.

In general, let $\sigma_i : X_i \rightarrow X_{i-1}$ ($1 \leq i \leq n - 2$) be the normalization of an (integral) irreducible component B_{i-1} of $\text{Supp } (\sigma_1 \circ \cdots \circ \sigma_{i-1})^*V_i$ ($\leq D_{i-1}$) intersecting $(\sigma_1 \circ \cdots \circ \sigma_{i-1})^*\Delta_f$ followed by an inclusion map. Then we get $f_i : X_i \rightarrow X_i$ polarized by the pull-back $(\sigma_1 \circ \cdots \circ \sigma_i)^*H$, $\dim(X_i) = n - i$ and $\deg(f_i) = q^{n-i}$ (cf. [NZ10, Lemma 2.1]). Let σ be the composition,

$$\sigma := \sigma_1 \circ \cdots \circ \sigma_{n-2} : X_{n-2} \longrightarrow X_0 = X.$$

Now, $S := X_{n-2}$ is a normal surface with following ample reduced divisors on S :

$$C_i := \text{Supp } \sigma^*V_i \quad (n - 1 \leq i \leq s).$$

After replacing f by its power, we may assume $f_{n-2}^*C_{ij} = qC_{ij}$ for each irreducible component C_{ij} of C_i ($i \geq n - 1$). On the one hand, using the log ramification divisor

formula for the pair $(S, C := \sum_{i \geq n-1} C_i)$, we have

$$K_S + C = f_{n-2}^*(K_S + C) + \Delta. \quad (7.2)$$

Here, Δ is an effective (integral) Weil divisor, sharing no common components with C . On the other hand, pulling back Equation (7.1) along the map σ , we get

$$K_S + D_{n-2} := \sigma^*(K_X + D_0) = f_{n-2}^*(K_S + D_{n-2}) + \sigma^*\Delta_f. \quad (7.3)$$

Comparing Equation (7.2) with (7.3), we get $C \leq D_{n-2}$ (cf. [Sho92, Corollary 3.11] and [Kol92, Corollary 16.7]), and thus $M := D_{n-2} - C \geq 0$. By assumption, the number of C_i , $r \geq s - (n - 2) \geq 3$.

We will show that $s = n + 1$. Indeed, note that each C_i ($n - 1 \leq i \leq s$) is connected and ample, so $(C_i \cdot C_j) > 0$ for $n - 1 \leq i, j \leq s$. Let $D = \sum_{j=n-1}^{n+1} C_j \leq C$. Then the dual graph of D contains a loop. By [Zha14, Lemma 2.8], $K_S + D \sim 0$. Back to Equation (7.3), we have the following:

$$M + C - D \equiv_w f_{n-2}^*(M + C - D) + \sigma^*\Delta_f. \quad (7.4)$$

Since $\sigma^*\Delta_f$ is an effective $\mathbb{Q}$ -divisor, by Lemmas 7.2.17 and 7.2.18, $M + C - D \equiv_w 0$ and $\sigma^*\Delta_f = 0$. Thus, $M = C - D = 0$ by the effectivity of M and $C - D$. As a result, we have $D_{n-2} = C = D$, which in turn implies that $s = n + 1$.

Recall our initial assumption that $\Delta_f \neq 0$. Then by our choice of each B_i and σ , $\sigma^*\Delta_f$ is a nonzero effective $\mathbb{Q}$ -divisor on S . However, by Equation (7.4) and Lemma 7.2.18, we have already got $\sigma^*\Delta_f = 0$, a contradiction. Therefore, $\Delta_f = 0$ and the ramification divisor of f , $R_f = (q - 1)D_0$. By the purity of branch loci, f is étale outside $(\bigcup V_j) \cup f^{-1}(\text{Sing}(X))$. Further, since $\Delta_f = 0$, our $K_X + D_0 \equiv 0$ (cf. Equation (7.1) and Lemma 7.2.17), which completes the proof of Proposition 7.2.15. $\square$

Remark 7.2.19. For Proposition 7.2.15, when the equality holds, i.e., $s = n + 1$, we claim that D_1 (and then each D_i) is reduced. With the same symbols given above, we follow the ideas and steps of [Zha14, Proof of Theorem 1.1] to prove it as follows.

Proof. Note that $\Delta_f = 0$. Suppose $\Gamma_1 = mF_1 \leq D_1$ is a non-reduced fractional component with $m < 1$. Since $\sigma_1^*V_2$ is ample on X_1 , the intersection number $(\Gamma_1 \cdot (\sigma_1^*V_2)^{\dim(X_1)-1}) > 0$. Therefore, there exists an (integral) irreducible component of $\text{Supp } \sigma_1^*V_2 (\leq D_1)$, intersecting F_1 . Let this component be our new B_1 as in Proof of Proposition 7.2.15. Then taking the normalization of B_1 followed by the inclusion map, we get

$$K_{X_2} + D_2 = f_2^*(K_{X_2} + D_2).$$

Then we get $0 \neq \Gamma'_2 := \sigma_2^*\Gamma_1 \leq D_2$ by the choice of B_1 . Fix an irreducible component Γ_2 of Γ'_2 , which has no common components with $\text{Supp } \sigma_1^*V_j (j \geq n-1)$.

In general, suppose we have fixed a component Γ_k of the pull-back of Γ_{k-1} . Consider the pull-back of V_{k+1} , which is ample on X_k . Choosing the (integral) irreducible component B_k which intersects Γ_k , we take the normalization of B_k followed by the inclusion.

Finally, we get a component $\Gamma_{n-2} \leq D_{n-2}$ on S . Since we have proved $\Delta_f = 0$ for the case when $s = n+1$ and it is independent of our choice of B_i and σ , we get

$$K_S + D_{n-2} = f_{n-2}^*(K_S + D_{n-2}).$$

Note that $K_S + D_{n-2}$ is the pull-back of $K_X + D_0$ under our new σ . Similarly, comparing the above equation with Equation (7.2) and (7.3), we have $M = D_{n-2} - C \geq 0$ and $f_{n-2}^*M = M$. Therefore, M is numerically trivial (cf. Lemma 7.2.17), which in turn gives $M = 0$. Therefore, $C = D_{n-2}$ and such Γ_{n-2} cannot exist since C is the support of the pull-back of $\sum_{j=n-1}^{n+1} V_j$ and thus there is no more space for Γ_{n-2} . As a result, our assumption is absurd. $\square$

We end up this section with the following lemma, which is known to Iitaka, Sommese, Fujimota and Nakayama.

Lemma 7.2.20 ([Nak08, Lemma 3.7.1]). *Let X be a normal projective variety of dimension n and $f : X \rightarrow X$ an endomorphism with $\deg(f) \geq 2$. Suppose the canonical divisor K_X is a pseudo-effective $\mathbb{Q}$ -Cartier divisor. Then f is étale in codimension one.*

7.3 Proof of Theorem 7.1.1

The whole section is devoted to the proof of Theorem 7.1.1. We follow the steps in [Zha14, Proof of Theorem 1.3]. The main idea is to run the MMP and use the induction on $\dim(X)$. In the beginning, we show a key proposition.

Proposition 7.3.1 (cf. [Zha14, Lemma 2.8]). *Let X be a rationally connected variety with only klt singularities. Then the Picard group $\text{Pic}(X)$ is torsion free and $\text{Pic}(X) \cong \text{NS}(X)$.*

To prove Proposition 7.3.1, we need the following well-known fact.

Lemma 7.3.2. *Suppose X is a normal projective variety. Then the set of $\mathbb{C}$ -points $X(\mathbb{C})$ is path-connected with respect to complex topology.*

Proof of Proposition 7.3.1. Take any resolution $g : S \rightarrow X$. Then since g is birational, our S is smooth and also rationally connected. Also, Theorem 7.2.10 tells us that S has a trivial fundamental group. By Lemma 7.2.8, the irregularity of S , $q(S) = 0$. Now, since X is klt, we have $q(X) = q(S) = 0$ (cf. [KM98, Theorem 5.22 and Definition 5.8]), which implies that $\text{Pic}^\circ(X) = 0$. Moreover, by Lemma 7.2.9, $\pi_1(X) \cong \pi_1(S) = \{1\}$.

Now, since $\text{Pic}^\circ(X) = 0$, $\text{Pic}(X) \cong \text{NS}(X)$, which is finitely generated. Suppose there exists some invertible sheaf $\mathcal{L} = \mathcal{O}(D)$, where D is Cartier such that $nD \sim 0$ for some (minimal) integer $n \geq 1$. Then there exists an unramified cyclic cover $p : X_{n,D} \rightarrow X$ of degree n , which is a finite étale morphism (cf. [KM98, Definition 2.49]). Since X is normal and projective, $X(\mathbb{C})$ is path-connected by Lemma 7.3.2. Then $\pi_1(X) = \{1\}$ implies that X is simply connected with respect to complex topology and thus algebraically simply connected, which means there does not exist any nontrivial connected finite étale cover. Therefore, $n = 1$ and $\mathcal{O}(D)$ is trivial. In conclusion, the Picard group of any rationally connected variety with only klt singularities over $\mathbb{C}$ is torsion free. This completes the proof of Proposition 7.3.1. $\square$

Now, we use the induction on $\dim(X)$ to prove Theorem 7.1.1.

Lemma 7.3.3. *Theorem 7.1.1 holds for the case when $\dim(X) = 1$.*

In this case, X being normal is equivalent to X being smooth. We give the proof of Lemma 7.3.3 as follows.

Proof. It's well known that a curve with the genus $g(X) > 1$ does not admit any non-isomorphic surjective endomorphisms. Since $f : X \rightarrow X$ is int-amplified, $\deg(f) > 1$ (cf. [Men20, Lemma 3,7]). Hence, there are two cases.

Case (a). The genus of X , $g(X) = 1$, i.e., X is an elliptic curve. In this case, Theorem 7.1.1 (1) naturally holds, since $c = 0$ by Hurwitz's Theorem. For Theorem 7.1.1 (2), since $\text{NS}(X) \cong \mathbb{Z}$ for the elliptic curve X (cf. [Mum74, pp.165 Corollary 2]), $f^*|_{\text{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{Q}$. Further, Theorem 7.1.1 (3) and (4) cannot happen in this case.

Case (b). The genus of X , $g(X) = 0$. Then $X \cong \mathbb{P}^1$. By Hurwitz's Theorem,

$$K_X + \sum V_i = f^*(K_X + \sum V_i) + R'_f,$$

with R'_f effective. Taking the degree of both sides, we have

$$-2 + c = (\deg(f))(-2 + c) + \deg(R'_f).$$

Therefore, $c \leq 2$. In this case, Theorem 7.1.1 (2),(3),(4) obviously hold since $\text{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$. For Theorem 7.1.1 (1), it follows from [KM98, Corollary 2.31] that $(X, \sum V_i)$ is log canonical. Moreover, $X \cong \mathbb{P}^1$ is rational, which is uniruled. Thus, we have completed the proof of Theorem 7.1.1 for the case when $\dim(X) = 1$. $\square$

Suppose Theorem 7.1.1 holds for those X' with $\dim(X') \leq n - 1$. From now on, we consider the case when $\dim(X) = n \geq 2$. We assume that V_j ($1 \leq j \leq c$) are all the prime divisors in $\text{TI}_f(X)$, which are contained in the ramification divisors of f (cf. Lemma 7.2.5).

According to the hypotheses in our theorem, we may assume $c \geq \rho(X) + n - 2 \geq 1$ (and hence K_X is not pseudo-effective by Lemma 7.2.20).

During the proof of Theorem 7.1.1, for (1) and (4), we only need one Fano contraction when running MMP while for (2) and (3), we need the end product Y of MMP in Theorem 7.2.2. The detailed proof is as follows.

7.3.1 The MMP for X .

By Theorem 7.2.2, if K_X is pseudo-effective, then X is $\mathbb{Q}$ -abelian. From now on, we assume that K_X is not pseudo-effective. Then the minimal model program of X will end with a Fano contraction $X_l \rightarrow Y$ (cf. [Ibid.]). For each $u_i : X_i \dashrightarrow X_{i+1}$, it is one of the following types: a divisorial contraction, a flip or a Fano contraction of a K_{X_i} -negative extremal ray. Furthermore, there exists a positive integer t , such that $g_0 = f^t$ descends to an int-amplified endomorphism g_i of X_i for each $1 \leq i \leq l + 1$.

By the log ramification divisor formula for the pair $(X, \sum V_j)$,

$$K_X + \sum V_j = f^*(K_X + \sum V_j) + \Delta_f, \quad (7.5)$$

with Δ_f effective, having no common components with $\bigcup \text{Supp} V_j$. Note that each V_j is f^{-1} -invariant and thus g_0^{-1} -invariant, i.e., $\text{TI}_f(X) \subseteq \text{TI}_{g_0}(X)$.

For each u_i , the Picard number will decrease by one if u_i is a divisorial contraction or a Fano contraction while the Picard number will be the same if u_i is a flip (cf. [KM98, Proposition 3.36, 3.37]).

We focus on a specific case. When running the MMP, once there appears a Fano contraction, one stops it immediately. In other words, we consider the following

$$X = X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_r \dashrightarrow X_{r+1} = Y_0,$$

where each $u_i : X_i \dashrightarrow X_{i+1}$ ($0 \leq i \leq r - 1$) is either a divisorial contraction or a flip and $u_r : X_r \rightarrow Y_0$ is a Fano contraction. Then each u_i ($0 \leq i \leq r - 1$) is a birational map and the dimensions of X_i ($0 \leq i \leq r$) are the same.

Let $V(1)_j \subseteq X_1$ be the strict transform of V_j under the birational map u_0 if V_j is not exceptional over X_1 . Then $V(1)_j$ is still a prime divisor. Since g_0 descends to an int-amplified endomorphism g_1 on X_1 by Theorem 7.2.2, $V(1)_j \in \text{TI}_{g_1}(X_1)$. Using Lemma 7.2.5, we get $g_1^*(V(1)_j) = \alpha_j V(1)_j$ for some $\alpha_j > 1$.

In general, for each $0 \leq i \leq r - 1$, let $V(i+1)_j$ be the strict transform of $V(i)_j$ under the birational map u_i if $V(i)_j$ is not exceptional over X_{i+1} . Then $V(i+1)_j \in \text{TI}_{g_{i+1}}(X_{i+1})$.

We use s_i ($0 \leq i \leq r$) to denote the cardinality of $\text{TI}_{g_i}(X_i)$. It is obvious that $s_0 \geq c$.

Also, for each i , the number of such $V(i)_j$ cannot exceed s_i . By the log ramification divisor formula for the pair $(X_i, \sum_j V(i)_j)$, we get

$$K_{X_i} + \sum_j V(i)_j = g_i^*(K_{X_i} + \sum_j V(i)_j) + \Delta(i). \quad (7.6)$$

Here, $\Delta(i)$ is an integral log ramification divisor having no common components with $\sum V(i)_j$.

Lemma 7.3.4. *With the same symbols given above, the equality $s_1 = s_0 - (\rho(X_0) - \rho(X_1))$ holds (and thus, $s_i = s_0 - (\rho(X_0) - \rho(X_i))$ for each $i \leq r$). In particular, if $c \geq \rho(X_0) + \dim(X_0) - k$ for some fixed integer k , then $s_r \geq \rho(X_r) + \dim(X_r) - k$.*

Proof. We use the induction to prove our claim. There are two cases for u_0 :

Case (1): Suppose u_0 is a divisorial contraction. Then, since u_0 is $g_0 = f^t$ equivariant, the exceptional divisor E of u_0 lies in $\text{TI}_{g_0}(X_0)$. Thus, $s_1 = s_0 - 1$ and also $\rho(X_1) = \rho(X_0) - 1$. In this case, we get $s_1 = s_0 - (\rho(X_0) - \rho(X_1))$.

Case (2): Suppose u_0 is a flip. Then u_0 is an isomorphism in codimension one. Therefore, u_0 does not contract or create any new divisors. Then $s_0 = s_1$, and $\rho(X_1) = \rho(X_0)$. In this case, $s_1 = s_0 - (\rho(X_0) - \rho(X_1))$ still holds.

In any case, we have the same equality for X_1 . Repeating the above discussion, we get the first part of our claim. For the second part, note that $s_0 \geq c$ and $\dim(X_i) = n$ for each $i \leq r$. Hence, the inequality follows from the above equality. $\square$

Lemma 7.3.5. *With the same symbols given above, suppose $\Delta(r) = 0$ for $i = r$ in Equation (7.6). Then the following statements hold.*

- (1) $K_{X_r} + \sum_j V(r)_j \equiv 0$.
- (2) If $\text{Pic}^\circ(X_r) = 0$ and every exceptional divisor of the composite map: $\delta : X_0 \dashrightarrow X_r$ is contained in $\text{TI}_f(X_0)$, where $\delta = u_{r-1} \circ \cdots \circ u_0$, then $\Delta_f = 0$ in Equation (7.5) and hence $K_X + \sum V_j \equiv 0$. Moreover, f is étale outside $(\bigcup V_j) \cup f^{-1}(\text{Sing}(X))$.

We will apply the above result to prove Theorem 7.1.1 (4). Before proving Lemma 7.3.5, we refer readers to the following remarks on the assumption of this lemma.

Remark 7.3.6. A prime divisor E on X_0 is δ -exceptional if it is exceptional over some X_i . Suppose $E \subseteq X_{i-1}$ is exceptional over X_i . Then for the divisorial contraction $u_{i-1} : X_{i-1} \rightarrow X_i$, the image $u_{i-1}(E)$ is a codimension ≥ 2 closed subset of X_i . So it cannot create new divisors when mapped into X_{i+1} no matter u_i is a divisorial contraction or a flip, since divisorial contractions are birational and flips are isomorphic in codimension one. Besides, we define the pull-back of δ (cf. Equation (7.8)) as the composite of $u_0^* \circ u_1^* \circ \cdots \circ u_{r-1}^*$. Moreover, $\text{NS}(X_0)$ is generated by the pull-back of generators of $\text{NS}(X_r)$ and all the δ -exceptional divisors (cf. [KM98, Proposition 3.36 and 3.37]).

The composite $\delta : X = X_0 \dashrightarrow X_r$ is always g_0 -equivariant by the choice of our g_0 , i.e., g_0 descends to an int-amplified endomorphism g_r on X_r . Hence, every exceptional divisor E of δ is contained in $\text{TI}_{g_0}(X_0)$. However, this is not enough to conclude our result, and the assumption of (2) in Lemma 7.3.5 is necessary. We will see later that the assumption of (2) holds when the cardinality c of $\text{TI}_f(X_0)$ achieves the upper bound $\dim(X_0) + \rho(X_0)$.

Proof of Lemma 7.3.5. Indeed, Lemma 7.3.5 (1) follows immediately from Lemma 7.2.5. Now, we prove (2). Since $\text{Pic}^\circ(X_r) = 0$, (1) implies that

$$K_{X_r} + \sum V(r)_j \sim_{\mathbb{Q}} 0. \quad (7.7)$$

Write down the well-defined δ -pull-back formula as follows:

$$K_X + \sum V_j = \delta^*(K_{X_r} + \sum V(r)_j) + E_2 - E_1. \quad (7.8)$$

Here, $E_k = \sum_l E_{kl}$ ($k = 1, 2$) are effective δ -exceptional divisors and each E_{kl} lies in $\text{TI}_f(X_0)$ by assumption. Hence, we may assume $f^*E_k = \sum_l a_{kl}E_{kl}$ with $a_{kl} > 1$. Recall the log ramification divisor formula for $(X, \sum V_j)$ (cf. Equation (7.5)):

$$K_X + \sum V_j = f^*(K_X + \sum V_j) + \Delta_f$$

with Δ_f effective, having no common components with $\bigcup V_i$. In summary, $\text{Supp } E_1$ and $\text{Supp } (E_2 + \Delta_f)$ have no common components and $\text{Supp } (E_1 + E_2) \subseteq \bigcup V_j$, since V_j ($1 \leq j \leq c$) are all the prime divisors in $\text{TI}_f(X)$. By Equation (7.5), (7.7) and (7.8),

$$E_2 - E_1 - \Delta_f \sim_{\mathbb{Q}} f^*(E_2 - E_1) = \sum a_{2l}E_{2l} - \sum a_{1l}E_{1l}.$$

Therefore, we get the following equation with both sides effective:

$$\Delta_f + \sum (a_{2l} - 1)E_{2l} \sim_{\mathbb{Q}} \sum (a_{1l} - 1)E_{1l}. \quad (7.9)$$

We claim that $E_1 = E_2 = \Delta_f = 0$. Suppose on the contrary that the claim does not hold. Let $F_k := \sum_l (a_{kl} - 1)E_{kl} \geq 0$ for $k = 1, 2$. On the one hand, since $\text{Supp } F_1$ and $\text{Supp } (\Delta_f + F_2)$ have no common components, Equation (7.9) gives us that $\kappa(X, F_1) \geq 1$. On the other hand, we take the graph of the birational map $\delta : X \dashrightarrow X_r$:

$$\begin{array}{ccc} & \Gamma & \\ p_1 \swarrow & & \searrow p_2 \\ X & \dashrightarrow & X_r \end{array}$$

Here, the projections p_1 and p_2 are birational morphisms. As we mentioned above, F_k is δ -exceptional for $k = 1, 2$; δ is a composite of flips or birational morphisms and thus δ does not extract divisors. Therefore, $p_{2*}p_1^*F_1 = 0$. As a result, we have the following:

$$\begin{aligned} H^0(X, \mathcal{O}_X(mF_1)) &= H^0(\Gamma, p_1^*\mathcal{O}_X(mF_1)) \\ &= H^0(X_r, p_{2*}p_1^*\mathcal{O}_X(mF_1)) = H^0(X_r, \mathcal{O}_{X_r}) = k \end{aligned}$$

for any $m > 0$. The first equality is due to projection formula. By definition, the Iitaka dimension $\kappa(X, F_1) = 0$, a contradiction. Therefore, our claim holds.

Since $\Delta_f = 0$ in Equation (7.5), applying Lemma 7.2.5 to Equation (7.5), $K_X + \sum V_j \equiv 0$. Moreover, $\Delta_f = 0$ implies that the ramification divisor of f consists only of these V_j 's. By the purity of branch loci, f is étale outside $(\bigcup V_j) \cup f^{-1}(\text{Sing}(X))$, which completes the proof of Lemma 7.3.5. $\square$

7.3.2 The case $\dim(Y_0) = 0$.

In this case, $\rho(X_r) = 1$ and $-K_{X_r}$ is ample. So X_r and hence X are rationally connected (cf. [Zha06a]). Then X is simply connected with respect to complex topology (cf. Theorem 7.2.10 and Lemma 7.2.9). Since X_r has only klt singularities (cf. [KM98, Corollary 3.42, 3.43]), by Proposition 7.3.1, $\text{Pic}^\circ(X_r) = \text{Pic}^\circ(X_0) = 0$, $\text{Pic}(X_0) \cong \text{NS}(X_0)$ and $\text{Pic}(X_r) \cong \text{NS}(X_r)$.

We fix an ample divisor H_{X_r} on X_r . Since $\rho(X_r) = 1$, we get $g_r^* H_{X_r} \equiv a H_{X_r}$ and also $a > 1$ (cf. Lemma 7.2.5). Hence g_r is polarized by [MZ18a, Lemma 2.3]. Further, since $\text{Pic}^\circ(X_r) = 0$, with H_{X_r} replaced by its power, we may assume $g_r^* H_{X_r} \sim a H_{X_r}$. By Remark 7.3.6, $\text{Pic}(X_0) \otimes_{\mathbb{Z}} \mathbb{Q} \cong \text{NS}_{\mathbb{Q}}(X_0)$ as a $\mathbb{Q}$ -vector space, is spanned by δ -exceptional divisors and $\delta^* H_{X_r}$ over $\mathbb{Q}$. On the one hand, $g_0^*(\delta^*(H_{X_r})) = \delta^*(g_r^*(H_{X_r})) \sim a \delta^*(H_{X_r})$ for some $a > 1$. On the other hand, for each δ -exceptional prime divisor E , it is contained in $\text{TI}_{g_0}(X_0)$ by the choice of g_0 and $g_0^* E = \beta E$ for some $\beta > 1$. Therefore, $g_0^*|_{\text{Pic} X}$ is diagonalizable over $\mathbb{Q}$. Since g_0 is a positive power of f , this completes the proof of Theorem 7.1.1 (2) and (3) for the case $\dim(Y_0) = 0$.

Suppose $c \geq \rho(X) + n$ from now on. Then $s_r \geq n + \rho(X_r) = n + 1$ (cf. Lemma 7.3.4). Since g_r is polarized and each $V(r)_j$ is ample, by Proposition 7.2.15, we get

$$s_r = n + 1, \quad K_{X_r} + \sum_{j=1}^{n+1} V(r)_j \equiv 0.$$

Therefore, by [BG+71, CH. XIII, Theorem 4.6] (and $\text{Pic}^\circ(X_r) = 0$), we also get Equation (7.7). Since $s_r = n + 1 = n + \rho(X_r)$, by Lemma 7.3.4, $s_i = n + \rho(X_i)$ for each $0 \leq i \leq r$. The equality for the case when $i = 0$ in turn forces $s_0 = c$. Thus, $\text{TI}_f(X_0) = \text{TI}_{g_0}(X_0)$, i.e., all of these g_0^{-1} -invariant prime divisors consist only of V_j 's. This proves Theorem 7.1.1 (1) for the case $\dim(Y_0) = 0$.

Now we prove Theorem 7.1.1 (4) for the case $\dim(Y_0) = 0$. Since $s_i = n + \rho(X_i)$ for each i , the exceptional divisor of u_i is contained in $\sum_j V(i)_j$ if u_i is a divisorial contraction. Since X_r and X are rationally connected with klt singularities, Theorem 7.1.1 (4) follows from Proposition 7.3.1 and Lemma 7.3.5.

7.3.3 Proof of Theorem 7.1.1 (1) for the case $\dim(Y_0) \geq 1$

We still assume that $\dim(Y_0) \geq 1$ and $c \geq \rho(X) + \dim(X)$. Actually, the second part of (1) follows immediately. Indeed, if $c \geq 1$, then the ramification divisor $R_f > 0$, which means that f is not étale in codimension one. By Lemma 7.2.20, K_X is not pseudo-effective and hence X is uniruled by Remark 7.2.7. Moreover, by Lemma 7.2.11, the pair $(X, \sum V_j)$ is log canonical.

Recall that Lemma 7.3.4 gives us the same inequality for each X_i ($i \leq r$): $s_i \geq n + \rho(X_i)$ if we assume $s_0 \geq n + \rho(X_0)$. We want to ask whether the inequality still holds for Y_0 . If the inequality holds, then we can use the induction on $\dim(Y_0)$, which proves Theorem 7.1.1 (1). Parallel to Lemma 7.3.4, we introduce the following lemma, from which, the first part of Theorem 7.1.1 follows immediately.

To emphasize the notation for the Fano contraction $u_r : X_r \rightarrow Y_0$, let s_{Y_0} be cardinality of $\text{TI}_{g_{Y_0}}(Y_0)$ and $g_{Y_0} := g_{r+1}$ the int-amplified endomorphism of Y_0 , which g_0 descends to.

Lemma 7.3.7. *With the same symbols given above, for the Fano contraction $u_r : X_r \rightarrow Y_0$, the inequality $s_r \leq s_{Y_0} + \dim(X_r) - \dim(Y_0) + 1$ holds. Further, if $s_r \geq n + \rho(X_r) - k$ for some fixed integer k , then $s_{Y_0} \geq \dim(Y_0) + \rho(Y_0) - k$. In particular, if $k = 0$, then $s_{Y_0} = \dim(Y_0) + \rho(Y_0)$ and also $s_r = n + \rho(X_r)$.*

To prove Lemma 7.3.7, we need to do some preparations. By Lemma 7.2.6, the periodic points of g_{Y_0} are dense. Let $y_0 \in Y_0$ be a general g_{Y_0} -periodic point. Replacing g_{Y_0} (and g_i) by its power, we may assume $g_{Y_0}(y_0) = y_0$. Let $W_0 := u_r^{-1}(y_0) \subseteq X_r$. Then W_0 , as a general fibre of u_r , is Fano of dimension $n - \dim(Y_0) \geq 1$, since the canonical divisor of a general fibre of a Fano contraction is anti-ample. Restricting g_r to W_0 , we get a surjective endomorphism of W_0 commuting with u_r and g_{Y_0} :

$$g_{W_0} := g_r|_{W_0} : W_0 \rightarrow W_0.$$

Claim 7.3.8. With the same symbols given above, g_{W_0} is polarized.

Proof. Since $u_r : X_r \rightarrow Y_0$ is a Fano contraction, the relative Picard number $\rho(X_r/Y_0) = 1$. Therefore, pulling back a fixed ample divisor H_r on X_r , we get

$$g_r^* H_r \equiv k H_r \pmod{(u_r^* N^1(Y_0))}. \quad (7.10)$$

Since g_r is int-amplified, each eigenvalue of $g_r^*|_{N^1(X_r)}$ is of modulus greater than 1 (cf. Lemma 7.2.5). Besides, the operation $g_r^*|_{N^1(X_r)}$ induces an operation on the space $N^1(X_r)/u_r^* N^1(Y_0)$, with all the eigenvalues of modulus greater than 1. Thus, $k > 1$.

Now, since $u_r^* D|_{W_0} = 0$ for any Cartier divisor D on Y_0 , restricting Equation (7.10) to a general fibre W_0 , we get $g_{W_0}^*(H_r|_{W_0}) \equiv kH_r|_{W_0}$, with $H_r|_{W_0}$ ample on W_0 . This together with $k > 1$ proves that g_{W_0} is polarized (cf. [NZ10, Lemma 2.3]). $\square$

Proof of Lemma 7.3.7. We divide these $V(r)_i (1 \leq i \leq s_r)$ into two groups.

Case (1): Suppose $u_r|_{V(r)_j} : V(r)_j \rightarrow Y_0$ ($j \leq s_r(1)$) is not surjective. Since u_r is projective and thus closed, the image $u_r(V(r)_j)$ is a codimension ≥ 1 closed subset. Take a general point $y_0 \in Y_0$ such that y_0 does not lie in the image of any $V(r)_j$ for $j \leq s_r(1)$. Then one may easily get $V(r)_j|_{W_0} = 0$. Since $\text{Pic}(X_r/Y_0) = 1$, any two contracted curves are proportional (under the numerical equivalence). Hence, for any curve C lying in a fibre of $X_r \rightarrow Y_0$, $(V(r)_j \cdot C) = 0$. Since $V(r)_j$ is $\mathbb{Q}$ -Cartier, we may take a suitable $k'_j \in \mathbb{N}$ such that $k'_j V(r)_j$ is Cartier.

By Cone Theorem (cf. [KM98, Theorem 3.7 (4)]), $k'_j V(r)_j = u_r^* G'_j$ for some effective Cartier divisor G'_j on Y_0 . Then the image $G_j := u_r(V(r)_j)$, which is the support of G'_j , is a prime divisor on Y_0 . Since Y_0 is normal, on the smooth locus of G_j , the pull-back of G_j is an integral divisor. Taking the closure, we get $u_r^* G_j = k_j V(r)_j$ for some $k_j \in \mathbb{N}$. Moreover, by the commutative diagram, $g_{Y_0}^{-1}(G_j) = G_j$ set-theoretically for each $j \leq s_r(1)$. Applying the inductive hypothesis on $\dim(Y_0)$, the following holds:

$$s_r(1) \leq s_{Y_0} \leq \dim(Y_0) + \rho(Y_0). \quad (7.11)$$

Here, the case when $s_r(1) = 0$ is allowed during our discussion.

Case (2): Suppose $u_r|_{V(r)_j} : V(r)_j \rightarrow Y_0$ ($s_r(1) < j \leq s_r$) is surjective. Note that the general fibre W_0 is not contained in any $V(r)_j$ for $s_r(1) < j \leq s_r$. Let $s_r(2) := s_r - s_r(1)$ and fix an ample divisor H_{X_r} on X_r , which does not lie in $u_r^* N^1(Y_0)$. Since $\rho(X_r/Y_0) = 1$, we have the following numerical property:

$$V(r)_j \equiv t_j H_{X_r} \pmod{(u_r^* N^1(Y_0))}$$

for some $t_j > 0$, and thus $V(r)_j|_{W_0}$ is ample on W_0 . Moreover, $V(r)_j|_{W_0}$ is $g_{W_0}^{-1}$ -invariant on W_0 for each $s_r(1) < j \leq s_r$. Note that we also allow the case $s_r(2) = 0$.

By Claim 7.3.8, g_{W_0} is polarized. If $\dim(W_0) \geq 2$, then by Proposition 7.2.15, we get $s_r(2) \leq \dim(W_0) + 1$; if $\dim(W_0) = 1$, then by Lemma 7.3.3, we get $s_r(2) \leq 2 =$

$\dim(W_0) + 1$. In any case, we get the following inequality:

$$s_r(2) \leq \dim(W_0) + 1. \quad (7.12)$$

Combining the inequality (7.12) with the inequality (7.11), the following holds:

$$s_r \leq s_{Y_0} + s_r(2) \leq s_{Y_0} + \dim(W_0) + 1 = s_{Y_0} + \dim(X_r) - \dim(Y_0) + 1. \quad (7.13)$$

This completes the first part of our lemma.

If $s_r \geq \dim(X_r) + \rho(X_r) - k$ for some integer k , then by Equation (7.13) and the fact that $\rho(Y_0) = \rho(X_r) - 1$, we have $s_{Y_0} \geq \dim(Y_0) + \rho(Y_0) - k$. In particular, if $k = 0$, then combining (7.12) with the second inequality of (7.11), we see that,

$$n + \rho(X_r) \leq s_r = s_r(1) + s_r(2) \leq \dim(Y_0) + \rho(Y_0) + \dim(W_0) + 1 = n + \rho(X_r).$$

Therefore, all the inequalities are equalities, and thus for each i , $s_i = n + \rho(X_i)$. This proves Lemma 7.3.7 and also Theorem 7.1.1 (1) for the case when $\dim(Y_0) \geq 1$ since $c \leq s_0$. $\square$

At the end of this subsection, we display the following, which is deduced from the inductive hypothesis on Y_0 :

$$s_r(1) = \dim(Y_0) + \rho(Y_0), \quad K_{Y_0} + \sum_{i=1}^{s_r(1)} G_i \sim_{\mathbb{Q}} 0. \quad (7.14)$$

7.3.4 Proofs of Theorem 7.1.1 (2), (3) for the case $\dim(Y_0) \geq 1$

In this part, we shall use Theorem 7.2.2 to prove Theorem 7.1.1 (2), (3). Recall that we have such a g_0 -equivariant relative MMP over Y

$$X = X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_r \dashrightarrow X_{r+1} = Y_0 \dashrightarrow Y.$$

Here, we continue running MMP from Y_0 ($X_r \rightarrow Y_0$ is the first Fano contraction) and terminate with the end product Y . By the above discussion, we first consider the case when $Y = Y_0$, i.e., assume the MMP has only one Fano contraction. Suppose

$$s_0 \geq c \geq \dim(X) + \rho(X) - 2.$$

Then by Lemmas 7.3.4 and 7.3.7, $s_{Y_0} \geq \dim(Y_0) + \rho(Y_0) - 2$. Similarly, if $c \geq \dim(X) + \rho(X) - 1$, then $s_{Y_0} \geq \dim(Y_0) + \rho(Y_0) - 1$. Now we may start from Y_0 and continuously run MMP as mentioned in Theorem 7.2.2.

Claim 7.3.9. For the case when $s_0 \geq \dim(X) + \rho(X) - 2$, there are only two choices for the end product Y of MMP: Y is either an elliptic curve or a point.

Proof. Suppose on the contrary that the claim does not hold. If $\dim(Y) \geq 2$, then $s_Y \geq \dim(Y) + \rho(Y) - 2 \geq 1$, which means the int-amplified endomorphism g_Y has ramification divisors. Since g_Y is not étale in codimension one, K_Y is not pseudo-effective by Lemma 7.2.20. However, our end product Y is Q -abelian and then $K_Y \sim_{\mathbb{Q}} 0$ (cf. Theorem 7.2.2), a contradiction. Therefore, $\dim(Y) = 1$ and by Lemma 7.3.3, Y is either elliptic or rational. If $Y \cong \mathbb{P}^1$, then K_Y is not pseudo-effective. By Theorem 7.2.2 again, we can continue to contract Y into a single point. This proves Claim 7.3.9. $\square$

Proof of Theorem 7.1.1 (3). If Y is elliptic, then $s_Y = 0$ (cf. Lemma 7.3.3), contradicting the deduced result $s_Y \geq \dim(Y) + \rho(Y) - 1$. Therefore, Y is a point (cf. Claim 7.3.9). By Theorem 7.2.2 (2), X is rationally connected since the whole X is a fibre. Then, by Proposition 7.3.1, $\text{Pic}^\circ(X) = 0$ and then we may identify the Picard group with the Néron-Severi group. Further, X is simply connected with respect to complex topology and $\text{Pic}(X)$ is torsion free.

Theorem 7.2.2 asserts that $g_0^*|_{\text{Pic}(X)}$ is diagonalizable over $\mathbb{C}$ if and only if so is $g_Y^*|_{\text{Pic}(Y)}$. Since Y is a point, $g_0^*|_{\text{Pic}(X)}$ is diagonalizable over $\mathbb{C}$. Furthermore, each eigenvalue λ of $g_0^*|_{\text{Pic}(X)}$ is either an eigenvalue of $g_Y^*|_{\text{Pic}(Y)}$ or an eigenvalue of $g_i^*|_{\text{Pic}(X_i)/u_i^*\text{Pic}(X_{i+1})}$ for some i . Since Y is a point, λ is an eigenvalue of $g_i^*|_{\text{Pic}(X_i)/u_i^*\text{Pic}(X_{i+1})}$ for some i . Since $\rho(X_i/X_{i+1}) = 1$, $\lambda \in \mathbb{Q}$. Indeed, all these eigenvalues are positive integers (cf. [MZ20a, Lemmas 5.1 and 5.2]). Thus, $g_0^*|_{\text{Pic}(X)}$ is diagonalizable over $\mathbb{Q}$. This completes the proof of Theorem 7.1.1 (3). $\square$

Proof of Theorem 7.1.1 (2). By Claim 7.3.9, there are two choices for the end product Y of MMP. If Y is a point, then similar to Theorem 7.1.1 (3), X is rationally connected and simply connected with respect to complex topology; if Y is an elliptic curve, then this

Y is our E in Theorem 7.1.1 (2). By Theorem 7.2.2, $\phi : X \rightarrow Y = E$ is holomorphic and equi-dimensional with every fibre irreducible. Then $\phi : X \rightarrow E$ is proper and surjective, the general fibre of which is connected, and thus ϕ is a fibration. Further, $g_0 = f^t$ descends to $g_E : E \rightarrow E$ of degree > 1 (cf. Theorem 7.2.2). Furthermore, $g_0^*|_{\text{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{C}$ if and only if $g_E^*|_{\text{NS}_{\mathbb{Q}}(E)}$ is diagonalizable over $\mathbb{C}$ (cf. [MZ18a, Lemma 9.2]). Since we have proved $g_E^*|_{\text{NS}_{\mathbb{Q}}(E)}$ is diagonalizable over $\mathbb{Q}$ by Lemma 7.3.3, $g_0^*|_{\text{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{C}$. As in the case of Theorem 7.1.1 (3), all the eigenvalues of $g_0^*|_{\text{NS}_{\mathbb{Q}}(X)}$ are rational numbers. Thus, $g_0^*|_{\text{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{Q}$.

Finally, we prove that every fibre of $\phi : X \rightarrow E$ is normal (cf. [Zha14, Proof of Theorem 1.3]). Let $\Sigma = \{e \in E \mid \phi^*e = X_e \text{ is not normal}\}$, a finite subset of E . By [NZ10, Proof of Lemma 4.7], $g_E^{-1}(\Sigma) \subseteq \Sigma$, which implies $g_E^{-1}(\Sigma) = \Sigma$ since Σ is a finite set. However, g_E is étale and it could not have any g_E^{-1} -invariant divisors (cf. Lemma 7.3.3). As a result, $\Sigma = \emptyset$ and Theorem 7.1.1 (2) holds. $\square$

7.3.5 Proof of Theorem 7.1.1 (4) for the case $\dim(Y_0) \geq 1$

We shall prove $K_X + \sum_{j=1}^{n+\rho} V_j \sim_{\mathbb{Q}} 0$ and f is étale outside $f^{-1}(\text{Sing}(X)) \cup (\bigcup V_j)$ for the case when $c = \dim(X) + \rho(X)$. In this subsection, the first Fano contraction is enough for our proof. As we proved in Subsection 7.3.3, if $s_0 \geq c \geq \dim(X) + \rho(X)$, then $s_0 = c = \dim(X) + \rho(X)$ and $s_r(2) = \dim(W_0) + 1$, where W_0 is a general fibre of $X_r \rightarrow Y_0$.

Claim 7.3.10. Suppose $c = \dim(X) + \rho(X)$. Then $K_{W_0} + \sum_{j=1}^{s_r} V(r)_j|_{W_0} \equiv 0$.

Proof. As we proved in Lemma 7.3.7, $V(r)_j|_{W_0}$ is ample on W_0 for each $j > s_r(1)$. When $c = \dim(X) + \rho(X)$, $s_r(2) = \dim(W_0) + 1$. Besides, by Claim 7.3.8, $g_{W_0} = g_r|_{W_0}$ is polarized by the restriction of an ample divisor on X_r . If $\dim(W_0) = 1$, then by Case (b) of Lemma 7.3.3, $K_{W_0} + \sum V(r)_j|_{W_0} \equiv 0$; if $\dim(W_0) \geq 2$, then applying Proposition 7.2.15 to the pair (W_0, g_{W_0}) with $g_{W_0}^{-1}$ -invariant ample divisors $V_j|_{W_0}$ ($j > s_r(1)$), we have

the following:

$$(K_{X_r} + \sum_{j=1}^{s_r} V(r)_j)|_{W_0} \equiv K_{W_0} + \sum_{j=s_r(1)+1}^{s_r(1)+s_r(2)} V(r)_j|_{W_0} \equiv 0. \quad (7.15)$$

Here, $V(r)_j|_{W_0} = 0$ when $j \leq s_r(1)$ (cf. Case (1) in Proof of Lemma 7.3.7). We have completed the proof of our claim. $\square$

We recall Equation (7.6) for the case when the index $i = r$ as follows:

$$K_{X_r} + \sum_{j=1}^{s_r} V(r)_j = g_r^*(K_{X_r} + \sum_{j=1}^{s_r} V(r)_j) + \Delta(r). \quad (7.16)$$

Restricting Equation (7.16) to a general fibre W_0 and comparing it with Equation (7.15), we get $\Delta(r)|_{W_0} \equiv 0$. Since $\Delta(r)$ is effective, by Cone Theorem, $\Delta(r)$ is the pull-back of some effective $\mathbb{Q}$ -divisor $\Delta_{Y_0} \subseteq Y_0$.

Claim 7.3.11. $\Delta_{Y_0} = 0$, i.e., $\Delta(r) = 0$.

Suppose Claim 7.3.11 holds for the time being. Then by Equation (7.16), we have

$$K_{X_r} + \sum_{j=1}^{s_r} V(r)_j = g_r^*(K_{X_r} + \sum_{j=1}^{s_r} V(r)_j). \quad (7.17)$$

Suppose $K_{X_r} + \sum_j V(r)_j \neq 0$. Then from Equation (7.17), 1 is an eigenvalue for the operator $g_r^*|_{\text{NS}_{\mathbb{Q}}(X_r)}$, a contradiction (cf. Lemma 7.2.5). This forces $K_{X_r} + \sum V(r)_j \equiv 0$ and thus $K_{X_r} + \sum V(r)_j \sim_{\mathbb{Q}} 0$, since $\text{Pic}^\circ(X_r) = \text{Pic}^\circ(X_0) = 0$ (cf. Proposition 7.3.1 and Proof of Theorem 7.1.1 (3)). Then Theorem 7.1.1 (4) follows from Lemma 7.3.5.

Now, we begin to prove Claim 7.3.11. We shall construct a generically finite and surjective morphism to Y_0 . Let $W(s_r + 1) := X_r$. We take four steps.

Step 1. First, let $\sigma(s_r) : W(s_r) \rightarrow X_r$ be the normalization $W(s_r) \rightarrow V(r)_{s_r}$ followed by an inclusion map $i : V(r)_{s_r} \hookrightarrow X_r$. Then $\dim(W(s_r)) = \dim(X_r) - 1$. It is easy to get Claim 7.3.12 by considering the commutative diagram and applying Lemma 7.2.4.

Claim 7.3.12. With the same symbols given above, there exists an int-amplified endomorphism $g(s_r)$ of $W(s_r)$ such that $\sigma(s_r) \circ g(s_r) = g_r \circ \sigma(s_r)$.

We return back to Step 1 of our proof for Claim 7.3.11. Pulling back Equation (7.16) along the morphism $\sigma(s_r)$, we get the following (cf. Claim 7.3.12):

$$K_{W(s_r)} + D(s_r) := \sigma(s_r)^*(K_{X_r} + \sum V(r)_j) = g(s_r)^*(K_{W(s_r)} + D(s_r)) + \Delta(r)_{s_r}.$$

Here, $\Delta(r)_{s_r}$ is the pull-back of $\Delta(r)$. Note that each $V(r)_j (j \leq s_r(1))$ intersects $V(r)_{s_r}$ (cf. Proof of Lemma 7.3.7). Indeed, since $k_j V(r)_j = u_r^* G_j$ for each $j \leq s_r(1)$, $V(r)_j$ contains all the fibres over G_j . Also, $V(r)_{s_r}$ dominates Y_0 and hence $V(r)_j$ intersects $V(r)_{s_r}$. Then the pull-back of each $V(r)_j (j \leq s_r(1))$ survives in $W(s_r)$. The next claim follows from [Sho92, Corollary 3.11] and [Kol92, Corollary 16.7].

Claim 7.3.13. The pair $(W(s_r), D(s_r))$ is log canonical and the support of the pull-back $\sum_{j=1}^{s_r(1)} \sigma(s_r)^* V(r)_j$ (as a reduced divisor) is contained in $D(s_r)$, i.e.,

$$D(s_r) \geq \sum_{j=1}^{s_r(1)} \text{Supp } \sigma(s_r)^* V(r)_j.$$

Proof. The first part follows from [Kaw07, Theorem 1]. For the second part, on the one hand, since $(W(s_r), D(s_r))$ is log canonical, the coefficient of each component in $D(s_r)$ cannot exceed 1. On the other hand, for each irreducible component $P(j)_i$ of the support of the intersection $V(r)_{s_r} \cap V(r)_j$, note that $(X_r, V(r)_{s_r})$ should be log terminal around $P(j)_i$ (cf. [KM98, Lemma 2.27]). Hence, the coefficient of $P(j)_i$ in $D(s_r)$ is given by (cf. [Sho92, Corollaries 3.10 and 3.11])

$$1 - \frac{1}{m} + \sum \frac{r_j}{m},$$

where r_j/m is the multiplicity of $\sigma(s_r)^* V(r)_j$ around $P(j)_i$ and $r_j \in \mathbb{Q}$. Since $V(r)_j (j \leq s_r(1))$ intersects $V(r)_{s_r}$ as we discussed above, $r_j \geq 1$ and thus the coefficient of $P(j)_i$ in $D(s_r)$ is no less than 1. Hence, the coefficient of each component of $\text{Supp } \sigma(s_r)^* V(r)_j (j \leq s_r(1))$ in $D(s_r)$ is 1 and our claim holds. $\square$

We come back to the proof of Claim 7.3.11. Suppose $\dim(X_r) - \dim(Y_0) = 1$. Then we go directly to **Step 4** with $k_0 = s_r$.

Step 2. Suppose $\dim(X_r) - \dim(Y_0) \geq 2$. Then $\dim(W_0) \geq 2$. Therefore, $V(r)_{s_r}|_{W_0}$ is connected since it is ample on W_0 (cf. [Har77, Corollary 7.9]). Further, for each $s_r(1) < j < s_r$, $V(r)_j|_{W_0}$ is also ample and thus nonzero on W_0 .

Since $V(r)_{s_r}$ also intersects $V(r)_j$ for each $s_r(1) < j < s_r$ by the ampleness of $V(r)_j$. With the same proof of Claim 7.3.13, the support of $\sigma(s_r)^*V(r)_j$ ($s_r(1) < j < s_r$), which is a reduced divisor, is contained in $D(s_r)$. We claim that each $\text{Supp } \sigma(s_r)^*V(r)_j$ ($s_r(1) < j < s_r$) still dominates Y_0 . Indeed, a general fibre of $V(r)_{s_r} \hookrightarrow X_r \rightarrow Y_0$ is as the form $W_0 \cap V(r)_{s_r}$. Since each $V(r)_j|_{W_0}$ ($s_r(1) < j < s_r$) is ample on W_0 , the restriction $V(r)_j|_{W_0 \cap V(r)_{s_r}}$ is still ample on $W_0 \cap V(r)_{s_r}$ for $s_r(1) < j < s_r$. Note that this can only be obtained under the condition that $\dim(W_0) \geq 2$. Therefore, the nonzero divisor $V(r)_j|_{V(r)_{s_r}}$ on $V(r)_{s_r}$ intersects the general fibre of $V(r)_{s_r} \rightarrow Y_0$ and thus its support dominates Y_0 .

Now that normalization is finite, $\sigma(s_r)^*V(r)_j$ is nonzero and ample when restricted to a general fibre of $W(s_r) \rightarrow Y_0$ for any $s_r(1) < j < s_r$. As in Proof of Proposition 7.2.15, we fix an irreducible component $\text{Supp } B_{s_r-1}$ ($\leq D(s_r)$) of $\text{Supp } \sigma(s_r)^*V(r)_{s_r-1}$ dominating Y_0 and take the normalization $W(s_r - 1)$ of B_{s_r-1} followed by the inclusion to get the next morphism $\sigma(s_r - 1) : W(s_r - 1) \rightarrow W(s_r)$.

Step 3. In general, for each $s_r \geq k \geq s_r - \dim(W_0) + 1 = s_r(1) + 2$, let $\sigma(k) : W(k) \rightarrow W(k + 1)$ be the normalization of a fixed (integral) irreducible component B_k of $\text{Supp } V(r)_k|_{W(k+1)}$ dominating Y_0 , followed by inclusion. Such B_k exists by the induction and the following condition. Since the dimension of the fibre of $W(k + 2) \rightarrow Y_0$ is $\dim(W_0) - s_r + k + 1$ which is at least 2 by assumption, $(\sigma(s_r) \circ \cdots \circ \sigma(k + 2))^*V(r)_k$ intersects B_{k+1} and hence the support of the pull-back $V(r)_k|_{W(k+1)}$ dominates Y_0 with the same proof as in Step 2.

Moreover, for each $k \geq s_r(1) + 2$, B_k intersects $V(r)_j|_{W(k+1)}$ ($j \leq s_r(1)$) and hence the support of the pull-back of $V(r)_j$ ($j \leq s_r(1)$) to $W(k)$ (as a reduced divisor) is contained in $D(k)$ (cf. Claim 7.3.13). Now, $\dim(W(k)) = \dim(X_r) - (s_r - k) - 1$ and similar to Claim 7.3.12, we get an int-amplified endomorphism $g(k)$ of $W(k)$, commuting with

each $g(l)$ for $l > k$. Therefore, we get the following equations:

$$K_{W(k)} + D(k) := \sigma(k)^*(K_{W(k+1)} + D(k+1)) = g(k)^*(K_{W(k)} + D(k)) + \Delta(r)_k \quad (7.18)$$

with $k = s_r, s_{r-1}, \dots, k_0 := s_r(1) + 2$. For each k , $\Delta(r)_k$ is the pull-back of $\Delta(r)$; the pair $(W(k), D(k))$ is log canonical with $D(k) \geq \sum_{j=1}^{s_r(1)} \text{Supp } V(r)_j|_{W(k)}$.

Step 4. Let R_j ($j \leq s_r(1)$) denote the support of the pull-back of $V(r)_j$ to $W(k_0)$, which are all reduced integral divisors (may not be irreducible). Then $D(k_0) \geq \sum_{j=1}^{s_r(1)} R_j$. Let

$$\tau_{k_0} := u_r \circ \sigma(s_r) \cdots \circ \sigma(k_0) : W(k_0) \rightarrow \cdots \rightarrow X_r \rightarrow Y_0.$$

By construction, τ_{k_0} is dominant and projective. Further, $\dim(W(k_0)) = \dim(Y_0)$, and thus τ_{k_0} is generically finite. By the commutative diagram, we have $\tau_{k_0} \circ g(k_0) = g_{Y_0} \circ \tau_{k_0}$. Also, taking the Stein factorization of τ_{k_0} , we get a birational morphism $\nu : W(k_0) \rightarrow S(k_0)$ with connected fibres and a finite morphism $\gamma : S(k_0) \rightarrow Y_0$ such that $\tau_{k_0} = \gamma \circ \nu$.

On the one hand, since γ is finite, by the log ramification divisor formula,

$$K_{S(k_0)} + \sum \text{Supp } \gamma^* G_j - R_\gamma = \gamma^*(K_{Y_0} + \sum G_j), \quad (7.19)$$

where R_γ is effective, having no common components with $\sum \text{Supp } \gamma^* G_j$. Since the right-hand side of Equation (7.19) is $\mathbb{Q}$ -Cartier, we can pull back Equation (7.19) to $W(k_0)$ under the birational morphism ν (cf. [BH14, Lemma 2.7]):

$$K_{W(k_0)} + \nu_*^{-1} \left(\sum_{j=1}^{s_r(1)} \text{Supp } \gamma^* G_j - R_\gamma \right) - F = \tau_{k_0}^*(K_{Y_0} + \sum_{j=1}^{s_r(1)} G_j). \quad (7.20)$$

Here, F is ν -exceptional, which may not be effective. On the other hand, we recall the following equations (cf. Equation (7.18) and Proof of Lemma 7.3.7):

$$K_{W(k_0)} + D(k_0) = g(k_0)^*(K_{W(k_0)} + D(k_0)) + \Delta(r)_{k_0}, \quad (7.21)$$

$$k_j V(r)_j = u_r^* G_j (1 \leq j \leq s_r(1)), \quad \Delta(r) = u_r^* \Delta_{Y_0}. \quad (7.22)$$

Putting Equations (7.20), (7.21) and (7.22) together, we get

$$\tau_{k_0}^*(K_{Y_0} + \sum G_j) + M + F = g(k_0)^*(\tau_{k_0}^*(K_{Y_0} + \sum G_j) + M + F) + \tau_{k_0}^* \Delta_{Y_0}. \quad (7.23)$$

Here, $M := D(k_0) - \nu_*^{-1}(\sum_{j=1}^{s_r(1)} \text{Supp } \gamma^* G_j) + \nu_*^{-1} R_\gamma$ and we claim that M is effective. Indeed, according to Equation (7.22) and the notation at the beginning of Step 4, we see that,

$$\nu_*^{-1} \sum_{j=1}^{s_r(1)} \text{Supp } \gamma^* G_j \leq \sum_{j=1}^{s_r(1)} \text{Supp } \tau_{k_0}^* G_j = \sum_{j=1}^{s_r(1)} R_j.$$

Therefore $M \geq D(k_0) - \sum R_j + \nu_*^{-1} R_\gamma \geq 0$ (cf. Claim 7.3.13), which completes the proof of our claim. Further, applying the inductive hypothesis on Y_0 (cf. Equation (7.14)) to Equation (7.23), we get

$$M + F \equiv g(k_0)^*(M + F) + \tau_{k_0}^* \Delta_{Y_0}. \quad (7.24)$$

End of Proof of Claim 7.3.11 and Theorem 7.1.1 (4). Now, suppose $\Delta_{Y_0} \neq 0$. Then $N := \tau_{k_0}^* \Delta_{Y_0}$ is some nonzero effective $\mathbb{Q}$ -divisor, which is not ν -exceptional on $W(k_0)$. We shall apply the method in Proof of Lemma 7.2.18 to find the contradiction. Multiplying Equation (7.24) by a positive integer, we may assume N is integral. Substituting the expression of $M + F$ to the RHS $(m - 1)$ -times, we get

$$M + F = (g(k_0)^m)^*(M + F) + \sum_{j=0}^{m-1} (g(k_0)^j)^* N.$$

According to the commutative diagram, $(g(k_0)^m)^* F$ is also ν -exceptional for each natural number m . Fixing an ample Cartier divisor H on $S(k_0)$ and pulling back to $W(k_0)$ along the birational morphism ν , we get the following inequality:

$$(M \cdot \nu^* H^{\dim(W(k_0))-1}) \geq \sum_{j=0}^{m-1} \left(\nu_* (g(k_0)^j)^* N \cdot H^{\dim(S(k_0))-1} \right).$$

Note that N is integral and not ν -exceptional, and thus $\nu_* (g(k_0)^j)^* N$ is an effective integral divisor on $S(k_0)$. So the RHS tends to infinity if we let $m \rightarrow \infty$, a contradiction. This completes the proof of Claim 7.3.11 and also Theorem 7.1.1 (4). $\square$

We give the following two remarks at the end of this chapter.

Remark 7.3.14. When extending the results in [Zha14, Theorem 1.3], we improve the original proof from the following points. First, when considering the Fano contraction

$u_r : X_r \rightarrow Y_0$, the pull-back of G_j ($j \leq s_r(1)$) is a multiple of $V(r)_j$ and thus may not be reduced, since $V(r)_j$ is only $\mathbb{Q}$ -Cartier. Second, when proving Proposition 7.2.15, D_i may not be reduced while it holds for the case when $s = n + 1$ (cf. Proof of Remark 7.2.19 and [Zha14, Proposition 2.12]). Finally, for each $V(r)_j$, the restriction to general fibre W_0 may be reducible (but still reduced) even though $V(r)_j$ is irreducible; thus we need to extend the result of [Ibid.] to the case when V_j are reduced (cf. Proposition 7.2.15).

Remark 7.3.15. We believe that, applying the same strategy and assume the density of periodic points of int-amplified endomorphisms on compact Kähler threefolds (cf. Question 2.1.7), one can further extend our main theorem to the Kähler setting and give a similar bound in terms of the dimension and the Picard number.

Invariant subvarieties with small dynamical degree

8.1 Introduction

We work over a fixed algebraically closed field k . Throughout this chapter, we assume the characteristic of k is zero unless otherwise indicated.

Let $f : X \rightarrow X$ be a dominant self-morphism of an algebraic variety X . Let $V \subseteq X$ be an (irreducible) closed subvariety. We say that V is *f-periodic* (resp. *f-invariant*) if $\overline{f^s(V)} = V$ for some $s > 0$ (resp. $\overline{f(V)} = V$). When X is projective or f is finite (and hence surjective), $\overline{f^s(V)} = V$ implies V is *f-periodic* in the usual sense, i.e., $f^s(V) = V$. We say that V is *f⁻¹-periodic* if $f^{-s}(V) = V$ for some $s > 0$.

An *f*-periodic closed subvariety V is said to be of *Small Dynamical Degree* (**SDD** for short) if the first dynamical degree $d_1(f^s|_V) < d_1(f^s)$ for some $s > 0$ such that $\overline{f^s(V)} = V$. Denote by Σ_{f^∞} the set of *f*-periodic (irreducible closed) subvarieties of small dynamical degree. For $V \in \Sigma_{f^\infty}$, we say that V is *maximal* if $V \subseteq V'$ and $V' \in \Sigma_{f^\infty}$ imply $V = V'$.

Denote by S_{f^∞} the set of maximal elements in Σ_{f^∞} . Denote by

$$S_f \quad (\text{resp. } P_f)$$

the subset of S_{f^∞} , consisting of all f -invariant elements (resp. prime divisors). In general, Σ_{f^∞} may have infinitely many non-maximal elements. So the question below is natural.

Question 8.1.1. Let $f : X \rightarrow X$ be a dominant self-morphism of an algebraic variety X .

- (1) Is S_f a finite set? Namely, are there only finitely many f -invariant maximal f -periodic subvarieties of small dynamical degree?
- (2) Is P_f a finite set? Namely, are there only finitely many prime divisors in S_f ?

Remark 8.1.2. Question 8.1.1 was first proposed in [MMSZ20, Question 8.4 (3)] and the motivation is to study the points with small arithmetic degrees. Indeed, together with positive answers to other two arithmetic questions in [MMSZ20, Question 8.4] (cf. [MMS⁺21, Question 3.3]), it is easy to show the equivalence of the (weak) Kawaguchi-Silverman Conjecture (cf. [KS16], [MMS⁺21, Conjecture 3.1]) and the (strong) small Arithmetic Non-Density (sAND) conjecture (cf. [MMSZ20, Conjecture 1.4]); see [MMS⁺21, Theorem 3.4]. In this chapter, we will mainly focus on Question 8.1.1 itself. We refer to [LS21, Theorem 2.33] for periodic SDD subvarieties of Hyperkähler varieties.

When studying many dynamical problems, we usually deal with the periodic subvarieties instead of simply invariant ones to observe some “growth” property of f . In general, the set S_{f^∞} may not be finite. For example, when X is projective and f is q -polarized with $q > 1$, the set S_{f^∞} coincides with the set of *periodic points* $\text{Per}(f)$, which is Zariski dense and countable by [Fak03, Proof of Theorem 5.1]. Therefore, it is reasonable to propose the following question.

Question 8.1.3. Let $f : X \rightarrow X$ be a dominant self-morphism of an algebraic variety. Does the upper bound (of cardinality)

$$\#S_{f^n} \leq (d_1(f))^{n \cdot \dim(X)}(1 + o(1))$$

hold as $n \rightarrow \infty$?

Remark 8.1.4. (1) Question 8.1.3 is sort of a generalization of [Zha06b, Conjecture 1.2.2] which claims that if X is projective and f is q -polarized, then $\#\text{Fix}(f^n) = q^{n \cdot \dim(X)}(1 +$

$o(1))$ as $n \rightarrow \infty$. In this case, the set of fixed points $\text{Fix}(f^n)$ is just S_{f^n} and Question 8.1.3 has a positive answer when X is smooth and general fixed points have multiplicity one; see the argument after [Zha06b, Conjecture 1.2.2] by using Lefschetz fixed point theorem and Serre's Theorem, i.e., [Zha06b, Theorem 1.1.2] (which are available for smooth X , but there are no easily applicable results even for singular surfaces) and noting that $\text{Fix}(f^n)$ is a finite set for each $n \geq 1$.

(2) Here is a concrete example showing that the upper bound of the inequality in Question 8.1.3 is optimal. Let A be an abelian variety and $f := (-q)_A$ the multiplication map by $-q$ (with $q > 1$) on A . Then f is q^2 -polarized and $d_1(f) = q^2$. Hence $\#S_{f^{2n+1}} = \#\text{Fix}(f^{2n+1})$ is the number of $(q^{2n+1} + 1)$ -torsion points. So we have

$$(q^2)^{(2n+1)\dim(A)} < \#S_{f^{2n+1}} = (q^{2n+1} + 1)^{2\dim(A)} = (q^2)^{(2n+1)\dim(A)}(1 + o(1))$$

as $n \rightarrow \infty$.

Theorems 8.1.5, 8.1.6 and 8.1.9 are our main results of this chapter.

The key idea of our first result concerning isogenies (and their translations) of algebraic groups is to find a *unique* maximal periodic SDD subgroup; see Definition 8.3.5. We leave it to the readers to consider more general cases, e.g., spherical varieties, (almost) homogeneous varieties. We use the fact $d_1(f^n) = d_1(f)^n$ for any $n > 0$ to get the following upper bound when applying theorems in Section 8.4.

Theorem 8.1.5 ([MMS⁺21, Theorem 1.5]). *Let f be a dominant self-morphism of a variety X . Then*

$$\#S_{f^n} \leq (d_1(f))^{n \cdot \dim(X)}(1 + o(1))$$

holds as $n \rightarrow \infty$, if (X, f) fits one of the following cases.

- (1) (cf. Theorem 8.4.7) X is an algebraic group; and $f = R_a \circ g$ where g is an isogeny and R_a is the right multiplication map by some $a \in X$ (when X is semi-abelian, the self-morphism f is automatically of this form).
- (2) (cf. Theorem 8.4.8) X is a toric variety with the big torus T being f -invariant.

Theorem 8.1.6 below is a simpler version of Theorem 8.7.1, and answers Question 8.1.1 (2) affirmatively: the finiteness of SDD prime divisors. The main idea is to apply the higher dimensional Hodge index theorem to construct sufficiently many pairwise disjoint prime divisors on a smooth model. In this way, we are able to construct an equivariant fibration to a curve which contracts those periodic SDD prime divisors; and then what left is simply to count the fixed points of a polarized endomorphism on a curve.

Theorem 8.1.6 ([MMS⁺21, Theorem 1.5]; cf. Theorem 8.7.1). *Let f be a surjective endomorphism of a projective variety. Then*

$$\sharp P_{f^n} \leq (d_1(f))^n (1 + o(1))$$

holds as $n \rightarrow \infty$.

The upper bound of the inequality in Theorem 8.1.6 above is optimal. Indeed, let $X = Y \times E$, where Y is any projective variety and E is an elliptic curve and let $f = \text{id}_Y \times f_E$ where $f_E := (-q)_E$ is the multiplication map by $-q$ (with $q > 1$) on E . In this case, by the product formula (cf. Notation 8.2.1), $d_1(f) = q^2$, and a prime divisor $P \in P_{f^n}$ if and only if it is an f^n -stable fibre of the natural projection $X \rightarrow E$. Thus $\sharp P_{f^{2n+1}} = \sharp \text{Fix}(f_E^{2n+1}) = (q^{2n+1} + 1)^2 = (d_1(f))^{2n+1} (1 + o(1))$ (cf. Remark 8.1.4).

When X is projective, Question 8.1.1 (2) can be reformulated as follows.

Theorem 8.1.7 ([MMS⁺21, Theorem 1.7]). *Let $f : X \rightarrow X$ be a surjective endomorphism of a projective variety X . Then the following assertions are equivalent.*

- (1) *The set $P_{f^\infty} := \bigcup_{n=1}^{\infty} P_{f^n}$ is infinite.*
- (2) *Replacing X by its normalization, there is an (iteration of) f -equivariant fibration $\pi : X \rightarrow C$ onto a smooth projective curve C such that $g := f|_C$ is $d_1(f)$ -polarized and the relative dynamical degree $d_1(f|_\pi) < d_1(f)$.*

In particular, the set of f^{-1} -periodic prime divisors of small dynamical degree is finite.

In Theorem 8.1.7 (2), the smooth curve C is either elliptic or rational, since it admits a non-isomorphic endomorphism g (and hence the Kodaira dimension $\kappa(C) \leq 0$).

Corollary 8.1.8 answers Question 8.1.1 affirmatively for projective surfaces, which is a direct consequence of Theorem 8.1.6 and Proposition 8.2.6. However, we are currently unable to show the upper bound in Question 8.1.3 in full generality except cases in Theorem 8.1.5.

Corollary 8.1.8 ([MMS⁺21, Corollary 1.8]). *S_f is finite for any surjective endomorphism f of a projective surface.*

Throughout this chapter, we try to make our argument as algebraic as we can so that most of our results can be generalized to positive characteristic. We leave such generalization to the careful readers which will not be mentioned too much in this chapter for the sake of a comfortable reading.

We end the introduction with Theorem 8.1.9, as an application of a technique used when considering algebraic groups (the commutative diagram in Proposition 8.3.3). This compound theorem has its own interest and extends the results in [CMZ20, Theorem 1.2 and Theorem 5.1], [MZ19b, Proposition 3.3], [NZ10, Lemma 2.3] and [San17, Remark 5.9] to the most general setting (without assumptions like smooth, characteristic 0, separable, int-amplified, etc.). Theorem 8.1.9 is a simpler version of Theorem 8.5.4.

Theorem 8.1.9 ([MMS⁺21, Theorem 1.9]; cf. Theorem 8.5.4). *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal projective variety over the field k of arbitrary characteristic. Assume $d_1(f) > 1$. Then we have:*

- (1) *$f^*D \sim_{\mathbb{R}} d_1(f)D$ for some nef $\mathbb{R}$ -Cartier divisor D .*
- (2) *Suppose $f^*B \equiv qB$ for some big divisor and $q > 1$. Then f is q -polarized.*
- (3) *Suppose f is q -polarized. Then any eigenvalue of $f^*|_{H_{\text{ét}}^1(X, \mathbb{Q}_\ell)}$ has modulus $\sqrt{q}$.*
- (4) *Suppose f is q -polarized (resp. int-amplified). Then so is $f|_{\text{Alb}(X)} : \text{Alb}(X) \rightarrow \text{Alb}(X)$.*

In Remark 8.1.4, the method of counting fixed points for polarized endomorphisms only holds in characteristic 0, because it applied Serre's theorem to compute the eigenvalues of $f^*|_{H^i(X, \mathbb{Z})}$ which heavily relied on a Kähler form. Serre's theorem is also a Kähler

analogue of Weil's conjecture on eigenvalues of the pull-back of (the purely inseparable q -polarized) Frobenius endomorphism F on cohomology. It was proved by Deligne's celebrated theorem: all eigenvalues of $F^*|_{H_{\text{ét}}^i(X, \mathbb{Q}_\ell)}$ have modulus $q^{i/2}$ (cf. [Del74, Théorème 1.6]). By our Theorem 8.1.9 (3), this holds for every polarized endomorphism when $i = 1$.

This chapter is based on the joint work with Yohsuke Matsuzawa, Sheng Meng, Takahiro Shibata and De-Qi Zhang [MMS⁺21]. For the organization, we shall skip [MMS⁺21, Section 3] in this thesis. Readers may refer to [MMS⁺21, Section 3] for the motivation from arithmetic geometry.

8.2 Preliminaries

Notation 8.2.1. We use the following notation throughout this chapter.

- Let X be a projective variety. Denote by $\text{NS}(X) := \text{Pic}(X)/\text{Pic}^\circ(X)$ the usual Néron-Severi group (in the sense of Cartier divisors). Let $\text{N}^1(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, and $\text{PE}^1(X)$ the closed cone of pseudo-effective classes in $\text{N}^1(X)$.
- An r -cycle C on a projective variety X is *pseudo-effective* if $C \cdot H_1 \cdots H_r \geq 0$ for all ample divisors H_i . Two r -cycles C_1 and C_2 are said to be *weakly numerically equivalent* if $(C_1 - C_2) \cdot L_1 \cdots L_r = 0$ for all Cartier divisors L_i ; cf. [MZ18a, Section 2] and the references therein for more information.
- The symbols $\sim$ (resp. $\sim_{\mathbb{Q}}$, $\sim_{\mathbb{R}}$) and $\equiv$ (resp. $\equiv_w$) denote the linear (resp. $\mathbb{Q}$ -linear, $\mathbb{R}$ -linear) equivalence and the numerical (resp. weak numerical) equivalence on $\mathbb{R}$ -divisors.
- A surjective endomorphism $f : X \rightarrow X$ of a projective variety is said to be *q -polarized* if $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$. The unique endomorphism on a point is q -polarized for any $q > 1$ by convention.
- A surjective endomorphism $f : X \rightarrow X$ of a projective variety is said to be *int-amplified* if $f^*H - H = L$ for some ample Cartier divisors H and L .

- Let X be a normal projective variety. Denote by $\text{Cl}(X)$ the Weil divisor class group on X , and by $\text{Cl}^\circ(X) \subseteq \text{Cl}(X)$ the subgroup of divisors algebraically equivalent to 0, which is a birational invariant (cf. [Kol18, Lemma 18]). The quotient

$$\text{Cl}^{\text{ns}}(X) := \text{Cl}(X)/\text{Cl}^\circ(X)$$

is called the *Néron-Severi* class group of X (in the sense of Weil divisors).

There are natural inclusions $\text{Pic}^\circ(X) \subseteq \text{Cl}^\circ(X)$ and $\text{NS}(X) \subseteq \text{Cl}^{\text{ns}}(X)$. Both of these are equalities when X is smooth.

- Denote by

$$\rho(X) := \dim_{\mathbb{R}} \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$$

the usual *Picard number* of X . Denote by

$$\rho^{\text{ns}}(X) := \dim_{\mathbb{R}} \text{Cl}^{\text{ns}}(X) \otimes_{\mathbb{Z}} \mathbb{R}$$

which is finite (cf. [Kol18, Theorem 17]). We refer readers to [Kol18, Section 3] for more information.

- Let $f : X \dashrightarrow X$ and $g : Y \dashrightarrow Y$ be dominant self-maps of projective varieties. Let $\pi : X \dashrightarrow Y$ be a dominant rational map such that $g \circ \pi = \pi \circ f$. We recall the following definitions and facts about the first dynamical degree.

- (1) The *(first) dynamical degree* of f is defined by

$$d_1(f) := \lim_{n \rightarrow \infty} ((f^n)^* H \cdot H^{\dim(X)-1})^{1/n}$$

where H is a nef and big Cartier divisor of X .

- (2) The *relative (first) dynamical degree* of f is defined by

$$d_1(f|_{\pi}) := \lim_{n \rightarrow \infty} ((f^n)^* H_X \cdot (\pi^* H_Y)^{\dim(Y)} \cdot H_X^{\dim(X)-\dim(Y)-1})^{1/n}$$

where H_X and H_Y are respectively nef and big Cartier divisors of X and Y .

- (3) Product formula: $d_1(f) = \max\{d_1(f|_{\pi}), d_1(g)\}$.

- (4) $d_1(f^n) = d_1(f)^n$ for any $n > 0$.
- (5) $d_1(f) = \rho(f^*|_{N^1(X)})$, the spectral radius of $f^*|_{N^1(X)}$, if f is a self-morphism.

(Relative) dynamical degrees can also be defined, by the above formulas, for (not necessarily projective) varieties over the field k (allowed to be of arbitrary characteristic) by taking any birational projective models since they are birational invariants and independent of the choices of nef and big divisors above; indeed, $d_1(f)$ is invariant under generically finite map; see [DS05] and [Dan20] for details.

- Let $f : X \rightarrow X$ be a surjective endomorphism of a smooth projective variety. Denote by

$$\chi_i(f) := \rho(f^*|_{H_{\text{ét}}^i(X, \mathbb{Q}_\ell)})$$

where ρ is the spectral radius, $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$ is the ℓ -adic cohomology, and ℓ is a prime $\neq$ the characteristic of the ground field k . When X is an abelian variety, a characteristic free proof by Hu (cf. [Hu19, Theorem 1.2 and (3.6)]) showed that

$$\chi_2(f) = d_1(f) = \chi_1(f)^2.$$

- A normal (not necessarily projective) variety X is a *toric variety* if X contains an algebraic torus $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on X .
- A commutative algebraic group S is *semi-abelian* if S is an extension of an abelian variety A by an algebraic torus T .
- Let $f : X \rightarrow X$ be a dominant self-morphism of a variety, which is not necessarily projective. A closed subvariety $Y \subseteq X$ is called *f -invariant* (resp. *f -periodic*) if $\overline{f(Y)} = Y$ (resp. $\overline{f^s(Y)} = Y$ for some $s > 0$). An f -periodic closed subvariety V is said to be of *Small Dynamical Degree* (**SDD** for short) if the first dynamical degree $d_1(f^s|_V) < d_1(f^s)$ for some $s > 0$ such that $\overline{f^s(V)} = V$.
- Denote by Σ_{f^∞} the set of f -periodic (irreducible closed) subvarieties of Small Dynamical Degree (**SDD** for short). We define the following subsets of Σ_{f^∞} :

- S_{f^∞} the subset of maximal f -periodic subvarieties in Σ_{f^∞} ,
- S_f the subset of S_{f^∞} , consisting of all f -invariant elements,
- P_f the subset of S_{f^∞} , consisting of all f -invariant prime divisors,
- P_{f^∞} the subset of (maximal) f -periodic prime divisors in Σ_{f^∞} ,
- $P_{f^\infty}^{-1}$ the subset of (maximal) f^{-1} -periodic prime divisors in Σ_{f^∞} .

- An element $D \in S_f$ is called an f -invariant maximal f -periodic SDD subvariety.
- Denote by $\text{Fix}(f)$ (resp. $\text{Per}(f)$) the set of fixed (resp. periodic) points of f .

The following lemma is basic, and is used in the proof of Proposition 8.7.5.

Lemma 8.2.2. *Let $\sigma : \tilde{X} \rightarrow X$ be a birational morphism between normal projective varieties. Let L be a nef $\mathbb{R}$ -Cartier divisor on X and denote by $\tilde{L} := \sigma^*L$. Let $\tilde{\pi} : \tilde{X} \rightarrow Y$ be a morphism to another projective variety Y such that $\tilde{L} \equiv \tilde{\pi}^*H$ with H an ample divisor on Y . Then $\tilde{\pi}$ factors through a well-defined morphism $\pi : X \rightarrow Y$ with $L \equiv \pi^*H$.*

Proof. By Zariski's main theorem, $\sigma_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$. For every curve C lying in a fibre of σ , it follows from the projection formula that $H \cdot \tilde{\pi}_*C = \tilde{\pi}^*H \cdot C = \sigma^*L \cdot C = 0$. Since H is ample on Y , $\tilde{\pi}(C)$ is a point and hence every fibre of σ will also be contracted by $\tilde{\pi}$. By the rigidity lemma (cf. [Deb01, Lemma 1.15]), $\tilde{\pi}$ factors through a well-defined morphism $\pi : X \rightarrow Y$. So $\pi^*H = \sigma_*\sigma^*\pi^*H \equiv \sigma_*\tilde{L} = L$. $\square$

Given an f -equivariant surjective morphism $\pi : X \rightarrow Y$ with $g := f|_Y$, it is natural to compare those $\sharp S_f, \sharp P_{f^\infty}^{-1}, \sharp P_f$ on f with the corresponding ones on g .

When π is finite, we formulate the following lemma to bound $\sharp S_g$ in terms of $\sharp S_{f^d!}$ with $d = \deg(\pi)$. It is applied to prove the finiteness of $\sharp S_g$ for Q -abelian case in Corollary 8.4.2. Our upper bound for $\sharp S_g$ in the proof of Lemma 8.2.3 is not optimal at all.

Lemma 8.2.3. *Let $\pi : X \rightarrow Y$ be a finite surjective morphism of degree d between projective varieties. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be two surjective endomorphisms such that $\pi \circ f = g \circ \pi$. If $\sharp S_{f^d!} < \infty$, then $\sharp S_g < \infty$. Conversely, if $\sharp S_g < \infty$, then $\sharp S_f < \infty$.*

Proof. Since $\pi : X \rightarrow Y$ is finite and $d_1(f)$ is invariant under finite map, π induces a surjective map $\pi : S_{f^\infty} \rightarrow S_{g^\infty}$ while π^{-1} induces a one-to-many surjective map $\pi^{-1} : S_{g^\infty} \rightarrow S_{f^\infty}$. Then the lemma follows. $\square$

When π is a birational morphism, we can only consider prime divisors rather than higher codimensional subvarieties due to the complexity of periodic subvarieties contained in the exceptional locus of π .

Lemma 8.2.4. *Let $\pi : X \dashrightarrow Y$ be a birational map between projective varieties. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be two surjective endomorphisms such that $\pi \circ f = g \circ \pi$. Then $\sharp P_{f^n} = \sharp P_{g^n} + O(1)$ as $n \rightarrow \infty$, and $\sharp P_{f^\infty}$ is finite if and only if so is $\sharp P_{g^\infty}$.*

Proof. By taking the graph $\Gamma \subseteq X \times Y$, our f lifts to a surjective endomorphism $(f \times g)|_\Gamma$. So it suffices for us to consider the case when π is a birational morphism. Let B be the closed subset of Y , over which π is not isomorphic. Let $\mathcal{B}_1$ be the set of prime divisors in B and $\mathcal{B}_2$ the set of prime divisors in $\pi^{-1}(B)$. For each $n > 0$, we have a natural map

$$\varphi_n : P_{f^n} \setminus \mathcal{B}_2 \rightarrow P_{g^n} \setminus \mathcal{B}_1,$$

sending D to $\pi(D)$. Note that φ_n admits an inverse, sending $D' \in P_{g^n} \setminus \mathcal{B}_1$ to its proper transform $\pi_*^{-1}(D')$. In particular, setting $K := \sharp \mathcal{B}_1 + \sharp \mathcal{B}_2$, we have:

$$\sharp P_{g^n} - K \leq \sharp P_{f^n} \leq \sharp P_{g^n} + K.$$

$\square$

By a similar proof, we have the following lemma.

Lemma 8.2.5. *Let $\pi : X \dashrightarrow Y$ be a birational map between projective varieties. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be two surjective endomorphisms such that $\pi \circ f = g \circ \pi$. Then $\sharp P_{f^\infty}^{-1}$ is finite if and only if so is $\sharp P_{g^\infty}^{-1}$.*

Proposition 8.2.6. *Let $f : X \rightarrow X$ be a surjective endomorphism of a projective variety. Then there are only finitely many points (as zero-dimensional subvarieties) in the set S_f .*

Proof. We may assume $d_1(f) > 1$. Let $Y \subseteq S_f$ be the subset of f -fixed closed points. Suppose on the contrary that $\#Y$ is infinite. Let Z be an irreducible component of the closure of Y in X with $\dim(Z) > 0$. Then $f|_Z = \text{id}_Z$ and $1 = d_1(f|_Z) < d_1(f)$, a contradiction to the maximality of the points in Y . $\square$

8.3 Isogenies of algebraic groups

We use the following notation throughout this section.

Notation 8.3.1. Let G be a connected algebraic group with the identity e . We focus on the isogeny (i.e., surjective group homomorphism with finite kernel)

$$g : G \rightarrow G$$

with $d_1(g) > 1$, and

$$f := R_a \circ g$$

where R_a is the right multiplication map by some $a \in G$. Note that there exists a G -equivariant embedding $G \hookrightarrow \tilde{G}$ to a (smooth) projective variety $\tilde{G}$ (cf. [Bri10, Theorem 2]). Then the same proof for [MS20, Lemma 5.6] shows that

$$d_1(f) = d_1(g) > 1.$$

We recall that g and hence f are finite surjective endomorphisms. So f -invariant (resp. f -periodic) closed subvarieties defined in Notation 8.2.1 are in the usual sense without taking closure.

Lemma 8.3.2. *Let H be a (connected closed) algebraic subgroup of G . Suppose $f(Hb) = Hb$ for some $b \in G$. Then $g(H) = H$ and $d_1(f|_{Hb}) = d_1(g|_H)$.*

Proof. Let $p := f(b)b^{-1} \in H$. We have

$$g(H) = g(Hb)g(b)^{-1} = f(Hb)a^{-1}(f(b)a^{-1})^{-1} = Hbf(b)^{-1} = Hp^{-1} = H.$$

Moreover, We have the following commutative diagram:

$$\begin{array}{ccc} H & \xrightarrow{(R_p \circ g)|_H} & H \\ R_b \downarrow & & \downarrow R_b \\ Hb & \xrightarrow{f|_{Hb}} & Hb \end{array}$$

where $R_b : H \rightarrow Hb$ is an isomorphism of varieties. So $d_1(f|_{Hb}) = d_1((R_p \circ g)|_H) = d_1(g|_H)$ (cf. [MS20, Lemma 5.6]). $\square$

The following result is a generalization of Lemma 8.3.2 to arbitrary closed subvariety of G , which plays a crucial role in many later results.

Proposition 8.3.3. *Let X be an f -invariant (irreducible closed) subvariety of G . Fix some $b \in X$ and denote by $X_e := Xb^{-1}$. Let H be the smallest algebraic subgroup of G containing X_e . Then $g(H) = H$, $f(Hb) = Hb$, and $d_1(g|_H) = d_1(f|_{Hb}) \leq d_1(f|_X)$.*

Proof. Let $p := f(b)b^{-1} \in X_e \subseteq H$. We have

$$g(H) \supseteq g(X_e) = g(X)g(b)^{-1} = f(X)f(b)^{-1} = X_ep^{-1}.$$

Then $X_e \subseteq X_ep^{-1}(X_ep^{-1})^{-1} \subseteq g(H)$ and hence $H \leq g(H)$ by the construction of H . Since g is an isogeny, $g(H) = H$.

Consider the alternate multiplication map $s_n : X^{\times(2n)} \rightarrow H$ defined by

$$\begin{aligned} s_n(x_1b, \dots, x_{2n}b) &= (x_1b)(x_2b)^{-1} \cdots (x_{2i+1}b)(x_{2i+2}b)^{-1} \cdots (x_{2n-1}b)(x_{2n}b)^{-1} \\ &= x_1x_2^{-1} \cdots x_{2i+1}x_{2i+2}^{-1} \cdots x_{2n-1}x_{2n}^{-1} \\ &\in (X_eX_e^{-1})^n \end{aligned}$$

Then the image of s_n is $(X_eX_e^{-1})^n = H$ when $n \gg 1$ by [Spr09, Proposition 2.2.6] and noting that $e \in X_e$. Consider the following diagram with $n \gg 1$:

$$\begin{array}{ccc} X^{\times(2n)} & \xrightarrow{(f|_X)^{\times(2n)}} & X^{\times(2n)} \\ s_n \downarrow & & \downarrow s_n \\ H & \xrightarrow{g|_H} & H \end{array} \quad (*)$$

Note that $f(x_i b) = g(x_i)pb$. Then

$$\begin{aligned} s_n(f(x_1 b), \dots, f(x_{2n} b)) &= s_n(g(x_1)pb, \dots, g(x_{2n})pb) \\ &= g(x_1)g(x_2)^{-1} \cdots g(x_{2n-1})g(x_{2n})^{-1} \\ &= (g \circ s_n)(x_1 b, \dots, x_{2n} b). \end{aligned}$$

So the diagram commutes. By the product formula, we have

$$d_1(g|_H) \leq d_1((f|_X)^{\times(2n)}) = d_1(f|_X).$$

On the other hand, note that

$$f(Hb) = g(Hb)a = Hg(b)a = Hf(b)b^{-1}b = (Hp)b = Hb.$$

By Lemma 8.3.2, we have $d_1(f|_{Hb}) = d_1(g|_H) \leq d_1(f|_X)$. □

Lemma 8.3.4. *There exists a unique $X \in S_{g^\infty}$ containing $e \in G$. Moreover, X is an algebraic subgroup and g -invariant.*

Proof. Take $X \in S_{g^\infty}$ containing e . It exists because $d_1(g) > 1$ and $d_1(g|_{\{e\}}) = 1$. Take $s > 0$ such that $g^s(X) = X$. Take the smallest (connected closed) algebraic subgroup H of G containing X . By Proposition 8.3.3, $g^s(H) = H$ and $d_1(g^s|_H) \leq d_1(g^s|_X) < d_1(g^s)$. By the maximality of X , we have $X = H$ and it is an algebraic subgroup.

Take another $Y \in S_{g^\infty}$ containing e and then take $t > 0$ such that $g^t(Y) = Y$. The argument in the previous paragraph shows that Y is a (connected closed) algebraic subgroup. Let Z be the smallest algebraic group containing the g^{st} -invariant irreducible closed subvariety XY . Apply the similar commutative diagram (*) and argument in Proof of Proposition 8.3.3,

$$\begin{array}{ccc} (X \times Y)^{\times(2n)} & \xrightarrow{(g^{st}|_X \times g^{st}|_Y)^{\times(2n)}} & (X \times Y)^{\times(2n)} \\ s_n \downarrow & & \downarrow s_n \\ Z & \xrightarrow{g^{st}|_Z} & Z \end{array}$$

where $s_n : (X \times Y)^{\times(2n)} \rightarrow Z = (XY)(XY)^{-1} \cdots (XY)(XY)^{-1}$ defined by

$$s_n(x_1, y_1, \dots, x_{2n}, y_{2n}) = (x_1 y_1)(x_2 y_2)^{-1} \cdots (x_{2n-1} y_{2n-1})(x_{2n} y_{2n})^{-1}$$

is surjective with $n \gg 1$. Then we have $d_1(g^{st}|_Z) < d_1(g^{st})$ and hence $Z = X = Y$ by the maximality of X and Y . This shows the uniqueness of X .

Next we show that X is g -invariant. Note that $g(X)$ is also g^s -invariant and contains e . By the product formula, the following diagram shows $d_1(g^s|_{g(X)}) \leq d_1(g^s|_X) < d_1(g^s)$:

$$\begin{array}{ccc} X & \xrightarrow{g^s|_X} & X \\ g \downarrow & & \downarrow g \\ g(X) & \xrightarrow{g^s|_{g(X)}} & g(X) \end{array}$$

Therefore, we have $g(X) \subseteq X$ by the maximality and uniqueness of X . In particular, X is g -invariant since g is an isogeny. This finishes our proof. $\square$

Definition 8.3.5. With the notation in Lemma 8.3.4, we denote by

$$G(f) = G(g) := X$$

the *unique* element of S_{g^∞} containing $e \in G$. Note that $G(f)$ is g -invariant and $G(f^m) = G(g^m) = G(f)$ for any $m > 0$.

We shall prove in the next section that $G(f)$ is actually a normal subgroup of G .

Proposition 8.3.6. A subvariety $X \subseteq G$ is a member of S_f if and only if $X = G(f)b$ with $f(b)b^{-1} \in G(f)$ ($= G(g)$).

Proof. Assume that $X \in S_f$. Fix some $b \in X$ and denote by $X_e := Xb^{-1}$ which contains $e \in G$. Let H be the smallest algebraic subgroup of G containing X_e . By Proposition 8.3.3, $g(H) = H$, $f(Hb) = Hb$, and $d_1(g|_H) \leq d_1(f|_X) < d_1(f)$. Then $f(b)b^{-1} \in X_e \subseteq H \leq G(g)$ by Lemma 8.3.4. Note that

$$f(G(f)b) = g(G(g))g(b)a = (G(g)f(b)b^{-1})b = G(f)b.$$

By Lemma 8.3.2,

$$d_1(f|_{G(f)b}) = d_1(g|_{G(g)}) < d_1(g) = d_1(f).$$

Since $X \subseteq Hb \subseteq G(f)b$, we have $X = G(f)b$ by the maximality of X .

Conversely, assume that $X = G(f)b$ with $f(b)b^{-1} \in G(f)$. Then $X \in \Sigma_f$ by the same argument above. Suppose that we have $Y \in S_{f^\infty}$ containing X . Take $s > 0$ such

that $Y \in S_{f^s}$. Then $Y = G(f^s)c = G(f)c$ for some $c \in G$ with $f^s(c)c^{-1} \in G(f)$ by the previous paragraph. But this implies that $X = Y$ since both are translations of the same group $G(f)$. So X is maximal. $\square$

The following modified product formula will be frequently used in the next Theorem and also next section.

Lemma 8.3.7. *Consider the commutative diagram of homomorphisms of connected algebraic groups:*

$$\begin{array}{ccccccc} 1 & \longrightarrow & B & \longrightarrow & A & \xrightarrow{\pi} & C \longrightarrow 1 \\ & & \downarrow g|_B & & \downarrow g & & \downarrow g|_C \\ 1 & \longrightarrow & B & \longrightarrow & A & \xrightarrow{\pi} & C \longrightarrow 1 \end{array}$$

where g is an isogeny. Then π is f -equivariant and the following hold.

- (1) $d_1(g|_\pi) = d_1(g|_B)$. In particular, $d_1(g) = \max\{d_1(g|_B), d_1(g|_C)\}$.
- (2) Suppose $d_1(g|_B) < d_1(g)$. Then $A(f) = \pi^{-1}(C(f|_C))$ and π induces a bijection between S_f and $S_{f|_C}$.

Proof. First, note that π is f -equivariant with $f|_C = R_{\pi(a)} \circ g|_C$.

Let A act on A and C by the right multiplication. By [Bri10, Theorem 2], we have A -equivariant open embeddings to smooth projective varieties $A \hookrightarrow \bar{A}$ and $C \hookrightarrow \bar{C}$ (taking A -equivariant resolution if necessary). Denote by $\bar{\pi} : \bar{A} \dashrightarrow \bar{C}$ the induced dominant rational map. Denote by $\bar{g} : \bar{A} \dashrightarrow \bar{A}$ and $\bar{g}|_{\bar{C}} : \bar{C} \dashrightarrow \bar{C}$ the induced dominant self-maps. Note that the graph of $\bar{\pi}$ is A -equivariant. Replacing $\bar{A}$ by the A -equivariant resolution of this graph, we may further assume that $\bar{\pi}$ is a well-defined algebraic fibration.

By the generic smoothness of $\bar{\pi}$, there exists a general point $c_1 = \pi(a_1) \in C$ such that the fibre $\bar{A}_{c_1} := \bar{\pi}^{-1}(c_1)$ is an (irreducible) smooth projective variety. For any $c_2 = \pi(a_2) \in C$, we have $\bar{A}_{c_2} = \bar{A}_{c_1 \cdot (c_1^{-1}c_2)} = \bar{A}_{c_1} \cdot (a_1^{-1}a_2)$. So all the fibres of $\bar{\pi}$ over C are (irreducible) smooth projective varieties isomorphic to each other.

We refer to [DN11] for the definitions of critical value and critical set. Now we have that each point of C is not a critical value of $\bar{\pi}$. Note that B is g -invariant and g is étale. Then $\bar{A}_e$ (the closure of B in $\bar{A}$) is not contained in the critical set of $\bar{g}$. Therefore, we

verified the assumptions for $\overline{A}_e$ in [DN11, Remark 3.4] and hence $d_1(\overline{g}|\overline{\pi}) = d_1(\overline{g}|\overline{A}_e) = d_1(g|_B)$. Together with the product formula, (1) is proved.

For (2), $d_1(g) = d_1(g|_C)$ by (1).

We apply the maximality and uniqueness of $A(g)$ and $C(g|_C)$ (cf. Lemma 8.3.4) and our new product formula (1) several times in this diagram. First, we have $B \leq A(g)$. Note that $d_1(g|_{\pi(A(g))}) = d_1(g|_{A(g)}) < d_1(g)$. So we have $\pi(A(g)) \leq C(g|_C)$. Conversely, $d_1(g|_{\pi^{-1}(C(g|_C))}) = d_1((g|_C)|_{C(g|_C)}) < d_1(g|_C) = d_1(g)$. So we have $A(g) = \pi^{-1}(C(g|_C))$.

Finally, we apply Proposition 8.3.6 for f and $f|_C$, then we see the induced map

$$\pi : S_f \rightarrow S_{f|_C}, \quad \text{via} \quad A(f)b \mapsto C(f|_C)\pi(b)$$

is a bijection. So (2) is proved. $\square$

We complete this section by considering the commutative algebraic groups first.

Theorem 8.3.8. *Let $f : G \rightarrow G$ be a dominant self-morphism of a connected **commutative** algebraic group G , such that $f = R_a \circ g$ where g is an isogeny and $a \in G$. Suppose $d_1(f) > 1$. Let $G(f)$ be as in Definition 8.3.5. Then*

$$\pi : G \rightarrow \overline{G} := G/G(f)$$

is f, g -equivariant and

$$0 < \#S_f = \#S_g = \#\text{Fix}(\overline{f}) = \#\text{Fix}(\overline{g}) < \infty$$

where $\overline{f} = f|_{\overline{G}} = R_{\pi(a)} \circ \overline{g}$.

Proof. We simply define $\overline{f}(x \cdot G(f)) := f(x) \cdot G(f)$. It is well-defined and $\overline{f} \circ \pi = \pi \circ f$ since $g(G(f)) = G(f)$. By Lemma 8.3.7 and Proposition 8.3.6, $S_{\overline{f}} = \text{Fix}(\overline{f})$ and π induces a bijection between S_f and $\text{Fix}(\overline{f})$.

Consider the case $\#\text{Fix}(\overline{g}) = \infty$. Then there is a positive dimensional (connected closed) algebraic subgroup $\overline{H}$ of $\overline{G}$ such that $\overline{g}|_{\overline{H}} = \text{id}_{\overline{H}}$. Let $H := \pi^{-1}(\overline{H})$ which is a g -invariant (connected closed) algebraic subgroup of G . By the product formula in Lemma

8.3.7, we have

$$d_1(g|_H) = \max\{d_1(g|_\pi), d_1(\bar{g}|_{\bar{H}})\} = d_1(g|_\pi) = d_1(g|_{G(g)}) < d_1(g).$$

This contradicts the maximality of $G(g)$.

Now we have shown $\sharp \text{Fix}(\bar{g}) < \infty$. Then $\phi : \bar{G} \rightarrow \bar{G} (\bar{x} \mapsto g(\bar{x}) \cdot \bar{x}^{-1})$ is an isogeny. Note that $\bar{f}(\bar{x}) = \bar{x}$ if and only if $\phi(\bar{x}) = \bar{a}^{-1}$. So we have

$$0 < \sharp \text{Fix}(\bar{f}) = \sharp \phi^{-1}(\bar{a}^{-1}) = \sharp \phi^{-1}(\bar{e}) = \sharp \text{Fix}(\bar{g}) < \infty$$

and the theorem is proved. $\square$

8.4 Proof of Theorem 8.1.5

In this section, we focus on the upper bound in Question 8.1.3. We first consider the case of abelian varieties.

Theorem 8.4.1. *Let f be a surjective endomorphism of an abelian variety A . Then*

$$\sharp S_f \leq (\sqrt{d_1(f)} + 1)^{2(\dim(A) - \dim(A(f)))}$$

(cf. Definition 8.3.5 for $A(f)$).

Proof. We may assume $d_1(f) > 1$. Write $f = R_a \circ g$ where g is an isogeny and $a \in A$. Since $d_1(f) = d_1(g)$ and $A(f) = A(g)$, applying Theorem 8.3.8, we may assume that f is an isogeny, i.e., $f = g$; further we have an f -equivariant quotient morphism

$$\pi : A \rightarrow A' := A/A(f)$$

such that $0 < \sharp S_f = \sharp \text{Fix}(\bar{f}) < \infty$ where $f' = f|_{A'}$. Denote by $\phi : A' \rightarrow A' (x' \mapsto f'(x') \cdot x'^{-1})$ (see the ending part of Proof of Theorem 8.3.8), which is an isogeny. Then

$$\sharp \text{Fix}(f') = \deg(\phi).$$

Set $m := \dim(A') = \dim(A) - \dim(A(f))$. We compute

$$\begin{aligned} \deg(\phi) &= \chi_{2m}(\phi) \\ &= \rho(\phi^*|_{H_{\text{ét}}^{2m}(A', \mathbb{Q}_\ell)}) \\ &= \rho\left((\wedge^{2m} \phi^*)|_{\wedge^{2m} H_{\text{ét}}^1(A', \mathbb{Q}_\ell)}\right) \\ &= |\det(\phi^*|_{H_{\text{ét}}^1(A', \mathbb{Q}_\ell)})|. \end{aligned}$$

Note that $\phi^*|_{H_{\text{ét}}^1(A', \mathbb{Q}_\ell)} = f'^*|_{H_{\text{ét}}^1(A', \mathbb{Q}_\ell)} - \text{id}_{H_{\text{ét}}^1(A', \mathbb{Q}_\ell)}$ (cf. [Hu19, Theorem 2.2]). So, letting $\alpha_1, \alpha_2, \dots, \alpha_{2m}$ be the eigenvalues of $f'^*|_{H_{\text{ét}}^1(A', \mathbb{Q}_\ell)}$, we have

$$\deg(\phi) = \prod_{i=1}^{2m} |\alpha_i - 1| \leq (\chi_1(f') + 1)^{2m}.$$

Note that $\chi_1(f')^2 = d_1(f') \leq d_1(f)$ by [Hu19, Formula (3.6)] and the product formula. Then the theorem follows. $\square$

Recall that a normal projective variety X is said to be **Q -abelian** (cf. [NZ10, Definition 2.13]) if there is a finite surjective morphism $A \rightarrow X$ from an abelian variety A , which is étale in codimension one.

Corollary 8.4.2. *S_f is finite for any surjective endomorphism f of a Q -abelian variety.*

Proof. By taking the Albanese closure $\pi : A \rightarrow X$ (cf. [CMZ20, Lemma 8.1]), f lifts to a surjective endomorphism $f_A : A \rightarrow A$ of an abelian variety A . Then our corollary follows from Lemma 8.2.3 and Theorem 8.4.1. $\square$

Next we treat the semi-abelian varieties. Let $f : X \rightarrow X$ be a (not necessarily dominant) self-morphism of a semi-abelian variety X . Then $R_{f(e)}^{-1} \circ f$ is a group homomorphism of X (cf. [Iit76, Theorem 2]).

When f is a group homomorphism, it induces the following commutative diagram where T is an algebraic torus and $A = \text{Alb}(X)$ is an abelian variety:

$$\begin{array}{ccccccc} 0 & \longrightarrow & T & \longrightarrow & X & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow f|_T & & \downarrow f & & \downarrow f|_A \\ 0 & \longrightarrow & T & \longrightarrow & X & \longrightarrow & A \longrightarrow 0 \end{array}$$

Theorem 8.4.3. *Let f be a dominant self-morphism of a semi-abelian variety X . Then*

$$\sharp S_f \leq (\sqrt{d_1(f)} + 1)^{2(\dim(X) - \dim(X(f)))}.$$

Proof. By the same argument as in the first paragraph of the proof of Theorem 8.4.1, we may replace f by g and X by $X/X(f)$ so that $\dim(X(f)) = 0$ and both f and $\phi : X \rightarrow X$ ($x \mapsto f(x) \cdot x^{-1}$) are isogenies.

Consider the commutative diagram before Theorem 8.4.3. Note that $f|_T$ is defined by a matrix M of size $\dim(T)$ with integer coefficients and a non-zero determinant. Then $d_1(f|_T)$ is equal to the spectral radius of M (cf. [Lin12, Theorem 1.1]). Let $\lambda_1, \dots, \lambda_{\dim(T)}$ be the diagonal entries of the Jordan canonical form of M .

We compute

$$\begin{aligned} \sharp \text{Fix}(f|_T) &= \deg(\phi|_T) = \det(M - \text{id}) \\ &= (\lambda_1 - 1) \cdots (\lambda_{\dim(T)} - 1) \\ &\leq (d_1(f|_T) + 1)^{\dim(T)}. \end{aligned}$$

Together with the computation in Proof of Theorem 8.4.1 for abelian varieties, we have

$$\begin{aligned} \sharp \text{Fix}(f) &= \sharp \text{Fix}(f|_T) \cdot \sharp \text{Fix}(f|_A) \\ &\leq (d_1(f|_T) + 1)^{\dim(T)} \cdot (\sqrt{d_1(f|_A)} + 1)^{2\dim(A)} \\ &\leq (\sqrt{d_1(f)} + 1)^{2\dim(X)} \end{aligned}$$

where the last inequality follows from $d_1(f|_T) \leq d_1(f)$ and $d_1(f|_A) \leq d_1(f)$; see e.g., [MS20, Theorem 1.6] or Lemma 8.3.7. So the theorem is proved. $\square$

Remark 8.4.4. In the proof of Theorem 8.4.3, we see that if $X = T$ is an algebraic torus, then the computation indeed shows that $\sharp S_f \leq (d_1(f) + 1)^{\dim(T) - \dim(T(f))}$.

We now prepare to deal with general algebraic groups. See, for instance, [MZ18b, §3] for several decomposition facts about algebraic groups. First, we need observations for unipotent groups and semisimple groups below.

Lemma 8.4.5. *Let $g : U \rightarrow U$ be an isogeny of a (connected) unipotent group U . Then $d_1(g) = 1$.*

Proof. Since $\text{Ker}(g)$ is a finite subgroup of the unipotent group U , it is trivial, so g is an automorphism. We show by induction on $\dim(U)$ and assume $\dim(U) > 0$. Note that U is solvable. So its commutator subgroup $U^{(1)}$ is strictly smaller than U . Note that $g(U^{(1)}) = U^{(1)}$. Consider the following commutative diagram:

$$\begin{array}{ccccccc} 1 & \longrightarrow & U^{(1)} & \hookrightarrow & U & \longrightarrow & U/U^{(1)} \longrightarrow 1 \\ & & \downarrow g|_{U^{(1)}} & & \downarrow g & & \downarrow g|_{U/U^{(1)}} \\ 1 & \longrightarrow & U^{(1)} & \hookrightarrow & U & \longrightarrow & U/U^{(1)} \longrightarrow 1 \end{array}$$

Suppose $U^{(1)}$ is not trivial. By Lemma 8.3.7 and induction, we have

$$d_1(g) = \max\{d_1(g|_{U^{(1)}}), d_1(g|_{U/U^{(1)}})\} = 1.$$

Suppose $U^{(1)}$ is trivial. Then U is commutative unipotent which is isomorphic to $k^{\times \dim(U)}$. One checks that $g \in \text{GL}(U)$ since g is additive. By [Bri10, Theorem 2], there is a $\text{GL}(U)$ -equivariant open embedding $U \hookrightarrow X$ to some smooth projective variety X (taking equivariant resolution if necessary). Thus the connected group $\text{GL}(U)$ and hence g act trivially on the lattice $\text{NS}(X)/(\text{torsion})$. So $d_1(g) = 1$. $\square$

Lemma 8.4.6. *Let $g : S \rightarrow S$ be an isogeny of a (connected) semisimple group S . Then $d_1(g) = 1$.*

Proof. First note that $(\deg(g))^n = \deg(g^n) = \#\text{Ker}(g^n) \leq \#Z(S) < \infty$ for any $n > 0$, where $Z(S)$ is the centre of S (a finite set). Thus $g \in \text{Aut}(S)$, the group of algebraic group automorphisms. Let $\pi : \tilde{S} \rightarrow S$ be the universal cover of S which is finite. Then g lifts to $\tilde{g} : \tilde{S} \rightarrow \tilde{S}$. By the product formula, we may replace S by $\tilde{S}$ and assume S is simply connected. By the homomorphisms theorem and [HM69, Proposition 2.2], $\text{Aut}(S) \cong \text{Aut}(\mathfrak{s})$ in the sense of algebraic groups via the differential map at e , where $\mathfrak{s}$ is the Lie algebra of S . By [FH91, Proposition D.40], $\text{Aut}(S)/\text{Aut}^\circ(S) \cong \text{Aut}(\mathfrak{s})/\text{Aut}^\circ(\mathfrak{s})$ is finite. Replacing g by a power, we may assume g is in the connected algebraic group $\text{Aut}^\circ(S)$. We are done by the same argument at the ending part of Lemma 8.4.5. $\square$

Finally, we are able to give the same upper bound for any algebraic group.

Theorem 8.4.7. *Let G be a connected algebraic group. Then $G(f)$ is normal in G and*

$$\sharp S_f \leq (\sqrt{d_1(f)} + 1)^{2(\dim(G) - \dim(G(f)))}$$

for any self-morphism $f = R_a \circ g$ where g is an isogeny and $a \in G$.

Proof. We may assume $1 < d_1(f)$ ($= d_1(g)$). Let G_a be the maximal connected affine normal subgroup of G , and U the unipotent radical of G_a . Then $g(G_a) = G_a$ and $g(U) = U$. Consider the following commutative diagram:

$$\begin{array}{ccccccc} 1 & \longrightarrow & U & \hookrightarrow & G & \xrightarrow{\pi} & G/U \longrightarrow 1 \\ & & \downarrow g|_U & & \downarrow g & & \downarrow g|_{G/U} \\ 1 & \longrightarrow & U & \hookrightarrow & G & \xrightarrow{\pi} & G/U \longrightarrow 1 \end{array}$$

By Lemma 8.4.5, we have $d_1(g|_U) = 1 < d_1(g)$. By Lemma 8.3.7, π is f -equivariant and $U \leq G(f) = \pi^{-1}(G(f|_{G/U}))$. Then $G(f)$ is normal in G if and only if $(G/U)(f|_{G/U})$ is normal in G/U . Therefore, we may replace G by G/U and assume that U is trivial.

Let $S := G^{(1)}$ be the commutator subgroup of G . Then $g(S) = S$, and $S = (G_a)^{(1)}$ which is a semisimple group since G_a is a reductive group now. Consider the following commutative diagram:

$$\begin{array}{ccccccc} 1 & \longrightarrow & S & \hookrightarrow & G & \xrightarrow{\pi} & G/S \longrightarrow 1 \\ & & \downarrow g|_S & & \downarrow g & & \downarrow g|_{G/S} \\ 1 & \longrightarrow & S & \hookrightarrow & G & \xrightarrow{\pi} & G/S \longrightarrow 1 \end{array}$$

By Lemma 8.4.6, we have $d_1(g|_S) = 1$. By Lemma 8.3.7 and an argument similar to the previous paragraph, we may replace G by G/S and assume that G is commutative.

Note that now G is a semi-abelian variety. In particular, $G(f)$ is normal in G , and the upper bound is as claimed, by Theorem 8.4.3 and Lemma 8.3.7. $\square$

At the end of this section, we treat the toric varieties. Let X be a toric variety and T the big torus. A subvariety of the form $O = x \cdot T$ is called a *torus orbit* on X . Note that $X = \coprod_{O \subseteq X} O$ is a finite disjoint union of all torus orbits on X .

Theorem 8.4.8. *Let X be a toric variety with the big torus $T \subseteq X$. Let $f: X \rightarrow X$ be a dominant self-morphism such that $f(T) \subseteq T$. Then*

$$\#S_f \leq \sum_{O \subseteq X} (d_1(f) + 1)^{\dim(O)}$$

where the sum runs over all the torus orbits on X .

Proof. By [Nak08, Lemma 3.4.3], $f = R_a \circ g$ such that $g|_T$ is an isogeny and $g(x \cdot t) = g(x) \cdot g(t)$; see [Nak08, Section 3.4] for details. Here R_a is the natural extension of the right multiplication map on T by $a \in T$. Since $g|_T$ is dominant, $f(T) = g(T) \cdot a = T$. Moreover, $f(x \cdot T) = g(x \cdot T) \cdot a = g(x) \cdot T$. In particular, f maps every torus orbit onto a torus orbit. Note that $d_1(f|_T) = d_1(f)$ since it is a birational invariant.

Let $V \in S_f$. Recall that $\overline{f(V)} = V$ and $d_1(f|_V) < d_1(f)$. Note that $V = \coprod_{O \subseteq X} V \cap O$. Then there is a torus orbit O such that $V \cap O$ contains an open dense subset of V . So $f(O) \cap O \supseteq f(V \cap O) \cap (V \cap O) \neq \emptyset$ and hence $f(O) = O$. Note that $d_1(f|_O) = d_1(g|_O)$ (cf. [MS20, Lemma 5.6]).

Now fixing any torus orbit O with $f(O) = O$, we compute the number of $V \in S_f$ such that $V \cap O$ contains an open dense subset of V . Fix any point $x_0 \in O$. Write $f(x_0) = x_0 \cdot b$ for some $b \in T$. Define $f_O : O \rightarrow O$ via $f_O := R_{b^{-1} \cdot a} \circ f|_O$. Then $d_1(f_O) = d_1(f|_O)$ (cf. [MS20, Lemma 5.6]). Define the natural surjective morphism $\pi : T \rightarrow O$ via $\pi(t) = x_0 \cdot t$. Note that

$$f_O \circ \pi(t) = f(x_0 \cdot t) \cdot b^{-1} \cdot a = g(x_0) \cdot g(t) \cdot a \cdot b^{-1} \cdot a = x_0 \cdot f(t) = \pi \circ f|_T(t).$$

By the product formula, $d_1(f|_O) = d_1(f_O) \leq d_1(f|_T) = d_1(f)$.

If $d_1(f|_O) < d_1(f)$, then $V = \overline{O}$ by the maximality. If $d_1(f|_O) \geq d_1(f)$, then $V \cap O \in S_{f|_O}$. Applying Remark 8.4.4 to $f|_O$, the number of such V is bounded by $(d_1(f) + 1)^{\dim(O)}$. This finishes the proof. $\square$

Remark 8.4.9. In Theorem 8.4.8, as we do not assume X is projective, f is not necessarily finite (nor surjective). Also during its proof, it is possible that $O \subseteq \text{Sing}(X)$, and then we cannot deduce $d_1(f|_O) \leq d_1(f)$ from [MS20, Lemma 5.3] where we need to take

(projective) resolution of X . Of course, if X (and hence $\overline{O}$) are projective, then there is no issue here.

8.5 Reduction to polarized case: Proof of Theorem 8.1.9

In this section, the ground field k is allowed to have arbitrary characteristic. We will prove Theorem 8.5.4 which includes Theorem 8.1.9 as a special case. The connection with polarized endomorphism will give us finiteness for $\sharp P_f$.

Proposition 8.5.1. *Let $\pi : X \dashrightarrow Y$ be a dominant rational map between projective varieties. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be surjective endomorphisms such that $g \circ \pi = \pi \circ f$, $\dim(Y) > 0$, and g is $d_1(f)$ -polarized. Then for $n \gg 1$,*

$$\sharp P_{f^n} \leq (d_1(f))^n (1 + o(1)).$$

Proof. By Lemma 8.2.4, replacing π with its graph and taking the normalizations, we may assume that π is a surjective morphism and both X and Y are normal.

Take the Stein factorization $X \rightarrow Y_1 \rightarrow Y$ such that $p_1 : X \rightarrow Y_1$ has connected fibres and $p_2 : Y_1 \rightarrow Y$ is finite. By [CMZ20, Lemma 5.2], f descends to a surjective endomorphism $h : Y_1 \rightarrow Y_1$ such that $p_1 \circ f = h \circ p_1$ and $p_2 \circ h = g \circ p_2$. Our h , like g , is also $d_1(f)$ -polarized. Hence we may assume further π has connected fibres.

For any $D \in P_{f^\infty} = \bigcup_{n=1}^\infty P_{f^n}$, say $D \in P_{f^s}$, $\pi(D)$ is a point; otherwise, $d_1(f^s) > d_1(f^s|_D) \geq d_1(g^s|_{\pi(D)}) = d_1(g^s) = d_1(f^s)$ since both g^s and $g^s|_{\pi(D)}$ are $(d_1(f)^s) = d_1(f^s)$ -polarized, a contradiction. For the upper bound, we may assume P_{f^∞} is an infinite set whose elements are contained in fibres of π . Then $\dim(Y) = 1$. Let $\mathcal{B}$ be the finite set of prime divisors in the (finitely many) reducible fibres of π . We have an injection

$$\varphi : P_{f^n} \setminus \mathcal{B} \rightarrow \text{Fix}(g^n),$$

sending D to $\pi(D)$. Then our argument follows from Proposition 8.5.2. $\square$

The following result is well known when the ground field k has $\text{char } k = 0$ and the proof is not that troublesome due to the Hodge decomposition. Our proof here works

when $\text{char } k$ is arbitrary, and is related to a generalized problem of Weil's Riemann Hypothesis.

Proposition 8.5.2. *Let $f : C \rightarrow C$ be a non-isomorphic endomorphism on a smooth projective curve C . Then $\# \text{Fix}(f) \leq \deg(f) + 2c(\deg(f))^{1/2} + 1$ where c is the genus of C .*

Proof. Let $d := d_1(f) = \deg(f) \geq 2$. Then f is d -polarized, so is its induced endomorphism $g : \text{Alb}(C) \rightarrow \text{Alb}(C)$ by Theorem 8.5.4. Note that f has at most finitely many fixed points. Applying Lefschetz fixed-point formula, we have

$$\# \text{Fix}(f) \leq \sum_{i=0}^2 (-1)^i \text{Tr}(f^*|_{H_{\text{ét}}^i(C, \mathbb{Q}_\ell)}).$$

Identifying $H_{\text{ét}}^1(C, \mathbb{Q}_\ell)$ with $H_{\text{ét}}^1(\text{Alb}(C), \mathbb{Q}_\ell)$, we have $f^*|_{H_{\text{ét}}^1(C, \mathbb{Q}_\ell)} = g^*|_{H_{\text{ét}}^1(\text{Alb}(C), \mathbb{Q}_\ell)}$. Since $\text{Alb}(C)$ is an abelian variety, we have $\chi_1(f)^2 = \chi_1(g)^2 = d_1(g) = d$ (cf. [Hu19, Formula (3.6)]). So the proposition is proved. $\square$

We are going to prove Theorem 8.5.4 (cf. Theorem 8.1.9). It has its own independent interest and various applications, and is also needed in proving Proposition 8.5.2 in positive characteristic. It generalizes [CMZ20, Theorem 1.2 and Theorem 5.1] by removing the separable assumption, [MZ19b, Proposition 3.3] by removing the surjective assumption, [NZ10, Lemma 2.3] without the polarized and characteristic 0 assumptions, and [San17, Remark 5.9] without the smooth and characteristic 0 assumptions.

We prepare some notation. Let f be a surjective endomorphism of a normal projective variety X over the field k which is allowed to be of **arbitrary characteristic**. Let

$$X^\vee := \text{Pic}^\circ(X)_{\text{red}},$$

$$f^\vee := f^*|_{\text{Pic}^\circ(X)_{\text{red}}} : X^\vee \rightarrow X^\vee$$

the dual of X and dual isogeny of f . Denote by

$$\text{Alb}(X) := \text{Pic}^\circ(\text{Pic}^\circ(X)_{\text{red}}) = X^{\vee\vee}$$

the Albanese variety of X . There are two canonical maps

$$\text{alb}_X : X \rightarrow \text{Alb}(X), \quad \text{alb}_X : X \dashrightarrow \mathfrak{Alb}(X)$$

called *Albanese morphism* and *Albanese map* satisfying the universal properties in the categories of morphisms and rational maps, respectively (cf. [FGI⁺05, Remark 9.5.25], [Lan83, Chapter II.3] and [CMZ20, Section 5]). In particular, f descends to surjective endomorphisms on $\text{Alb}(X)$ and $\mathfrak{Alb}(X)$.

We need the following in proving Theorem 8.5.4.

Proposition 8.5.3. *Let $f : X \rightarrow X$ be a (not necessarily separable) surjective endomorphism of a normal projective variety X over an algebraically closed field k of arbitrary characteristic. Let λ be an eigenvalue of $f^*|_{\text{Pic}^\circ(X)_{\text{red}}}$, equivalently of $(f^\vee)^*|_{H^1(X^\vee, \mathbb{Z}_\ell)}$. Then*

$$\iota_f \leq |\lambda|^2 \leq d_1(f)$$

where ι_f is the minimal modulus of eigenvalues of $f^*|_{N^1(X)}$.

Proof. Let $P \in \mathbb{Z}[t]$ be the minimal polynomial of $f^\vee : X^\vee \rightarrow X^\vee$, which is the minimal polynomial of $(f^\vee)^*|_{H^1(X^\vee, \mathbb{Z}_\ell)}$. Then $P(f^\vee) = 0$ (cf. [Hu19, Section 2]). Note also that $P(f^{\vee\vee}) = P(f^\vee)^\vee = 0$. By a dual argument, P is also the minimal polynomial of $f^{\vee\vee}$, i.e., the minimal polynomial of $(f^{\vee\vee})^*|_{H^1(X^{\vee\vee}, \mathbb{Z}_\ell)}$. Let a and b be the minimal and maximal modulus of roots of P , respectively. By [Hu19, Formula (3.6)], $b^2 = d_1(f^\vee)$. The same proof for [Hu19, Formula (3.6)] shows that $a^2 = \iota_{f^\vee}$. A similar argument for $f^{\vee\vee}$ says $b^2 = d_1(f^{\vee\vee})$ and $a^2 = \iota_{f^{\vee\vee}}$. Moreover, note that $f|_{\text{Alb}(X)} = f^{\vee\vee} + t$ for some $t \in X^{\vee\vee} = \text{Alb}(X)$. Then we have $\iota_{f|_{\text{Alb}(X)}} = \iota_{f^{\vee\vee}} = \iota_{f^\vee}$ and $d_1(f|_{\text{Alb}(X)}) = d_1(f^{\vee\vee}) = d_1(f^\vee)$. Let λ be an eigenvalue of $f^\vee$. Then λ is a root of P and hence $a \leq |\lambda| \leq b$.

We may assume $0 \in \text{alb}_X(X)$. Note that the smallest abelian subvariety of A containing $\text{alb}_X(X)$ is A itself by the universal property. Applying the similar technique used in the proof of Proposition 8.3.3, we have the following commutative diagram

$$\begin{array}{ccc} X^{\times(2n)} & \xrightarrow{f^{\times(2n)}} & X^{\times(2n)} \\ \downarrow s_n & & \downarrow s_n \\ \text{Alb}(X) & \xrightarrow{f|_{\text{Alb}(X)}} & \text{Alb}(X) \end{array} \quad (**)$$

where $n \gg 1$ and $s_n : X^{\times(2n)} \rightarrow \text{Alb}(X)$ is defined by

$$s_n(x_1, \dots, x_{2n}) = \sum_{i=1}^n \text{alb}_X(x_i) - \sum_{j=n+1}^{2n} \text{alb}_X(x_j).$$

Since $s_n^* : N^1(\text{Alb}(X)) \rightarrow N^1(X^{\times(2n)})$ is (equivariant) injective, we conclude:

$$\iota_f = \iota_{f^{\times(2n)}} \leq \iota_{f|_{\text{Alb}(X)}} \leq |\lambda|^2 \leq d_1(f|_{\text{Alb}(X)}) \leq d_1(f^{\times(2n)}) = d_1(f).$$

So the proposition is proved. $\square$

Theorem 8.5.4. *Let $f : X \rightarrow X$ be a (not necessarily separable) surjective endomorphism of a normal projective variety X over an algebraically closed field k of arbitrary characteristic. Assume $d_1(f) > 1$. Then we have:*

- (1) *Set $\mathbb{K} := \mathbb{Q}$ or $\mathbb{R}$. Suppose $f^*D \equiv d_1(f)D$ for some $\mathbb{K}$ -Cartier divisor D . Then there exists another $\mathbb{K}$ -Cartier divisor $D' \equiv D$ such that $f^*D' \sim_{\mathbb{K}} d_1(f)D'$. In particular, $f^*D_1 \sim_{\mathbb{R}} d_1(f)D_1$ for some nef $\mathbb{R}$ -Cartier divisor D_1 .*
- (2) *Suppose $f^*B \equiv qB$ for some big $\mathbb{R}$ -Cartier divisor and $q > 1$ a rational number. Then f is q -polarized.*
- (3) *Suppose f is q -polarized. Then any eigenvalue of $f^*|_{H^1(X, \mathbb{Q}_\ell)}$ has modulus $\sqrt{q}$.*
- (4) *Suppose f is q -polarized (resp. int-amplified). Then so are the induced endomorphisms $f|_{\text{Alb}(X)} : \text{Alb}(X) \rightarrow \text{Alb}(X)$ and $f|_{\mathfrak{Alb}(X)} : \mathfrak{Alb}(X) \rightarrow \mathfrak{Alb}(X)$ via the Albanese morphism and Albanese map respectively.*

Proof. We follow the notation in Proof of Proposition 8.5.3.

(1) Denote by $(f^\vee)_{\mathbb{K}} := f^\vee \otimes_{\mathbb{Z}} \mathbb{K}$ and $(X^\vee)_{\mathbb{K}} := X^\vee \otimes_{\mathbb{Z}} \mathbb{K}$. Note that $d_1(f) \in \mathbb{K}$. If $f^*D \equiv d_1(f)D$, then $f^*D - d_1(f)D =: \Delta \in (X^\vee)_{\mathbb{K}}$. Note that although $(X^\vee)_{\mathbb{K}}$ is in general an infinite dimensional $\mathbb{K}$ -vector space, the Jordan canonical form of $(f^\vee)_{\mathbb{K}}$ has all the Jordan blocks of size $\leq \deg(P)$ (recall that P is the minimal polynomial of $f^\vee$). By Proposition 8.5.3, any diagonal entry λ further satisfies $|\lambda| \leq \sqrt{d_1(f)} < d_1(f)$. In particular, $(f^\vee)_{\mathbb{K}} - d_1(f) \text{id}_{(X^\vee)_{\mathbb{K}}}$ is invertible. Thus, there is a $\mathbb{K}$ -Cartier divisor $L \in X^\vee$

such that $f^*L - d_1(f)L \sim_{\mathbb{K}} \Delta$. Then $f^*D' \sim_{\mathbb{K}} d_1(f)D'$ for $D' = D - L$. For the second half of (1), applying Perron-Frobenius theorem to the nef cone (cf. [Bir67]), there is a nef $\mathbb{R}$ -Cartier divisor D_1 such that $f^*D_1 \equiv d_1(f)D_1$; we then apply the first half of (1).

(2) Suppose $f^*B \equiv qB$ for some big $\mathbb{R}$ -Cartier divisor and $q > 1$ a rational number. Then algebraic integer q is an integer and $f^*H \equiv qH$ for some ample Cartier divisor H by the norm criterion; see [MZ18a, Propositions 1.1 and 2.9]. By (1), $f^*H' \sim_{\mathbb{Q}} qH'$ for some ample $\mathbb{Q}$ -Cartier divisor $H' \equiv H'$. Replacing H' by a multiple, (2) is proved.

(3) just follows from Proposition 8.5.3.

(4) By the commutative diagram (**) in Proof of Proposition 8.5.3, if f is q -polarized (resp. int-amplified), then so are $f^{\times(2n)}$ and $f|_{\text{Alb}(X)}$ (cf. [MZ18a, Lemma 3.10, Theorem 3.11] and [Men20, Lemma 3.4]). For the Albanese map $\text{alb}_X : X \dashrightarrow \text{Alb}(X)$, we may replace X by the normalization of the graph of alb_X and argue like above (cf. [CMZ20, Proof of Theorem 1.2]). So (4) is proved. $\square$

Remark 8.5.5. In Theorem 8.5.4 (1), we cannot replace $d_1(f)$ by ι_f because it is possible that some eigenvalue λ of $f^\vee$ is ι_f . By the intersection pairing and the projection formula, one can show in Theorem 8.5.4 (3) that any eigenvalue of $f^*|_{H^{2\dim(X)-1}(X, \mathbb{Q}_\ell)}$ has modulus $q^{\dim(X)-1/2}$.

8.6 Higher-dimensional Hodge Index Theorem

In this section, we recall the higher-dimensional Hodge index theorem and give an application (cf. Proposition 8.6.2) which will be crucially used in Section 8.7.

The following lemma is known in [DS04, Corollaire 3.5] for the case of compact Kähler manifolds. See [MZ20b, Lemma 4.3]=Lemma 4.4.3 for its full algebraic version. The proof below is taken from [MZ20b, Lemma 4.3]=Lemma 4.4.3 with slight simplification for the convenience of the readers.

Lemma 8.6.1. *Let X be a normal projective variety. Let $L \not\equiv 0$ and $D \not\equiv 0$ be two pseudo-effective $\mathbb{R}$ -Cartier divisors such that L^2 and D^2 are pseudo-effective (cf. Notation 8.2.1). Suppose $L \cdot D \equiv_w 0$. Then $L \equiv aD$ for some $a > 0$.*

Proof. We may assume $n := \dim(X) \geq 2$. Fix very ample Cartier divisors $A_1, \dots, A_{n-1}$ on X . Denote by

$$V := \mathbb{R}D + \mathbb{R}L, \quad \text{and}$$

$$W := \{x \in N^1(X) \mid x \cdot A_1 \cdots A_{n-1} = 0\}$$

subspaces of $N^1(X)$. By [Zha16, Lemma 3.2], $L \not\equiv_w 0$ and $D \not\equiv_w 0$. Then $a := L \cdot A_1 \cdots A_{n-1} / D \cdot A_1 \cdots A_{n-1} > 0$, and $V \cap W$ is of dimension ≤ 1 and spanned by

$$\tilde{D} := L - aD \in V \cap W.$$

For each $1 \leq i \leq n-1$, consider the following bilinear form on $N^1(X)$:

$$q_i(x, y) := x \cdot y \cdot A_1 \cdots A_{i-1} \cdot A_{i+1} \cdots A_{n-1}.$$

Choose a general normal projective surface S on X such that $S \equiv_w A_1 \cdots A_{i-1} \cdot A_{i+1} \cdots A_{n-1}$.

Then $q_i(x, y) = x|_S \cdot y|_S$. By the Hodge index theorem on S , q_i is semi-negative on W .

Note that q_i is semi-positive on V since L^2, D^2 are pseudo-effective and $q_i(L, D) = 0$.

Hence $q_i(\tilde{D}, \tilde{D}) = 0$. For any $w \in W$ and $\lambda \in \mathbb{R}$, we have $q_i(\lambda\tilde{D} - w, \lambda\tilde{D} - w) \leq 0$.

Then the inequality

$$q_i(w, w) - 2\lambda q_i(\tilde{D}, w) \leq 0$$

holds for any $\lambda \in \mathbb{R}$ and $w \in W$. This happens only when $q_i(\tilde{D}, w) = 0$ for any $w \in W$.

Note that W and A_i span $N^1(X)$ because W is a hyperplane of $N^1(X)$ and $A_i \notin W$.

Note also that $q_i(\tilde{D}, A_i) = 0$. Then

$$q_i(\tilde{D}, x) = \tilde{D} \cdot A_1 \cdots A_{i-1} \cdot x \cdot A_{i+1} \cdots A_{n-1} = 0$$

for any $x \in N^1(X)$. This implies that $V \cap W$ is independent of the choice of each A_i . So

$\tilde{D} \cdot x_1 \cdots x_{n-1} = 0$ for any divisors $x_1, \dots, x_{n-1} \in N^1(X)$, which means $\tilde{D} \equiv_w 0$. Note

that $\tilde{D}$ is $\mathbb{R}$ -Cartier. Therefore, $\tilde{D} \equiv 0$ by [Zha16, Lemma 3.2]. $\square$

Proposition 8.6.2. *Let X be a normal projective variety of dimension n , L a nef $\mathbb{R}$ -Cartier divisor, and D_1, D_2 two non-zero effective $\mathbb{R}$ -Cartier divisors on X . Suppose D_1 and D_2 have no common components and $D_1 \equiv D_2$. Suppose further that $L \cdot D_1 \cdot H_1 \cdots H_{n-2} = 0$ for some nef and big divisors $H_1, \dots, H_{n-2}$. Then $L \equiv aD_1 \equiv aD_2$ for some $a > 0$ and $L^2 \equiv_w D_1 \cdot D_2 \equiv_w 0$. In particular, $\text{Supp } D_1$ and $\text{Supp } D_2$ are disjoint.*

Proof. For each H_i , we write $H_i \sim A_i + E_i$ for some ample divisor A_i and effective divisor E_i . By the assumption, the intersection of (the numerically movable) D_1 with any prime divisor is pseudo-effective, hence so is $D_1 \cdot E_i$ for each i . Then $L \cdot D_1 \cdot H_1 \cdots H_{n-2} = 0$ implies that

$$L \cdot D_1 \cdot A_1 \cdot H_2 \cdots H_{n-2} = 0.$$

Repeating the same process, we have $L \cdot D_1 \cdot A_1 \cdots A_{n-2} = 0$ and thus $L \cdot D_1 \equiv_w 0$. By Lemma 8.6.1 and noting $D_1^2 \equiv_w D_1 D_2$, we get $L \equiv a D_1 (\equiv a D_2)$ with $a > 0$. So we have

$$L^2 \equiv_w a^2 D_1 \cdot D_2 \equiv_w a L \cdot D_1 \equiv_w 0.$$

If $(\text{Supp } D_1) \cap (\text{Supp } D_2) \neq \emptyset$, then $D_1 \cdot D_2$ is a non-zero effective cycle and hence we get a contradiction with $D_1 \cdot D_2 \not\equiv_w 0$. $\square$

8.7 Proofs of Theorems 8.1.6 and 8.1.7

In this section, we will prove Theorems 8.1.7 and 8.7.1 below which includes Theorem 8.1.6 as a simpler version and bounds the number $\sharp P_f$ in terms of the Néron-Severi class group rank $\rho^{\text{ns}}(X)$, the Picard number $\rho(\tilde{X})$ of any projective resolution $\tilde{X}$ of X , and the first dynamical degree $d_1(f)$.

Theorem 8.7.1. *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal projective variety. Let $\tilde{X} \rightarrow X$ be any projective resolution. Then*

$$\sharp P_f \leq \max\{(1 + \sqrt{d_1(f)})^2 + a, (\rho^{\text{ns}}(X) + 1) \cdot (\lfloor \frac{\rho(\tilde{X})}{2} \rfloor + 1)\},$$

where a is a positive constant depending only on X . In particular, for $n \gg 1$,

$$\sharp P_{f^n} \leq (d_1(f))^n \cdot (1 + o(1)).$$

We now prepare to prove Theorem 8.7.1.

Notation 8.7.2. Let $f : X \rightarrow X$ be a surjective endomorphism on a normal projective variety X of dimension n . We fix a projective resolution

$$\sigma : \tilde{X} \rightarrow X$$

with $E_1, \dots, E_k$ its exceptional prime divisors. For $D \in P_f$, denote by $\widetilde{D} := \sigma_*^{-1}(D)$, the proper transform of D . Denote by

$$\widetilde{P}_f := \{\widetilde{D} = \sigma_*^{-1}(D) \mid D \in P_f\}.$$

By Perron-Frobenius theorem (cf. [Bir67]), fix a nef $\mathbb{R}$ -Cartier divisor $L_f \not\equiv 0$ on X with

$$f^*L_f \equiv d_1(f)L_f.$$

Denote by

$$\widetilde{L}_f := \sigma^*L_f.$$

Lemma 8.7.3. $L_f \cdot D \equiv_w 0$ for any $D \in P_f$.

Proof. Note that $(f|_D)^*(L_f|_D) \equiv d_1(f)(L_f|_D)$ and $d_1(f|_D) < d_1(f)$. Then $L_f|_D \equiv 0$ and hence $L_f \cdot D \equiv_w 0$. $\square$

Lemma 8.7.4. There exist at least $2\lfloor \#P_f/(\rho^{\text{ns}}(X) + 1) \rfloor$ disjoint prime divisors $\widetilde{D}_u$ in $\widetilde{P}_f$ with $\widetilde{L}_f \cdot \widetilde{D}_u \equiv_w 0$.

Proof. Let $m := \lfloor \#P_f/(\rho^{\text{ns}}(X) + 1) \rfloor$. Then we have m pairwise disjoint subsets $S_1, \dots, S_m$ of $\widetilde{P}_f$ with $\#S_t = \rho^{\text{ns}}(X) + 1$ for each $1 \leq t \leq m$.

Note that $\rho(\widetilde{X}) = \rho^{\text{ns}}(X) + k$ where k is the number of σ -exceptional prime divisors (cf. [Kol18, Lemma 18]). By the assumption, we have the equality

$$\#(S_t \cup \{E_1, \dots, E_k\}) = \rho(\widetilde{X}) + 1.$$

For each t , we then have a numerical equivalence of two non-zero effective divisors which are combinations of prime divisors in S_t and σ -exceptional prime divisors such that they have no common component. We write them in the following form

$$\sum_{i \in I_t} a_i \widetilde{D}_i + \widetilde{E}_t \equiv \sum_{j \in J_t} b_j \widetilde{D}_j + \widetilde{F}_t$$

where I_t and J_t are two (possibly empty) disjoint subsets of S_t , $a_i > 0$, $b_j > 0$, and $\widetilde{E}_t, \widetilde{F}_t$ are effective σ -exceptional divisors. Here without ambiguity, we identify $\widetilde{D}_i$ with the index i in I_t .

Let H be an ample divisor on X . By Lemma 8.7.3 and the projection formula,

$$\widetilde{L}_f \cdot \left(\sum_{i \in I_t} a_i \widetilde{D}_i + \widetilde{E}_t \right) \cdot (\sigma^* H)^{n-2} = L_f \cdot \left(\sum_{i \in I_t} a_i D_i \right) \cdot H^{n-2} = 0$$

where $D_i = \sigma_*(\widetilde{D}_i) \in P_f$. Applying Proposition 8.6.2, for each t , we may assume (after some scalar multiplication)

$$\widetilde{L}_f \equiv \sum_{i \in I_t} a_i \widetilde{D}_i + \widetilde{E}_t \equiv \sum_{j \in J_t} b_j \widetilde{D}_j + \widetilde{F}_t$$

and we also have

$$\widetilde{L}_f^2 \equiv_w \widetilde{L}_f \cdot \left(\sum_{i \in I_t} a_i \widetilde{D}_i + \widetilde{E}_t \right) \equiv_w \widetilde{L}_f \cdot \left(\sum_{j \in J_t} b_j \widetilde{D}_j + \widetilde{F}_t \right) \equiv_w 0.$$

This further implies that

$$\widetilde{L}_f \cdot \widetilde{D}_i \equiv_w \widetilde{L}_f \cdot \widetilde{D}_j \equiv_w \widetilde{L}_f \cdot \widetilde{E} \equiv_w 0 \quad (8.1)$$

for any $i \in I_t, j \in J_t$ and any prime divisor $\widetilde{E}$ in $\widetilde{E}_t$ and $\widetilde{F}_t$.

We claim that I_t and J_t are non-empty. Suppose on the contrary, for example, that I_t is empty. Then $\widetilde{L}_f \equiv \widetilde{E}_t \neq 0$. However, this contradicts with the negativity lemma (cf. [KM98, Lemma 3.39]).

Consider the pair $(I_t, J_{t'})$ for any t, t' . Note that

$$\sum_{i \in I_t} a_i \widetilde{D}_i + \widetilde{E}_t \equiv \sum_{j \in J_{t'}} b_j \widetilde{D}_j + \widetilde{F}_{t'} (\equiv \widetilde{L}_f) \quad (8.2)$$

We claim that $\widetilde{E}_t$ and $\widetilde{F}_{t'}$ have no common components. Readjust it a bit in the following way, where Δ is a σ -exceptional effective $\mathbb{R}$ -divisor,

$$\sum_{i \in I_t} a_i \widetilde{D}_i + (\widetilde{E}_t - \Delta) \equiv \sum_{j \in J_{t'}} b_j \widetilde{D}_j + (\widetilde{F}_{t'} - \Delta),$$

so that $\widetilde{E}_t - \Delta$ and $\widetilde{F}_{t'} - \Delta$ are effective but with no common components. Note that

$$\widetilde{L}_f \cdot \left(\sum_{i \in I_t} a_i \widetilde{D}_i + (\widetilde{E}_t - \Delta) \right) \equiv_w 0.$$

By Proposition 8.6.2, for some $e > 0$, we have

$$e\widetilde{L}_f \equiv \sum_{i \in I_t} a_i \widetilde{D}_i + (\widetilde{E}_t - \Delta).$$

Then $(1-e)\widetilde{L}_f \equiv \Delta$, and hence $e \leq 1$ since both $\widetilde{L}_f$ and Δ are pseudo-effective and X is projective. By the negativity lemma again (cf. [KM98, Lemma 3.39]), $e = 1$ and $\Delta \equiv 0$, so $\Delta = 0$ and the claim is proved. By Proposition 8.6.2 and the display (8.2) above and since $\widetilde{L}_f \cdot \widetilde{L}_f \equiv_w 0$, we have $\widetilde{D}_i \cap \widetilde{D}_j = \emptyset$ for any $i \in I_t$ and $j \in J_{t'}$.

The same argument works for the pairs $(I_t, I_{t'})$ and $(J_t, J_{t'})$ with $t \neq t'$. In particular, we pick one prime divisor in I_t and one prime divisor in J_t for each t , and all these $2m$ prime divisors are pairwise disjoint. This completes the proof. $\square$

With $\sharp P_f$ large enough, we can construct an f -equivariant fibration over a curve.

Proposition 8.7.5. *Suppose $\sharp P_f \geq (\lfloor \frac{\rho(\widetilde{X})}{2} \rfloor + 1) \cdot (\rho^{\text{ns}}(X) + 1)$. Then there exists a fibration $\widetilde{\pi} : \widetilde{X} \rightarrow C$ onto a smooth curve C such that the following hold.*

- (1) $\widetilde{L}_f \equiv \widetilde{\pi}^* H_C$ for some ample $\mathbb{R}$ -Cartier divisor H_C on C .
- (2) The fibration $\widetilde{\pi}$ factors through a fibration $\pi : X \rightarrow C$ such that $L_f \equiv \pi^* H_C$.
- (3) There exists a $d_1(f)$ -polarized endomorphism $g : C \rightarrow C$ such that $\pi \circ f = g \circ \pi$.

Proof. (1) By Lemma 8.7.4, there are $\rho(\widetilde{X}) + 1$ pairwise disjoint prime divisors $\widetilde{D}_u$ in $\widetilde{P}_f$ with $\widetilde{L}_f \cdot \widetilde{D}_u \equiv_w 0$. By [BPS16, Theorem 1.1], there is a fibration $\widetilde{\pi} : \widetilde{X} \rightarrow C$ onto a smooth projective curve such that $\widetilde{\pi}(\widetilde{D}_u)$ is a point for each u and $\widetilde{D}_u = \widetilde{\pi}^{-1}(\widetilde{\pi}(\widetilde{D}_u))$ for $1 \leq u \leq 3$. Now $\widetilde{D}_1 \equiv a\widetilde{D}_2$ for some $a > 0$. By Proposition 8.6.2, $\widetilde{L}_f \equiv t\widetilde{D}_1$ for some $t > 0$. Note that $\widetilde{\pi}^*(\widetilde{\pi}(\widetilde{D}_1)) = b\widetilde{D}_1$ for some $b > 0$. Letting $H_C := t\widetilde{\pi}(\widetilde{D}_1)/b$, our (1) follows.

(2) follows from (1) and Lemma 8.2.2.

(3) For every curve B lying in a fibre of π , by (2) and the projection formula, we have

$$\pi_*(f_*B) \cdot H_C = f_*B \cdot L_f = B \cdot f^*L_f = d_1(f)B \cdot L_f = d_1(f)\pi_*B \cdot H_C = 0$$

So every fibre of π will also be contracted by $\pi \circ f$. By the rigidity lemma (cf. [Deb01, Lemma 1.15]), $\pi \circ f$ factors through π and thus f descends to a surjective endomorphism $g = f|_C$. Moreover, if we assume $g^*H_C \equiv qH_C$, then

$$f^*L_f \equiv f^*\pi^*H_C = \pi^*g^*H_C \equiv q\pi^*H_C \equiv qL_f,$$

which implies $q = d_1(f)$. Hence g is $d_1(f)$ -polarized. This proves (3). $\square$

With all the preparations done, we are now ready to prove Theorems 8.7.1, 8.1.6 and 8.1.7.

Proof of Theorem 8.7.1. If $d_1(f) = 1$, then $\sharp P_f = 0$ and the theorem is trivial. From now on we assume that $d_1(f) > 1$.

Suppose $\sharp P_f \geq (\rho^{\text{ns}}(X) + 1) \cdot (\lfloor \frac{\rho(\tilde{X})}{2} \rfloor + 1)$. By Proposition 8.7.5, there is an f -equivariant fibration $\pi : X \rightarrow C$ onto a smooth curve C such that $g := f|_C$ is $d_1(f)$ -polarized. Let $\mathcal{B}$ be the finite set of prime divisors in the reducible fibres of π . By the same argument as in Proof of Proposition 8.5.1, we have an injection $\varphi : P_f \setminus \mathcal{B} \rightarrow \text{Fix}(g)$, sending D to $\pi(D)$. Let $a := \sharp \mathcal{B}$. By Lemma 8.5.2, we can conclude the theorem:

$$\sharp P_f \leq \sharp \text{Fix}(g) + a \leq (1 + \sqrt{d_1(f)})^2 + a.$$

$\square$

Proof of Theorem 8.1.6. Apply Theorem 8.7.1 to the normalization of X (cf. Lemma 8.2.4).

$\square$

Proof of Theorem 8.1.7. By taking normalization of X , we may assume that X is normal (cf. Lemmas 8.2.4 and 8.2.5).

“(2) $\Rightarrow$ (1).” By [Fak03, Theorem 5.1], $\text{Per}(g)$ is Zariski dense in C . Since π is a fibration, it has irreducible and reduced general fibres. Therefore $X_c := \pi^{-1}(c) = \pi^*c$ is an f -periodic prime divisor for general $c \in \text{Per}(g)$. Suppose $f^s(X_c) = X_c$. Then $d_1(f^s|_{X_c}) = d_1(f^s|_\pi) < d_1(f^s)$ by the assumption. So $X_c \in P_{f^\infty}$ and $\sharp P_{f^\infty}$ is infinite.

“(1) $\Rightarrow$ (2).” After a sufficient iteration of f , we may assume that $\sharp P_f \gg 1$. Applying Proposition 8.7.5, there is an f -equivariant fibration $\pi : X \rightarrow C$ onto a smooth curve

C such that $g := f|_C$ is $d_1(f)$ -polarized. We may also assume that $D = \pi^{-1}(\pi(D))$ for some $D \in P_f$. Then $d_1(f|_\pi) = d_1(f|_D) < d_1(f)$, which completes the first part of our theorem.

Now we prove the finiteness of $P_{f^\infty}^{-1}$. Suppose on the contrary that $P_{f^\infty}^{-1}$ is infinite. By the above equivalent condition, there is an f -equivariant (after iteration) fibration $\pi : X \rightarrow C$ onto a smooth curve C such that $g := f|_C$ is $d_1(f)$ -polarized. Let $\mathcal{B}$ be the finite set of prime divisors in the reducible fibres of π . By the same argument as in Proof of Proposition 8.5.1 and [CMZ20, Lemma 7.5], we have an injection $\varphi : P_{f^\infty}^{-1} \setminus \mathcal{B} \rightarrow P_{g^\infty}^{-1}$, sending D to $\pi(D)$. Note that there are at most finitely many g^{-1} -periodic points (cf. [MZ20a, Corollary 3.8]). So the theorem is proved. $\square$

Corollary 8.7.6. *Let $f : X \rightarrow X$ be a surjective endomorphism of a normal projective variety X with $\sharp P_{f^\infty} = \infty$. Then D is either $\mathbb{Q}$ -Cartier or f^{-1} -periodic for any $D \in P_{f^\infty}$.*

Proof. We apply Theorem 8.1.7 and use the notation there. If $D = \pi^{-1}(\pi(D))$, then D is $\mathbb{Q}$ -Cartier. Otherwise, $\pi^{-1}(\pi(D))$ is reducible. Denote by Σ the finite subset of C over which π has reducible fibres. Then $g^{-1}(\Sigma) = \Sigma$ by [CMZ20, Lemma 7.4]. Therefore D is an irreducible component of $\pi^{-1}(\Sigma)$ and hence f^{-1} -periodic. $\square$

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